https://stmiraclesacademy.com/

Strictly Confidential — (For Internal and Restricted Use Only)

Senior School Certificate Examination

March 2019

Marking Scheme — Mathematics (041) 65(B)

General Instructions:

1. You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully. **Evaluation is a 10-12 days mission for all of us. Hence, it is necessary that you put in your best efforts in this process.**
2. Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one's own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. **However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and marks be awarded to them.**
3. The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
4. If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled.
5. If a question does not have any parts, marks must be awarded in the left hand margin and encircled.
6. If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out.
7. No marks to be deducted for the cumulative effect of an error. It should be penalized only once.
8. A full scale of marks 0 to 100 has to be used. Please do not hesitate to award full marks if the answer deserves it.
9. Every examiner has to necessarily do evaluation work for full working hours i.e. 8 hours every day and evaluate 25 answer books per day.
10. Ensure that you do not make the following common types of errors committed by the Examiner in the past:-
    - Leaving answer or part thereof unassessed in an answer book.
    - Giving more marks for an answer than assigned to it.
    - Wrong transfer of marks from the inside pages of the answer book to the title page.
    - Wrong question wise totaling on the title page.
    - Wrong totaling of marks of the two columns on the title page.
    - Wrong grand total.
    - Marks in words and figures not tallying.
    - Wrong transfer of marks from the answer book to online award list.
    - Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.)
    - Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
11. While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as (X) and awarded zero (0) Marks.
12. Any unassessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
13. The Examiners should acquaint themselves with the guidelines given in the Guidelines for spot Evaluation before starting the actual evaluation.
14. Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
15. The Board permits candidates to obtain photocopy of the Answer Book on request in an RTI application and also separately as a part of the re-evaluation process on payment of the processing charges.

QUESTION PAPER CODE 65(B)
EXPECTED ANSWER/VALUE POINTS

SECTION A

1. $AB = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ 1

2. $x = -3$ or $\frac{1}{2}$ (Any one value) 1

OR

Let $u = e^x$, $v = \sqrt{x}$

$\frac{du}{dx} = e^x, \frac{dv}{dx} = \frac{1}{2\sqrt{x}}$ $\frac{1}{2}$

$\frac{du}{dv} = 2\sqrt{x}\, e^x$ $\frac{1}{2}$

3. $dy = \left(\frac{e^x - e^{-x}}{e^x + e^{-x}}\right) dx$ $\frac{1}{2}$

Integrating, $y = \log (e^x + e^{-x}) + c$ $\frac{1}{2}$

4. d·r′s are $-2, 1, 2$ $\frac{1}{2}$

d·c′s are $\frac{-2}{3}, \frac{1}{3}, \frac{2}{3}$ $\frac{1}{2}$

SECTION B

5. R is not reflexive as $\left(\frac{1}{2}, \frac{1}{2}\right) \notin R$ $\frac{1}{2}+\frac{1}{2}$

R is not transitive as

$(9, 4) \in R$, $(4, 2) \in R$ but $(9, 2) \notin R$ $\frac{1}{2}+\frac{1}{2}$

OR

For $a, b \in Q$, $\frac{a+b}{2} \in Q \Rightarrow a * b \in Q \Rightarrow *$ is a binary operation $\frac{1}{2}+\frac{1}{2}$

$a * b = \frac{a+b}{2} = \frac{b+a}{2} = b * a \Rightarrow *$ is commutative $\forall\ a, b \in Q$ $\frac{1}{2}+\frac{1}{2}$

6. $\begin{bmatrix} 2 & -3 \\ -5 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A$ $\frac{1}{2}$

$R_2 \to R_2 + \frac{5}{2} R_1$

$\begin{bmatrix} 2 & -3 \\ 0 & \frac{-1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{5}{2} & 1 \end{bmatrix} A$

$R_2 \to -2R_2$

$\begin{bmatrix} 2 & -3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -5 & -2 \end{bmatrix} A$ 1

$R_1 \to R_1 + 3R_2$

$\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -14 & -6 \\ -5 & -2 \end{bmatrix} A$

$R_2 \to \frac{R_2}{2}$

$\therefore \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -7 & -3 \\ -5 & -2 \end{bmatrix} A$

$\therefore \quad A^{-1} = \begin{bmatrix} -7 & -3 \\ -5 & -2 \end{bmatrix}$ $\frac{1}{2}$

7. $\int \frac{1 + \tan x}{1 - \tan x} = \int \frac{\cos x + \sin x}{\cos x - \sin x} dx$ 1

$= -\log |\cos x - \sin x| + c$ 1

OR

$\int_0^{\pi/4} \sqrt{1 + \cos 2x}\, dx = \sqrt{2} \int_0^{\pi/4} \cos x\, dx = \sqrt{2} [\sin x]_0^{\pi/4}$ $\frac{1}{2} + \frac{1}{2}$

$= \sqrt{2} \left(\sin \frac{\pi}{4} - \sin 0 \right) = 1$ 1

8. $\int \left(\frac{1}{x} - \frac{1}{2x^2} \right) e^{2x}\, dx = \frac{1}{2} \int \left(\frac{2}{t} - \frac{2}{t^2} \right) e^t\, dt$ (Put $2x = t$) $\frac{1}{2} + \frac{1}{2}$

$= \frac{1}{2} e^t \left(\frac{2}{t} \right) + c$ $\frac{1}{2}$

$= \frac{e^{2x}}{2x} + c$ $\frac{1}{2}$

9. $\frac{dy}{dx} = e^{(2x+3y)} = e^{2x}\ e^{3y}$ $\frac{1}{2}$

$\Rightarrow \quad e^{-3y}\ dy = e^{2x}\ dx$ $\frac{1}{2}$

Integrating,

$\frac{e^{-3y}}{-3} = \frac{e^{2x}}{2} + c$ $\frac{1}{2}+\frac{1}{2}$

10. $\vec{b} + \vec{c} = 3\hat{i} + \hat{j} + 2\hat{k}$ $\frac{1}{2}$

Projection of $(\vec{b} + \vec{c})$ on $\vec{a} = \frac{(\vec{b} + \vec{c})\cdot\vec{a}}{|\vec{a}|} = \frac{6 - 2 + 2}{3}$ $\frac{1}{2}+\frac{1}{2}$

$= 2$ $\frac{1}{2}$

11. A = Experiment succeds, B = Experiment Fail's

$P(A) = \frac{3}{4},\ P(B) = \frac{1}{4},\ n = 5,\ X =$ number of successes $\frac{1}{2}$

$P(X \geq 4) = P(X = 4) + P(X = 5)$ $\frac{1}{2}$

$= {}^5C_4\left(\frac{3}{4}\right)^4\left(\frac{1}{4}\right) + {}^5C_5\left(\frac{3}{4}\right)^5$ $\frac{1}{2}$

$= \left(\frac{3}{4}\right)^4\left(\frac{5}{4} + \frac{3}{4}\right) = \frac{162}{256}$ or $\frac{81}{128}$ $\frac{1}{2}$

12. $P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(A)\ P(B)$ $\frac{1}{2}+\frac{1}{2}$

$= P(A) + P(B)(1 - P(A))$

$= P(A) + P(B)\ P(A')$

$= 1 - P(A') + P(B)\ P(A')$

$= 1 - P(A')\ (1 - P(B))$ $\frac{1}{2}$

$= 1 - P(A')\ P(B')$ $\frac{1}{2}$

OR

Let A: the sum of numbers on the dice is 10

B: numbers on two dice are different

$A \cap B = \{(4, 6), (6, 4)\}$ $\frac{1}{2}$

$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{2/36}{30/36} = \frac{1}{15}$ $\frac{1}{2}+1$

SECTION C

13. Let $x_1, x_2 \in [-1, 1]$ such that $\frac{1}{2}$

$f(x_1) = f(x_2) \Rightarrow \frac{x_1}{x_1 + 2} = \frac{x_2}{x_2 + 2}$ $\frac{1}{2}$

$x_1x_2 + 2x_1 = x_1x_2 + 2x_2$

$\Rightarrow \quad x_1 = x_2 \Rightarrow f$ is one-one 1

Let $y \in$ Range of f

$\frac{x}{x + 2} = y \Rightarrow x = y(x + 2)$ $\frac{1}{2}$

$x = \frac{2y}{1 - y}, \ x \in [-1, 1]$ 1

$f^{-1}(y) = \frac{2y}{1 - y}$ or $f^{-1}(x) = \frac{2x}{1 - x}$ $\frac{1}{2}$

14. $\tan^{-1}\left(\frac{\frac{(x - 1)(x + 2) + (x + 1)(x - 2)}{x^2 - 4}}{\frac{-3}{x^2 - 4}}\right) = \frac{\pi}{4}$ 1

$\tan^{-1}\left(\frac{2x^2 - 4}{-3}\right) = \frac{\pi}{4}$ 1

$2x^2 - 4 = -3$ 1

$x = \pm\frac{1}{\sqrt{2}}$ 1

OR

Put $x = \cos 2\theta$ $\frac{1}{2}$

$$\tan^{-1}\left(\frac{\sqrt{2}\cos\theta - \sqrt{2}\sin\theta}{\sqrt{2}\cos\theta + \sqrt{2}\sin\theta}\right) = \tan^{-1}\left(\frac{\cos\theta - \sin\theta}{\cos\theta + \sin\theta}\right) \quad 1+\frac{1}{2}$$

$$= \tan^{-1}\left(\tan\left(\frac{\pi}{4} - \theta\right)\right) \quad 1$$

$$= \frac{\pi}{4} - \theta = \frac{\pi}{4} - \frac{\cos^{-1}x}{2} \quad \frac{1}{2}+\frac{1}{2}$$

15. $\begin{vmatrix} x & x^2 & 1+ax^3 \\ y & y^2 & 1+ay^3 \\ z & z^2 & 1+az^3 \end{vmatrix}$

$$= \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + \begin{vmatrix} x & x^2 & ax^3 \\ y & y^2 & ay^3 \\ z & z^2 & az^2 \end{vmatrix} \quad \frac{1}{2}$$

$$= (-1)^2\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} + axyz\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} \quad 1$$

$$= (1+axyz)\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} \quad \frac{1}{2}$$

$R_2 \to R_2 - R_1$, $R_3 \to R_3 - R_1$

$$= (1+axyz)\begin{vmatrix} 1 & x & x^2 \\ 0 & y-x & y^2-x^2 \\ 0 & z-x & z^2-x^2 \end{vmatrix} \quad 1$$

$$= (1+axyz)(y-x)(z-x)\begin{vmatrix} 1 & y+x \\ 1 & z+x \end{vmatrix} \quad \frac{1}{2}$$

$$= (1+axyz)(x-y)(y-z)(z-x) \quad \frac{1}{2}$$

16. Put $x = \sin\theta$, $\frac{5}{13} = \cos\phi$, $\frac{12}{13} = \sin\phi$ $1\frac{1}{2}$

$y = \sin^{-1}(\sin\theta\cos\phi + \cos\theta\sin\phi) = \sin^{-1}(\sin(\theta + \phi))$ 1

$= \theta + \phi$ $\frac{1}{2}$

$= \sin^{-1}x + \cos^{-1}\left(\frac{5}{13}\right)$ $\frac{1}{2}$

$\Rightarrow \quad \frac{dy}{dx} = \frac{1}{\sqrt{1 - x^2}}$ $\frac{1}{2}$

OR

$y\sqrt{x^2 + 1} = \log\left(\sqrt{x^2 + 1} - x\right)$

Diff. w.r.t. x

$\frac{xy}{\sqrt{x^2 + 1}} + \sqrt{x^2 + 1}\frac{dy}{dx} = \frac{1}{\sqrt{x^2 + 1} - x}\left(\frac{x}{\sqrt{x^2 + 1}} - 1\right)$ 2

$xy + (x^2 + 1)\frac{dy}{dx} = -1$ $1\frac{1}{2}$

$(x^2 + 1)\frac{dy}{dx} + xy + 1 = 0$ $\frac{1}{2}$

17. $y = e^x(\sin x + \cos x)$

$\frac{dy}{dx} = e^x(\cos x - \sin x) + e^x(\sin x + \cos x)$ 1

$= e^x(\cos x - \sin x) + y$ $\frac{1}{2}$

$\frac{d^2y}{dx^2} = e^x(-\sin x - \cos x) + (\cos x - \sin x)e^x + \frac{dy}{dx}$ $1\frac{1}{2}$

$= -y + \frac{dy}{dx} - y + \frac{dy}{dx}$ $\frac{1}{2}$

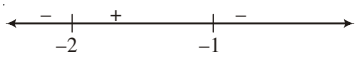
$\frac{d^2y}{dx^2} - \frac{2dy}{dx} + 2y = 0$ $\frac{1}{2}$

18. $f'(x) = -6x^2 - 18x - 12$ 1

$= -6(x^2 + 3x + 2) = -6(x + 1)(x + 2)$ $\frac{1}{2}$

Critical points are $x = -2, -1$ $\frac{1}{2}$

(sin x of $f'(x)$)



$\therefore$ Function is strictly increasing in $(-2, -1)$ $\frac{1}{2}$

or $[-2, -1]$

Function is strictly decreasing in $(-\infty, -2), (-1, \infty)$ $\frac{1}{2}$

or $[-\infty, -2], [-1, \infty]$

Note: Stritctly increasing and strictly decreasing mentioned in a closed interval should be given full marks.

19. $2x + 1 = A\frac{d}{dx}(4 - 3x - x^2) + B$

$= A(-2x - 3) + B$

$\Rightarrow$ $A = -1, B = -2$ 1

$$\int \frac{(2x+1)\,dx}{\sqrt{4-3x-x^2}} = -\int \frac{-2x-3\,dx}{\sqrt{4-3\,x-x^2}} - 2\int \frac{dx}{\sqrt{4-3x-x^2}}$$ 1

$$= -2\sqrt{4-3x-x^2} - 2\int \frac{dx}{\sqrt{\left(\frac{5}{2}\right)^2 - \left(x+\frac{3}{2}\right)^2}}$$ 1

$$= -2\sqrt{4-3x-x^2} - 2\sin^{-1}\left(\frac{2x+3}{5}\right) + c$$ 1

20. $$\frac{17}{(2x+1)(x^2+4)} = \frac{A}{2x+1} + \frac{Bx+C}{x^2+4}$$ 1

Getting $A = 4, B = -2, C = 1$ 1

$\therefore$ $$\int \frac{17\,dx}{(2x+1)(x^2+4)} = 4\int \frac{dx}{2x+1} + \int \frac{(1-2x)\,dx}{x^2+4}$$ $\frac{1}{2}$

$$= 2\log|2x+1| + \frac{1}{2}\tan^{-1}\frac{x}{2} + -\log|x^2+4| + C$$ $1\frac{1}{2}$

OR

Let $I = \int_0^{\pi} \frac{x \sin dx}{1 + \cos^2 x}$

$I = \int_0^{\pi} \frac{(\pi - x) \sin (\pi - x)\, dx}{1 + \cos^2(\pi - x)} = \int_0^{\pi} \frac{(\pi - x)\sin x}{1 + \cos^2 x} dx$ 1

$2I = \pi \int_0^{\pi} \frac{\sin x\, dx}{1 + \cos^2 x}$ 1

Put $\cos x = t$
$\therefore \quad -\sin x\, dx = dt$ $\frac{1}{2}$

$2I = -\pi \int_1^{-1} \frac{dt}{1 + t^2}$

$= \pi \int_{-1}^{1} \frac{dt}{1 + t^2}$

$= \pi \left[\tan^{-1} t\right]_{-1}^{1}$ $\frac{1}{2}$

$= \pi(\tan^{-1}(1) - \tan^{-1}(-1))$ $\frac{1}{2}$

$\Rightarrow \quad I = \frac{\pi^4}{4}$ $\frac{1}{2}$

21. Writing $\frac{dy}{dx} = \frac{y}{x} - \sin^2\left(\frac{y}{x}\right)$ 1

Put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ 1

Getting $\frac{dx}{x} = \frac{-dv}{\sin^2 v}$

Integrating,

$\log |x| = \cot v + c \Rightarrow \log |x| = \cot\left(\frac{y}{x}\right) + c$ 1

$y = \frac{\pi}{4}$, $x = 1$ gives $c = -1$. $\therefore$ Required solution is $\log |x| = \cot\left(\frac{y}{x}\right) - 1$ 1

22. $\overrightarrow{AB} = -\hat{i} - 2\hat{j} - 6\hat{k}$, $\overrightarrow{BC} = 2\hat{i} - \hat{j} + \hat{k}$, $\overrightarrow{AC} = \hat{i} - 3\hat{j} - 5\hat{k}$ $1\frac{1}{2}$

$\overrightarrow{BC} \cdot \overrightarrow{AC} = 0 \Rightarrow \Delta ABC$ is right angled triangle at C. $\frac{1}{2}$

$$\cos B = \frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{|\overrightarrow{BA}||\overrightarrow{BC}|} = \frac{6}{\sqrt{41} \times \sqrt{6}} \Rightarrow B = \cos^{-1}\left(\sqrt{\frac{6}{41}}\right) \quad 1$$

$$\cos A = \frac{\overrightarrow{AB} \cdot \overrightarrow{AC}}{|\overrightarrow{AB}||\overrightarrow{AC}|} = \frac{35}{\sqrt{41} \times \sqrt{35}} \Rightarrow A = \cos^{-1}\left(\sqrt{\frac{35}{41}}\right) \quad 1$$

23. $\frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7} = \lambda$

$\therefore$ The general point on first line is $x = 3\lambda - 1$, $y = 5\lambda - 3$, $z = 7\lambda - 5$. $\frac{1}{2}$

$$\frac{x-2}{1} = \frac{y-4}{3} = \frac{z-6}{5} = \mu$$

The general point on second line is $x = \mu + 2$, $y = 3\mu + 4$, $z = 5\mu + 6$ $\frac{1}{2}$

$\therefore$ $3\lambda - 1 = \mu + 2$ and $5\lambda - 3 = 3\mu + 4$ gives $\lambda = \frac{1}{2}$, $\mu = \frac{-3}{2}$ 1

$$\left.\begin{array}{l} z-\text{coordinate of first line} = \frac{-3}{2} \\ z-\text{coordinate of second line} = \frac{-3}{2} \end{array}\right\} \Rightarrow \text{lines are intersecting} \quad 1$$

$\therefore$ Point of intersection is $\left(\frac{1}{2}, \frac{-1}{2}, \frac{-3}{2}\right)$ 1

SECTION D

24. $A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 4 & -5 \\ 2 & -1 & 3 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $B = \begin{bmatrix} 7 \\ -5 \\ 12 \end{bmatrix}$ $\therefore$ Given system can be written as $AX = B$

$|A| = 1(7) + 1(19) + 2(-11) = 4 \neq 0$ $\therefore$ A^{-1} will exist 1

$X = A^{-1}B$ 1

$$\text{adj } A = \begin{bmatrix} 7 & 1 & -3 \\ -19 & -1 & 11 \\ -11 & -1 & 7 \end{bmatrix} \quad 2$$

$$A^{-1} = \frac{1}{4}\begin{bmatrix} 7 & 1 & -3 \\ -19 & -1 & 11 \\ -11 & -1 & 7 \end{bmatrix}$$ $\frac{1}{2}$

$$\therefore \quad A^{-1}B = \frac{1}{4}\begin{bmatrix} 7 & 1 & -3 \\ -19 & -1 & 11 \\ -11 & -1 & 7 \end{bmatrix}\begin{bmatrix} 7 \\ -5 \\ 12 \end{bmatrix}$$

$$\therefore \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$ 1

$$\Rightarrow \quad x = 2,\ y = 1,\ z = 3$$ $\frac{1}{2}$

25. Let 'x' be the side of the square and 'r' be the radius of the circle.

$$P = 4x + 2\pi r \Rightarrow r = \frac{P - 4x}{2\pi}$$ 1

$$A = x^2 + \pi r^2 = x^2 + \pi\left(\frac{P - 4x}{2\pi}\right)^2$$

$$= x^2 + \frac{(P - 4x)^2}{4\pi}$$ 1

$$\frac{dA}{dx} = 2x + \frac{2}{4\pi}(P - 4x)(-4)$$ 1

$$\frac{dA}{dx} = 0 \ \Rightarrow \ x = \frac{P}{\pi + 4}$$ 1

$$\frac{d^2A}{dx^2} = 2 + \frac{8}{\pi} > 0, \text{ gives } x = \frac{P}{\pi + 4} \text{ is minima}$$ 1

So, $$r = \frac{P - 4\left(\frac{P}{\pi + 4}\right)}{2\pi} = \frac{P}{2(\pi + 4)}$$ $\frac{1}{2}$

$$\therefore \quad x = 2r$$ $\frac{1}{2}$

OR

$9y^2 = x^3 \therefore 18y\frac{dy}{dx} = 3x^2 \Rightarrow \frac{dy}{dx} = \frac{x^2}{6y}$ 1

$\left(\frac{dy}{dx}\right)_{(h, k)} = \frac{h^2}{6k}$ $\frac{1}{2}$

Solpe of the normal $= \frac{-6k}{h^2} = \pm 1 \Rightarrow -6k = \pm h^2$ $1\frac{1}{2}$

also $9k^2 = h^3$ $\frac{1}{2}$

gives $h = 4$ as $h \neq 0$ 1

and $k = \pm\frac{8}{3}$ 1

Points are $\left(4, \pm\frac{8}{3}\right)$ $\frac{1}{2}$

26. Area of ellipse $= 4\left(\frac{b}{a}\int_0^a \sqrt{a^2 - x^2}\, dx\right)$ 3

$= 4\left|\left(\frac{b}{a}\left(\frac{x}{2}\sqrt{(a^2 - x^2)} + \frac{a^2}{2}\sin^{-1}\frac{x}{a}\right)\right)\right|_0^a$ 2

$= 4\frac{b}{a}\left(\frac{\pi a^2}{4}\right)$

$= \pi ab$ 1

OR

$a = 1, b = 3, nh = 2$ 1

$\int_3^3 (x^2 + x + e^x)\, dx = \lim_{h\to 0} h(f(1) + f(1 + h) + ... + f(1 + (n - 1)h)$ 1

$= \lim_{h\to 0} = h(2 + e + (1 + h)^2 + (1 + h) + e^{1+h} + ... + (1 + (n - 1)h)^2 + (1 + (n - 1)h) + e^{1+(n-1)h})$ $\frac{1}{2}$

$= \lim_{h\to 0} = h(2 + e + 2 + 3h + h^2 + e^{1+h} + ... + (1)^2 + (n - 1)^2 h^2 + 2(n - 1)h + 1 + (n - 1)h + e^{1+(n-1)h})$ $\frac{1}{2}$

$= \lim_{h\to 0} h\left(2n + 3h\left(\frac{(n-1)n}{2}\right) + h^2\left(\frac{(n-1)n(2n-1)}{6}\right) + e(1 + e^h + ... + e^{(n-1)h})\right)$ 1

$= \lim_{n\to 0}\left(2nh + \frac{3(nh-h)(nh)}{2} + \frac{(nh-h)(nh)(2nh-h)}{6} + eh\left(\frac{e^{nh}-1}{e^h-1}\right)\right)$ $\frac{1}{2}$

$= 2(2) + 3\frac{(2-0)(2)}{2} + \frac{(2-0)(2)(2(2)-0)}{6} + e(e^2-1) \times \lim_{h\to 0}\frac{h}{e^h-1}$ 1

$= \frac{38}{3} + e^3 - e$ $\frac{1}{2}$

27. Equation of the line is $\frac{x-3}{-1} = \frac{y+4}{1} = \frac{z+5}{6}$ $1\frac{1}{2}$

Equation of the plane is $\begin{vmatrix} x-2 & y-2 & z-1 \\ 1 & -2 & 0 \\ 2 & -3 & -1 \end{vmatrix} = 0$ $1\frac{1}{2}$

$\Rightarrow \quad 2x + y + z = 7$ 1

$\frac{x-3}{-1} = \frac{y+4}{1} = \frac{z+5}{6} = \lambda$ gives general point on the line as

$x = 3 - \lambda,\ y = \lambda - 4,\ z = 6\lambda - 5$ $\frac{1}{2}$

as it lies on the plane,

$2(3 - \lambda) + (\lambda - 4) + 6\lambda - 5 = 7$ $\frac{1}{2}$

$\Rightarrow \quad \lambda = 2$ $\frac{1}{2}$

Point is $(1, -2, 7)$ $\frac{1}{2}$

OR

Equation of the plane through intersection of planes will be

$(x + y + z - 1) + \lambda(2x + 3y + 4z - 5) = 0$ 1

$(1 + 2\lambda)x + (1 + 3\lambda)y + (1 + 4\lambda)z - (1 + 5\lambda) = 0 \qquad ...(1)$ 1

as (1) is perpendicular to $x - y + z = 0$

$(1 + 2\lambda)(1) + (1 + 3\lambda)(-1) + (1 + 4\lambda)(1) = 0$ 1

$\Rightarrow \quad \lambda = \frac{-1}{3}$ 1

Equation of the plane is $x - z + 2 = 0$ 1

distance of the plane from origin = $\frac{2}{\sqrt{2}} = \sqrt{2}$ 1

28. Let E_1 : Selecting a bag of group I
E_2 : Selecting a bag of group II
H : Getting a red ball 1

$P(E_1) = \frac{3}{5} \qquad P(E_2) = \frac{2}{5}$ $1\frac{1}{2}$

$P(H|E_1) = \frac{5}{8} \qquad P(H|E_2) = \frac{2}{6} = \frac{1}{3}$ $1\frac{1}{2}$

$P(H \mid E_1) = \frac{\frac{3}{5} \times \frac{5}{8}}{\frac{3}{5} \times \frac{5}{8} + \frac{2}{5} \times \frac{1}{3}}$ $1\frac{1}{2}$

$= \frac{45}{61}$ $\frac{1}{2}$

29. Let the number of units of X = x
Let the number of units of Y = y 1

Minimize $P = 10x + 4y$ 1

Subject to $4x + y \geq 80$ $1\frac{1}{2}$

$2x + y \geq 60$ $1\frac{1}{2}$

$x, y \geq 0$ 1

Strictly Confidential — (For Internal and Restricted Use Only)

Senior School Certificate Examination

March 2019

Marking Scheme — Mathematics (041) 65/1/1, 65/1/2, 65/1/3

General Instructions:

1. You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully. **Evaluation is a 10-12 days mission for all of us. Hence, it is necessary that you put in your best efforts in this process.**
2. Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one's own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. **However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and marks be awarded to them.**
3. The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
4. If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled.
5. If a question does not have any parts, marks must be awarded in the left hand margin and encircled.
6. If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out.
7. No marks to be deducted for the cumulative effect of an error. It should be penalized only once.
8. A full scale of marks 0 to 100 has to be used. Please do not hesitate to award full marks if the answer deserves it.
9. Every examiner has to necessarily do evaluation work for full working hours i.e. 8 hours every day and evaluate 25 answer books per day.
10. Ensure that you do not make the following common types of errors committed by the Examiner in the past:-
    - Leaving answer or part thereof unassessed in an answer book.
    - Giving more marks for an answer than assigned to it.
    - Wrong transfer of marks from the inside pages of the answer book to the title page.
    - Wrong question wise totaling on the title page.
    - Wrong totaling of marks of the two columns on the title page.
    - Wrong grand total.
    - Marks in words and figures not tallying.
    - Wrong transfer of marks from the answer book to online award list.
    - Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.)
    - Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
11. While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as (X) and awarded zero (0) Marks.
12. Any unassessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
13. The Examiners should acquaint themselves with the guidelines given in the Guidelines for spot Evaluation before starting the actual evaluation.
14. Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
15. The Board permits candidates to obtain photocopy of the Answer Book on request in an RTI application and also separately as a part of the re-evaluation process on payment of the processing charges.

QUESTION PAPER CODE 65/1/1
EXPECTED ANSWER/VALUE POINTS

SECTION A

1. $AB = 2I \Rightarrow |AB| = |2I| \Rightarrow |A| \cdot |B| = 2^3|I|$ $\frac{1}{2}$

$\Rightarrow 2 \times |B| = 8 \Rightarrow |B| = 4$ $\frac{1}{2}$

2. $(fof)(x) = f(x + 1) = x + 2$ $\frac{1}{2}$

$\frac{d}{dx}(fof)(x) = 1$ $\frac{1}{2}$

3. order = 2, degree = 1 $\frac{1}{2}+\frac{1}{2}$

4. d.c.'s = <cos 90°, cos 135°, cos 45°> $\frac{1}{2}$

$= <0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}>$ $\frac{1}{2}$

OR

$\vec{r} = (3\hat{i}+4\hat{j}+5\hat{k})+\lambda(2\hat{i}+2\hat{j}-3\hat{k})$ 1

SECTION B

5. As $a, b \in R \Rightarrow ab \in R \Rightarrow ab + 1 \in R \Rightarrow a*b \in R \Rightarrow *$ is binary. 1

For associative $(a*b)*c = (ab+1)*c = (ab+1)c+1 = abc+c+1$

also, $a*(b*c) = a*(bc+1) = a.(bc+1)+1 = abc+a+1$

In general $(a*b)*c \neq a*(b*c) \Rightarrow *$ is not associative. 1

6. $2A-\begin{bmatrix}-6 & 6 & 0\\ 9 & 3 & 12\end{bmatrix}+\begin{bmatrix}10 & 0 & -10\\ 35 & 5 & 30\end{bmatrix}=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$ 1

$\Rightarrow A=\begin{bmatrix}-8 & 3 & 5\\ -13 & -1 & -9\end{bmatrix}$ 1

7. Put $\tan x = t \Rightarrow \sec^2 x\, dx = dt$ $\frac{1}{2}$

$$I = \int \frac{\sec^2 x}{\sqrt{\tan^2 x + 4}}\, dx = \int \frac{dt}{\sqrt{t^2 + 4}} = \log | t + \sqrt{t^2 + 4} | + C$$ 1

$$= \log | \tan x + \sqrt{\tan^2 x + 4} | + C$$ $\frac{1}{2}$

8. Let $I = \int \sqrt{1 - \sin 2x}\, dx$

$= \int (\sin x - \cos x) dx$ as $\sin x > \cos x$ when $x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ 1

$= -\cos x - \sin x + C$ 1

OR

$$I = \int \sin^{-1}(2x).1\, dx$$

$$= x.\sin^{-1}(2x) - \int \frac{2x}{\sqrt{1 - 4x^2}}\, dx$$ 1

$$= x.\sin^{-1}(2x) + \frac{1}{4}\int \frac{-8x}{\sqrt{1 - 4x^2}}\, dx = x \sin^{-1}(2x) + \frac{1}{2}.\sqrt{1 - 4x^2} + C$$ 1

9. $y' = be^{2x} + 2y \Rightarrow b = \frac{y' - 2y}{e^{2x}}$ $\frac{1}{2}$

differentiating again

$$\frac{e^{2x}.(y'' - 2y') - (y' - 2y).2x^{2x}}{(e^{2x})^2} = 0$$ 1

$$\Rightarrow y'' - 4y' + 4y = 0 \text{ or } \frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$$ $\frac{1}{2}$

10. Given $| \hat{a} + \hat{b} | = 1$

As $| \hat{a} + \hat{b} |^2 + | \hat{a} - \hat{b} |^2 = 2(| \hat{a} |^2 + | \hat{b} |^2)$ 1

$\Rightarrow 1 + | \vec{a} - \vec{b} |^2 = 2(1 + 1)$

$\Rightarrow |\hat{a}-\hat{b}|^2=3 \Rightarrow |\hat{a}-\hat{b}|=\sqrt{3}$ 1

OR

$[\vec{a} \quad \vec{b} \quad \vec{c}]=\begin{vmatrix} 2 & 3 & 1 \\ 1 & -2 & 1 \\ -3 & 1 & 2 \end{vmatrix}$ 1

$= -30$ 1

11. $A = \{2, 4, 6\}$, $B = \{1, 2, 3\}$, $A \cap B = \{2\}$

Now, $P(A)=\frac{1}{2}$, $P(B)=\frac{1}{2}$, $P(A \cap B)=\frac{1}{6}$ 1

as $P(A)\times P(B)=\frac{1}{2}\times\frac{1}{2}=\frac{1}{4}\neq P(A \cap B)$ $\frac{1}{2}$

$\Rightarrow$ A and B are not independent. $\frac{1}{2}$

12. Let X: getting an odd number

$p=\frac{1}{2}$, $q=\frac{1}{2}$, $n=6$ $\frac{1}{2}$

(i) $P(X=5)={}^6C_5\left(\frac{1}{2}\right)^6=\frac{3}{32}$ $\frac{1}{2}$

(ii) $P(X\leq 5)=1-P(X=6)=1-\frac{1}{64}=\frac{63}{64}$ 1

OR

$k + 2k + 3k = 1$ 1

$\Rightarrow k=\frac{1}{6}$ 1

13. Clearly $a \leq a \ \forall\ a \in \mathbb{R} \Rightarrow (a, a) \in R \Rightarrow R$ is reflexive. 1

For transitive:

Let (a, b) $\in$ R and (b, c) $\in$ R, a, b, c $\in \mathbb{R}$

$\Rightarrow a \le b$ and $b \le c \Rightarrow a \le c \Rightarrow (a, c) \in R$

$\Rightarrow$ R is transitive. $1\frac{1}{2}$

For non-symmetric:

Let $a = 1, b = 2$. As $1 \le 2 \Rightarrow (1, 2) \in R$ but $2 \not\le 1 \Rightarrow (2,1) \notin R$

$\Rightarrow$ R is non-symmetric. $1\frac{1}{2}$

OR

For one-one. Let $x_1, x_2 \in N$.

$f(x_1) = f(x_2) \Rightarrow x_1^2 + x_1 + 1 = x_2^2 + x_2 + 1$

$\Rightarrow (x_1 - x_2)(x_1 + x_2 + 1) = 0$

$\Rightarrow x_1 = x_2$ as $x_1 + x_2 + 1 \neq 0$ $(\because x_1, x_2 \in N)$ $1\frac{1}{2}$

$\Rightarrow$ f is one-one.

For not onto.

for $y = 1 \in N$, there is no $x \in N$ for which $f(x) = 1$ $1\frac{1}{2}$

For f^{-1}: $y = f(x) \Rightarrow y = x^2 + x + 1 \Rightarrow y = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4}$

$$\Rightarrow x = \frac{\sqrt{4y-3}-1}{2}$$

$\therefore f^{-1}(y) = \frac{\sqrt{4y-3}-1}{2}$ or $f^{-1}(x) = \frac{\sqrt{4x-3}-1}{2}$ 1

14. $\tan^{-1}\left(\frac{4x+6x}{1-(4x)(6x)}\right) = \frac{\pi}{4}$ 1

$\Rightarrow \frac{10x}{1-24x^2} = 1 \Rightarrow 24x^2 + 10x - 1 = 0$ $1\frac{1}{2}$

$\Rightarrow x = \frac{1}{12}$ or $-\frac{1}{2}$ 1

as $x = -\frac{1}{2}$ does not satisfy the given equation, so x = $\frac{1}{12}$ $\frac{1}{2}$

15. LHS = $\begin{vmatrix} a^2+2a & 2a+1 & 1 \\ 2a+1 & a+2 & 1 \\ 3 & 3 & 1 \end{vmatrix}$

$R_1 \to R_1 - R_2$, $R_2 \to R_2 - R_3$

$= \begin{vmatrix} a^2-1 & a-1 & 0 \\ 2(a-1) & a-1 & 0 \\ 3 & 3 & 1 \end{vmatrix}$ 2

$= (a-1)^2 \begin{vmatrix} a+1 & 1 & 0 \\ 2 & 1 & 0 \\ 3 & 3 & 1 \end{vmatrix}$ 1

Expanding along C_3,

$= (a-1)^2.(a-1) = (a-1)^3 = \text{RHS}.$ 1

16. $\log(x^2+y^2) = 2\tan^{-1}\left(\frac{y}{x}\right)$

differentiating both sides w.r.t. x,

$$\frac{1}{x^2+y^2}\left(2x+2y\frac{dy}{dx}\right) = 2\cdot\frac{1}{1+\left(\frac{y}{x}\right)^2}\cdot\left(\frac{x\cdot\frac{dy}{dx}-y\cdot 1}{x^2}\right)$$ 2

$$\Rightarrow \frac{2}{x^2+y^2}\left(x+y\frac{dy}{dx}\right) = \frac{2x^2}{x^2+y^2}\cdot\frac{1}{x^2}\cdot\left(x\frac{dy}{dx}-y\right)$$ 1

$$\Rightarrow (x+y) = (x-y)\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{x+y}{x-y}$$ 1

OR

Let $u = x^y$, $v = y^x$. Then $u - v = a^b$

$$\Rightarrow \frac{du}{dx} - \frac{dv}{dx} = 0 \qquad ...(1)$$ 1

Now, log u = y.logx

$$\Rightarrow \frac{1}{u}\cdot\frac{du}{dx} = y\cdot\frac{1}{x} + \log x\cdot\frac{dy}{dx} \Rightarrow \frac{du}{dx} = x^y\left(\frac{y}{x} + \log x\cdot\frac{dy}{dx}\right) \qquad ...(2)$$ 1

Again, log v = x · log y

$$\Rightarrow \frac{1}{v}\cdot\frac{dv}{dx} = x\cdot\frac{1}{y}\cdot\frac{dy}{dx} + \log y\cdot 1 \Rightarrow \frac{dv}{dx} = y^x\left(\frac{x}{y}\frac{dy}{dx} + \log y\right) \qquad ...(3)$$ 1

From (1), (2) and (3)

$$x^y\left(\frac{y}{x} + \log x\frac{dy}{dx}\right) - y^x\left(\frac{x}{y}\frac{dy}{dx} + \log y\right) = 0$$ $\frac{1}{2}$

$$\Rightarrow \frac{dy}{dx} = \frac{y^x\cdot\log y - x^{y-1}\cdot y}{x^y\cdot\log x - y^{x-1}\cdot x}$$ $\frac{1}{2}$

17. $y = (\sin^{-1} x)^2$

$$\Rightarrow y' = 2\cdot\sin^{-1} x\cdot\frac{1}{\sqrt{1-x^2}}$$ 1

$$\Rightarrow \sqrt{1-x^2}\cdot y' = 2\sin^{-1} x$$

$$\Rightarrow \sqrt{1-x^2}\cdot y'' + y'\cdot\frac{1}{2\sqrt{1-x^2}}(-2x) = \frac{2}{\sqrt{1-x^2}}$$ 2

$$\Rightarrow (1-x^2)\cdot y'' - xy' = 2 \text{ or } (1-x^2)\cdot\frac{d^2y}{dx^2} - x\frac{dy}{dx} - 2 = 0.$$ 1

18. Let the point of contact be $P(x_1, y_1)$

$$\frac{dy}{dx} = \frac{3}{2\sqrt{3x-2}}$$ (slope of tangent)

$$\Rightarrow m_1 = \left.\frac{dy}{dx}\right]_{(x_1, y_1)} = \frac{3}{2\sqrt{3x_1-2}}$$ 1

also, slope of given line = 2 = m_2

$m_1 = m_2 \Rightarrow x_1 = \frac{41}{48}$ 1

when $x_1 = \frac{41}{48}, y_1 = \sqrt{\frac{41}{16} - 2} = \frac{3}{4}$ $\therefore P\left(\frac{41}{48}, \frac{3}{4}\right)$ $\frac{1}{2}$

Equation of tangent is: $y - \frac{3}{4} = 2\left(x - \frac{41}{48}\right)$

$\Rightarrow 48x - 24y = 23$ 1

and, Equation of normal is: $y - \frac{3}{4} = \frac{-1}{2}\left(x - \frac{41}{48}\right)$

$\Rightarrow 48x + 96y = 113$ $\frac{1}{2}$

19. $I = \int \frac{3x+5}{x^2+3x-18}\,dx = \frac{3}{2}\int \frac{2x+3}{x^2+3x-18}dx + \frac{1}{2}\int \frac{1}{x^2+3x-18}\,dx$ 1

$= \frac{3}{2}\int \frac{2x+3}{x^2+3x-18}dx + \frac{1}{2}\int \frac{1}{\left(x+\frac{3}{2}\right)^2 - \left(\frac{9}{2}\right)^2}\,dx$ 1

$= \frac{3}{2}\log|x^2+3x-18| + \frac{1}{18}\log\left|\frac{x-3}{x+6}\right| + C$ 1 + 1

20. Let $I = \int_0^a f(a-x)\,dx$

Put $a - x = t \Rightarrow -dx = dt$ $\frac{1}{2}$

$I = -\int_a^0 f(t)\,dt = \int_0^a f(t)\,dt = \int_0^a f(x)\,dx$ $\frac{1}{2}$

II part.

$I = \int_0^{\pi} \frac{x \sin x}{1+\cos^2 x}\,dx$

$\Rightarrow \quad I = \int_0^{\pi} \frac{(\pi - x) \cdot \sin x}{1 + \cos^2 x} dx$

$\Rightarrow \quad 2I = \int_0^{\pi} \frac{\pi \cdot \sin x}{1 + \cos^2 x} dx$ $1\frac{1}{2}$

Put $\cos x = t \Rightarrow -\sin x \, dx = dt$

$\Rightarrow I = -\frac{\pi}{2} \cdot \int_1^{-1} \frac{dt}{1+t^2} = \frac{\pi}{2} \times 2 \times \int_0^1 \frac{dt}{1+t^2}$

$= \pi[\tan^{-1} t]_0^1 = \frac{\pi^2}{4}$ $1\frac{1}{2}$

21. Writing $\frac{dy}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x} = \frac{y}{x} + \sqrt{1 + \left(\frac{y}{x}\right)^2}$ $\frac{1}{2}$

Put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ $\frac{1}{2}$

Differential equation becomes $v + x\frac{dv}{dx} = v + \sqrt{1 + v^2}$

$\Rightarrow \int \frac{dv}{\sqrt{1+v^2}} = \int \frac{dx}{x}$ 1

$\Rightarrow \log\left|v + \sqrt{1+v^2}\right| = \log|x| + \log c$ 1

$\Rightarrow v + \sqrt{1+v^2} = cx \Rightarrow y + \sqrt{x^2 + y^2} = cx^2$

when $x = 1, y = 0 \Rightarrow c = 1$ $\frac{1}{2}$

$\therefore \quad y + \sqrt{x^2 + y^2} = x^2$ $\frac{1}{2}$

OR

Given equation is $\frac{dy}{dx} + \frac{2x}{1+x^2} \cdot y = \frac{4x^2}{1+x^2}$ $\frac{1}{2}$

I.F. $= e^{\int \frac{2x}{1+x^2}\,dx} = 1+x^2$ 1

Solution is given by,

$y\cdot(1+x^2) = \int \frac{4x^2}{1+x^2}\cdot(1+x^2)\,dx = \int 4x^2\,dx$ 1

$\Rightarrow\ y\cdot(1+x^2) = \frac{4x^3}{3}+c$ $\frac{1}{2}$

when x = 0, y = 0 $\Rightarrow$ c = 0 $\frac{1}{2}$

$y\cdot(1+x^2) = \frac{4x^3}{3}$ or $y = \frac{4x^3}{3(1+x^2)}$ $\frac{1}{2}$

22. $\overrightarrow{AB} = \hat{i}+4\hat{j}-\hat{k}$ 1

$\overrightarrow{CD} = -2\hat{i}-8\hat{j}+2\hat{k}$ 1

Let required angle be θ.

Then $\cos\theta = \frac{\overrightarrow{AB}\cdot\overrightarrow{CD}}{|\overrightarrow{AB}||\overrightarrow{CD}|} = \frac{-2-32-2}{\sqrt{18}\sqrt{72}} = -1$ 1

$\Rightarrow$ θ = 180° or π $\frac{1}{2}$

Since θ = π so $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are collinear. $\frac{1}{2}$

23. Given lines are: $\frac{x-1}{-3} = \frac{y-2}{\left(\frac{\lambda}{7}\right)} = \frac{z-3}{2}$ and $\frac{x-1}{\left(\frac{-3\lambda}{7}\right)} = \frac{y-5}{1} = \frac{z-6}{-5}$ 1

As lines are perpendicular,

$(-3)\left(\frac{-3\lambda}{7}\right)+\left(\frac{\lambda}{7}\right)(1)+2(-5) = 0 \Rightarrow \lambda = 7$ 1

So, lines are

$\frac{x-1}{-3} = \frac{y-2}{1} = \frac{x-3}{2}$ and $\frac{x-1}{-3} = \frac{y-5}{1} = \frac{z-6}{-5}$ $\frac{1}{2}$

Consider $\Delta = \begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = \begin{vmatrix} 0 & 3 & 3 \\ -3 & 1 & 2 \\ -3 & 1 & -5 \end{vmatrix} = -63$ 1

as $\Delta \neq 0 \Rightarrow$ lines are not intersecting. $\frac{1}{2}$

SECTION D

24. $|A| = 4 \neq 0 \Rightarrow A^{-1}$ exists. 1

$$\text{adj } A = \begin{bmatrix} -2 & 0 & 2 \\ 5 & -2 & -1 \\ 1 & 2 & -1 \end{bmatrix}$$ 2

$$\therefore \ A^{-1} = \frac{1}{|A|} \cdot \text{adj } A = \frac{1}{4}\begin{bmatrix} -2 & 0 & 2 \\ 5 & -2 & -1 \\ 1 & 2 & -1 \end{bmatrix}$$ $\frac{1}{2}$

Given system of equations can be written as AX = B where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 6 \\ 7 \\ 12 \end{bmatrix}$

$\therefore \ X = A^{-1} \cdot B$ 1

$$= \frac{1}{4}\begin{bmatrix} -2 & 0 & 2 \\ 5 & -2 & -1 \\ 1 & 2 & -1 \end{bmatrix}\begin{bmatrix} 6 \\ 7 \\ 12 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$ 1

$\Rightarrow x = 3, y = 1, z = 2$ $\frac{1}{2}$

OR

A = I.A

$$\Rightarrow \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot A$$ 1

$R_2 \to R_2 + R_1$

$$\Rightarrow \begin{bmatrix} 1 & 2 & -2 \\ 0 & 5 & -2 \\ 0 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot A$$

$R_2 \to \frac{R_2}{5}$

$$\Rightarrow \begin{bmatrix} 1 & 2 & -2 \\ 0 & 1 & -2/5 \\ 0 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1/5 & 1/5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot A$$

$R_1 \to R_1 - 2R_2, R_3 \to R_3 + 2R_2$

$$\Rightarrow \begin{bmatrix} 1 & 0 & -6/5 \\ 0 & 1 & -2/5 \\ 0 & 0 & 1/5 \end{bmatrix} = \begin{bmatrix} 3/5 & -2/5 & 0 \\ 1/5 & 1/5 & 0 \\ 2/5 & 2/5 & 1 \end{bmatrix} \cdot A$$

$R_3 \to 5R_3$

$$\Rightarrow \begin{bmatrix} 1 & 0 & -6/5 \\ 0 & 1 & -2/5 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3/5 & -2/5 & 0 \\ 1/5 & 1/5 & 0 \\ 2 & 2 & 5 \end{bmatrix} \cdot A$$

$R_1 \to R_1 + \frac{6}{5}R_3, R_2 \to R_2 + \frac{2}{5}R_3$

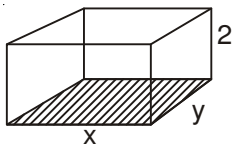
$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix} \cdot A$$

4

$$\Rightarrow A^{-1} = \begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix}$$

1

25.



$V = 2xy \Rightarrow 2xy = 8$ (given) 1

$\Rightarrow y = \frac{4}{x}$

Now, cost, $C = 70xy + 45 \times 2 \times (2x + 2y)$ 1

$= 280 + 180x + \frac{720}{x}$ 1

$\frac{dC}{dx} = 180 - \frac{720}{x^2}$ 1

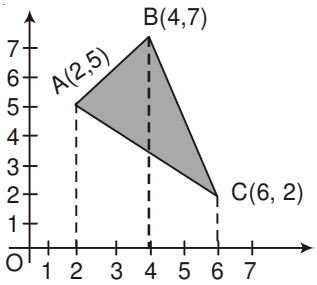
$\frac{dC}{dx} = 0 \Rightarrow x = 2m$ $\frac{1}{2}$

$\frac{d^2C}{dx^2} = \frac{1440}{x^3} = 180 > 0$ at $x = 2$ $\frac{1}{2}$

$\Rightarrow$ C is minimum at $x = 2m$. $\frac{1}{2}$

Minimum cost $= 280 + 180(2) + \frac{720}{2} = ₹\ 1{,}000$ $\frac{1}{2}$

26.



Correct Figure 1

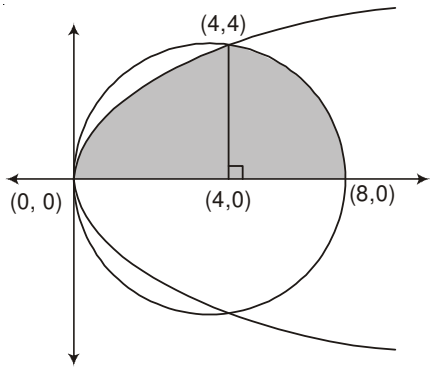
$$\left.\begin{array}{l} \text{Equation of AB}: y = x + 3 \\ \text{Equation of BC}: y = \frac{-5x}{2} + 17 \\ \text{Equation of AC}: y = \frac{-3x}{4} + \frac{13}{2} \end{array}\right\}$$ $1\frac{1}{2}$

Required Area $= \int_2^4 (x+3)\,dx + \int_4^6 \left(\frac{-5x}{2} + 17\right) dx - \int_2^6 \left(\frac{-3x}{4} + \frac{13}{2}\right) dx$ $1\frac{1}{2}$

$= \left[\frac{(x+3)^2}{2}\right]_2^4 + \left[\frac{-5x^2}{4} + 17x\right]_4^6 - \left[\frac{-3x^2}{8} + \frac{13x}{2}\right]_2^6$ $1\frac{1}{2}$

$= 7$ $\frac{1}{2}$

OR



Correct Figure 1

Given circle $x^2 - 8x + y^2 = 0$

or $(x-4)^2 + y^2 = 4^2$

Point of intersection (0, 0) and (4, 4) 1

Required Area $= \int_0^4 2\sqrt{x}\,dx + \int_4^8 \sqrt{4^2-(x-4)^2}\,dx$ $1\frac{1}{2}$

$= \left[\frac{4}{3}x^{3/2}\right]_0^4 + \left[\frac{x-4}{2}\sqrt{16-(x-4)^2} + \frac{16}{2}\sin^{-1}\left(\frac{x-4}{4}\right)\right]_4^8$ $1\frac{1}{2}$

$= \left(4\pi + \frac{32}{3}\right)$ 1

27. Equation of plane is $\begin{vmatrix} x-2 & y-2 & z+1 \\ 1 & 2 & 3 \\ 5 & -2 & 7 \end{vmatrix} = 0$ 2

$\Rightarrow 5x + 2y - 3z = 17$ (Cartesian equation) 1

Vector equation is $\vec{r}.(5\hat{i}+2\hat{j}-3\hat{k}) = 17$ 1

Equation of required parallel plane is

$\vec{r}\cdot(5\hat{i}+2\hat{j}-3\hat{k}) = (4\hat{i}+3\hat{j}+\hat{k})\cdot(5\hat{i}+2\hat{j}-3\hat{k})$ 1

$\Rightarrow \vec{r}\cdot(5\hat{i}+2\hat{j}-3\hat{k}) = 23$ 1

OR

Let required plane be $a(x + 1) + b(y - 3) + c(z + 4) = 0$ ...(1) 1

Plane contains the given line, so it will also contain the point (1, 1, 0).

So, $2a - 2b + 4c = 0$ or $a - b + 2c = 0$ ...(2) 1

Also, $a + 2b - c = 0$ ...(3) 1

From (2) and (3),

$\frac{a}{-3} = \frac{b}{3} = \frac{c}{3}$ 1

$\therefore$ Required plane is $-3(x + 1) + 3(y - 3) + 3(z + 4) = 0$

$\therefore$ $-x + y + z = 0$

Also vector equation is: $\vec{r}.(-\hat{i} + \hat{j} + \hat{k}) = 0$ 1

Length of perpendicular from (2, 1, 4) = $\frac{|-2+1+4|}{\sqrt{(-1)^2+1^2+1^2}} = \sqrt{3}$ 1

28.

Let E_1 : item is produced by A
E_2 : item is produced by B
E_3 : item is produced by C
A : defective item is found. 1

$P(E_1) = \frac{50}{100}, P(E_2) = \frac{30}{100}, P(E_3) = \frac{20}{100}$ 1

$P(A / E_1) = \frac{1}{100}, P(A / E_2) = \frac{5}{100}, P(A | E_3) = \frac{7}{100}$ 1

$$P(E_1 | A) = \frac{\frac{50}{100} \times \frac{1}{100}}{\frac{50}{100} \times \frac{1}{100} + \frac{30}{100} \times \frac{5}{100} + \frac{20}{100} \times \frac{7}{100}}$$ 2

$= \frac{5}{34}$ 1

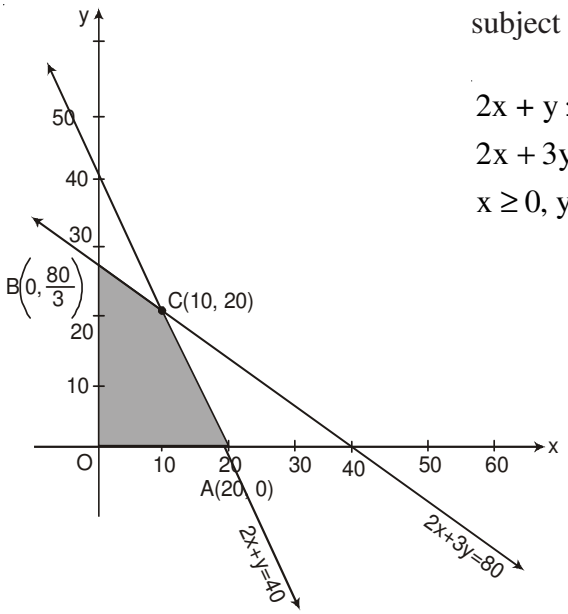
29. Let number of items produced of model A be x and that of model B be y.

LPP is:

Maximize, profit $z = 15x + 10y$ 1

subject to

$$\left.\begin{array}{l} 2x + y \le 5(8) \quad \text{i.e.,} \quad 2x + y \le 40 \\ 2x + 3y \le 10(8) \quad \text{i.e.,} \quad 2x + 3y \le 80 \\ x \ge 0,\ y \ge 0 \end{array}\right\}$$ 2



Correct Figure 2

Corner point	$z = 15x + 10y$	
A(20, 0)	300	
$B\left(0, \frac{80}{3}\right)$	$\frac{800}{3} \approx 266.6$	$\frac{1}{2}$
C(10, 20)	350 ← maximum	

Maximum profit = ₹ 350

when x = 10, y = 20. $\frac{1}{2}$

If a student has interpreted the language of the question in a different way, then the LPP will be of the type:

Maximise profit $z = 15x + 10y$

Subject to $2x + y \le 8$

$2x + 3y \le 8$

$x \ge 0,\ y \ge 0$

This is be accepted and marks may be given accordingly.

QUESTION PAPER CODE 65/1/2
EXPECTED ANSWER/VALUE POINTS

SECTION A

1. order = 2, degree = 1 $\frac{1}{2}+\frac{1}{2}$

2. (fog) (x) = f(x – 7) = x $\frac{1}{2}$

$\Rightarrow \frac{d}{dx}[(fog)(x)]=1$ $\frac{1}{2}$

3. $\begin{bmatrix} 2+y & 6 \\ 1 & 2x+2 \end{bmatrix}=\begin{bmatrix} 5 & 6 \\ 1 & 8 \end{bmatrix}$ $\frac{1}{2}$

$\Rightarrow$ x = 3, y = 3 $\quad\quad$ $\therefore$ x – y = 0 $\frac{1}{2}$

4. d.c.'s = <cos 90°, cos 135°, cos 45°> $\frac{1}{2}$

$= <0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}>$ $\frac{1}{2}$

OR

$\vec{r}=(3\hat{i}+4\hat{j}+5\hat{k})+\lambda(2\hat{i}+2\hat{j}-3\hat{k})$ 1

SECTION B

5. As a, b $\in$ R $\Rightarrow$ ab $\in$ R $\Rightarrow$ ab + 1 $\in$ R $\Rightarrow$ a*b $\in$ R $\Rightarrow$ * is binary. 1

For associative $(a*b)*c=(ab+1)*c=(ab+1)c+1=abc+c+1$

also, $a*(b*c)=a*(bc+1)=a.(bc+1)+1=abc+a+1$

In general (a*b)*c $\neq$ a*(b*c) $\Rightarrow$ * is not associative. 1

6. $A^2=\begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix}$ 1

$$A^2 - 5A = \begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix} - \begin{bmatrix} 10 & 0 & 5 \\ 10 & 5 & 15 \\ 5 & -5 & 0 \end{bmatrix} = \begin{bmatrix} -5 & -1 & -3 \\ -1 & -7 & -10 \\ -5 & 4 & -2 \end{bmatrix}$$ 1

7. Let $I = \int \sqrt{1 - \sin 2x}\, dx$

$= \int (\sin x - \cos x) dx$ as $\sin x > \cos x$ when $x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ 1

$= -\cos x - \sin x + C$ 1

OR

$I = \int \sin^{-1}(2x).1\, dx$

$= x.\sin^{-1}(2x) - \int \frac{2x}{\sqrt{1-4x^2}} dx$ 1

$= x.\sin^{-1}(2x) + \frac{1}{4}\int \frac{-8x}{\sqrt{1-4x^2}} dx = x\sin^{-1}(2x) + \frac{1}{2}.\sqrt{1-4x^2} + C$ 1

8. $y' = be^{2x} + 2y \Rightarrow b = \frac{y' - 2y}{e^{2x}}$ $\frac{1}{2}$

differentiating again

$\frac{e^{2x}.(y'' - 2y') - (y' - 2y).2x^{2x}}{(e^{2x})^2} = 0$ 1

$\Rightarrow y'' - 4y' + 4y = 0$ or $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$ $\frac{1}{2}$

9. Let X: getting an odd number

$p = \frac{1}{2},\ q = \frac{1}{2},\ n = 6$ $\frac{1}{2}$

(i) $P(X = 5) = {}^6C_5\left(\frac{1}{2}\right)^6 = \frac{3}{32}$ $\frac{1}{2}$

(ii) $P(X \le 5) = 1 - P(X = 6) = 1 - \frac{1}{64} = \frac{63}{64}$ 1

OR

$k + 2k + 3k = 1$ 1

$\Rightarrow k = \frac{1}{6}$ 1

10. $A = \{2, 4, 6\}$, $B = \{1, 2, 3\}$, $A \cap B = \{2\}$

Now, $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{2}$, $P(A \cap B) = \frac{1}{6}$ 1

as $P(A) \times P(B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} \neq P(A \cap B)$ $\frac{1}{2}$

$\Rightarrow$ A and B are not independent. $\frac{1}{2}$

11. Given $|\hat{a} + \hat{b}| = 1$

As $|\hat{a} + \hat{b}|^2 + |\hat{a} - \hat{b}|^2 = 2(|\hat{a}|^2 + |\hat{b}|^2)$ 1

$\Rightarrow 1 + |\vec{a} - \vec{b}|^2 = 2(1+1)$

$\Rightarrow |\hat{a} - \hat{b}|^2 = 3 \Rightarrow |\hat{a} - \hat{b}| = \sqrt{3}$ 1

OR

$$[\vec{a} \quad \vec{b} \quad \vec{c}] = \begin{vmatrix} 2 & 3 & 1 \\ 1 & -2 & 1 \\ -3 & 1 & 2 \end{vmatrix}$$ 1

$= -30$ 1

12. $I = \int \frac{\tan^2 x . \sec^2 x}{1 - (\tan^3 x)^2} dx$

Put $\tan^3 x = t \Rightarrow I = \frac{1}{3} \int \frac{dt}{1 - t^2}$ 1

$= \frac{1}{6} \log \left| \frac{1+t}{1-t} \right| + C = \frac{1}{6} \log \left| \frac{1 + \tan^3 x}{1 - \tan^3 x} \right| + C$ $\frac{1}{2} + \frac{1}{2}$

SECTION C

13. $\tan^{-1}\left(\frac{2x+3x}{1-(2x)(3x)}\right)=\frac{\pi}{4}$ 1

$\Rightarrow \frac{5x}{1-6x^2}=\tan\frac{\pi}{4}=1 \Rightarrow 6x^2+5x-1=0$ $1\frac{1}{2}$

$\Rightarrow x=-1$ or $x=\frac{1}{6}$ 1

as x = –1 does not satisfy the given equation,

$\therefore x=\frac{1}{6}$ $\frac{1}{2}$

14. $\log(x^2+y^2)=2\tan^{-1}\left(\frac{y}{x}\right)$

differentiating both sides w.r.t. x,

$$\frac{1}{x^2+y^2}\left(2x+2y\frac{dy}{dx}\right)=2\cdot\frac{1}{1+\left(\frac{y}{x}\right)^2}\cdot\left(\frac{x\cdot\frac{dy}{dx}-y\cdot 1}{x^2}\right)$$ 2

$\Rightarrow \frac{2}{x^2+y^2}\left(x+y\frac{dy}{dx}\right)=\frac{2x^2}{x^2+y^2}\cdot\frac{1}{x^2}\cdot\left(x\frac{dy}{dx}-y\right)$ 1

$\Rightarrow (x+y)=(x-y)\frac{dy}{dx}\Rightarrow\frac{dy}{dx}=\frac{x+y}{x-y}$ 1

OR

Let $u = x^y$, $v = y^x$. Then $u - v = a^b$

$\Rightarrow \frac{du}{dx}-\frac{dv}{dx}=0$ ...(1) 1

Now, log u = y.logx

$\Rightarrow \frac{1}{u}\cdot\frac{du}{dx}=y\cdot\frac{1}{x}+\log x\cdot\frac{dy}{dx}\Rightarrow\frac{du}{dx}=x^y\left(\frac{y}{x}+\log x\cdot\frac{dy}{dx}\right)$ ...(2) 1

Again, $\log v = x \cdot \log y$

$$\Rightarrow \frac{1}{v}\cdot\frac{dv}{dx} = x\cdot\frac{1}{y}\cdot\frac{dy}{dx} + \log y\cdot 1 \Rightarrow \frac{dv}{dx} = y^x\left(\frac{x}{y}\frac{dy}{dx} + \log y\right) \quad ...(3)$$ 1

From (1), (2) and (3)

$$x^y\left(\frac{y}{x} + \log x\frac{dy}{dx}\right) - y^x\left(\frac{x}{y}\frac{dy}{dx} + \log y\right) = 0$$ $\frac{1}{2}$

$$\Rightarrow \frac{dy}{dx} = \frac{y^x\cdot\log y - x^{y-1}\cdot y}{x^y\cdot\log x - y^{x-1}\cdot x}$$ $\frac{1}{2}$

15. $$I = \int\frac{3x+5}{x^2+3x-18}\,dx = \frac{3}{2}\int\frac{2x+3}{x^2+3x-18}dx + \frac{1}{2}\int\frac{1}{x^2+3x-18}\,dx$$ 1

$$= \frac{3}{2}\int\frac{2x+3}{x^2+3x-18}dx + \frac{1}{2}\int\frac{1}{\left(x+\frac{3}{2}\right)^2 - \left(\frac{9}{2}\right)^2}\,dx$$ 1

$$= \frac{3}{2}\log|x^2+3x-18| + \frac{1}{18}\log\left|\frac{x-3}{x+6}\right| + C$$ 1 + 1

16. Let $I = \int_0^a f(a-x)\,dx$

Put $a - x = t \Rightarrow -dx = dt$ $\frac{1}{2}$

$$I = -\int_a^0 f(t)\,dt = \int_0^a f(t)\,dt = \int_0^a f(x)\,dx$$ $\frac{1}{2}$

II part.

$$I = \int_0^\pi \frac{x\sin x}{1+\cos^2 x}\,dx$$

$$\Rightarrow \quad I = \int_0^\pi \frac{(\pi - x)\cdot\sin x}{1+\cos^2 x}\,dx$$

$\Rightarrow \quad 2I = \int_0^{\pi} \frac{\pi \cdot \sin x}{1+\cos^2 x}\, dx$ $1\frac{1}{2}$

Put $\cos x = t \Rightarrow -\sin x\, dx = dt$

$\Rightarrow I = -\frac{\pi}{2} \cdot \int_1^{-1} \frac{dt}{1+t^2} = \frac{\pi}{2} \times 2 \times \int_0^1 \frac{dt}{1+t^2}$

$= \pi[\tan^{-1} t]_0^1 = \frac{\pi^2}{4}$ $1\frac{1}{2}$

17. $\overrightarrow{AB} = \hat{i} + 4\hat{j} - \hat{k}$ 1

$\overrightarrow{CD} = -2\hat{i} - 8\hat{j} + 2\hat{k}$ 1

Let required angle be θ.

Then $\cos\theta = \frac{\overrightarrow{AB} \cdot \overrightarrow{CD}}{|\overrightarrow{AB}||\overrightarrow{CD}|} = \frac{-2-32-2}{\sqrt{18}\sqrt{72}} = -1$ 1

$\Rightarrow \theta = 180^\circ$ or π $\frac{1}{2}$

Since $\theta = \pi$ so $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are collinear. $\frac{1}{2}$

18. LHS $= \begin{vmatrix} a+b+c & -c & -b \\ -c & a+b+c & -a \\ -b & -a & a+b+c \end{vmatrix}$

$C_2 \to C_2 + C_1, C_3 \to C_3 + C_1$

$= \begin{vmatrix} a+b+c & a+b & a+c \\ -c & a+b & -(a+c) \\ -b & -(a+b) & (a+c) \end{vmatrix}$ $1\frac{1}{2}$

$= (a+b)(a+c) \begin{vmatrix} a+b+c & 1 & 1 \\ -c & 1 & -1 \\ -b & -1 & 1 \end{vmatrix}$ $\frac{1}{2}$

$C_3 \to C_3 + C_2$

$$= (a+b)(a+c)\begin{vmatrix} a+b+c & 1 & 2 \\ -c & 1 & 0 \\ -b & -1 & 0 \end{vmatrix}$$ 1

$= 2(a + b)(b + c)(c + a) = \text{RHS}.$ 1

19. $\dfrac{dx}{dt} = -\sin t + \dfrac{1}{\tan\frac{t}{2}} \times \left(\sec^2\dfrac{t}{2} \times \dfrac{1}{2}\right) = \dfrac{\cos^2 t}{\sin t}$ 1

$\dfrac{dy}{dt} = \cos t$ $\frac{1}{2}$

$\dfrac{d^2y}{dt^2} = -\sin t \Rightarrow \dfrac{d^2y}{dt^2}\Big]_{t=\frac{\pi}{4}} = -\dfrac{1}{\sqrt{2}}$ 1

$\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt} = \tan t$ $\frac{1}{2}$

$\dfrac{d^2y}{dx^2} = \sec^2 t \cdot \dfrac{dt}{dx} = \sec^4 t \cdot \sin t$

$\Rightarrow \dfrac{d^2y}{dx^2}\Big]_{t=\frac{\pi}{4}} = 2\sqrt{2}$ 1

20. Clearly $a \le a \ \forall\ a \in \mathbb{R} \Rightarrow (a, a) \in R \Rightarrow R$ is reflexive. 1

For transitive:

Let (a, b) $\in$ R and (b, c) $\in$ R, a, b, c $\in \mathbb{R}$

$\Rightarrow a \le b$ and $b \le c \Rightarrow a \le c \Rightarrow (a, c) \in R$

$\Rightarrow$ R is transitive. $1\frac{1}{2}$

For non-symmetric:

Let a = 1, b = 2. As $1 \le 2 \Rightarrow (1, 2) \in R$ but $2 \not\le 1 \Rightarrow (2,1) \notin R$

$\Rightarrow$ R is non-symmetric. $1\frac{1}{2}$

OR

For one-one. Let $x_1, x_2 \in N$.

$f(x_1) = f(x_2) \Rightarrow x_1^2 + x_1 + 1 = x_2^2 + x_2 + 1$

$\Rightarrow (x_1 - x_2)(x_1 + x_2 + 1) = 0$

$\Rightarrow x_1 = x_2$ as $x_1 + x_2 + 1 \neq 0$ $(\because x_1, x_2 \in N)$ $1\frac{1}{2}$

$\Rightarrow$ f is one-one.

For not onto.

for $y = 1 \in N$, there is no $x \in N$ for which $f(x) = 1$ $1\frac{1}{2}$

For f^{-1}: $y = f(x) \Rightarrow y = x^2 + x + 1 \Rightarrow y = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4}$

$$\Rightarrow x = \frac{\sqrt{4y-3}-1}{2}$$

$\therefore f^{-1}(y) = \frac{\sqrt{4y-3}-1}{2}$ or $f^{-1}(x) = \frac{\sqrt{4x-3}-1}{2}$ 1

21. Let the point of contact be $P(x_1, y_1)$

$$\frac{dy}{dx} = \frac{3}{2\sqrt{3x-2}} \qquad \text{(slope of tangent)}$$

$$\Rightarrow m_1 = \left.\frac{dy}{dx}\right]_{(x_1, y_1)} = \frac{3}{2\sqrt{3x_1-2}}$$ 1

also, slope of given line = $2 = m_2$

$m_1 = m_2 \Rightarrow x_1 = \frac{41}{48}$ 1

when $x_1 = \frac{41}{48}, y_1 = \sqrt{\frac{41}{16} - 2} = \frac{3}{4}$ $\therefore P\left(\frac{41}{48}, \frac{3}{4}\right)$ $\frac{1}{2}$

Equation of tangent is: $y - \frac{3}{4} = 2\left(x - \frac{41}{48}\right)$

$\Rightarrow 48x - 24y = 23$ 1

and, Equation of normal is: $y - \frac{3}{4} = \frac{-1}{2}\left(x - \frac{41}{48}\right)$

$\Rightarrow 48x + 96y = 113$ $\frac{1}{2}$

22. Writing $\frac{dy}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x} = \frac{y}{x} + \sqrt{1 + \left(\frac{y}{x}\right)^2}$ $\frac{1}{2}$

Put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ $\frac{1}{2}$

Differential equation becomes $v + x\frac{dv}{dx} = v + \sqrt{1 + v^2}$

$\Rightarrow \int \frac{dv}{\sqrt{1 + v^2}} = \int \frac{dx}{x}$ 1

$\Rightarrow \log\left|v + \sqrt{1 + v^2}\right| = \log|x| + \log c$ 1

$\Rightarrow v + \sqrt{1 + v^2} = cx \Rightarrow y + \sqrt{x^2 + y^2} = cx^2$

when $x = 1, y = 0 \Rightarrow c = 1$ $\frac{1}{2}$

$\therefore y + \sqrt{x^2 + y^2} = x^2$ $\frac{1}{2}$

OR

Given equation is $\frac{dy}{dx} + \frac{2x}{1 + x^2} \cdot y = \frac{4x^2}{1 + x^2}$ $\frac{1}{2}$

I.F. = $e^{\int \frac{2x}{1+x^2} dx} = 1 + x^2$ 1

Solution is given by,

$y \cdot (1 + x^2) = \int \frac{4x^2}{1 + x^2} \cdot (1 + x^2)\, dx = \int 4x^2\, dx$ 1

$\Rightarrow \; y \cdot (1+x^2) = \frac{4x^3}{3} + c$ $\frac{1}{2}$

when $x = 0, y = 0 \Rightarrow c = 0$ $\frac{1}{2}$

$y \cdot (1+x^2) = \frac{4x^3}{3}$ or $y = \frac{4x^3}{3(1+x^2)}$ $\frac{1}{2}$

23. Given lines are: $\frac{x-1}{-3} = \frac{y-2}{\left(\frac{\lambda}{7}\right)} = \frac{z-3}{2}$ and $\frac{x-1}{\left(\frac{-3\lambda}{7}\right)} = \frac{y-5}{1} = \frac{z-6}{-5}$ 1

As lines are perpendicular,

$(-3)\left(\frac{-3\lambda}{7}\right) + \left(\frac{\lambda}{7}\right)(1) + 2(-5) = 0 \Rightarrow \lambda = 7$ 1

So, lines are

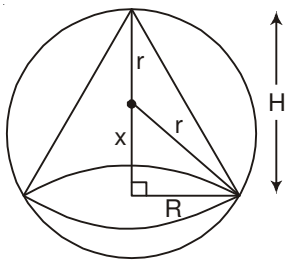
$\frac{x-1}{-3} = \frac{y-2}{1} = \frac{x-3}{2}$ and $\frac{x-1}{-3} = \frac{y-5}{1} = \frac{z-6}{-5}$ $\frac{1}{2}$

Consider $\Delta = \begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = \begin{vmatrix} 0 & 3 & 3 \\ -3 & 1 & 2 \\ -3 & 1 & -5 \end{vmatrix} = -63$ 1

as $\Delta \neq 0 \Rightarrow$ lines are not intersecting. $\frac{1}{2}$

SECTION D

24.



Correct Figure 1

$r^2 = x^2 + R^2$

Now, $V = \frac{1}{3}\pi R^2 H$

$= \frac{1}{3}\pi(r^2 - x^2)(r + x)$

$= \frac{1}{3}\pi(r+x)^2 (r - x)$ 1

$$\frac{dV}{dx} = \frac{1}{3}\pi\left[(r+x)^2(-1) + (r-x).2(r+x)\right]$$

$$= \frac{1}{3}\pi(r+x)(r-3x) \quad 1$$

$$\frac{dV}{dx} = 0 \Rightarrow x = -r \text{ or } x = \frac{r}{3} \quad \frac{1}{2}$$

(Rejected)

$$\frac{d^2V}{dx^2} = \frac{1}{3}\pi\left[(r+x)(-3) + (r-3x)\right] = -\pi H < 0 \quad 1$$

$\Rightarrow$ V is maximum when $x = \frac{r}{3}$.

$$H = r + x = r + \frac{r}{3} = \frac{4r}{3} \quad \frac{1}{2}$$

Maximum volume $V = \frac{1}{3}\pi\left(r + \frac{r}{3}\right)^2\left(r - \frac{r}{3}\right) = \frac{32}{81}\pi r^3$ 1

25. $|A| = -1 \neq 0 \Rightarrow A^{-1}$ exists. 1

$$\text{adj } A = \begin{bmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{bmatrix} \quad 2$$

$$A^{-1} = \frac{1}{|A|} \cdot \text{adj } A = \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix} \quad \frac{1}{2}$$

Given system of equations can be written as AX = B where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 11 \\ -5 \\ -3 \end{bmatrix}$

Now, $X = A^{-1}B$ 1

$$= \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix}\begin{bmatrix} 11 \\ -5 \\ -3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad 1$$

$\Rightarrow x = 1, y = 2, z = 3$ $\frac{1}{2}$

OR

$A = I.A$

$$\Rightarrow \begin{bmatrix} -1 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot A$$ 1

$$R_1 \leftrightarrow R_2$$
$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ -1 & 1 & 2 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot A$$
$$R_2 \to R_2 + R_1, R_3 \to R_3 - 3R_1$$
$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & 5 \\ 0 & -5 & -8 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} \cdot A$$
$$R_2 \to \frac{R_2}{3}$$
$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 5/3 \\ 0 & -5 & -8 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1/3 & 1/3 & 0 \\ 0 & -3 & 1 \end{bmatrix} \cdot A$$
$$R_1 \to R_1 - 2R_2, R_3 \to R_3 + 5R_2$$
$$\Rightarrow \begin{bmatrix} 1 & 0 & -1/3 \\ 0 & 1 & 5/3 \\ 0 & 0 & 1/3 \end{bmatrix} = \begin{bmatrix} -2/3 & 1/3 & 0 \\ 1/3 & 1/3 & 0 \\ 5/3 & -4/3 & 1 \end{bmatrix} \cdot A$$ 4
$$R_3 \to 3R_3$$
$$\Rightarrow \begin{bmatrix} 1 & 0 & -1/3 \\ 0 & 1 & 5/3 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2/3 & 1/3 & 0 \\ 1/3 & 1/3 & 0 \\ 5 & -4 & 3 \end{bmatrix} \cdot A$$
$$R_1 \to R_1 + \frac{1}{3}R_3, R_2 \to R_2 - \frac{5}{3}R_3$$
$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 \\ -8 & 7 & -5 \\ 5 & -4 & 3 \end{bmatrix} \cdot A$$

$\Rightarrow A^{-1} = \begin{bmatrix} 1 & -1 & 1 \\ -8 & 7 & -5 \\ 5 & -4 & 3 \end{bmatrix}$ 1

26. Let E_1 : item is produced by A
E_2 : item is produced by B
E_3 : item is produced by C
A : defective item is found. 1

$P(E_1) = \frac{50}{100}, P(E_2) = \frac{30}{100}, P(E_3) = \frac{20}{100}$ 1

$P(A / E_1) = \frac{1}{100}, P(A / E_2) = \frac{5}{100}, P(A \mid E_3) = \frac{7}{100}$ 1

$$P(E_1 \mid A) = \frac{\frac{50}{100} \times \frac{1}{100}}{\frac{50}{100} \times \frac{1}{100} + \frac{30}{100} \times \frac{5}{100} + \frac{20}{100} \times \frac{7}{100}}$$ 2

$= \frac{5}{34}$ 1

27. Equation of plane is $\begin{vmatrix} x-2 & y-2 & z+1 \\ 1 & 2 & 3 \\ 5 & -2 & 7 \end{vmatrix} = 0$ 2

$\Rightarrow 5x + 2y - 3z = 17$ (Cartesian equation) 1

Vector equation is $\vec{r}.(5\hat{i} + 2\hat{j} - 3\hat{k}) = 17$ 1

Equation of required parallel plane is

$\vec{r} \cdot (5\hat{i} + 2\hat{j} - 3\hat{k}) = (4\hat{i} + 3\hat{j} + \hat{k}) \cdot (5\hat{i} + 2\hat{j} - 3\hat{k})$ 1

$\Rightarrow \vec{r} \cdot (5\hat{i} + 2\hat{j} - 3\hat{k}) = 23$ 1

OR

Let required plane be $a(x + 1) + b(y - 3) + c(z + 4) = 0$ ...(1) 1

Plane contains the given line, so it will also contain the point (1, 1, 0).

So, $2a - 2b + 4c = 0$ or $a - b + 2c = 0$ ...(2) 1

Also, $a + 2b - c = 0$ ...(3) 1

From (2) and (3),

$\frac{a}{-3}=\frac{b}{3}=\frac{c}{3}$ 1

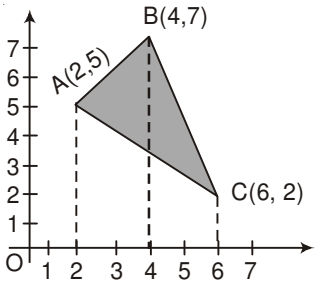
$\therefore$ Required plane is $-3(x+1)+3(y-3)+3(z+4)=0$

$\therefore\ -x+y+z=0$

Also vector equation is: $\vec{r}.(-\hat{i}+\hat{j}+\hat{k})=0$ 1

Length of perpendicular from (2, 1, 4) $=\frac{|-2+1+4|}{\sqrt{(-1)^2+1^2+1^2}}=\sqrt{3}$ 1

28.



Correct Figure 1

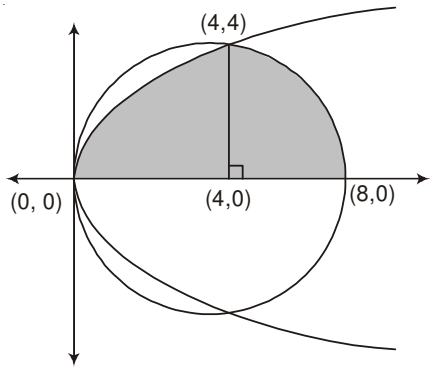
$$\left.\begin{aligned}&\text{Equation of AB}: y=x+3\\&\text{Equation of BC}: y=\frac{-5x}{2}+17\\&\text{Equation of AC}: y=\frac{-3x}{4}+\frac{13}{2}\end{aligned}\right\}$$ $1\frac{1}{2}$

$$\text{Required Area}=\int_2^4(x+3)\,dx+\int_4^6\left(\frac{-5x}{2}+17\right)dx-\int_2^6\left(\frac{-3x}{4}+\frac{13}{2}\right)dx$$ $1\frac{1}{2}$

$$=\left[\frac{(x+3)^2}{2}\right]_2^4+\left[\frac{-5x^2}{4}+17x\right]_4^6-\left[\frac{-3x^2}{8}+\frac{13x}{2}\right]_2^6$$ $1\frac{1}{2}$

$= 7$ $\frac{1}{2}$

OR



Correct Figure 1

Given circle $x^2-8x+y^2=0$

or $(x-4)^2+y^2=4^2$

Point of intersection (0, 0) and (4, 4) 1

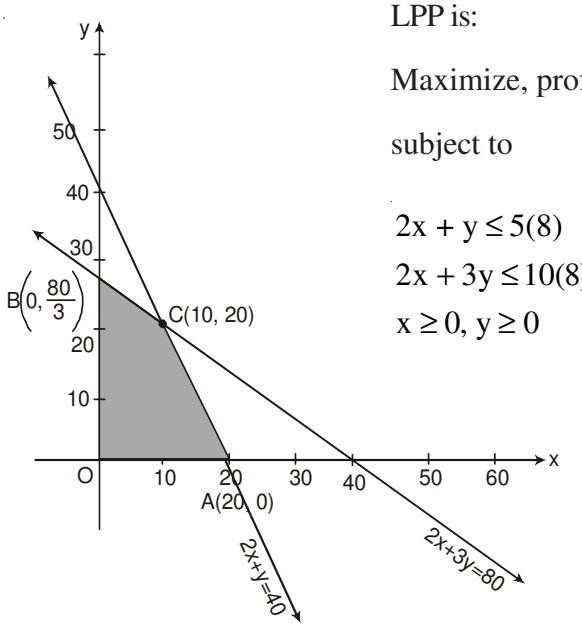
$$\text{Required Area} = \int_0^4 2\sqrt{x}\,dx + \int_4^8 \sqrt{4^2-(x-4)^2}\,dx \qquad 1\frac{1}{2}$$

$$= \left[\frac{4}{3}x^{3/2}\right]_0^4 + \left[\frac{x-4}{2}\sqrt{16-(x-4)^2} + \frac{16}{2}\sin^{-1}\left(\frac{x-4}{4}\right)\right]_4^8 \qquad 1\frac{1}{2}$$

$$= \left(4\pi + \frac{32}{3}\right) \qquad 1$$

Note: A student may also arrive at the answer $\left(8\pi + \frac{64}{3}\right)$ which is double $\left(4\pi + \frac{32}{3}\right)$ because of 'about x-axis'. He/she may be given full marks.

29. Let number of items produced of model A be x and that of model B be y.



LPP is:

Maximize, profit $z = 15x + 10y$ 1

subject to

$$\left.\begin{array}{l} 2x + y \le 5(8) \quad \text{i.e.,} \quad 2x + y \le 40 \\ 2x + 3y \le 10(8) \quad \text{i.e.,} \quad 2x + 3y \le 80 \\ x \ge 0,\ y \ge 0 \end{array}\right\} \qquad 2$$

Correct Figure 2

Corner point	$z = 15x + 10y$	
A(20, 0)	300	
$B\left(0, \frac{80}{3}\right)$	$\frac{800}{3} \approx 266.6$	$\frac{1}{2}$
C(10, 20)	350 ← maximum	

Maximum profit = ₹ 350

when x = 10, y = 20. $\frac{1}{2}$

If a student has interpreted the language of the question in a different way, then the LPP will be of the type:

Maximise profit $z = 15x + 10y$

Subject to $2x + y \le 8$

$2x + 3y \le 8$

$x \ge 0,\ y \ge 0$

This is be accepted and marks may be given accordingly.

QUESTION PAPER CODE 65/1/3
EXPECTED ANSWER/VALUE POINTS

SECTION A

1. $3A = \begin{bmatrix} 9 & 3 \\ 3 & 6 \end{bmatrix} \Rightarrow A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ $\frac{1}{2}+\frac{1}{2}$

2. Order = 2, degree = 2 $\frac{1}{2}+\frac{1}{2}$

3. (fof) (x) = f(x + 1) = x + 2 $\frac{1}{2}$

$\frac{d}{dx}(fof)(x) = 1$ $\frac{1}{2}$

4. d.c.'s = <cos 90°, cos 135°, cos 45°> $\frac{1}{2}$

$= <0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}>$ $\frac{1}{2}$

OR

$\vec{r} = (3\hat{i}+4\hat{j}+5\hat{k})+\lambda(2\hat{i}+2\hat{j}-3\hat{k})$ 1

SECTION B

5. $I = \int \sin x \cdot \log(\cos x)\, dx$

$\cos x = t \Rightarrow I = -\int \log t \cdot 1 dt$ 1

$= -\left[t.\log t - \int \frac{1}{t}.tdt \right]$ $\frac{1}{2}$

$= t(1 - \log t) + C = \cos x(1 - \log(\cos x)) + C$ $\frac{1}{2}$

6. Let $f(x) = (1 - x^2) . \sin x \cos^2 x$

as $f(-x) = -f(x) \Rightarrow$ f is odd function. 1

$\therefore I = 0$ 1

OR

$I = \int_{-1}^{2} \frac{|x|}{x} dx = \int_{-1}^{0} -1 dx + \int_{0}^{2} 1 dx$ 1

$= -1 + 2 = 1$ 1

7. As a, b $\in$ R $\Rightarrow$ ab $\in$ R $\Rightarrow$ ab + 1 $\in$ R $\Rightarrow$ a*b $\in$ R $\Rightarrow$ * is binary. 1

For associative $(a*b)*c = (ab+1)*c = (ab+1)c+1 = abc+c+1$

also, $a*(b*c) = a*(bc+1) = a.(bc+1)+1 = abc+a+1$

In general (a*b)*c $\neq$ a*(b*c) $\Rightarrow$ * is not associative. 1

8. $2A - \begin{bmatrix} -6 & 6 & 0 \\ 9 & 3 & 12 \end{bmatrix} + \begin{bmatrix} 10 & 0 & -10 \\ 35 & 5 & 30 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ 1

$\Rightarrow A = \begin{bmatrix} -8 & 3 & 5 \\ -13 & -1 & -9 \end{bmatrix}$ 1

9. A = {2, 4, 6}, B = {1, 2, 3}, A $\cap$ B = {2}

Now, $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{2}$, $P(A \cap B) = \frac{1}{6}$ 1

as $P(A) \times P(B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} \neq P(A \cap B)$ $\frac{1}{2}$

$\Rightarrow$ A and B are not independent. $\frac{1}{2}$

10. $y' = be^{2x} + 2y \Rightarrow b = \frac{y' - 2y}{e^{2x}}$ $\frac{1}{2}$

differentiating again

$\frac{e^{2x}.(y'' - 2y') - (y' - 2y).2x^{2x}}{(e^{2x})^2} = 0$ 1

$\Rightarrow y'' - 4y' + 4y = 0$ or $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$ $\frac{1}{2}$

11. Let X: getting an odd number

$p = \frac{1}{2},\ q = \frac{1}{2},\ n = 6$ $\frac{1}{2}$

(i) $P(X = 5) = {}^6C_5\left(\frac{1}{2}\right)^6 = \frac{3}{32}$ $\frac{1}{2}$

(ii) $P(X \le 5) = 1 - P(X = 6) = 1 - \frac{1}{64} = \frac{63}{64}$ 1

OR

$k + 2k + 3k = 1$ 1

$\Rightarrow k = \frac{1}{6}$ 1

12. Given $|\hat{a} + \hat{b}| = 1$

As $|\hat{a} + \hat{b}|^2 + |\hat{a} - \hat{b}|^2 = 2(|\hat{a}|^2 + |\hat{b}|^2)$ 1

$\Rightarrow 1 + |\vec{a} - \vec{b}|^2 = 2(1+1)$

$\Rightarrow |\hat{a} - \hat{b}|^2 = 3 \Rightarrow |\hat{a} - \hat{b}| = \sqrt{3}$ 1

OR

$$[\vec{a}\ \ \vec{b}\ \ \vec{c}] = \begin{vmatrix} 2 & 3 & 1 \\ 1 & -2 & 1 \\ -3 & 1 & 2 \end{vmatrix}$$ 1

$= -30$ 1

SECTION C

13. LHS $= \begin{vmatrix} a & b & c \\ a-b & b-c & c-a \\ b+c & c+a & a+b \end{vmatrix}$

$R_1 \to R_1 + R_3$ and taking $a + b + c$ common from R_1

$$= (a+b+c)\begin{vmatrix} 1 & 1 & 1 \\ a-b & b-c & c-a \\ b+c & c+a & a+b \end{vmatrix}$$ 1

$C_1 \to C_1 - C_2, C_2 \to C_2 - C_3$

$$= (a+b+c)\begin{vmatrix} 0 & 0 & 1 \\ a-2b+c & b-2c+a & c-a \\ b-a & c-b & a+b \end{vmatrix}$$ 1

$= (a + b + c)\ [(a - 2b + c)\ (c - b) - (b - 2c + a)\ (b - a)]$ 1

$= (a + b + c)\ (a^2 + b^2 + c^2 - ab - bc - ca)$

$= a^3 + b^3 + c^3 - 3abc = RHS.$ 1

14. $\tan^{-1}\left(\dfrac{4x+6x}{1-(4x)(6x)}\right)=\dfrac{\pi}{4}$ 1

$\Rightarrow \dfrac{10x}{1-24x^2}=1 \Rightarrow 24x^2+10x-1=0$ $1\frac{1}{2}$

$\Rightarrow x=\dfrac{1}{12}$ or $-\dfrac{1}{2}$ 1

as $x=-\dfrac{1}{2}$ does not satisfy the given equation, so $x = \dfrac{1}{12}$ $\frac{1}{2}$

15. Clearly $a \le a\ \forall\ a \in \mathbb{R} \Rightarrow (a, a) \in R \Rightarrow R$ is reflexive. 1

For transitive:

Let $(a, b) \in R$ and $(b, c) \in R$, $a, b, c \in \mathbb{R}$

$\Rightarrow a \le b$ and $b \le c \Rightarrow a \le c \Rightarrow (a, c) \in R$

$\Rightarrow$ R is transitive. $1\frac{1}{2}$

For non-symmetric:

Let $a = 1, b = 2$. As $1 \le 2 \Rightarrow (1, 2) \in R$ but $2 \not\le 1 \Rightarrow (2,1) \notin R$

$\Rightarrow$ R is non-symmetric. $1\frac{1}{2}$

OR

For one-one. Let $x_1, x_2 \in N$.

$f(x_1) = f(x_2) \Rightarrow x_1^2 + x_1 + 1 = x_2^2 + x_2 + 1$

$\Rightarrow (x_1 - x_2)(x_1 + x_2 + 1) = 0$

$\Rightarrow x_1 = x_2$ as $x_1 + x_2 + 1 \neq 0$ ($\because x_1, x_2 \in N$) $1\frac{1}{2}$

$\Rightarrow$ f is one-one.

For not onto.

for $y = 1 \in N$, there is no $x \in N$ for which $f(x) = 1$ $1\frac{1}{2}$

For f^{-1}: $y = f(x) \Rightarrow y = x^2 + x + 1 \Rightarrow y = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4}$

$$\Rightarrow x = \frac{\sqrt{4y-3}-1}{2}$$

$\therefore f^{-1}(y) = \frac{\sqrt{4y-3}-1}{2}$ or $f^{-1}(x) = \frac{\sqrt{4x-3}-1}{2}$ 1

16. Let the point of contact be $P(x_1, y_1)$

$$\frac{dy}{dx} = \frac{3}{2\sqrt{3x-2}}$$ (slope of tangent)

$$\Rightarrow m_1 = \left.\frac{dy}{dx}\right]_{(x_1, y_1)} = \frac{3}{2\sqrt{3x_1-2}}$$ 1

also, slope of given line = $2 = m_2$

$m_1 = m_2 \Rightarrow x_1 = \frac{41}{48}$ 1

when $x_1 = \frac{41}{48}, y_1 = \sqrt{\frac{41}{16} - 2} = \frac{3}{4}$ $\therefore P\left(\frac{41}{48}, \frac{3}{4}\right)$ $\frac{1}{2}$

Equation of tangent is: $y-\frac{3}{4}=2\left(x-\frac{41}{48}\right)$

$\Rightarrow 48x - 24y = 23$ 1

and, Equation of normal is: $y-\frac{3}{4}=\frac{-1}{2}\left(x-\frac{41}{48}\right)$

$\Rightarrow 48x + 96y = 113$ $\frac{1}{2}$

17. $\log(x^2+y^2)=2\tan^{-1}\left(\frac{y}{x}\right)$

differentiating both sides w.r.t. x,

$$\frac{1}{x^2+y^2}\left(2x+2y\frac{dy}{dx}\right)=2\cdot\frac{1}{1+\left(\frac{y}{x}\right)^2}\cdot\left(\frac{x\cdot\frac{dy}{dx}-y\cdot 1}{x^2}\right)$$ 2

$$\Rightarrow \frac{2}{x^2+y^2}\left(x+y\frac{dy}{dx}\right)=\frac{2x^2}{x^2+y^2}\cdot\frac{1}{x^2}\cdot\left(x\frac{dy}{dx}-y\right)$$ 1

$$\Rightarrow (x+y)=(x-y)\frac{dy}{dx}\Rightarrow\frac{dy}{dx}=\frac{x+y}{x-y}$$ 1

OR

Let $u = x^y$, $v = y^x$. Then $u - v = a^b$

$$\Rightarrow \frac{du}{dx}-\frac{dv}{dx}=0 \quad ...(1)$$ 1

Now, log u = y.logx

$$\Rightarrow \frac{1}{u}\cdot\frac{du}{dx}=y\cdot\frac{1}{x}+\log x\cdot\frac{dy}{dx}\Rightarrow\frac{du}{dx}=x^y\left(\frac{y}{x}+\log x\cdot\frac{dy}{dx}\right) \quad ...(2)$$ 1

Again, $\log v = x \cdot \log y$

$$\Rightarrow \frac{1}{v}\cdot\frac{dv}{dx}=x\cdot\frac{1}{y}\cdot\frac{dy}{dx}+\log y\cdot 1\Rightarrow\frac{dv}{dx}=y^x\left(\frac{x}{y}\frac{dy}{dx}+\log y\right) \quad ...(3)$$ 1

From (1), (2) and (3)

$$x^y\left(\frac{y}{x}+\log x\frac{dy}{dx}\right)-y^x\left(\frac{x}{y}\frac{dy}{dx}+\log y\right)=0 \qquad \frac{1}{2}$$

$$\Rightarrow \frac{dy}{dx}=\frac{y^x\cdot\log y-x^{y-1}\cdot y}{x^y\cdot\log x-y^{x-1}\cdot x} \qquad \frac{1}{2}$$

18. $y=(\sin^{-1}x)^2$

$$\Rightarrow y'=2\cdot\sin^{-1}x\cdot\frac{1}{\sqrt{1-x^2}} \qquad 1$$

$$\Rightarrow \sqrt{1-x^2}\cdot y'=2\sin^{-1}x$$

$$\Rightarrow \sqrt{1-x^2}\cdot y''+y'\cdot\frac{1}{2\sqrt{1-x^2}}(-2x)=\frac{2}{\sqrt{1-x^2}} \qquad 2$$

$$\Rightarrow (1-x^2)\cdot y''-xy'=2 \text{ or } (1-x^2)\cdot\frac{d^2y}{dx^2}-x\frac{dy}{dx}-2=0. \qquad 1$$

19. Let $I=\int_0^a f(a-x)\,dx$

Put $a - x = t \Rightarrow -dx = dt$ $\qquad \frac{1}{2}$

$$I=-\int_a^0 f(t)\,dt=\int_0^a f(t)\,dt=\int_0^a f(x)\,dx \qquad \frac{1}{2}$$

II part.

$$I=\int_0^{\pi}\frac{x\sin x}{1+\cos^2 x}dx$$

$$\Rightarrow \quad I=\int_0^{\pi}\frac{(\pi-x)\cdot\sin x}{1+\cos^2 x}dx$$

$$\Rightarrow \quad 2I=\int_0^{\pi}\frac{\pi\cdot\sin x}{1+\cos^2 x}dx \qquad 1\frac{1}{2}$$

Put cos x = t $\Rightarrow$ –sinx dx = dt

$$\Rightarrow I = -\frac{\pi}{2}\cdot\int_{1}^{-1}\frac{dt}{1+t^2} = \frac{\pi}{2}\times 2\times\int_{0}^{1}\frac{dt}{1+t^2}$$

$$= \pi[\tan^{-1} t]_0^1 = \frac{\pi^2}{4} \qquad 1\frac{1}{2}$$

20. $I = \int \frac{\cos x}{(1+\sin x)(2+\sin x)}\,dx.$ Put sin x = t $\frac{1}{2}$

$$= \int \frac{dt}{(1+t)(2+t)} = \int\left(\frac{1}{1+t} - \frac{1}{2+t}\right)dt \qquad 2$$

$$= \log\left|\frac{1+t}{2+t}\right| + c = \log\left|\frac{1+\sin x}{2+\sin x}\right| + c \qquad 1+\frac{1}{2}$$

21. I.F. = $e^{-\int\frac{2x}{1+x^2}dx} = \frac{1}{1+x^2}$ 1

Solution is given by,

$$y\cdot\left(\frac{1}{1+x^2}\right) = \int\frac{x^2+2}{1+x^2}\,dx \qquad 1\frac{1}{2}$$

$$y\cdot\frac{1}{1+x^2} = \int\left(1+\frac{1}{1+x^2}\right)dx = x + \tan^{-1} x + c \qquad 1\frac{1}{2}$$

or y = (1 + x^2) (x + $\tan^{-1}$x + c)

OR

Given equation can be written as

$$\int\frac{dy}{2e^{-y}-1} = \int\frac{dx}{x+1}$$

$$\Rightarrow \int\frac{e^y}{2-e^y}dy = \int\frac{dx}{x+1} \qquad 1$$

$$\Rightarrow -\log|2-e^y| + \log c = \log|x+1| \qquad 1\frac{1}{2}$$

$\Rightarrow$ (2 – e^y) (x + 1) = c

When $x = 0, y = 0 \Rightarrow c = 1$ 1

$\therefore$ Solution is $(2 - e^y)(x + 1) = 1$ $\frac{1}{2}$

22. $\overrightarrow{AB} = \hat{i} + 4\hat{j} - \hat{k}$ 1

$\overrightarrow{CD} = -2\hat{i} - 8\hat{j} + 2\hat{k}$ 1

Let required angle be θ.

Then $\cos\theta = \dfrac{\overrightarrow{AB} \cdot \overrightarrow{CD}}{|\overrightarrow{AB}||\overrightarrow{CD}|} = \dfrac{-2-32-2}{\sqrt{18}\sqrt{72}} = -1$ 1

$\Rightarrow \theta = 180^\circ$ or π $\frac{1}{2}$

Since $\theta = \pi$ so $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are collinear. $\frac{1}{2}$

23. Given lines are: $\dfrac{x-1}{-3} = \dfrac{y-2}{\left(\frac{\lambda}{7}\right)} = \dfrac{z-3}{2}$ and $\dfrac{x-1}{\left(\frac{-3\lambda}{7}\right)} = \dfrac{y-5}{1} = \dfrac{z-6}{-5}$ 1

As lines are perpendicular,

$(-3)\left(\dfrac{-3\lambda}{7}\right) + \left(\dfrac{\lambda}{7}\right)(1) + 2(-5) = 0 \Rightarrow \lambda = 7$ 1

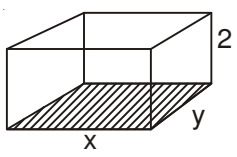
So, lines are

$\dfrac{x-1}{-3} = \dfrac{y-2}{1} = \dfrac{x-3}{2}$ and $\dfrac{x-1}{-3} = \dfrac{y-5}{1} = \dfrac{z-6}{-5}$ $\frac{1}{2}$

Consider $\Delta = \begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = \begin{vmatrix} 0 & 3 & 3 \\ -3 & 1 & 2 \\ -3 & 1 & -5 \end{vmatrix} = -63$ 1

as $\Delta \neq 0 \Rightarrow$ lines are not intersecting. $\frac{1}{2}$

24. $V = 2xy \Rightarrow 2xy = 8$ (given) 1



$\Rightarrow y = \dfrac{4}{x}$

Now, cost, $C = 70xy + 45 \times 2 \times (2x + 2y)$ 1

$= 280 + 180x + \frac{720}{x}$ 1

$\frac{dC}{dx} = 180 - \frac{720}{x^2}$ 1

$\frac{dC}{dx} = 0 \Rightarrow x = 2m$ $\frac{1}{2}$

$\frac{d^2C}{dx^2} = \frac{1440}{x^3} = 180 > 0$ at $x = 2$ $\frac{1}{2}$

$\Rightarrow$ C is minimum at $x = 2m$. $\frac{1}{2}$

Minimum cost $= 280 + 180(2) + \frac{720}{2} = ₹\ 1{,}000$ $\frac{1}{2}$

25. $|A| = 4 \neq 0 \Rightarrow A^{-1}$ exists. 1

$$\text{adj } A = \begin{bmatrix} -2 & 0 & 2 \\ 5 & -2 & -1 \\ 1 & 2 & -1 \end{bmatrix}$$ 2

$$\therefore\ A^{-1} = \frac{1}{|A|} \cdot \text{adj } A = \frac{1}{4}\begin{bmatrix} -2 & 0 & 2 \\ 5 & -2 & -1 \\ 1 & 2 & -1 \end{bmatrix}$$ $\frac{1}{2}$

Given system of equations can be written as $AX = B$ where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 6 \\ 7 \\ 12 \end{bmatrix}$

$\therefore\ X = A^{-1} \cdot B$ 1

$$= \frac{1}{4}\begin{bmatrix} -2 & 0 & 2 \\ 5 & -2 & -1 \\ 1 & 2 & -1 \end{bmatrix}\begin{bmatrix} 6 \\ 7 \\ 12 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$ 1

$\Rightarrow x = 3,\ y = 1,\ z = 2$ $\frac{1}{2}$

OR

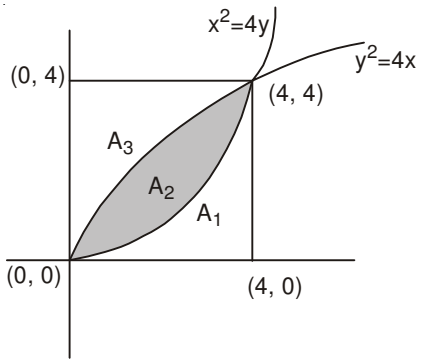
$A = I.A$

$$\Rightarrow \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot A$$ 1

$$R_2 \to R_2 + R_1$$
$$\Rightarrow \begin{bmatrix} 1 & 2 & -2 \\ 0 & 5 & -2 \\ 0 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot A$$
$$R_2 \to \frac{R_2}{5}$$
$$\Rightarrow \begin{bmatrix} 1 & 2 & -2 \\ 0 & 1 & -2/5 \\ 0 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1/5 & 1/5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot A$$
$$R_1 \to R_1 - 2R_2, R_3 \to R_3 + 2R_2$$
$$\Rightarrow \begin{bmatrix} 1 & 0 & -6/5 \\ 0 & 1 & -2/5 \\ 0 & 0 & 1/5 \end{bmatrix} = \begin{bmatrix} 3/5 & -2/5 & 0 \\ 1/5 & 1/5 & 0 \\ 2/5 & 2/5 & 1 \end{bmatrix} \cdot A$$
$$R_3 \to 5R_3$$
$$\Rightarrow \begin{bmatrix} 1 & 0 & -6/5 \\ 0 & 1 & -2/5 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3/5 & -2/5 & 0 \\ 1/5 & 1/5 & 0 \\ 2 & 2 & 5 \end{bmatrix} \cdot A$$
$$R_1 \to R_1 + \frac{6}{5}R_3, R_2 \to R_2 + \frac{2}{5}R_3$$
$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix} \cdot A$$ 4

$$\Rightarrow A^{-1} = \begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix}$$ 1

26. Correct Figure 1



Point of intersection are (0, 0) and (4, 4) 1

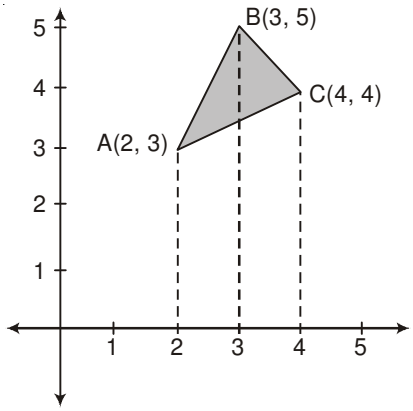
here, $A_1 = \int_0^4 \frac{x^4}{4} dx = \frac{16}{3}$ ...(1) 1

$A_2 = \int_0^4 \left(2\sqrt{x} - \frac{x^2}{4} \right) dx = \frac{16}{3}$ ..(2) $1\frac{1}{2}$

$A_3 = \int_0^4 \frac{y^2}{4} dy = \frac{16}{3}$ ...(3) 1

From (1), (2) and (3), $A_1 = A_2 = A_3$. $\frac{1}{2}$

OR



Correct Figure 1

$$\left.\begin{array}{l} \text{Equation of AB}: y = 2x - 1 \\ \text{Equation of BC}: y = -x + 8 \\ \text{Equation of AC}: y = \frac{1}{2}(x + 4) \end{array}\right\}$$ $1\frac{1}{2}$

Required Area $= \int_2^3 (2x - 1) dx + \int_3^4 (-x + 8) dx - \int_2^4 \left(\frac{x + 4}{2} \right) dx$ $1\frac{1}{2}$

$= \left[x^2 - x \right]_2^3 + \left[\frac{-x^2}{2} + 8x \right]_3^4 - \frac{1}{2} \left[\frac{x^2}{2} + 4x \right]_2^4$ $1\frac{1}{2}$

$= 4 + \frac{9}{2} - 7 = \frac{3}{2}$ $\frac{1}{2}$

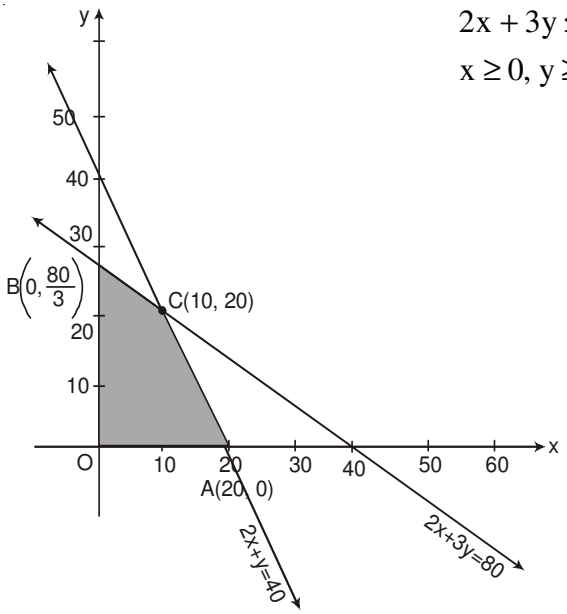
27. Let number of items produced of model A be x and that of model B be y.

LPP is:

Maximize, profit $z = 15x + 10y$ 1

subject to

$2x + y \le 5(8)$ i.e., $2x + y \le 40$
$2x + 3y \le 10(8)$ i.e., $2x + 3y \le 80$
$x \ge 0, y \ge 0$ } 2



Correct Figure 2

Corner point	$z = 15x + 10y$	
A(20, 0)	300	
$B\left(0, \frac{80}{3}\right)$	$\frac{800}{3} \approx 266.6$	$\frac{1}{2}$
C(10, 20)	350 ← maximum	

Maximum profit = ₹ 350

when x = 10, y = 20. $\frac{1}{2}$

If a student has interpreted the language of the question in a different way, then the LPP will be of the type:

Maximise profit $z = 15x + 10y$

Subject to $2x + y \le 8$

$2x + 3y \le 8$

$x \ge 0, y \ge 0$

This is be accepted and marks may be given accordingly.

28. Equation of plane is $\begin{vmatrix} x-2 & y-2 & z+1 \\ 1 & 2 & 3 \\ 5 & -2 & 7 \end{vmatrix} = 0$ 2

$\Rightarrow 5x + 2y - 3z = 17$ (Cartesian equation) 1

Vector equation is $\vec{r}.(5\hat{i}+2\hat{j}-3\hat{k}) = 17$ 1

Equation of required parallel plane is

$\vec{r} \cdot (5\hat{i} + 2\hat{j} - 3\hat{k}) = (4\hat{i} + 3\hat{j} + \hat{k}) \cdot (5\hat{i} + 2\hat{j} - 3\hat{k})$ 1

$\Rightarrow \vec{r} \cdot (5\hat{i} + 2\hat{j} - 3\hat{k}) = 23$ 1

OR

Let required plane be $a(x + 1) + b(y - 3) + c(z + 4) = 0$ ...(1) 1

Plane contains the given line, so it will also contain the point (1, 1, 0).

So, $2a - 2b + 4c = 0$ or $a - b + 2c = 0$ ...(2) 1

Also, $a + 2b - c = 0$ ...(3) 1

From (2) and (3),

$\frac{a}{-3} = \frac{b}{3} = \frac{c}{3}$ 1

$\therefore$ Required plane is $-3(x + 1) + 3(y - 3) + 3(z + 4) = 0$

$\therefore$ $-x + y + z = 0$

Also vector equation is: $\vec{r}.(-\hat{i} + \hat{j} + \hat{k}) = 0$ 1

Length of perpendicular from (2, 1, 4) = $\frac{|-2+1+4|}{\sqrt{(-1)^2+1^2+1^2}} = \sqrt{3}$ 1

29. X = no. of kings = 0, 1, 2 $\frac{1}{2}$

$P(X = 0) = P(\text{no king}) = \frac{48}{52} \times \frac{47}{51} = \frac{188}{221}$ 1

$P(X = 1) = P(\text{one king and one non-king}) = \frac{4}{52} \times \frac{48}{51} \times 2 = \frac{32}{221}$ 1

$P(X = 2) = P(\text{two kings}) = \frac{4}{52} \times \frac{3}{51} = \frac{1}{221}$ 1

Probability distribution is given by

X	0	1	2
P(X)	$\frac{188}{221}$	$\frac{32}{221}$	$\frac{1}{221}$

$\frac{1}{2}$

Now, Mean = $\Sigma X \cdot P(X) = \frac{34}{221}$ or $\frac{2}{13}$ 1

and $\text{Var}(X) = \Sigma X^2 \cdot P(X) - [\Sigma X \cdot P(X)]^2$

$= \frac{36}{221} - \left(\frac{34}{221}\right)^2 = \frac{6800}{48841}$ or $\frac{400}{2873}$ 1

Strictly Confidential — (For Internal and Restricted Use Only)

Senior School Certificate Examination

March 2019

Marking Scheme — Mathematics (041) 65/2/1, 65/2/2, 65/2/3

General Instructions:

1. You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully. **Evaluation is a 10-12 days mission for all of us. Hence, it is necessary that you put in your best efforts in this process.**
2. Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one's own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. **However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and marks be awarded to them.**
3. The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
4. If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled.
5. If a question does not have any parts, marks must be awarded in the left hand margin and encircled.
6. If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out.
7. No marks to be deducted for the cumulative effect of an error. It should be penalized only once.
8. A full scale of marks 0 to 100 has to be used. Please do not hesitate to award full marks if the answer deserves it.
9. Every examiner has to necessarily do evaluation work for full working hours i.e. 8 hours every day and evaluate 25 answer books per day.
10. Ensure that you do not make the following common types of errors committed by the Examiner in the past:-
    - Leaving answer or part thereof unassessed in an answer book.
    - Giving more marks for an answer than assigned to it.
    - Wrong transfer of marks from the inside pages of the answer book to the title page.
    - Wrong question wise totaling on the title page.
    - Wrong totaling of marks of the two columns on the title page.
    - Wrong grand total.
    - Marks in words and figures not tallying.
    - Wrong transfer of marks from the answer book to online award list.
    - Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.)
    - Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
11. While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as (X) and awarded zero (0) Marks.
12. Any unassessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
13. The Examiners should acquaint themselves with the guidelines given in the Guidelines for spot Evaluation before starting the actual evaluation.
14. Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
15. The Board permits candidates to obtain photocopy of the Answer Book on request in an RTI application and also separately as a part of the re-evaluation process on payment of the processing charges.

QUESTION PAPER CODE 65/2/1
EXPECTED ANSWER/VALUE POINTS

SECTION A

1. $|A'|\,|A| = |I| \Rightarrow |A|^2 = 1$ $\frac{1}{2}$

$\therefore \quad |A| = 1$ or $|A| = -1$ $\frac{1}{2}$

2. For $x < 0$, $y = x\,|x| = -x^2$ $\frac{1}{2}$

$\therefore \quad \frac{dy}{dx} = -2x$ $\frac{1}{2}$

3. Order = 2, Degree not defined $\frac{1}{2} + \frac{1}{2}$

4. D. Rs are 1, 1, 1
$\therefore$ Direction cosines of the line are:

$$\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$$ 1

OR

Equation of the line is:

$$\frac{x-2}{1} = \frac{y+1}{1} = \frac{z-4}{-2}$$ 1

SECTION B

5. $\left.\begin{array}{l} \forall\ a, b \in \mathbb{R},\ \sqrt{a^2+b^2} \in \mathbb{R} \\ \therefore\ * \text{ is a binary operation on } \mathbb{R} \end{array}\right\}$ 1

Also,

$$\left.\begin{array}{l} a*(b*c) = a*\sqrt{b^2+c^2} = \sqrt{a^2+b^2+c^2} \\ (a*b)*c = \sqrt{a^2+b^2}*c = \sqrt{a^2+b^2+c^2} \end{array}\right\} \Rightarrow a*(b*c) = (a*b)*c$$
$$\therefore \quad * \text{ is Associative}$$ 1

6. $(A - 2I)\,(A - 3I) = \begin{bmatrix} 2 & 2 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$ 1

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$$ 1

7. $\int \sqrt{3-2x-x^2}\,dx = \int \sqrt{2^2-(x+1)^2}\,dx$ 1

$= \frac{x+1}{2}\sqrt{3-2x-x^2} + 2\sin^{-1}\left(\frac{x+1}{2}\right) + c$ 1

8. $\int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x}dx = \int (\sec x \cdot \tan x + \operatorname{cosec} x \cdot \cot x)\,dx$ 1

$= \sec x - \operatorname{cosec} x + c$ 1

OR

$\int \frac{x-3}{(x-1)^3}e^x dx = \int e^x\{(x-1)^{-2} - 2(x-1)^{-3}\}dx$ 1

$= e^x(x-1)^{-2} + c$
or
$\frac{e^x}{(x-1)^2} + c$ 1

9. Differentiating $y = Ae^{2x} + Be^{-2x}$, we get

$\frac{dy}{dx} = 2Ae^{2x} - 2Be^{-2x}$, differentiate again to get, 1

$\frac{d^2y}{dx^2} = 4Ae^{2x} + 4Be^{-2x} = 4y$ or $\frac{d^2y}{dx^2} - 4y = 0$ 1

10. Let θ be the angle between $\vec{a}$ & $\vec{b}$, then

$\sin\theta = \frac{|\vec{a}\times\vec{b}|}{|\vec{a}||\vec{b}|} = \frac{7}{2.7} = \frac{1}{2}$ $1\frac{1}{2}$

$\Rightarrow \quad \theta = \frac{\pi}{6}$ or $\frac{5\pi}{6}$ $\frac{1}{2}$

OR

$(-3\hat{i}+7\hat{j}+5\hat{k})\cdot\{(-5\hat{i}+7\hat{j}-3\hat{k})\times(7\hat{i}-5\hat{j}-3\hat{k})\} = \begin{vmatrix} -3 & 7 & 5 \\ -5 & 7 & -3 \\ 7 & -5 & -3 \end{vmatrix}$ 1

$= -264$ $\frac{1}{2}$

$\therefore$ Volume of cuboid = 264 cubic units $\frac{1}{2}$

11. $P(\bar{A}) = 0.7 \Rightarrow 1 - P(A) = 0.7 \Rightarrow P(A) = 0.3$ $\frac{1}{2}$

$P(A \cap B) = P(A) \cdot P(B|A) = 0.3 \times 0.5 = 0.15$ $\frac{1}{2}$

$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.15}{0.7} = \frac{15}{70}$ or $\frac{3}{14}$ 1

12. (i) $P(3 \text{ heads}) = {}^5C_3\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^2 = \frac{5}{16}$ 1

(ii) P(At most 3 heads) = P ($r \leq 3$)

$$\left.\begin{aligned} &= 1 - P(4 \text{ heads or } 5 \text{ heads}) \\ &= 1 - \left\{ {}^5C_4\left(\frac{1}{2}\right)^4\left(\frac{1}{2}\right) + {}^5C_5\left(\frac{1}{2}\right)^5 \right\} \\ &= \frac{26}{32} \text{ or } \frac{13}{16} \end{aligned}\right\}$$ 1

OR

X = No. of heads in simultaneous toss of two coins.

X:	0	1	2	$\frac{1}{2}$
P(x):	1/4	1/2	1/4	$1\frac{1}{2}$

SECTION C

13. R = {(1, 2), (2, 3), (3, 4), (4, 5), (5, 6)}

For $1 \in A$, $(1, 1) \notin R \Rightarrow$ R is not reflexive 1

For $1, 2 \in A$, $(1, 2) \in R$ but $(2, 1) \notin R \Rightarrow$ R is not symmetric $1\frac{1}{2}$

For $1, 2, 3 \in A$, $(1, 2), (2, 3) \in R$ but $(1, 3) \notin R \Rightarrow$ R is not transitive $1\frac{1}{2}$

OR

One-One: Let for $x_1, x_2 \in N$, $f(x_1) = f(x_2) \Rightarrow 4x_1 + 3 = 4x_2 + 3 \Rightarrow x_1 = x_2$ 2

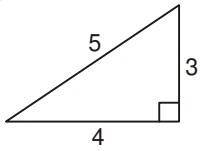
$\therefore$ 'f ' is one-one

Onto: co-domain of f = Range of f = Y 1

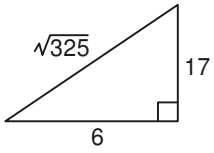
$\therefore$ 'f ' is onto

$\therefore$ f is invertible with, $f^{-1}: Y \to N$ and $f^{-1}(y) = \frac{y-3}{4}$ or $f^{-1}(x) = \frac{x-3}{4}, x \in Y$ 1

14. $\sin\left[\cos^{-1}\left(\frac{4}{5}\right) + \tan^{-1}\left(\frac{2}{3}\right)\right]$



$= \sin\left[\tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{2}{3}\right)\right]$ 1



$= \sin\left[\tan^{-1}\left(\frac{3/4 + 2/3}{1 - 3/4 \cdot 2/3}\right)\right] = \sin\left[\tan^{-1}\left(\frac{17}{6}\right)\right]$ $1\frac{1}{2}$

$= \sin\left[\sin^{-1}\left(\frac{17}{\sqrt{325}}\right)\right] = \frac{17}{\sqrt{325}}$ $1\frac{1}{2}$

15. $\begin{vmatrix} 3a & -a+b & -a+c \\ -b+a & 3b & -b+c \\ -c+a & -c+b & 3c \end{vmatrix}$

$= \begin{vmatrix} a+b+c & -a+b & -a+c \\ a+b+c & 3b & -b+c \\ a+b+c & -c+b & 3c \end{vmatrix}$ (By applying, $C_1 \to C_1 + C_2 + C_3$) 1

$= \begin{vmatrix} a+b+c & -a+b & -a+c \\ 0 & 2b+a & a-b \\ 0 & a-c & 2c+a \end{vmatrix}$ (By applying, $R_2 \to R_2 - R_1$; $R_3 \to R_3 - R_1$) 2

$= (a + b + c)\ \{4bc + 2ab + 2ac + a^2 - (a^2 - ac - ba + bc)\}$

$= 3(a + b + c)\ (ab + bc + ca)$ 1

16. $x\sqrt{1+y} + y\sqrt{1+x} = 0 \Rightarrow x\sqrt{1+y} = -y\sqrt{1+x}$ $\frac{1}{2}$

Squaring to get: $x^2(1 + y) = y^2(1 + x)$ $\frac{1}{2}$

Simplifying to get: $(x - y)(x + y + xy) = 0$ 1

As, $x \neq y \quad \therefore \quad y = -\frac{x}{1+x}$ 1

Differentiating w.r.t. 'x', we get:

$$\frac{dy}{dx} = \frac{-1(1+x) - (-x)\cdot 1}{(1+x)^2} = \frac{-1}{(1+x)^2}$$ 1

OR

$(\cos x)^y = (\sin y)^x \Rightarrow y \cdot \log(\cos x) = x \cdot \log(\sin y)$ 1

Differentiating w.r.t 'x'

$$\Rightarrow \frac{dy}{dx} \cdot \log(\cos x) + y(-\tan x) = \log(\sin y) + x \cdot \cot y \cdot \frac{dy}{dx}$$ 2

$$\Rightarrow \frac{dy}{dx} = \frac{y \cdot \tan x + \log(\sin y)}{\log(\cos x) - x \cot y}$$ 1

17. $(x - a)^2 + (y - b)^2 = c^2,\ c > 0$

Differentiating both sides with respect to 'x', we get

$$2(x - a) + 2(y - b) \cdot \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x-a}{y-b}$$ $1\frac{1}{2}$

Differentiating again with respect to 'x', we get;

$$\frac{d^2y}{dx^2} = -\frac{(y-b) - (x-a)\cdot\frac{dy}{dx}}{(y-b)^2} = \frac{-c^2}{(y-b)^3}$$ (By substituting '$\frac{dy}{dx}$') $1+\frac{1}{2}$

$$\frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}} = \frac{\left[1+\frac{(x-a)^2}{(y-b)^2}\right]^{3/2}}{-\frac{c^2}{(y-b)^3}} = \frac{\frac{c^3}{(y-b)^3}}{-\frac{c^2}{(y-b)^3}} = -c$$ 1

Which is a constant independent of 'a' & 'b'.

18. Let (α, β) be the point on the curve where normal

passes through $(-1, 4)$ $\therefore \alpha^2 = 4\beta$, also $\frac{dy}{dx} = \frac{x}{2}$

Slope of normal at $(\alpha, \beta) = \frac{-1}{\left.\frac{dy}{dx}\right]_{(\alpha,\beta)}} = \frac{-1}{\frac{\alpha}{2}} = \frac{-2}{\alpha}$ 1

Equation of normal: $y - 4 = \frac{-2}{\alpha}(x+1)$ $\frac{1}{2}$

(α, β) lies on the normal $\Rightarrow \beta - 4 = \frac{-2}{\alpha}(\alpha+1)$

Putting $\beta = \frac{\alpha^2}{4}$, we get; $\alpha^3 - 8\alpha + 8 = 0 \Rightarrow (\alpha - 2)(\alpha^2 + 2\alpha - 4) = 0$ 1

For $\alpha = 2$ Equation of normal is: $x + y - 3 = 0$ 1

For $\alpha = \pm\sqrt{5} - 1$; Equation of normal is: $y - 4 = \frac{-2}{\pm\sqrt{5} - 1}(x+1)$ $\frac{1}{2}$

19. $\int \frac{x^2 + x + 1}{(x+2)(x^2+1)} dx = \frac{3}{5}\int \frac{1}{x+2} dx + \frac{1}{5}\int \frac{2x}{x^2+1} dx + \frac{1}{5}\int \frac{1}{x^2+1} dx$ 2

$= \frac{3}{5}\log|x+2| + \frac{1}{5}\log|x^2+1| + \frac{1}{5}\tan^{-1} x + c$ 2

20. $\int_0^a f(x)\,dx = -\int_a^0 f(a-t)dx$ Put $x = a - t, dx = -dt$

Upper limit $= t = a - x = a - a = 0$

Lower limit $= t = a - x = a - 0 = a$

$= \int_0^a f(a-t)\,dt = \int_0^a f(a-x)\,dx$ 1

Let $I = \int_0^{\pi/2} \frac{x}{\sin x + \cos x} dx$ ...(i)

$\Rightarrow I = \int_0^{\pi/2} \frac{\pi/2 - x}{\sin\left(\frac{\pi}{2} - x\right) + \cos\left(\frac{\pi}{2} - x\right)} dx$ $\therefore I = \int_0^{\pi/2} \frac{\pi/2 - x}{\cos x + \sin x} dx$ ...(ii) 1

Adding, (i) and (ii) we get

$$2I = \frac{\pi}{2}\int_0^{\pi/2} \frac{1}{\cos x + \sin x}\,dx = \frac{\pi}{2\sqrt{2}}\int_0^{\pi/2} \frac{1}{\cos\left(x - \frac{\pi}{4}\right)}\,dx$$

$$= \frac{\pi}{2\sqrt{2}}\int_0^{\pi/2} \sec\left(x - \frac{\pi}{4}\right)dx \qquad \frac{1}{2}$$

$$\Rightarrow\ 2I = \frac{\pi}{2\sqrt{2}}\left\{\log\left|\sec\left(x - \frac{\pi}{4}\right) + \tan\left(x - \frac{\pi}{4}\right)\right|\right\}_0^{\frac{\pi}{2}} \qquad \frac{1}{2}$$

$$= \frac{\pi}{2\sqrt{2}}\{\log(\sqrt{2}+1) - \log(\sqrt{2}-1)\} \qquad \frac{1}{2}$$

$$\Rightarrow\ I = \frac{\pi}{4\sqrt{2}}\{\log(\sqrt{2}+1) - \log(\sqrt{2}-1)\} \text{ or } \frac{\pi}{4\sqrt{2}}\log\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right) \qquad \frac{1}{2}$$

21. The given differential equation can be written as:

$$\frac{dy}{dx} = \frac{y}{x} - \tan\left(\frac{y}{x}\right) \qquad \frac{1}{2}$$

Put $\frac{y}{x} = v$ and $\frac{dy}{dx} = v + x\frac{dv}{dx}$, to get 1

$$v + x\frac{dv}{dx} = v - \tan v \Rightarrow x\frac{dv}{dx} = -\tan v$$

$$\Rightarrow\ \cot v\ dv = -\frac{1}{x}dx, \qquad 1$$

Integrating both sides we get,

$$\left.\begin{aligned} &\log|\sin v| = -\log|x| + \log c \\ \Rightarrow\ &\log|\sin v| = \log\left|\frac{c}{x}\right| \end{aligned}\right\} \qquad 1$$

$\therefore$ Solution of differential equation is

$$\sin\left(\frac{y}{x}\right) = \frac{c}{x} \text{ or } x\cdot\sin\left(\frac{y}{x}\right) = c \qquad \frac{1}{2}$$

OR

The given differential equation can be written as:

$$\frac{dy}{dx}+\frac{\cos x}{1+\sin x}\cdot y=\frac{-x}{1+\sin x};$$ $1\frac{1}{2}$

$$\text{I.F.} = e^{\int\frac{\cos x}{1+\sin x}dx}=e^{\log(1+\sin x)}=1+\sin x$$ 1

$\therefore$ Solution of the given differential equation is:

$$y(1+\sin x)=\int\frac{-x}{1+\sin x}\times(1+\sin x)\,dx+c$$ 1

$$\Rightarrow\ y(1+\sin x)=\frac{-x^2}{2}+c \text{ or } y=\frac{-x^2}{2(1+\sin x)}+\frac{c}{1+\sin x}$$ $\frac{1}{2}$

22. $\vec{a}\cdot\frac{(\vec{b}+\vec{c})}{|\vec{b}+\vec{c}|}=1$ 1

$$\Rightarrow\ (\hat{i}+\hat{j}+\hat{k})\cdot\{(2+\lambda)\hat{i}+6\hat{j}-2\hat{k}\}=\sqrt{(2+\lambda)^2+36+4}$$ $1\frac{1}{2}$

$$\Rightarrow\ \lambda+6=\sqrt{(2+\lambda)^2+40}$$

Squaring to get

$$\lambda^2+12\lambda+36=\lambda^2+4\lambda+44\ \Rightarrow\ \lambda=1$$ $\frac{1}{2}$

$\therefore$ Unit vector along $(\vec{b}+\vec{c})$ is $\frac{3}{7}\hat{i}+\frac{6}{7}\hat{j}-\frac{2}{7}\hat{k}$ 1

23. Lines are perpendicular

$\therefore$ $-3(3\lambda)+2\lambda(2)+2(-5)=0\Rightarrow\lambda=-2$ 2

$$\begin{vmatrix}x_2-x_1 & y_2-y_1 & z_2-z_1\\ a_1 & b_1 & c_1\\ a_2 & b_2 & c_3\end{vmatrix}=\begin{vmatrix}1-1 & 1-2 & 6-3\\ -3 & 2(-2) & 2\\ 3(-2) & 2 & -5\end{vmatrix}=-63\neq0$$ $1\frac{1}{2}$

$\therefore$ Lines are not intersecting $\frac{1}{2}$

SECTION D

24. $|A| = 11$; Adj $(A) = \begin{bmatrix} -1 & 1 & 2 \\ 8 & -19 & 6 \\ -3 & 14 & -5 \end{bmatrix}$ 1+2

$\therefore \quad A^{-1} = \frac{1}{|A|} \cdot \text{Adj}\, A = \frac{1}{11}\begin{bmatrix} -1 & 1 & 2 \\ 8 & -19 & 6 \\ -3 & 14 & -5 \end{bmatrix}$ $\frac{1}{2}$

Taking; $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$; $B = \begin{bmatrix} 8 \\ 5 \\ 7 \end{bmatrix}$

The system of equations in matrix form is

$A \cdot X = B \qquad \therefore X = A^{-1} \cdot B$ 1

$\therefore$ Solution is:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{11}\begin{bmatrix} -1 & 1 & 2 \\ 8 & -19 & 6 \\ -3 & 14 & -5 \end{bmatrix}\begin{bmatrix} 8 \\ 5 \\ 7 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$ 1

$\therefore \quad x = 1,\ y = 1,\ z = 1$ $\frac{1}{2}$

OR

We know: $A = IA$

$$\Rightarrow \begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$ 1

$R_2 \to R_2 - 2R_1$

$$\Rightarrow \begin{bmatrix} 2 & 0 & -1 \\ 1 & 1 & 2 \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$R_1 \leftrightarrow R_2$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 2 & 0 & -1 \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$R_2 \to R_2 - 2R_1$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & -2 & -5 \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 5 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$R_2 \to R_2 + 2R_3$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 5 & -2 & 2 \\ 0 & 0 & 1 \end{bmatrix} A$$

$R_2 \leftrightarrow R_3$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 0 & 0 & 1 \\ 5 & -2 & 2 \end{bmatrix} A$$

$R_2 \to R_2 - 3R_3$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix} A$$

$R_1 \to R_1 - R_2 - 2R_3$

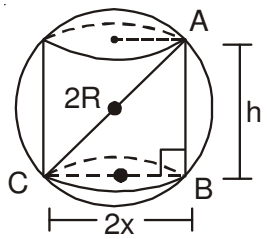
$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix} A$$

4

$$\therefore \quad A^{-1} = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$$

1

25. Correct Figure 1



ln rt. ΔABC; $4x^2 + h^2 = 4R^2$, $x^2 = \frac{4R^2 - h^2}{4}$ 1

V(Volume of cylinder) = $\pi x^2 h = \frac{\pi}{4}(4R^2 h - h^3)$ 1

$V'(h) = \frac{\pi}{4}(4R^2 - 3h^2)$; $V''(h) = \frac{\pi}{4}(-6h)$ $\frac{1}{2} + \frac{1}{2}$

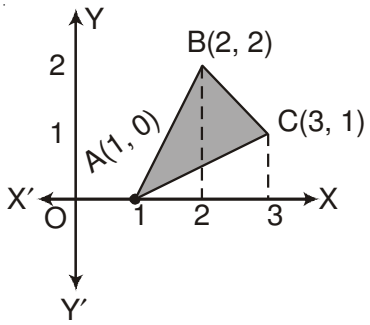
$V'(h) = 0 \Rightarrow h = \frac{2R}{\sqrt{3}}$ 1

$V''\left(\frac{2R}{\sqrt{3}}\right) = \frac{-6\pi}{4}\left(\frac{2R}{\sqrt{3}}\right) < 0 \Rightarrow$ Volume 'V' is max. $\frac{1}{2}$

for $h = \frac{2R}{\sqrt{3}}$

Max. Volume: $V = \frac{4}{3\sqrt{3}}\pi R^3$ $\frac{1}{2}$

26. Correct Figure 1



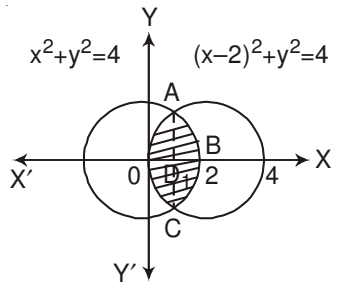
Equation of line AB : $y = 2(x - 1)$
Equation of line BC : $y = 4 - x$
Equation of line AC : $y = \frac{1}{2}(x - 1)$ $1\frac{1}{2}$

$\text{ar}(\Delta ABC) = 2\int_1^2 (x - 1)\,dx + \int_2^3 (4 - x)\,dx - \frac{1}{2}\int_1^3 (x - 1)\,dx$ $1\frac{1}{2}$

$= (x - 1)^2\Big]_1^2 - \frac{1}{2}(4 - x)^2\Big]_2^3 - \frac{1}{4}(x - 1)^2\Big]_1^3$ $1\frac{1}{2}$

$= 1 + \frac{3}{2} - 1 = \frac{3}{2}$ $\frac{1}{2}$

OR



Correct Figure 1

Getting the point of intersection as x = 1 1

Area (OABCO) = 4 × ar(ABD)

$= 4\int_1^2 \sqrt{2^2 - x^2}\, dx$ 2

$= 4\left\{\frac{x\sqrt{4-x^2}}{2} + 2\sin^{-1}\left(\frac{x}{2}\right)\right\}_1^2$ 1

$= \left(\frac{8\pi}{3} - 2\sqrt{3}\right)$ 1

27. Let $\vec{a} = \hat{i} + \hat{j} - 2\hat{k},\ \vec{b} = 2\hat{i} - \hat{j} + \hat{k},\ \vec{c} = \hat{i} + 2\hat{j} + \hat{k}$

$$(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 3 \\ 0 & 1 & 3 \end{vmatrix} = -9\hat{i} - 3\hat{j} + \hat{k}$$ 2

Vector equation of plane is:

$$\{\vec{r} - (\hat{i} + \hat{j} - 2\hat{k})\} \cdot (-9\hat{i} - 3\hat{j} + \hat{k}) = 0$$

$\Rightarrow \quad \vec{r} \cdot (-9\hat{i} - 3\hat{j} + \hat{k}) + 14 = 0$ 1

Cartesian Equation of plane is: $-9x - 3y + z + 14 = 0$ 1

Equation of plane through (2, 3, 7) and parallel to above plane is:

$-9(x - 2) - 3(y - 3) + (z - 7) = 0$

$\Rightarrow \quad -9x - 3y + z + 20 = 0$ 1

Distance between parallel planes = $\left|\frac{-14 + 20}{\sqrt{91}}\right| = \frac{6}{\sqrt{91}}$ 1

OR

Equation of line: $\frac{x-2}{3}=\frac{y+1}{4}=\frac{z-2}{2}$ 1

Equation of plane: $\begin{vmatrix} x-2 & y & z-3 \\ -1 & 1 & 2 \\ 1 & 2 & 1 \end{vmatrix}=0$ 1

$\Rightarrow$ $-3(x-2)+3y-3(z-3)=0$

$\Rightarrow$ $x-y+z-5=0$ 1

General point on line: $\frac{x-2}{3}=\frac{y+1}{4}=\frac{z-2}{2}$

is: P(3k + 2, 4k – 1, 2k + 2); Putting in the equation of plane 1

we get, $3k+2-4k+1+2k+2=5 \Rightarrow k=0$ 1

$\therefore$ Point of intersection is: (2, –1, 2) 1

28. Let E_1 = Event that two-headed coin is chosen
E_2 = Event that biased coin is chosen
E_3 = Event that unbiased coin is chosen
A = Event that coin tossed shows head 1

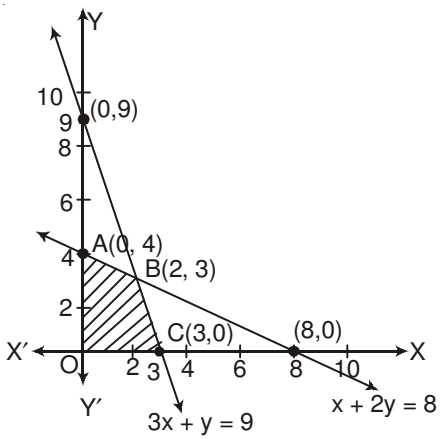
Then, $P(E_1) = P(E_2) = P(E_3) = 1/3$ 1

$P(A|E_1) = 1,\ P(A \mid E_2)=\frac{75}{100}=\frac{3}{4},\ P(A \mid E_3)=\frac{1}{2}$ $1\frac{1}{2}$

$$P(E_1|A) = \frac{P(E_1)\cdot P(A \mid E_1)}{P(E_1)\cdot P(A \mid E_1)+P(E_2)\cdot P(A \mid E_2)+P(E_3)\cdot P(A \mid E_3)}$$

$$=\frac{\frac{1}{3}\cdot 1}{\frac{1}{3}\cdot 1+\frac{1}{3}\cdot\frac{3}{4}+\frac{1}{3}\cdot\frac{1}{2}}=\frac{4}{9}$$ $2+\frac{1}{2}$

29.



let the company produce: Goods A = x units

Goods B = y units

then, the linear programming problem is:

Maximize profit: $z = 40x + 50y$ (In ₹) $\frac{1}{2}$

Subject to constraints:

$$\left.\begin{aligned} 3x + y &\le 9 \\ x + 2y &\le 8 \\ x, y &\ge 0 \end{aligned}\right\} \quad 2\frac{1}{2}$$

Correct graph: 2

Corner point	Value of z (₹)	
A(0, 4)	200	
B(2, 3)	230 (Max)	$\frac{1}{2}$
C(3, 0)	120	

$\therefore$ Maximum profit = ₹ 230 at:

Goods A produced = 2 units, Goods B produced = 3 units $\frac{1}{2}$

QUESTION PAPER CODE 65/2/2
EXPECTED ANSWER/VALUE POINTS

SECTION A

1. $|A| = |B| = 0$ $\frac{1}{2}$

$\Rightarrow |AB| = 0$ $\frac{1}{2}$

2. $\frac{d}{dx}(e^{\sqrt{3x}}) = \frac{\sqrt{3}}{2\sqrt{x}} e^{\sqrt{3x}}$ 1

3. Order = 2, Degree not defined $\frac{1}{2} + \frac{1}{2}$

4. D. Rs are 1, 1, 1
$\therefore$ Direction cosines of the line are:

$\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$ 1

OR

Equation of the line is:

$\frac{x-2}{1} = \frac{y+1}{1} = \frac{z-4}{-2}$ 1

SECTION B

5. $\int \sqrt{3-2x-x^2}\, dx = \int \sqrt{2^2-(x+1)^2}\, dx$ 1

$= \frac{x+1}{2}\sqrt{3-2x-x^2} + 2\sin^{-1}\left(\frac{x+1}{2}\right) + c$ 1

6. $|A| = p^2 - 4$ $\frac{1}{2}$

$|A^3| = 125 \Rightarrow |A|^3 = 125 \Rightarrow |A| = 5$ 1

$\therefore \; p^2 - 4 = 5 \Rightarrow p = \pm 3$ $\frac{1}{2}$

7. $\forall\, a, b \in \mathbb{R}, \sqrt{a^2+b^2} \in \mathbb{R}$
$\therefore *$ is a binary operation on $\mathbb{R}$ 1

Also,

$$\left.\begin{array}{l} a*(b*c)=a*\sqrt{b^2+c^2}=\sqrt{a^2+b^2+c^2} \\ (a*b)*c=\sqrt{a^2+b^2}*c=\sqrt{a^2+b^2+c^2} \end{array}\right\} \Rightarrow a*(b*c)=(a*b)*c \left.\begin{array}{l} \\ \therefore \quad *\text{ is Associative} \end{array}\right\} \quad 1$$

8. Let θ be the angle between $\vec{a}$ & $\vec{b}$, then

$$\sin\theta=\frac{|\vec{a}\times\vec{b}|}{|\vec{a}||\vec{b}|}=\frac{7}{2.7}=\frac{1}{2} \qquad 1\frac{1}{2}$$

$$\Rightarrow \quad \theta=\frac{\pi}{6} \text{ or } \frac{5\pi}{6} \qquad \frac{1}{2}$$

OR

$$(-3\hat{i}+7\hat{j}+5\hat{k})\cdot\{(-5\hat{i}+7\hat{j}-3\hat{k})\times(7\hat{i}-5\hat{j}-3\hat{k})\} = \begin{vmatrix} -3 & 7 & 5 \\ -5 & 7 & -3 \\ 7 & -5 & -3 \end{vmatrix} \qquad 1$$

$$= -264 \qquad \frac{1}{2}$$

$\therefore$ Volume of cuboid = 264 cubic units $\qquad \frac{1}{2}$

9. (i) P(3 heads) = $^5C_3\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^2=\frac{5}{16}$ 1

(ii) P(At most 3 heads) = P ($r \le 3$)

$$\left.\begin{array}{l} =1-P(4 \text{ heads or } 5 \text{ heads}) \\ =1-\left\{{}^5C_4\left(\frac{1}{2}\right)^4\left(\frac{1}{2}\right)+{}^5C_5\left(\frac{1}{2}\right)^5\right\} \\ =\frac{26}{32} \text{ or } \frac{13}{16} \end{array}\right\} \quad 1$$

OR

X = No. of heads in simultaneous toss of two coins.

X:	0	1	2	$\frac{1}{2}$
P(x):	1/4	1/2	1/4	$1\frac{1}{2}$

10. $\int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x} dx = \int (\sec x \cdot \tan x + \text{cosec } x \cdot \cot x) dx$ 1

$= \sec x - \text{cosec } x + c$ 1

OR

$\int \frac{x-3}{(x-1)^3} e^x dx = \int e^x \{(x-1)^{-2} - 2(x-1)^{-3}\} dx$ 1

$= e^x (x-1)^{-2} + c$

or

$\frac{e^x}{(x-1)^2} + c$ 1

11. $P(\bar{A}) = 0.7 \Rightarrow 1 - P(A) = 0.7 \Rightarrow P(A) = 0.3$ $\frac{1}{2}$

$P(A \cap B) = P(A) \cdot P(B/A) = 0.3 \times 0.5 = 0.15$ $\frac{1}{2}$

$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{0.15}{0.7} = \frac{15}{70}$ or $\frac{3}{14}$ 1

12. Given differential equation can be written as:

$\frac{dy}{dx} = e^x \cdot e^y \Rightarrow e^{-y} dy = e^x dx$ 1

Integrating both sides, we get

$-e^{-y} = e^x + c$ 1

SECTION C

13. $\sin\left[\cos^{-1}\left(\frac{4}{5}\right) + \tan^{-1}\left(\frac{2}{3}\right)\right]$

$= \sin\left[\tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{2}{3}\right)\right]$ 1

$= \sin\left[\tan^{-1}\left(\frac{3/4 + 2/3}{1 - 3/4 \cdot 2/3}\right)\right] = \sin\left[\tan^{-1}\left(\frac{17}{6}\right)\right]$ $1\frac{1}{2}$

$= \sin\left[\sin^{-1}\left(\frac{17}{\sqrt{325}}\right)\right] = \frac{17}{\sqrt{325}}$ $1\frac{1}{2}$

14. $\begin{vmatrix} 3a & -a+b & -a+c \\ -b+a & 3b & -b+c \\ -c+a & -c+b & 3c \end{vmatrix}$

$= \begin{vmatrix} a+b+c & -a+b & -a+c \\ a+b+c & 3b & -b+c \\ a+b+c & -c+b & 3c \end{vmatrix}$ (By applying, $C_1 \to C_1 + C_2 + C_3$) 1

$= \begin{vmatrix} a+b+c & -a+b & -a+c \\ 0 & 2b+a & a-b \\ 0 & a-c & 2c+a \end{vmatrix}$ (By applying, $R_2 \to R_2 - R_1$; $R_3 \to R_3 - R_1$) 2

$= (a + b + c)\ \{4bc + 2ab + 2ac + a^2 - (a^2 - ac - ba + bc)\}$

$= 3(a + b + c)\ (ab + bc + ca)$ 1

15. $R = \{(1, 2), (2, 3), (3, 4), (4, 5), (5, 6)\}$

For $1 \in A$, $(1, 1) \notin R \Rightarrow$ R is not reflexive 1

For $1, 2 \in A$, $(1, 2) \in R$ but $(2, 1) \notin R \Rightarrow$ R is not symmetric $1\frac{1}{2}$

For $1, 2, 3 \in A$, $(1, 2), (2, 3) \in R$ but $(1, 3) \notin R \Rightarrow$ R is not transitive $1\frac{1}{2}$

OR

One-One: Let for $x_1, x_2 \in N$, $f(x_1) = f(x_2) \Rightarrow 4x_1 + 3 = 4x_2 + 3 \Rightarrow x_1 = x_2$ 2

$\therefore$ 'f ' is one-one

Onto: co-domain of f = Range of f = Y 1

$\therefore$ 'f ' is onto

$\therefore$ f is invertible with, $f^{-1}: Y \to N$ and $f^{-1}(y) = \frac{y-3}{4}$ or $f^{-1}(x) = \frac{x-3}{4}, x \in Y$ 1

16. $x\sqrt{1+y} + y\sqrt{1+x} = 0 \Rightarrow x\sqrt{1+y} = -y\sqrt{1+x}$ $\frac{1}{2}$

Squaring to get: $x^2 (1 + y) = y^2(1 + x)$ $\frac{1}{2}$

Simplifying to get: $(x - y) (x + y + xy) = 0$ 1

As, $x \neq y \quad \therefore \quad y = -\frac{x}{1+x}$ 1

Differentiating w.r.t. 'x', we get:

$$\frac{dy}{dx} = \frac{-1(1+x) - (-x)\cdot 1}{(1+x)^2} = \frac{-1}{(1+x)^2}$$ 1

OR

$(\cos x)^y = (\sin y)^x \Rightarrow y \cdot \log(\cos x) = x \cdot \log(\sin y)$ 1

Differentiating w.r.t 'x'

$$\Rightarrow \frac{dy}{dx} \cdot \log(\cos x) + y(-\tan x) = \log(\sin y) + x \cdot \cot y \cdot \frac{dy}{dx}$$ 2

$$\Rightarrow \frac{dy}{dx} = \frac{y \cdot \tan x + \log(\sin y)}{\log(\cos x) - x \cot y}$$ 1

17. $\int_0^a f(x)\,dx = -\int_a^0 f(a-t)dx$ Put $x = a - t, dx = -dt$

Upper limit = $t = a - x = a - a = 0$

Lower limit = $t = a - x = a - 0 = a$

$= \int_0^a f(a-t)\,dt = \int_0^a f(a-x)\,dx$ 1

Let $I = \int_0^{\pi/2} \frac{x}{\sin x + \cos x} dx$ ...(i)

$$\Rightarrow I = \int_0^{\pi/2} \frac{\pi/2 - x}{\sin\left(\frac{\pi}{2} - x\right) + \cos\left(\frac{\pi}{2} - x\right)} dx \qquad \therefore I = \int_0^{\pi/2} \frac{\pi/2 - x}{\cos x + \sin x} dx \quad ...(ii)$$ 1

Adding (i) and (ii) we get

$$2I = \frac{\pi}{2}\int_0^{\pi/2} \frac{1}{\cos x + \sin x} dx = \frac{\pi}{2\sqrt{2}}\int_0^{\pi/2} \frac{1}{\cos\left(x - \frac{\pi}{4}\right)} dx$$

$$= \frac{\pi}{2\sqrt{2}}\int_0^{\pi/2} \sec\left(x - \frac{\pi}{4}\right) dx$$ $\frac{1}{2}$

$\Rightarrow \quad 2I = \frac{\pi}{2\sqrt{2}}\left\{\log\left|\sec\left(x-\frac{\pi}{4}\right)+\tan\left(x-\frac{\pi}{4}\right)\right|\right\}_0^{\pi/2}$ $\frac{1}{2}$

$= \frac{\pi}{2\sqrt{2}}\{\log|\sqrt{2}+1|-\log|\sqrt{2}-1|\}$ $\frac{1}{2}$

$\Rightarrow \quad I = \frac{\pi}{4\sqrt{2}}\{\log(\sqrt{2}+1)-\log(\sqrt{2}-1)\}$ or $\frac{\pi}{4\sqrt{2}}\log\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right)$ $\frac{1}{2}$

18. $\int\frac{x^2+x+1}{(x+2)(x^2+1)}dx = \frac{3}{5}\int\frac{1}{x+2}dx+\frac{1}{5}\int\frac{2x}{x^2+1}dx+\frac{1}{5}\int\frac{1}{x^2+1}dx$ 2

$= \frac{3}{5}\log|x+2|+\frac{1}{5}\log|x^2+1|+\frac{1}{5}\tan^{-1}x+c$ 2

19. The given differential equation can be written as:

$\frac{dy}{dx}=\frac{y}{x}-\tan\left(\frac{y}{x}\right)$ $\frac{1}{2}$

Put $\frac{y}{x}=v$ and $\frac{dy}{dx}=v+x\frac{dv}{dx}$, to get 1

$v+x\frac{dv}{dx}=v-\tan v \Rightarrow x\frac{dv}{dx}=-\tan v$

$\Rightarrow \quad \cot v\, dv = -\frac{1}{x}dx,$ 1

Integrating both sides we get,

$\left.\begin{aligned}&\log|\sin v|=-\log|x|+\log c\\ \Rightarrow\quad &\log|\sin v|=\log\left|\frac{c}{x}\right|\end{aligned}\right\}$ 1

$\therefore$ Solution of differential equation is

$\sin\left(\frac{y}{x}\right)=\frac{c}{x}$ or $x\cdot\sin\left(\frac{y}{x}\right)=c$ $\frac{1}{2}$

OR

The given differential equation can be written as:

$\frac{dy}{dx}+\frac{\cos x}{1+\sin x}\cdot y=\frac{-x}{1+\sin x};$ $1\frac{1}{2}$

I.F. $= e^{\int \frac{\cos x}{1+\sin x} dx} = e^{\log(1+\sin x)} = 1+\sin x$ 1

$\therefore$ Solution of the given differential equation is:

$y(1 + \sin x) = \int \frac{-x}{1+\sin x} \times (1+\sin x)\,dx + c$ 1

$\Rightarrow$ $y(1 + \sin x) = \frac{-x^2}{2} + c$ or $y = \frac{-x^2}{2(1+\sin x)} + \frac{c}{(1+\sin x)}$ $\frac{1}{2}$

20. $\vec{a} \cdot \frac{(\vec{b}+\vec{c})}{|\vec{b}+\vec{c}|} = 1$ 1

$\Rightarrow$ $(\hat{i}+\hat{j}+\hat{k}) \cdot \{(2+\lambda)\hat{i} + 6\hat{j} - 2\hat{k}\} = \sqrt{(2+\lambda)^2 + 36 + 4}$ $1\frac{1}{2}$

$\Rightarrow$ $\lambda + 6 = \sqrt{(2+\lambda)^2 + 40}$

Squaring to get

$\lambda^2 + 12\lambda + 36 = \lambda^2 + 4\lambda + 44 \Rightarrow \lambda = 1$ $\frac{1}{2}$

$\therefore$ Unit vector along $(\vec{b}+\vec{c})$ is $\frac{3}{7}\hat{i} + \frac{6}{7}\hat{j} - \frac{2}{7}\hat{k}$ 1

21. $e^{y/x} = \frac{x}{a+bx}$, taking log, on both sides, we get $\frac{y}{x} = \log x - \log(a + bx)$ 1

Differentiating with respect to 'x'

$$\left.\begin{aligned} &\frac{x \cdot y' - y}{x^2} = \frac{1}{x} - \frac{b}{a+bx} = \frac{a}{(a+bx)x} \\ \Rightarrow\ &x \cdot y' - y = \frac{ax}{a+bx} \qquad \text{...(i)} \end{aligned}\right\}$$ $1\frac{1}{2}$

Differentiating with respect to 'x'

$\Rightarrow$ $x \cdot y'' + y' - y' = \frac{(a+bx) \cdot a - ax \cdot b}{(a+bx)^2} = \left(\frac{a}{a+bx}\right)^2$ 1

$$\left.\begin{aligned} &\Rightarrow\ x \cdot y'' = \left(\frac{a}{a+bx}\right)^2 \Rightarrow\ x^3 \cdot y'' = \left(\frac{ax}{a+bx}\right)^2 \\ &\Rightarrow\ x^3 \frac{d^2y}{dx^2} = \left\{x \cdot \frac{dy}{dx} - y\right\}^2 \quad \text{(Using (i))} \end{aligned}\right\}$$ $\frac{1}{2}$

22. Let Edge = x cm, then

V(Volume of cube) = x^3, S(Surface area) = $6x^2$ 1

$$\frac{dv}{dt} = 8\,cm^3/s \Rightarrow 3x^2\frac{dx}{dt} = 8 \Rightarrow \frac{dx}{dt} = \frac{8}{3x^2}$$ 1

$$\frac{ds}{dt} = 12x\frac{dx}{dt} = 12x \cdot \frac{8}{3x^2} = \frac{32}{x}$$ 1

$$\therefore \quad \left.\frac{ds}{dt}\right]_{x=12} = \frac{32}{12} = \frac{8}{3}\,cm^2/s$$ 1

23. Equation of plane:

$$\left.\begin{array}{l} \begin{vmatrix} x-2 & y-5 & z+3 \\ -2-2 & -3-5 & 5+3 \\ 5-2 & 3-5 & -3+3 \end{vmatrix} = 0 \\ \Rightarrow \begin{vmatrix} x-2 & y-5 & z+3 \\ -4 & -8 & 8 \\ 3 & -2 & 0 \end{vmatrix} = 0 \end{array}\right\}$$ 2

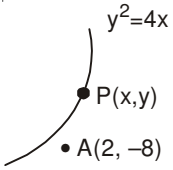
$\Rightarrow$ (x – 2)(16) – (y – 5)(–24) + (z + 3)(32) = 0

$\Rightarrow$ 2x + 3y + 4z – 7 = 0 1

Vector form: $\vec{r} \cdot (2\hat{i} + 3\hat{j} + 4\hat{k}) - 7 = 0$ 1

SECTION D

24. Let P(x, y) be any point on the curve $y^2 = 4x$



$$z = AP = \sqrt{(x-2)^2 + (y+8)^2}$$ 1

$$\text{let } s = z^2 = \left(\frac{y^2}{4} - 2\right)^2 + (y+8)^2$$ 1

$$\frac{ds}{dy} = 2\left(\frac{y^2}{4} - 2\right)\left(\frac{y}{2}\right) + 2(y+8) = \frac{y^3}{4} + 16$$ 1

$$\frac{d^2s}{dy^2} = \frac{3y^2}{4}$$ $\frac{1}{2}$

Let $\frac{ds}{dy} = 0 \Rightarrow y^3 = -64 \Rightarrow y = -4$ 1

$$\left.\frac{d^2s}{dy^2}\right]_{y=-4} = \frac{3(16)}{4} > 0$$ $\frac{1}{2}$

$\therefore$ s or z is minimum at y = –4; x = $\frac{y^2}{4} = 4$

$\therefore$ The nearest point is P(4, –4) 1

25. Let $f(x) = x^2 + 2 + e^{2x}$, a = 1, b = 3, nh = 2 1

then,

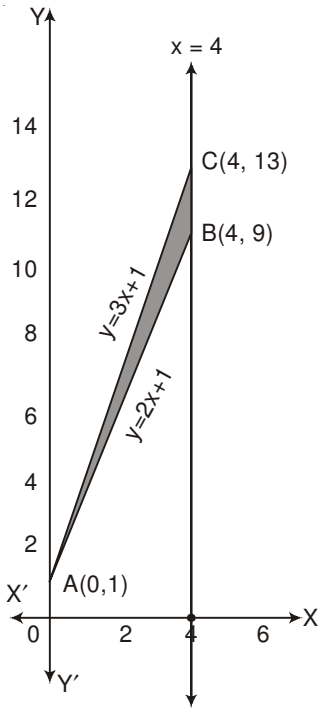
$$f(1) + f(1+h) + f(1+2h) + ... + f(1+\overline{n-1}\,h)$$

$$= 3n + h^2[1^2 + 2^2 + ... + (n-1)^2] + 2h[1 + 2 + ... + (n-1)] + e^2[1 + e^{2h} + e^{4h} + ... + e^{2(n-1)h}]$$ 1

$$= 3n + h^2\frac{(n-1)(n)(2n-1)}{6} + \frac{2h(n-1)(n)}{2} + \frac{e^2(e^{2nh}-1)}{e^{2h}-1}$$ 2

$$\int_1^3 (x^2 + 2 + e^{2x})dx = \lim_{h\to 0}\left[3(nh) + \frac{(nh-h)(nh)(2nh-h)}{6} + (nh)(nh-h) + \frac{e^2}{2}\cdot\frac{2h}{e^{2h}-1}(e^{2nh}-1)\right]$$ 1

$$= 6 + \frac{2\times 2\times 4}{6} + 2\times 2 + \frac{e^2}{2}\times 1\times(e^4-1)$$
$$= \frac{38}{3} + \frac{e^2(e^4-1)}{2}$$ 1



OR

Correct figure 1

Points of intersection of given lines

are A(0, 1), B(4, 9), C(4, 13) $1\frac{1}{2}$

$\therefore$ Req. Area $= \int_0^4 (3x+1)\,dx - \int_0^4 (2x+1)\,dx$ 2

$= \int_0^4 x\,dx = \frac{1}{2}x^2\Big]_0^4 = 8$ $1\frac{1}{2}$

26. $|A| = 11$; Adj $(A) = \begin{bmatrix} -1 & 1 & 2 \\ 8 & -19 & 6 \\ -3 & 14 & -5 \end{bmatrix}$ 1+2

$\therefore \quad A^{-1} = \frac{1}{|A|} \cdot \text{Adj}\, A = \frac{1}{11}\begin{bmatrix} -1 & 1 & 2 \\ 8 & -19 & 6 \\ -3 & 14 & -5 \end{bmatrix}$ $\frac{1}{2}$

Taking; $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$; $B = \begin{bmatrix} 8 \\ 5 \\ 7 \end{bmatrix}$

The system of equations in matrix form is

$A \cdot X = B \qquad \therefore X = A^{-1} \cdot B$ 1

$\therefore$ Solution is:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{11}\begin{bmatrix} -1 & 1 & 2 \\ 8 & -19 & 6 \\ -3 & 14 & -5 \end{bmatrix}\begin{bmatrix} 8 \\ 5 \\ 7 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$ 1

$\therefore \quad x = 1,\ y = 1,\ z = 1$ $\frac{1}{2}$

OR

We know: $A = IA$

$$\Rightarrow \begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$ 1

$R_2 \to R_2 - 2R_1$

$$\Rightarrow \begin{bmatrix} 2 & 0 & -1 \\ 1 & 1 & 2 \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$R_1 \leftrightarrow R_2$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 2 & 0 & -1 \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$R_2 \to R_2 - 2R_1$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & -2 & -5 \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 5 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$R_2 \to R_2 + 2R_3$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 5 & -2 & 2 \\ 0 & 0 & 1 \end{bmatrix} A$$ 4

$R_2 \leftrightarrow R_3$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 0 & 0 & 1 \\ 5 & -2 & 2 \end{bmatrix} A$$

$R_2 \to R_2 - 3R_3$

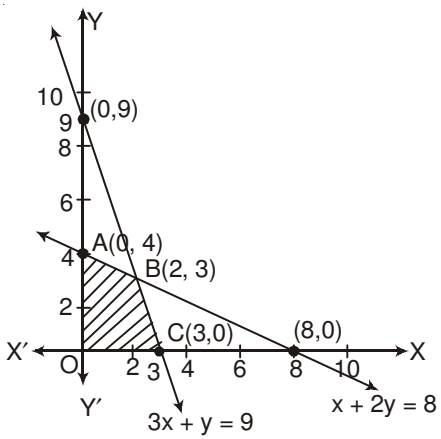
$$\Rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix} A$$

$R_1 \to R_1 - R_2 - 2R_3$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix} A$$

$$\therefore \quad A^{-1} = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$$ 1

27.



let the company produce: Goods A = x units

Goods B = y units

then, the linear programming problem is:

Maximize profit: z = 40x + 50y (In ₹) $\frac{1}{2}$

Subject to constraints:

$$\left.\begin{aligned} 3x + y \le 9 \\ x + 2y \le 8 \\ x, y \ge 0 \end{aligned}\right\}$$ $2\frac{1}{2}$

Correct graph. 2

Corner point	Value of z (₹)	
A(0, 4)	200	
B(2, 3)	230 (Max)	$\frac{1}{2}$
C(3, 0)	120	

∴ Maximum profit = ₹ 230 at:

Goods A produced = 2 units, Goods B produced = 3 units $\frac{1}{2}$

28. Let E_1 = Event that two-headed coin is chosen
E_2 = Event that biased coin is chosen
E_3 = Event that unbiased coin is chosen
A = Event that coin tossed shows head

1

Then, $P(E_1) = P(E_2) = P(E_3) = 1/3$ 1

$P(A|E_1) = 1,\ P(A \mid E_2) = \frac{75}{100} = \frac{3}{4},\ P(A \mid E_3) = \frac{1}{2}$ $1\frac{1}{2}$

$$P(E_1|A) = \frac{P(E_1) \cdot P(A \mid E_1)}{P(E_1) \cdot P(A \mid E_1) + P(E_2) \cdot P(A \mid E_2) + P(E_3) \cdot P(A \mid E_3)}$$

$$= \frac{\frac{1}{3} \cdot 1}{\frac{1}{3} \cdot 1 + \frac{1}{3} \cdot \frac{3}{4} + \frac{1}{3} \cdot \frac{1}{2}} = \frac{4}{9}$$ $2+\frac{1}{2}$

29. Let $\vec{a} = \hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + 2\hat{j} + \hat{k}$

$$(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 3 \\ 0 & 1 & 3 \end{vmatrix} = -9\hat{i} - 3\hat{j} + \hat{k}$$ 2

Vector equation of plane is:

$$\{\vec{r} - (\hat{i} + \hat{j} - 2\hat{k})\} \cdot (-9\hat{i} - 3\hat{j} + \hat{k}) = 0$$

$\Rightarrow \quad \vec{r} \cdot (-9\hat{i} - 3\hat{j} + \hat{k}) + 14 = 0$ 1

Cartesian Equation of plane is: $-9x - 3y + z + 14 = 0$ 1

Equation of plane through (2, 3, 7) and parallel to above plane is:

$-9(x - 2) - 3(y - 3) + (z - 7) = 0$

$\Rightarrow \quad -9x - 3y + z + 20 = 0$ 1

Distance between parallel planes $= \left| \frac{-14 + 20}{\sqrt{91}} \right| = \frac{6}{\sqrt{91}}$ 1

OR

Equation of line: $\frac{x - 2}{3} = \frac{y + 1}{4} = \frac{z - 2}{2}$ 1

Equation of plane: $\begin{vmatrix} x - 2 & y & z - 3 \\ -1 & 1 & 2 \\ 1 & 2 & 1 \end{vmatrix} = 0$ 1

$\Rightarrow \quad -3(x - 2) + 3y - 3(z - 3) = 0$

$\Rightarrow \quad x - y + z - 5 = 0$ 1

General point on line: $\frac{x - 2}{3} = \frac{y + 1}{4} = \frac{z - 2}{2}$

is: $P(3k + 2, 4k - 1, 2k + 2)$; Putting in the equation of plane 1

we get, $3k + 2 - 4k + 1 + 2k + 2 = 5 \Rightarrow k = 0$ 1

$\therefore$ Point of intersection is: $(2, -1, 2)$ 1

QUESTION PAPER CODE 65/2/3
EXPECTED ANSWER/VALUE POINTS

SECTION A

1. $\frac{dy}{dx} = 2ae^{2x} \Rightarrow \frac{dy}{dx} = 2(y-5)$ $\frac{1}{2}+\frac{1}{2}$

2. $\frac{dy}{dx} = -\frac{\sqrt{3}\sin(\sqrt{3x})}{2\sqrt{x}}$ 1

3. $|A'|\,|A| = |I| \Rightarrow |A|^2 = 1$ $\frac{1}{2}$

   $\Rightarrow |A| = 1$ or $|A| = -1$ $\frac{1}{2}$

4. D. Rs are 1, 1, 1
∴ Direction cosines of the line are:

$$\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$$ 1

OR

Equation of the line is:

$$\frac{x-2}{1} = \frac{y+1}{1} = \frac{z-4}{-2}$$ 1

SECTION B

5. $\overrightarrow{AB} = 3\hat{i} - \hat{j} - 2\hat{k};\ \overrightarrow{AC} = 9\hat{i} - 3\hat{j} - 6\hat{k}$ 1

Clearly, $\overrightarrow{AC} = 3\cdot\overrightarrow{AB} \Rightarrow \overrightarrow{AC} \parallel \overrightarrow{AB}$, ∴ A, B & C are Collinear 1

OR

$$\vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 3 \\ 3 & 5 & -2 \end{vmatrix} = -17\hat{i} + 13\hat{j} + 7\hat{k}$$ $1\frac{1}{2}$

∴ $|\vec{a}\times\vec{b}| = \sqrt{289+169+49} = \sqrt{507}$ $\frac{1}{2}$

6. $\int \frac{x-5}{(x-3)^3} \cdot e^x \, dx = \int e^x \left[\frac{(x-3)-2}{(x-3)^3} \right] dx$ $\frac{1}{2}$

$= \int e^x [(x-3)^{-2} - 2(x-3)^{-3}] dx$ $\frac{1}{2}$

$= e^x (x-3)^{-2} + c$ or $\frac{e^x}{(x-3)^2} + c$ 1

7. $\forall\, a, b \in \mathbb{R}, \sqrt{a^2+b^2} \in \mathbb{R}$
$\therefore *$ is a binary operation on $\mathbb{R}$ 1

Also,

$a*(b*c) = a*\sqrt{b^2+c^2} = \sqrt{a^2+b^2+c^2}$
$(a*b)*c = \sqrt{a^2+b^2}*c = \sqrt{a^2+b^2+c^2}$
$\Rightarrow a*(b*c) = (a*b)*c$
$\therefore$ * is Associative 1

8. $(A - 2I)(A - 3I) = \begin{bmatrix} 2 & 2 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$ 1

$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$ 1

9. $\int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x} dx = \int (\sec x \cdot \tan x + \operatorname{cosec} x \cdot \cot x) dx$ 1

$= \sec x - \operatorname{cosec} x + c$ 1

OR

$\int \frac{x-3}{(x-1)^3} e^x dx = \int e^x \{(x-1)^{-2} - 2(x-1)^{-3}\} dx$ 1

$= e^x (x-1)^{-2} + c$
or
$\frac{e^x}{(x-1)^2} + c$ 1

10. (i) P(3 heads) $= {}^5C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 = \frac{5}{16}$ 1

(ii) P(At most 3 heads) = P ($r \le 3$)

$$\left.\begin{aligned} &= 1 - P(4 \text{ heads or } 5 \text{ heads}) \\ &= 1 - \left\{ {}^5C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right) + {}^5C_5 \left(\frac{1}{2}\right)^5 \right\} \\ &= \frac{26}{32} \text{ or } \frac{13}{16} \end{aligned}\right\} \quad 1$$

OR

X = No. of heads in simultaneous toss of two coins.

X:	0	1	2	$\frac{1}{2}$
P(x):	1/4	1/2	1/4	$1\frac{1}{2}$

11. $P(\bar{A}) = 0.7 \Rightarrow 1 - P(A) = 0.7 \Rightarrow P(A) = 0.3$ $\frac{1}{2}$

$P(A \cap B) = P(A) \cdot P(B/A) = 0.5 \times 0.3 = 0.15$ $\frac{1}{2}$

$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{0.15}{0.7} = \frac{15}{70}$ or $\frac{3}{14}$ 1

12. Differenatiae $y = Ae^{2x} + Be^{-2x}$, we get

$\frac{dy}{dx} = 2Ae^{2x} - 2Be^{-2x}$, differenatiate again to get, 1

$\frac{d^2y}{dx^2} = 4Ae^{2x} + 4Be^{-2x} = 4y$ or $\frac{d^2y}{dx^2} - 4y = 0$ 1

SECTION C

13. $\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1}\frac{8}{31}$

$\Rightarrow \tan^{-1}\left\{\frac{2x}{1-(x^2-1)}\right\} = \tan^{-1}\frac{8}{31}$ 2

$\Rightarrow 4x^2 + 31x - 8 = 0 \Rightarrow x = \frac{1}{4}$ or $x = -8$ 1

$x = -8$ does not satisty the given equation so $x = \frac{1}{4}$ 1

14. $x = ae^t (\sin t + \cos t)$; $y = ae^t (\sin t - \cos t)$

then

$$\left.\begin{aligned}\frac{dy}{dt} &= ae^t(\sin t - \cos t) + ae^t(\cos t + \sin t)\\ &= y + x\end{aligned}\right\} \quad 1\frac{1}{2}$$

$$\left.\begin{aligned}\frac{dx}{dt} &= ae^t(\sin t + \cos t) + ae^t(\cos t - \sin t)\\ &= x - y\end{aligned}\right\} \quad 1\frac{1}{2}$$

$\therefore \quad \frac{dy}{dx} = \frac{y+x}{x-y}$ or $\frac{x+y}{x-y}$ 1

OR

Let, $y = x^{\sin x} + (\sin x)^{\cos x} = u + v$; $\therefore \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$ 1

$u = x^{\sin x} \Rightarrow \log u = \sin x \cdot \log x \Rightarrow \frac{du}{dx} = x^{\sin x}\left\{\cos x \cdot \log x + \frac{\sin x}{x}\right\}$ $1\frac{1}{2}$

$v = (\sin x)^{\cos x} \Rightarrow \log v = \cos x \cdot \log(\sin x) \Rightarrow \frac{dv}{dx} = (\sin x)^{\cos x}\{\cos x \cdot \cot x - \sin x \cdot \log \sin x\}$ 1

$\therefore \quad \frac{dy}{dx} = x^{\sin x}\left\{\cos x \cdot \log x + \frac{\sin x}{x}\right\} + (\sin x)^{\cos x}\{\cos x \cdot \cot x - \sin x \,.\, \log \sin x\}$ $\frac{1}{2}$

15. $\int \frac{2\cos x}{(1-\sin x)(2-\cos^2 x)}dx = \int \frac{2\cos x}{(1-\sin x)(1+\sin^2 x)}dx = \int \frac{2}{(1-t)(1+t^2)}dt$, where $\sin x = t$, $\cos x\, dx = dt$

1

$$\int \frac{2}{(1-t)(1+t^2)}dt = \int \frac{1}{1-t}dt + \int \frac{t+1}{t^2+1}dt \quad 1\frac{1}{2}$$

$$= -\log(1-t) + \frac{1}{2}\log(t^2+1) + \tan^{-1}(t) + c \quad 1$$

$$= -\log|1-\sin x| + \frac{1}{2}\log(\sin^2 x + 1) + \tan^{-1}(\sin x) + c \quad \frac{1}{2}$$

16. R = {(1, 2), (2, 3), (3, 4), (4, 5), (5, 6)}

For $1 \in A$, $(1, 1) \notin R \Rightarrow$ R is not reflexive 1

For $1, 2 \in A$, $(1, 2) \in R$ but $(2, 1) \notin R \Rightarrow$ R is not symmetric $1\frac{1}{2}$

For $1, 2, 3 \in A$, $(1, 2), (2, 3) \in R$ but $(1, 3) \notin R \Rightarrow$ R is not transitive $1\frac{1}{2}$

OR

One-One: Let for $x_1, x_2 \in N$, $f(x_1) = f(x_2) \Rightarrow 4x_1 + 3 = 4x_2 + 3 \Rightarrow x_1 = x_2$ 2

$\therefore$ 'f ' is one-one

Onto: co-domain of f = Range of f = Y 1

$\therefore$ 'f ' is onto

$\therefore$ f is invertible with, $f^{-1}: Y \to N$ and $f^{-1}(y) = \frac{y-3}{4}$ or $f^{-1}(x) = \frac{x-3}{4}, x \in Y$ 1

17. $\begin{vmatrix} 3a & -a+b & -a+c \\ -b+a & 3b & -b+c \\ -c+a & -c+b & 3c \end{vmatrix}$

$= \begin{vmatrix} a+b+c & -a+b & -a+c \\ a+b+c & 3b & -b+c \\ a+b+c & -c+b & 3c \end{vmatrix}$ (By applying, $C_1 \to C_1 + C_2 + C_3$) 1

$= \begin{vmatrix} a+b+c & -a+b & -a+c \\ 0 & 2b+a & a-b \\ 0 & a-c & 2c+a \end{vmatrix}$ (By applying, $R_2 \to R_2 - R_1$; $R_3 \to R_3 - R_1$) 2

$= (a + b + c)\ \{4bc + 2ab + 2ac + a^2 - (a^2 - ac - ba + bc)\}$

$= 3(a + b + c)\ (ab + bc + ca)$ 1

18. $(x - a)^2 + (y - b)^2 = c^2$, $c > 0$

Differentiating both sides with respect to 'x', we get

$2(x - a) + 2(y - b) \cdot \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x-a}{y-b}$ $1\frac{1}{2}$

Differentiating again with respect to 'x', we get;

$$\frac{d^2y}{dx^2} = -\frac{(y-b)-(x-a)\cdot\frac{dy}{dx}}{(y-b)^2} = \frac{-c^2}{(y-b)^3} \quad \text{(By substituting '}\frac{dy}{dx}\text{')} \qquad 1+\frac{1}{2}$$

$$\frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}} = \frac{\left[1+\frac{(x-a)^2}{(y-b)^2}\right]^{3/2}}{-\frac{c^2}{(y-b)^3}} = \frac{\frac{c^3}{(y-b)^3}}{-\frac{c^2}{(y-b)^3}} = -c \qquad 1$$

Which is a constant independent of 'a' & 'b'.

19. Let (α, β) be the point on the curve where normal

passes through $(-1, 4)$ $\therefore$ $\alpha^2 = 4\beta$, also $\frac{dy}{dx} = \frac{x}{2}$

Slope of normal at $(\alpha, \beta) = \frac{-1}{\left[\frac{dy}{dx}\right]_{(\alpha,\beta)}} = \frac{-1}{\frac{\alpha}{2}} = \frac{-2}{\alpha}$ 1

Equation of normal: $y-4 = \frac{-2}{\alpha}(x+1)$ $\frac{1}{2}$

(α, β) lies on normal $\Rightarrow \beta - 4 = \frac{-2}{\alpha}(\alpha+1)$

Putting $\beta = \frac{\alpha^2}{4}$, we get; $\alpha^3 - 8\alpha + 8 = 0 \Rightarrow (\alpha-2)(\alpha^2+2\alpha-4) = 0$ 1

For $\alpha = 2$ Equation of normal is: $x + y - 3 = 0$ 1

For $\alpha = \pm\sqrt{5}-1$; Equation of normal is: $y-4 = \frac{-2}{\pm\sqrt{5}-1}(x+1)$ $\frac{1}{2}$

20. $\int_0^a f(x)\,dx = -\int_a^0 f(a-t)dx$ Put $x = a - t, dx = -dt$

Upper limit = $t = a - x = a - a = 0$

Lower limit = $t = a - x = a - 0 = a$

$= \int_0^a f(a-t)\,dt = \int_0^a f(a-x)\,dx$ 1

Let $I = \int_0^{\pi/2} \frac{x}{\sin x + \cos x} dx$ ...(i)

$\Rightarrow I = \int_0^{\pi/2} \frac{\pi/2 - x}{\sin\left(\frac{\pi}{2} - x\right) + \cos\left(\frac{\pi}{2} - x\right)} dx$

$\therefore I = \int_0^{\pi/2} \frac{\pi/2 - x}{\cos x + \sin x} dx$ ...(ii) 1

Adding (i) and (ii) we get

$2I = \frac{\pi}{2} \int_0^{\pi/2} \frac{1}{\cos x + \sin x} dx = \frac{\pi}{2\sqrt{2}} \int_0^{\pi/2} \frac{1}{\cos\left(x - \frac{\pi}{4}\right)} dx$

$= \frac{\pi}{2\sqrt{2}} \int_0^{\pi/2} \sec\left(x - \frac{\pi}{4}\right) dx$ $\frac{1}{2}$

$\Rightarrow 2I = \frac{\pi}{2\sqrt{2}} \left\{ \log \left| \sec\left(x - \frac{\pi}{4}\right) + \tan\left(x - \frac{\pi}{4}\right) \right| \right\}_0^{\pi/2}$ $\frac{1}{2}$

$= \frac{\pi}{2\sqrt{2}} \{\log|\sqrt{2} + 1| - \log|\sqrt{2} - 1|\}$ $\frac{1}{2}$

$\Rightarrow I = \frac{\pi}{4\sqrt{2}} \{\log(\sqrt{2} + 1) - \log(\sqrt{2} - 1)$ or $\frac{\pi}{4\sqrt{2}} \log\left(\frac{\sqrt{2} + 1}{\sqrt{2} - 1}\right)$ $\frac{1}{2}$

21. The given differential equation can be written as:

$\frac{dy}{dx} = \frac{y}{x} - \tan\left(\frac{y}{x}\right)$ $\frac{1}{2}$

Put $\frac{y}{x} = v$ and $\frac{dy}{dx} = v + x\frac{dv}{dx}$, to get 1

$v + x\frac{dv}{dx} = v - \tan v \Rightarrow x\frac{dv}{dx} = -\tan v$

$\Rightarrow \cot v \, dv = -\frac{1}{x} dx,$ 1

Integrating both sides we get,

$$\left.\begin{array}{l} \log|\sin v| = -\log|x| + \log c \\ \Rightarrow \quad \log|\sin v| = \log\left|\dfrac{c}{x}\right| \end{array}\right\} \quad 1$$

$\therefore$ Solution of differential equation is

$$\sin\left(\frac{y}{x}\right) = \frac{c}{x} \text{ or } x \cdot \sin\left(\frac{y}{x}\right) = c \quad \frac{1}{2}$$

OR

The given differential equation can be written as:

$$\frac{dy}{dx} + \frac{\cos x}{1+\sin x} \cdot y = \frac{-x}{1+\sin x}; \quad 1\frac{1}{2}$$

$$\text{I.F.} = e^{\int \frac{\cos x}{1+\sin x} dx} = e^{\log(1+\sin x)} = 1+\sin x \quad 1$$

$\therefore$ Solution of the given differential equation is:

$$y(1 + \sin x) = \int \frac{-x}{1+\sin x} \times (1+\sin x)\,dx + c \quad 1$$

$$\Rightarrow \quad y(1 + \sin x) = \frac{-x^2}{2} + c \text{ or } y = \frac{-x^2}{2(1+\sin x)} + \frac{c}{1+\sin x} \quad \frac{1}{2}$$

22. Lines are perpendicular

$\therefore$ $-3(3\lambda) + 2\lambda(2) + 2(-5) = 0 \Rightarrow \lambda = -2$ 2

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_3 \end{vmatrix} = \begin{vmatrix} 1-1 & 1-2 & 6-3 \\ -3 & 2(-2) & 2 \\ 3(-2) & 2 & -5 \end{vmatrix} = -63 \neq 0 \quad 1\frac{1}{2}$$

$\therefore$ Lines are not intersecting $\frac{1}{2}$

23. $\vec{a} \cdot \dfrac{(\vec{b}+\vec{c})}{|\vec{b}+\vec{c}|} = 1$ 1

$$\Rightarrow \quad (\hat{i}+\hat{j}+\hat{k}) \cdot \{(2+\lambda)\hat{i} + 6\hat{j} - 2\hat{k}\} = \sqrt{(2+\lambda)^2 + 36 + 4} \quad 1\frac{1}{2}$$

$$\Rightarrow \quad \lambda + 6 = \sqrt{(2+\lambda)^2 + 40}$$

Squaring to get

$\lambda^2 + 12\lambda + 36 = \lambda^2 + 4\lambda + 44 \Rightarrow \lambda = 1$ $\frac{1}{2}$

$\therefore$ Unit vector along $(\vec{b} + \vec{c})$ is $\frac{3}{7}\hat{i} + \frac{6}{7}\hat{j} - \frac{2}{7}\hat{k}$ 1

SECTION D

24. $A^2 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix}$ $1\frac{1}{2}$

$A^3 = \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} = \begin{bmatrix} 8 & 7 & 1 \\ -23 & 27 & -69 \\ 32 & -13 & 58 \end{bmatrix}$ $1\frac{1}{2}$

LHS = $A^3 - 6A^2 + 5A + 11I$

$= \begin{bmatrix} 8 & 7 & 1 \\ -23 & 27 & -69 \\ 32 & -13 & 58 \end{bmatrix} - \begin{bmatrix} 24 & 12 & 6 \\ -18 & 48 & -84 \\ 42 & -18 & 84 \end{bmatrix} + \begin{bmatrix} 5 & 5 & 5 \\ 5 & 10 & -15 \\ 10 & -5 & 15 \end{bmatrix} + \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix}$ 1

$= \begin{bmatrix} 24 & 12 & 6 \\ -18 & 48 & -84 \\ 42 & -18 & 84 \end{bmatrix} - \begin{bmatrix} 24 & 12 & 6 \\ -18 & 48 & -84 \\ 42 & -18 & 84 \end{bmatrix} = O = \text{R.H.S.}$

$A^3 - 6A^2 + 5A + 11I = O$, Pre-multiplying by A^{-1}

$\Rightarrow A^2 - 6A + 5I + 11A^{-1} = O \Rightarrow A^{-1} = -\frac{1}{11}(A^2 - 6A + 5I)$ 1

$\therefore A^{-1} = \begin{bmatrix} -3/11 & 4/11 & 5/11 \\ 9/11 & -1/11 & -4/11 \\ 5/11 & -3/11 & -1/11 \end{bmatrix}$ 1

OR

Let,

$A = \begin{bmatrix} 3 & -2 & 3 \\ 2 & 1 & -1 \\ 4 & -3 & 2 \end{bmatrix}; X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}; B = \begin{bmatrix} 8 \\ 1 \\ 4 \end{bmatrix}$

System of equation in Matrix form: $A \cdot X = B$

$|A| = 3(2 - 3) + 2(4 + 4) + 3(-6 - 4) = -17 \neq 0$ 1

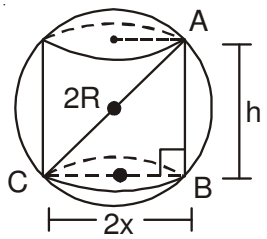
Solution matrix, $X = A^{-1} \cdot B$. 1

$$(\text{adj } A) = \begin{bmatrix} -1 & -5 & -1 \\ -8 & -6 & 9 \\ -10 & 1 & 7 \end{bmatrix}$$ 2

$$\therefore \quad A^{-1} = \frac{-1}{17}\begin{bmatrix} -1 & -5 & -1 \\ -8 & -6 & 9 \\ -10 & 1 & 7 \end{bmatrix}$$ $\frac{1}{2}$

$$\therefore \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = -\frac{1}{17}\begin{bmatrix} -1 & -5 & -1 \\ -8 & -6 & 9 \\ -10 & 1 & 7 \end{bmatrix}\begin{bmatrix} 8 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \Rightarrow x = 1, y = 2, x = 3$$ $1\frac{1}{2}$

25. Correct Figure 1



ln rt. ΔABC; $4x^2 + h^2 = 4R^2$, $x^2 = \frac{4R^2 - h^2}{4}$ 1

V(Volume of cylinder) = $\pi x^2 h = \frac{\pi}{4}(4R^2 h - h^3)$ 1

$V'(h) = \frac{\pi}{4}(4R^2 - 3h^2)$; $V''(h) = \frac{\pi}{4}(-6h)$ $\frac{1}{2} + \frac{1}{2}$

$V'(h) = 0 \Rightarrow h = \frac{2R}{\sqrt{3}}$ 1

$V''\left(\frac{2R}{\sqrt{3}}\right) = \frac{-6\pi}{4}\left(\frac{2R}{\sqrt{3}}\right) < 0 \Rightarrow$ Volume 'V' is max. $\frac{1}{2}$

for $h = \frac{2R}{\sqrt{3}}$

Max. Volume: $V = \frac{4}{3\sqrt{3}}\pi R^3$ $\frac{1}{2}$

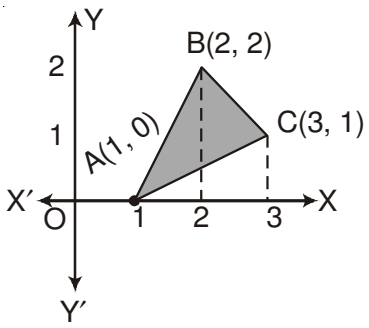
26. E_1 = Event of selecting first bag
E_2 = Event of selecting second bag
A = Event both balls drawn are red. } 1

$$P(E_1) = P(E_2) = \frac{1}{2}; P(A|E_1) = \frac{{}^5C_2}{{}^9C_2} = \frac{20}{72}; P(A|E_2) = \frac{{}^3C_2}{{}^9C_2} = \frac{6}{72} \quad 1+2$$

$$P(E_2|A) = \frac{P(E_2)\cdot P(A|E_2)}{P(E_1)\cdot P(A|E_1) + P(E_2)\cdot P(A|E_2)}$$

$$= \frac{\frac{1}{2}\cdot\frac{6}{72}}{\frac{1}{2}\cdot\frac{20}{72} + \frac{1}{2}\cdot\frac{6}{72}} = \frac{6}{26} = \frac{3}{13} \quad 1\frac{1}{2}+\frac{1}{2}$$

27. Correct Figure 1



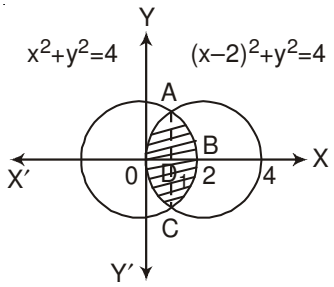
Equation of line AB : $y = 2(x-1)$
Equation of line BC : $y = 4 - x$
Equation of line AC : $y = \frac{1}{2}(x-1)$ } $1\frac{1}{2}$

$$\text{ar}(\Delta ABC) = 2\int_1^2 (x-1)\,dx + \int_2^3 (4-x)\,dx - \frac{1}{2}\int_1^3 (x-1)\,dx \quad 1\frac{1}{2}$$

$$= (x-1)^2\Big]_1^2 - \frac{1}{2}(4-x)^2\Big]_2^3 - \frac{1}{4}(x-1)^2\Big]_1^3 \quad 1\frac{1}{2}$$

$$= 1 + \frac{3}{2} - 1 = \frac{3}{2}\text{ sq. units} \quad \frac{1}{2}$$

OR



Correct Figure 1

Getting the point of intersection as x = 1 1

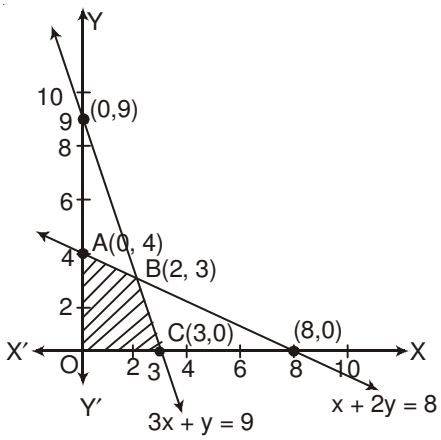
Area (OABCO) = 4 × ar(ABD)

$= 4\int_1^2 \sqrt{2^2 - x^2}\, dx$ 2

$= 4\left\{\frac{x\sqrt{4-x^2}}{2} + 2\sin^{-1}\left(\frac{x}{2}\right)\right\}_1^2$ 1

$= \left(\frac{8\pi}{3} - 2\sqrt{3}\right)$ 1

28.



let the company produce: Goods A = x units

Goods B = y units

then, the linear programming problem is:

Maximize profit: z = 40x + 50y (In ₹) $\frac{1}{2}$

Subject to constraints:

$$\left.\begin{aligned} 3x + y &\le 9 \\ x + 2y &\le 8 \\ x, y &\ge 0 \end{aligned}\right\}$$ $2\frac{1}{2}$

Correct graph. 2

Corner point	Value of z (₹)	
A(0, 4)	200	
B(2, 3)	230 (Max)	$\frac{1}{2}$
C(3, 0)	120	

∴ Maximum profit = ₹ 230 at:

Goods A produced = 2 units, Goods B produced = 3 units $\frac{1}{2}$

29. Let $\vec{a} = \hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + 2\hat{j} + \hat{k}$

$$(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 3 \\ 0 & 1 & 3 \end{vmatrix} = -9\hat{i} - 3\hat{j} + \hat{k}$$ 2

Vector equation of plane is:

$$\{\vec{r} - (\hat{i} + \hat{j} - 2\hat{k})\} \cdot (-9\hat{i} - 3\hat{j} + \hat{k}) = 0$$

$\Rightarrow \quad \vec{r} \cdot (-9\hat{i} - 3\hat{j} + \hat{k}) + 14 = 0$ 1

Cartesian Equation of plane is: $-9x - 3y + z + 14 = 0$ 1

Equation of plane through (2, 3, 7) and parallel to above plane is:

$$-9(x - 2) - 3(y - 3) + (z - 7) = 0$$

$\Rightarrow \quad -9x - 3y + z + 20 = 0$ 1

Distance between parallel planes $= \left| \frac{-14 + 20}{\sqrt{91}} \right| = \frac{6}{\sqrt{91}}$ 1

OR

Equation of line: $\frac{x - 2}{3} = \frac{y + 1}{4} = \frac{z - 2}{2}$ 1

Equation of plane: $\begin{vmatrix} x - 2 & y & z - 3 \\ -1 & 1 & 2 \\ 1 & 2 & 1 \end{vmatrix} = 0$ 1

$\Rightarrow \quad -3(x - 2) + 3y - 3(z - 3) = 0$

$\Rightarrow \quad x - y + z - 5 = 0$ 1

General point on line: $\frac{x - 2}{3} = \frac{y + 1}{4} = \frac{z - 2}{2} = k$ (say)

is: $P(3k + 2, 4k - 1, 2k + 2)$; Putting in the equation of plane 1

we get, $3k + 2 - 4k + 1 + 2k + 2 = 5 \Rightarrow k = 0$ 1

$\therefore$ Point of intersection is: $(2, -1, 2)$ 1

Strictly Confidential — (For Internal and Restricted Use Only)

Senior School Certificate Examination

March 2019

Marking Scheme — Mathematics (041) 65/3/1, 65/3/2, 65/3/3

General Instructions:

1. You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully. **Evaluation is a 10-12 days mission for all of us. Hence, it is necessary that you put in your best efforts in this process.**
2. Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one's own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. **However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and marks be awarded to them.**
3. The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
4. If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled.
5. If a question does not have any parts, marks must be awarded in the left hand margin and encircled.
6. If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out.
7. No marks to be deducted for the cumulative effect of an error. It should be penalized only once.
8. A full scale of marks 0 to 100 has to be used. Please do not hesitate to award full marks if the answer deserves it.
9. Every examiner has to necessarily do evaluation work for full working hours i.e. 8 hours every day and evaluate 25 answer books per day.
10. Ensure that you do not make the following common types of errors committed by the Examiner in the past:-
    - Leaving answer or part thereof unassessed in an answer book.
    - Giving more marks for an answer than assigned to it.
    - Wrong transfer of marks from the inside pages of the answer book to the title page.
    - Wrong question wise totaling on the title page.
    - Wrong totaling of marks of the two columns on the title page.
    - Wrong grand total.
    - Marks in words and figures not tallying.
    - Wrong transfer of marks from the answer book to online award list.
    - Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.)
    - Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
11. While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as (X) and awarded zero (0) Marks.
12. Any unassessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
13. The Examiners should acquaint themselves with the guidelines given in the Guidelines for spot Evaluation before starting the actual evaluation.
14. Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
15. The Board permits candidates to obtain photocopy of the Answer Book on request in an RTI application and also separately as a part of the re-evaluation process on payment of the processing charges.

QUESTION PAPER CODE 65/3/1
EXPECTED ANSWER/VALUE POINTS

SECTION A

1. $|-2A| = (-2)^3 \cdot |A|$ $\frac{1}{2}$

$= -8 \cdot 4 = -32$ $\frac{1}{2}$

2. $y = \frac{\pi}{2} \Rightarrow \frac{dy}{dx} = 0$ $\frac{1}{2}+\frac{1}{2}$

3. order 4, degree 2 $\frac{1}{2}+\frac{1}{2}$

4. $\sqrt{(-18)^2+(12)^2+(-4)^2} = 22$ $\frac{1}{2}$

$\therefore$ DC's are $\frac{-18}{22}, \frac{12}{22}, \frac{-4}{22}$ or $\frac{-9}{11}, \frac{6}{11}, \frac{-2}{11}$ $\frac{1}{2}$

OR

D·R's of required line are 3, –5, 6 $\frac{1}{2}$

Equation of line is $\frac{x+2}{3} = \frac{y-4}{-5} = \frac{z+5}{6}$ $\frac{1}{2}$

SECTION B

5. Let $e \in \mathbb{R}$ be the identity element.

then $a*e = e*a = a$ 1

$\Rightarrow a^2 + e^2 = e^2 + a^2 = a^2 \Rightarrow e^2 = 0 \Rightarrow e = 0.$ 1

$\therefore$ Identity element is $0 \in \mathbb{R}$

6. $kA = k\begin{bmatrix} 0 & 2 \\ 3 & -4 \end{bmatrix} = \begin{bmatrix} 0 & 2k \\ 3k & -4k \end{bmatrix}$

$\therefore \begin{bmatrix} 0 & 2k \\ 3k & -4k \end{bmatrix} = \begin{bmatrix} 0 & 3a \\ 2b & 24 \end{bmatrix} \Rightarrow 2k = 3a, 3k = 2b \text{ and } -4k = 24$ $\frac{1}{2}$

$\Rightarrow k = -6,\ a = \frac{-12}{3} = -4,\ b = \frac{-18}{2} = -9$ $1\frac{1}{2}$

7. $\int \frac{\sin x - \cos x}{\sqrt{1+\sin 2x}} dx = \int \frac{\sin x - \cos x}{\sin x + \cos x} dx$ 1

$= -\log|\sin x + \cos x| + c$ 1

8. $\int \frac{\sin(x-a)}{\sin(x+a)}dx = \int \frac{\sin(x+a-2a)}{\sin(x+a)}dx$ $\frac{1}{2}$

$= \int \left[\frac{\sin(x+a)\cos 2a}{\sin(x+a)} - \frac{\cos(x+a)\sin 2a}{\sin(x+a)}\right]dx$ $\frac{1}{2}$

$= x \cdot \cos 2a - \sin 2a \cdot \log|\sin(x+a)| + c$ $\frac{1}{2}+\frac{1}{2}$

OR

$\int (\log x)^2 \cdot 1dx = (\log x)^2 \cdot x - \int 2 \cdot \log x \cdot \frac{1}{x} \cdot x\, dx$ 1

$= x \cdot (\log x)^2 - \left\{\log x \cdot 2x - \int \frac{1}{x} \cdot 2x dx\right\}$ $\frac{1}{2}$

$= x(\log x)^2 - 2x \log x + 2x + c$ $\frac{1}{2}$

9. $y^2 = m(a^2 - x^2) \Rightarrow 2y\frac{dy}{dx} = -2mx$ $\frac{1}{2}$

or $y\frac{dy}{dx} = -mx$ ...(i)

$y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = -m$ ...(ii) $\frac{1}{2}$

form (i) and (ii) we get $y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = \frac{y}{x}\frac{dy}{dx}$ 1

or $xy\frac{d^2y}{dx^2} + x\left(\frac{dy}{dx}\right)^2 - y\frac{dy}{dx} = 0$

10. A vector perpendicular to both $\vec{a}$ and $\vec{b} = \vec{a} \times \vec{b} = 19\hat{j} + 19\hat{k}$ or $\hat{j} + \hat{k}$ 1

$\therefore$ Unit vector perpendicular to both $\vec{a}$ and $\vec{b} = \frac{1}{\sqrt{2}}(\hat{j} + \hat{k})$ 1

OR

Let $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}, \vec{b} = -2\hat{i} + 3\hat{j} - 4\hat{k}, \vec{c} = \hat{i} - 3\hat{j} + 5\hat{k}$

$\vec{a}, \vec{b}, \vec{c}$ are coplanar if $\vec{a} \cdot \vec{b} \times \vec{c} = 0$ $\frac{1}{2}$

$$\vec{a}\cdot\vec{b}\times\vec{c}=\begin{vmatrix}1 & -2 & 3\\ -2 & 3 & -4\\ 1 & -3 & 5\end{vmatrix}=1(3)+2(-6)+3(3)$$
$$=3-12+9=0$$ $1+\frac{1}{2}$

Hence $\vec{a},\vec{b},\vec{c}$ are coplanar

11. A = {(S, F, M), (S, M, F), (M, F, S), (F, M, S)}
B = {(S, F, M), (M, F, S)} 1

Total number of possible arrangements = 6

$$P(B|A) = \frac{P(B\cap A)}{P(A)}$$

$$= \frac{2/6}{4/6}=\frac{1}{2}$$ 1

12. Given 2 $P(X = x_1) = 3P(X = x_2) = P(X = x_3) = 5P(X = x_4)$

Let $P(X = x_3) = k$, then $P(X = x_1) = \frac{k}{2}, P(X = x_2) = \frac{k}{3}$ and $P(X = x_4) = \frac{k}{5}$ $\frac{1}{2}$

$\therefore\ k+\frac{k}{2}+\frac{k}{3}+\frac{k}{5}=1\Rightarrow k=\frac{30}{61}$ 1

$\therefore$ Probability distribution is

X	x_1	x_2	x_3	x_4
P(X)	$\frac{15}{61}$	$\frac{10}{61}$	$\frac{30}{61}$	$\frac{6}{61}$

$\frac{1}{2}$

OR

(i) P(at least 4 heads) = $P(r \geq 4) = P(4) + P(5)$

$$= {}^5C_4\left(\frac{1}{2}\right)^1\left(\frac{1}{2}\right)^4 + {}^5C_5\left(\frac{1}{2}\right)^5 = 6\left(\frac{1}{2}\right)^5 = \frac{6}{32} \text{ or } \frac{3}{16}$$ 1

(ii) P(at most 4 heads) = $P(r \leq 4) = 1 - P(5)$

$$= 1-\left(\frac{1}{2}\right)^5 = \frac{31}{32}$$ 1

SECTION C

13. (i) For $a \in Z, (a,a) \in R \because a - a = 0$ is divisible by 2

$\therefore$ R is reflexive ...(i) 1

Let (a, b) $\in$ R for a, b $\in$ Z, then a – b is divisible by 2

$\Rightarrow$ (b – a) is also divisible by 2

$\therefore$ (b, a) $\in$ R $\Rightarrow$ R is symmetric ...(ii) 1

For a, b, c, $\in$ Z, Let (a, b) $\in$ R and (b, c) $\in$ R

$\therefore$ a – b = 2p, p $\in$ Z, and b – c = 2q, q $\in$ Z,

adding, a – c = 2 (p + q) $\Rightarrow$ (a – c) is divisible by 2

$\Rightarrow$ (a, c) $\in$ R, so R is transitive ...(iii) $1\frac{1}{2}$

(i), (ii), and (iii) $\Rightarrow$ R is an equivalence relation. $\frac{1}{2}$

OR

$$fof(x) = f\left(\frac{4x+3}{6x-4}\right)$$ 1

$$= \frac{4\left[\frac{4x+3}{6x-4}\right]+3}{6\left[\frac{4x+3}{6x-4}\right]-4}$$ 1

$$\Rightarrow fof(x) = \frac{4(4x+3)+3(6x-4)}{6(4x+3)-4(6x-4)} = \frac{34x}{34} = x$$ 1

Since fof(x) = x $\Rightarrow$ fof = I $\Rightarrow f^{-1} = f$ 1

14. Given $\tan^{-1} x - \cot^{-1} x = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right), x > 0$

$$\Rightarrow \tan^{-1} x - \left(\frac{\pi}{2} - \tan^{-1} x\right) = \frac{\pi}{6}$$ 1

$$\Rightarrow 2\tan^{-1} x = \frac{2\pi}{3} \Rightarrow \tan^{-1} x = \frac{\pi}{3}$$ 1

$$\Rightarrow x = \tan\left(\frac{\pi}{3}\right) = \sqrt{3} \qquad \therefore \sec^{-1}\frac{2}{x} = \sec^{-1}\frac{2}{\sqrt{3}} = \frac{\pi}{6}$$ 1+1

15. Let $\Delta = \begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$

$R_1 \to R_1 - (R_2 + R_3) \Rightarrow \Delta = \begin{vmatrix} 0 & -2c & -2b \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$ 1

$\therefore \Delta = -2\begin{vmatrix} 0 & c & b \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = -\frac{2}{b}\begin{vmatrix} 0 & bc & b \\ b & bc+ab & b \\ c & bc & a+b \end{vmatrix}$ $1+\frac{1}{2}$

$C_2 \to C_2 - cC_3$

$\Rightarrow \Delta = -\frac{2}{b}\begin{vmatrix} 0 & 0 & b \\ b & ab & b \\ c & -ac & a+b \end{vmatrix}$ 1

$= -\frac{2}{b} \cdot b \cdot (-abc - abc) = 4abc.$ $\frac{1}{2}$

16. $\sin y = x \cdot \sin(a+y) \Rightarrow x = \frac{\sin y}{\sin(a+y)}$ $\frac{1}{2}$

differentiating w.r.t. y, we get

$\frac{dx}{dy} = \frac{\sin(a+y)\cos y - \sin y \cos(a+y)}{\sin^2(a+y)}$ $1\frac{1}{2}$

$\frac{dx}{dy} = \frac{\sin(a+y-y)}{\sin^2(a+y)} = \frac{\sin a}{\sin^2(a+y)}$ $1\frac{1}{2}$

$\therefore \frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$ $\frac{1}{2}$

OR

$(\sin x)^y = (x+y) \Rightarrow y \cdot \log \sin x = \log(x+y)$ 1

differentiating w.r.t. x, we get

$y \cdot \cot x + \log \sin x \cdot \frac{dy}{dx} = \frac{1}{x+y}\left(1 + \frac{dy}{dx}\right)$ $1\frac{1}{2}$

$$\Rightarrow \frac{dy}{dx} = \frac{\frac{1}{x+y} - y\cot x}{\log \sin x - \frac{1}{x+y}}$$ 1

$$= \frac{1 - y(x+y)\cot x}{(x+y)\log \sin x - 1}$$ $\frac{1}{2}$

17. $y = (\sec^{-1} x)^2,\ x > 0$

$$\frac{dy}{dx} = 2\sec^{-1} x \cdot \frac{1}{x\sqrt{x^2-1}}$$ 1

$$\Rightarrow x\sqrt{x^2-1}\frac{dy}{dx} = 2\sqrt{y}$$ $\frac{1}{2}$

squaring both sides, we get

$$x^2(x^2-1)\left(\frac{dy}{dx}\right)^2 = 4y \quad \text{or} \quad (x^4 - x^2)\left(\frac{dy}{dx}\right)^2 = 4y$$ $\frac{1}{2}$

differentiating w.r.t. x.

$$(x^4 - x^2)2\cdot\frac{dy}{dx}\cdot\frac{d^2y}{dx^2} + (4x^3 - 2x)\left(\frac{dy}{dx}\right)^2 = 4\cdot\frac{dy}{dx}$$ $1\frac{1}{2}$

$$\Rightarrow x^2(x^2-1)\frac{d^2y}{dx^2} + (2x^3 - x)\frac{dy}{dx} - 2 = 0$$ $\frac{1}{2}$

18. Curve $y = \frac{x-7}{(x-2)(x-3)}$ cuts at x – axis at the point x = 7, y = 0 i.e. (7, 0) $\frac{1}{2}$

$$\frac{dy}{dx} = \frac{(x^2 - 5x + 6)\cdot 1 - (x-7)(2x-5)}{(x^2-5x+6)^2}$$ $\frac{1}{2}$

at (7, 0) $\frac{dy}{dx} = \frac{20}{(20)^2} = \frac{1}{20}$ $\frac{1}{2}$

$\therefore$ Slope of tangent at (7, 0) is $\frac{1}{20}$ $\frac{1}{2}$

and slope of Normal at (7, 0) is –20 $\frac{1}{2}$

Equation of tangent at (7, 0) is $y - 0 = \frac{1}{20}(x - 7)$

or $x - 20y - 7 = 0$ 1

Equation of Normal at (7, 0) is $y - 0 = -20(x - 7)$

or $20x + y = 140$. $\frac{1}{2}$

19. $I = \int \frac{\sin 2x}{(\sin^2 x + 1)(\sin^2 x + 3)} dx$

Put $\sin^2 x = t \Rightarrow \sin 2x\, dx = dt$ $\frac{1}{2}$

$\therefore I = \int \frac{dt}{(t+1)(t+3)} = \int \left(\frac{1/2}{t+1} + \frac{-1/2}{t+3}\right) dt$ $1\frac{1}{2}$

$= \frac{1}{2}\log|t+1| - \frac{1}{2}\log|t+3| + c$ $1\frac{1}{2}$

$= \frac{1}{2}\log(\sin^2 x + 1) - \frac{1}{2}\log(\sin^2 + 3) + c.$ $\frac{1}{2}$

20. RHS $= \int_a^b f(a+b-x)\, dx = -\int_b^a f(t)\, dt$, where $a + b - x = t$, $dx = -dt$ $\frac{1}{2}$

$= \int_a^b f(t)\, dt = \int_a^b f(x)\, dx = LHS$ $\frac{1}{2}$

Let $I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{1+\sqrt{\tan x}} = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx$ ...(i) $\frac{1}{2}$

$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\cos(\pi/2 - x)}}{\sqrt{\cos(\pi/2-x)} + \sqrt{\sin(\pi/2-x)}} dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$ ...(ii) $1\frac{1}{2}$

adding (i) and (ii) to get $2I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 1 \cdot dx = x]_{\pi/6}^{\pi/3} = \pi/6.$ $\frac{1}{2}$

$\Rightarrow I = \frac{\pi}{12}$ $\frac{1}{2}$

21. $\frac{dy}{dx} = \frac{x+y}{x-y} = \frac{1+\frac{y}{x}}{1-\frac{y}{x}}$

Put y/x = v so that $\frac{dy}{dx} = v + x\frac{dv}{dx}$ $\frac{1}{2}$

$\therefore \; v + x\frac{dv}{dx} = \frac{1+v}{1-v} \Rightarrow x\frac{dv}{dx} = \frac{1+v}{1-v} - v = \frac{1+v-v+v^2}{1-v}$

$\Rightarrow \int \frac{1-v}{1+v^2}dv = \int \frac{dx}{x} \Rightarrow \int \frac{1}{1+v^2}dv - \frac{1}{2}\int \frac{2v}{1+v^2}dv = \int \frac{dx}{x}$ $1+\frac{1}{2}$

$\Rightarrow \tan^{-1} v = \frac{1}{2}\log|1+v^2| + \log|x| + c$ 1

$\Rightarrow \tan^{-1}\left(\frac{y}{x}\right) = \frac{1}{2}\log\left|\frac{x^2+y^2}{x^2}\right| + \log|x| + c$ 1

or $\tan^{-1}\left(\frac{y}{x}\right) = \frac{1}{2}\log|x^2+y^2| + c$

OR

$(1+x^2)dy + 2xy\,dx = \cot x.dx.$

$\Rightarrow \frac{dy}{dx} + \frac{2x}{1+x^2}\cdot y = \frac{\cot x}{1+x^2}$ 1

I.F. = $e^{\int \frac{2x}{1+x^2}dx} = e^{\log(1+x^2)} = (1+x^2)$ 1

$\therefore$ Solution is, $y\cdot(1+x^2) = \int \cot x\,dx = \log|\sin x| + c$ 1+1

or $y = \frac{1}{1+x^2}\cdot\log|\sin x| + \frac{c}{1+x^2}$

22. Given $\frac{\vec{b}\cdot\vec{a}}{|\vec{a}|} = \frac{\vec{c}\cdot\vec{a}}{|\vec{a}|} \therefore \vec{b}\cdot\vec{a} = \vec{c}\cdot\vec{a}$ ...(i) $\frac{1}{2}$

$\vec{b} \perp \vec{c} \Rightarrow \vec{b}\cdot\vec{c} = 0$ ...(ii) $\frac{1}{2}$

$(|3\vec{a}-2\vec{b}+2\vec{c}|)^2 = 9|\vec{a}|^2+4|\vec{b}|^2+4|\vec{c}|^2-12\,\vec{a}\cdot\vec{b}-8\,\vec{b}\cdot\vec{c}+12\,\vec{a}\cdot\vec{c}$ 1

$= 9(1)^2+4(2)^2+4(3)^2$ [using (i) and (ii)] 1

$= 9 + 16 + 36 = 61$

$\Rightarrow |3\vec{a}-2\vec{b}+2\vec{c}| = \sqrt{61}$ 1

23. Writing the equations of given lines in standard form, as

$$\frac{x-5}{5\lambda+2}=\frac{y-2}{-5}=\frac{z-1}{1};\ \frac{x}{1}=\frac{y+\frac{1}{2}}{2\lambda}=\frac{z-1}{3}$$ $\frac{1}{2}$

lines are perpendicular to each other,

$\Rightarrow (5\lambda+2)\cdot 1+(-5)(2\lambda)+1(3)=0$ $\frac{1}{2}$

$-5\lambda+5=0 \Rightarrow \lambda=1$ $\frac{1}{2}$

$\therefore$ lines are $\frac{x-5}{7}=\frac{y-2}{-5}-\frac{z-1}{1};\ \frac{x}{1}=\frac{y+\frac{1}{2}}{2}=\frac{z-1}{3}$ $\frac{1}{2}$

Shortest distance between these lines $= \frac{|(\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2)|}{|\vec{b}_1\times\vec{b}_2|} = \frac{\left|\left(5\hat{i}+\frac{5}{2}\hat{j}\right)\cdot(-17\hat{i}-20\hat{j}+19\hat{k})\right|}{|\vec{b}_1\times\vec{b}_2|}$ 1

$= \frac{135}{|\vec{b}_1\times\vec{b}_2|} \neq 0$ $\frac{1}{2}$

$\therefore$ lines are not intersecting. $\frac{1}{2}$

SECTION D

24. $|A| = 1(7)-1(-3)+1(-1) = 9$ 1

$$(\text{adj A}) = \begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix}$$ 2

$$\Rightarrow A^{-1} = \frac{1}{9}\begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix}$$ $\frac{1}{2}$

Given equations can be written as $\begin{bmatrix}1 & 1 & 1\\ 0 & 1 & 3\\ 1 & -2 & 1\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}6\\ 11\\ 0\end{bmatrix}$

or $AX = B \Rightarrow X = A^{-1}B$ $\frac{1}{2}$

$\therefore \begin{bmatrix}x\\ y\\ z\end{bmatrix}=\frac{1}{9}\begin{bmatrix}7 & -3 & 2\\ 3 & 0 & -3\\ -1 & 3 & 1\end{bmatrix}\begin{bmatrix}6\\ 11\\ 0\end{bmatrix}=\begin{bmatrix}1\\ 2\\ 3\end{bmatrix}$ $1\frac{1}{2}$

$\therefore$ x = 1, y = 2, z = 3 $\frac{1}{2}$

OR

Let: $\begin{bmatrix}2 & 3 & 1\\ 2 & 4 & 1\\ 3 & 7 & 2\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}A$ 1

$$R_1 \leftrightarrow R_3 \begin{bmatrix}3 & 7 & 2\\ 2 & 4 & 1\\ 2 & 3 & 1\end{bmatrix}=\begin{bmatrix}0 & 0 & 1\\ 0 & 1 & 0\\ 1 & 0 & 0\end{bmatrix}A$$

$$\begin{matrix}R_1 \to R_1 - R_3\\ R_2 \to R_2 - R_3\end{matrix}\begin{bmatrix}1 & 4 & 1\\ 0 & 1 & 0\\ 2 & 3 & 1\end{bmatrix}=\begin{bmatrix}-1 & 0 & 1\\ -1 & 1 & 0\\ 1 & 0 & 0\end{bmatrix}A$$

$$R_3 \to R_3 - 2R_1 \begin{bmatrix}1 & 4 & 1\\ 0 & 1 & 0\\ 0 & -5 & -1\end{bmatrix}=\begin{bmatrix}-1 & 0 & 1\\ -1 & 1 & 0\\ 3 & 0 & -2\end{bmatrix}A$$

$$R_3 \to R_3 + 5R_2 \begin{bmatrix}1 & 4 & 1\\ 0 & 1 & 0\\ 0 & 0 & -1\end{bmatrix}=\begin{bmatrix}-1 & 0 & 1\\ -1 & 1 & 0\\ -2 & 5 & -2\end{bmatrix}A$$

$$\begin{matrix}R_1 \to R_1 - 4R_2\\ R_3 \to -R_3\end{matrix}\begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}3 & -4 & 1\\ -1 & 1 & 0\\ 2 & -5 & 2\end{bmatrix}A$$

$$R_1 \to R_1 - R_3 \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 1 & -1\\ -1 & 1 & 0\\ 2 & -5 & 2\end{bmatrix}A$$

4

$$\Rightarrow A^{-1} = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 0 \\ 2 & -5 & 2 \end{bmatrix}$$ 1

25. Let Given surface area of open cylinder be S.

Then $S = 2\pi rh + \pi r^2$

$$\Rightarrow h = \frac{S - \pi r^2}{2\pi r}$$ 1

Volume $V = \pi r^2 h$ $\frac{1}{2}$

$$V = \pi r^2\left[\frac{S-\pi r^2}{2\pi r}\right] = \frac{1}{2}[Sr - \pi r^3]$$ 1

$$\frac{dV}{dr} = \frac{1}{2}[S - 3\pi r^2]$$ $\frac{1}{2}$

$$\frac{dV}{dr} = 0 \Rightarrow S = 3\pi r^2 \text{ or } 2\pi rh + \pi r^2 = 3\pi r^2$$ 1

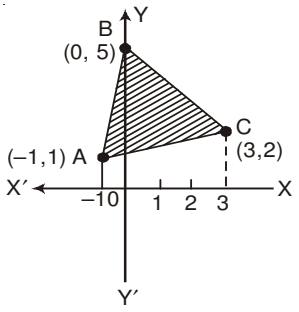
$$\Rightarrow 2\pi rh = 2\pi r^2 \qquad \Rightarrow h = r$$ 1

$$\frac{d^2V}{dr^2} = -6\pi r < 0$$ 1

$\therefore$ For volume to be maximum, height = radius

26. Let the points be A (–1, 1), B (0, 5) and C (3, 2)

Correct Figure 1



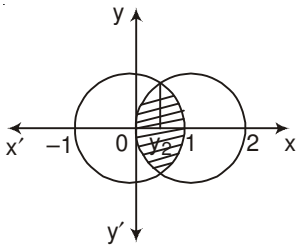
$$\left.\begin{array}{c} \text{Equation of AB} : y = 4x + 5 \\ BC : y = 5 - x \\ AC : y = \frac{1}{4}(x+5) \end{array}\right\}$$ $1\frac{1}{2}$

$$\text{Req. Area} = \int_{-1}^{0}(4x+5)dx + \int_{0}^{3}(5-x)dx - \int_{-1}^{3}\frac{1}{4}(x+5)dx$$ $1\frac{1}{2}$

$$\therefore A = \left[\frac{(4x+5)^2}{8}\right]_{-1}^{0} + \left[\frac{(5-x)^2}{-2}\right]_{0}^{3} - \frac{1}{4}\left[\frac{(x+5)^2}{2}\right]_{-1}^{3}$$ 1

$$= 3 + \frac{21}{2} - 6 = \frac{15}{2}$$ 1

OR



Correct Figure 1

$$(x-1)^2 + y^2 = 1$$

and $x^2 + y^2 = 1 \Rightarrow (x-1)^2 = x^2$

$$\Rightarrow x = \frac{1}{2}$$ 1

$$\therefore \text{ Required area } = 2\left[\int_0^{\frac{1}{2}} \sqrt{1-(x-1)^2}\,dx + \int_{\frac{1}{2}}^{1} \sqrt{1-x^2}\,dx\right]$$ 2

$$= 2\left[\frac{x-1}{2}\sqrt{1-(x-1)^2} + \frac{1}{2}\sin^{-1}(x-1)\right]_0^{\frac{1}{2}} + 2\left[\frac{x}{2}\sqrt{1-x^2} + \frac{1}{2}\sin^{-1}x\right]_{\frac{1}{2}}^{1}$$ 1

$$= 2\left[\frac{-\sqrt{3}}{8} + \frac{\pi}{6}\right] + 2\left[\frac{-\sqrt{3}}{8} + \frac{\pi}{6}\right] = \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right)$$ 1

27. Equation of plane passing through (2, 5, –3), (–2, –3, 5) and (5, 3, –3) is

$$\begin{vmatrix} x-2 & y-5 & z+3 \\ -4 & -8 & 8 \\ 3 & -2 & 0 \end{vmatrix} = 0$$ 1

$$\Rightarrow 16(x-2) + 24(y-5) + 32(z+3) = 0$$

i.e. $2x + 3y + 4z - 7 = 0$ ...(i) 1

which in vector form can be written as $\vec{r} \cdot (2\hat{i} + 3\hat{j} + 4\hat{k}) = 7$ 1

Equation of line passing through (3, 1, 5) and (–1, –3, –1) is

$$\frac{x-3}{4} = \frac{y-1}{4} = \frac{z-5}{6} \text{ or } \frac{x-3}{2} = \frac{y-1}{2} = \frac{z-5}{3}$$ ...(ii) 1

Any point on (ii) is $(2\lambda + 3, 2\lambda + 1, 3\lambda + 5)$ $\frac{1}{2}$

If this is point of intersection with plane (i), then

$2(2\lambda+3)+3(2\lambda+1)+4(3\lambda+5)-7=0$ $\frac{1}{2}$

$22\lambda+22=0 \Rightarrow \lambda=-1$ $\frac{1}{2}$

$\therefore$ Point of intersection is (1, –1, 2) $\frac{1}{2}$

OR

Equation of plane through the intersection of planes

$\vec{r}\cdot(\hat{i}+\hat{j}+\hat{k})-1=0$ and $\vec{r}\cdot(2\hat{i}+3\hat{j}-\hat{k})+4=0,$ is

$[\vec{r}\cdot(\hat{i}+\hat{j}+\hat{k})-1]+\lambda[\vec{r}\cdot(2\hat{i}+3\hat{j}-\hat{k})+4] = 0$ 1

$\Rightarrow \vec{r}\cdot[(1+2\lambda)\hat{i}+(1+3\lambda)\hat{j}+(1-\lambda)\hat{k}]-1+4\lambda = 0$ ...(i) 1

Plane (i) is || to x–axis $\Rightarrow 1+2\lambda=0 \Rightarrow \lambda=\frac{-1}{2}$ $1\frac{1}{2}$

$\therefore$ Equation of plane is $\vec{r}\cdot\left(-\frac{1}{2}\hat{j}+\frac{3}{2}\hat{k}\right)-3=0$ $1\frac{1}{2}$

or $\vec{r}\cdot(-\hat{j}+3\hat{k})-6 = 0$

Distance of this plane from x–axis

$= \frac{|-6|}{\sqrt{(-1)^2+(3)^2}} = \frac{6}{\sqrt{10}}$ units 1

28. Let the events be

E_1 : bag I is selected
E_2 : bag II is selected
A : getting a red ball
} 1

$P(E_1) = P(E_2) = \frac{1}{2}$ $\frac{1}{2}$

$P(A/E_1)=\frac{3}{9}=\frac{1}{3};\ P(A/E_2)=\frac{5}{5+n}$ $\frac{1}{2}+1$

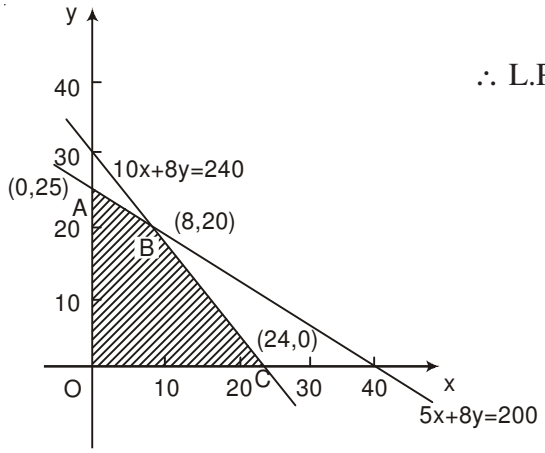
$$P(E_2/A) = \frac{3}{5} = \frac{\frac{1}{2}\cdot\frac{5}{5+n}}{\frac{1}{2}\cdot\frac{1}{3}+\frac{1}{2}\cdot\frac{5}{5+n}}$$ 2

$$\Rightarrow \frac{3}{5} = \frac{15}{5+n+15} \Rightarrow n = 5.$$ 1

29. Let number of Souvenirs of type A be x, and that of type B be y.

$\therefore$ L.P.P is maximise $P = 50x + 60y$ $\frac{1}{2}$

such that $5x + 8y \le 200$
$10x + 8y \le 240$
$x, y \ge 0$ $2\frac{1}{2}$



Correct Graph 2

P(at A) = ₹1500

P(at B) = ₹(400 + 1200) = ₹1600

P(at C) = ₹(1200)

$\therefore$ Max Profit = ₹ 1600, when number of Souvenirs of type A = 8 and number of Souvenirs of type B = 20. 1

QUESTION PAPER CODE 65/3/2
EXPECTED ANSWER/VALUE POINTS

SECTION A

1. $\frac{dy}{dx} - \frac{2}{x} \cdot y = 2x \Rightarrow \text{I.F.} = e^{-2\log x} = \frac{1}{x^2}$ $\frac{1}{2}+\frac{1}{2}$

2. $y^2 + 2xy\frac{dy}{dx} - 2x = 0 \Rightarrow \frac{dy}{dx} = \frac{2x - y^2}{2xy}$ $\frac{1}{2}+\frac{1}{2}$

3. $|-2A| = (-2)^3 \cdot |A|$ $\frac{1}{2}$

$= -8 \times 4 = -32$ $\frac{1}{2}$

4. $\sqrt{(-18)^2 + (12)^2 + (-4)^2} = 22$ $\frac{1}{2}$

$\therefore$ DC's are $\frac{-18}{22}, \frac{12}{22}, \frac{-4}{22}$ or $\frac{-9}{11}, \frac{6}{11}, \frac{-2}{11}$ $\frac{1}{2}$

OR

D·R's of required line are 3, –5, 6 $\frac{1}{2}$

Equation of line is $\frac{x+2}{3} = \frac{y-4}{-5} = \frac{z+5}{6}$ $\frac{1}{2}$

SECTION B

5. Let a = 2, b = 3 $\Rightarrow 2*3 = \frac{2}{4} = \frac{1}{2}$, $3*2 = \frac{3}{3} = 1 \Rightarrow 2*3 \neq 3*2$. 1

$(2*3)*4 = \frac{1}{2}*4 = \frac{\frac{1}{2}}{4+1} = \frac{1}{10}, 2*(3*4) = 2*\frac{3}{5} = \frac{2}{8/5} = \frac{5}{4}$

$\Rightarrow (2*3)*4 \neq 2*(3*4)$ 1

6. $A^2 = \begin{bmatrix} -3 & 6 \\ -2 & 4 \end{bmatrix}\begin{bmatrix} -3 & 6 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} -3 & 6 \\ -2 & 4 \end{bmatrix} = A$ 1

$\Rightarrow A^3 = A^2 \cdot A = A \cdot A = A^2 = A$ 1

7. $y^2 = m(a^2 - x^2) \Rightarrow 2y\frac{dy}{dx} = -2mx$ $\frac{1}{2}$

or $y\frac{dy}{dx} = -mx$ ...(i)

$y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2 = -m$ ...(ii) $\frac{1}{2}$

form (i) and (ii) we get $y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2 = \frac{y}{x}\frac{dy}{dx}$ 1

or $xy\frac{d^2y}{dx^2}+x\left(\frac{dy}{dx}\right)^2 - y\frac{dy}{dx} = 0$

8. $\int\frac{\sin x-\cos x}{\sqrt{1+\sin 2x}}dx = \int\frac{\sin x-\cos x}{\sin x+\cos x}dx$ 1

$= -\log|\sin x+\cos x|+c$ 1

9. $\int\frac{\sin(x-a)}{\sin(x+a)}dx = \int\frac{\sin(x+a-2a)}{\sin(x+a)}dx$ $\frac{1}{2}$

$= \int\left[\frac{\sin(x+a)\cdot\cos 2a}{\sin(x+a)}-\frac{\cos(x+a)\sin 2a}{\sin(x+a)}\right]dx$ $\frac{1}{2}$

$= x\cdot\cos 2a-\sin 2a\cdot\log|\sin(x+a)|+c$ $\frac{1}{2}+\frac{1}{2}$

OR

$\int(\log x)^2\cdot 1dx = (\log x)^2\cdot x-\int 2\cdot\log x\cdot\frac{1}{x}\cdot x\,dx$ 1

$= x\cdot(\log x)^2-\left\{\log x\cdot 2x-\int\frac{1}{x}\cdot 2xdx\right\}$ $\frac{1}{2}$

$= x(\log x)^2-2x\log x+2x+c$ $\frac{1}{2}$

10. A = {(S, F, M), (S, M, F), (M, F, S), (F, M, S)}
B = {(S, F, M), (M, F, S)} 1

Total number of possible arrangements = 6

$P(B|A) = \frac{P(B\cap A)}{P(A)}$

$= \frac{2/6}{4/6} = \frac{1}{2}$ 1

11. Given $2\,P(X = x_1) = 3P(X = x_2) = P(X = x_3) = 5P(X = x_4)$

Let $P(X = x_3) = k$, then $P(X = x_1) = \frac{k}{2}, P(X = x_2) = \frac{k}{3}$ and $P(X = x_4) = \frac{k}{5}$ $\frac{1}{2}$

$\therefore\ k + \frac{k}{2} + \frac{k}{3} + \frac{k}{5} = 1 \Rightarrow k = \frac{30}{61}$ 1

$\therefore$ Probability distribution is

X	x_1	x_2	x_3	x_4
P(X)	$\frac{15}{61}$	$\frac{10}{61}$	$\frac{30}{61}$	$\frac{6}{61}$

$\frac{1}{2}$

OR

(i) P(at least 4 heads) = $P(r \geq 4) = P(4) + P(5)$

$= {}^5C_4\left(\frac{1}{2}\right)^1\left(\frac{1}{2}\right)^4 + {}^5C_5\left(\frac{1}{2}\right)^5 = 6\left(\frac{1}{2}\right)^5 = \frac{6}{32}$ or $\frac{3}{16}$ 1

(ii) P(at most 4 heads) = $P(r \leq 4) = 1 - P(5)$

$= 1 - \left(\frac{1}{2}\right)^5 = \frac{31}{32}$ 1

12. A vector perpendicular to both $\vec{a}$ and $\vec{b} = \vec{a} \times \vec{b} = 19\hat{j} + 19\hat{k}$ or $\hat{j} + \hat{k}$ 1

$\therefore$ Unit vector perpendicular to both $\vec{a}$ and $\vec{b} = \frac{1}{\sqrt{2}}(\hat{j} + \hat{k})$ 1

OR

Let $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}, \vec{b} = -2\hat{i} + 3\hat{j} - 4\hat{k}, \vec{c} = \hat{i} - 3\hat{j} + 5\hat{k}$

$\vec{a}, \vec{b}, \vec{c}$ are coplanar if $\vec{a} \cdot \vec{b} \times \vec{c} = 0$ $\frac{1}{2}$

$$\left.\begin{aligned} \vec{a} \cdot \vec{b} \times \vec{c} &= \begin{vmatrix} 1 & -2 & 3 \\ -2 & 3 & -4 \\ 1 & -3 & 5 \end{vmatrix} = 1(3) + 2(-6) + 3(3) \\ &= 3 - 12 + 9 = 0 \end{aligned}\right\}$$ $1 + \frac{1}{2}$

Hence $\vec{a}, \vec{b}, \vec{c}$ are coplanar

SECTION C

13. (i) For $a \in Z, (a, a) \in R \because a - a = 0$ is divisible by 2

$\therefore$ R is reflexive ...(i) 1

Let (a, b) $\in$ R for a, b $\in$ Z, then a – b is divisible by 2

$\Rightarrow$ (b – a) is also divisible by 2

$\therefore$ (b, a) $\in$ R $\Rightarrow$ R is symmetric ...(ii) 1

For a, b, c, $\in$ Z, Let (a, b) $\in$ R and (b, c) $\in$ R

$\therefore$ a – b = 2p, p $\in$ Z, and b – c = 2q, q $\in$ Z,

adding, a – c = 2 (p + q) $\Rightarrow$ (a – c) is divisible by 2

$\Rightarrow$ (a, c) $\in$ R, so R is transitive ...(iii) $1\frac{1}{2}$

(i), (ii), and (iii) $\Rightarrow$ R is an equivalence relation. $\frac{1}{2}$

OR

$$fof(x) = f\left(\frac{4x+3}{6x-4}\right)$$ 1

$$= \frac{4\left[\frac{4x+3}{6x-4}\right]+3}{6\left[\frac{4x+3}{6x-4}\right]-4}$$ 1

$$\Rightarrow fof(x) = \frac{4(4x+3)+3(6x-4)}{6(4x+3)-4(6x-4)} = \frac{34x}{34} = x$$ 1

Since fof(x) = x $\Rightarrow$ fof = I $\Rightarrow f^{-1} = f$ 1

14. $\sin y = x \cdot \sin(a+y) \Rightarrow x = \frac{\sin y}{\sin(a+y)}$ $\frac{1}{2}$

differentiating w.r.t. y, we get

$$\frac{dx}{dy} = \frac{\sin(a+y)\cos y - \sin y \cos(a+y)}{\sin^2(a+y)}$$ $1\frac{1}{2}$

$$\frac{dx}{dy} = \frac{\sin(a+y-y)}{\sin^2(a+y)} = \frac{\sin a}{\sin^2(a+y)} \qquad 1\frac{1}{2}$$

$$\therefore \frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a} \qquad \frac{1}{2}$$

OR

$(\sin x)^y = (x+y) \Rightarrow y \cdot \log \sin x = \log(x+y)$ 1

differentiating w.r.t. x, we get

$$y \cdot \cot x + \log \sin x \cdot \frac{dy}{dx} = \frac{1}{x+y}\left(1+\frac{dy}{dx}\right) \qquad 1\frac{1}{2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\frac{1}{x+y} - y\cot x}{\log \sin x - \frac{1}{x+y}} \qquad 1$$

$$= \frac{1-y(x+y)\cot x}{(x+y)\log \sin x - 1} \qquad \frac{1}{2}$$

15. $\sin^{-1}\left(\frac{3}{x}\right) + \sin^{-1}\left(\frac{4}{x}\right) = \frac{\pi}{2} \Rightarrow \sin^{-1}\left(\frac{3}{x}\right) = \frac{\pi}{2} - \sin^{-1}\frac{4}{x} = \cos^{-1}\frac{4}{x}$ 1

$$\Rightarrow \sin^{-1}\left(\frac{3}{x}\right) = \sin^{-1}\left(\sqrt{1-\frac{16}{x^2}}\right) \Rightarrow \left(\frac{3}{x}\right)^2 = \frac{x^2-16}{x^2} \qquad 1\frac{1}{2}$$

$\Rightarrow x^2 = 25 \Rightarrow x = \pm 5$, $x = -5$ (rejected) $\therefore x = 5$ $\frac{1}{2}+1$

16. LHS $= \begin{vmatrix} a^2+1 & ab & ac \\ ab & b^2+1 & bc \\ ac & bc & c^2+1 \end{vmatrix} = \frac{1}{abc}\begin{vmatrix} a(a^2+1) & a^2b & a^2c \\ ab^2 & b(b^2+1) & b^2c \\ ac^2 & bc^2 & c(c^2+1) \end{vmatrix}$ $\begin{cases} \text{Applying} \\ R_1 \to aR_1 \\ R_2 \to bR_2 \\ R_3 \to cR_3 \end{cases}$ 1

$$= \begin{vmatrix} a^2+1 & a^2 & a^2 \\ b^2 & b^2+1 & b^2 \\ c^2 & c^2 & c^2+1 \end{vmatrix} = (1+a^2+b^2+c^2)\begin{vmatrix} 1 & 1 & 1 \\ b^2 & b^2+1 & b^2 \\ c^2 & c^2 & c^2+1 \end{vmatrix} \{R_1 \to R_1+R_2+R_3 \qquad \frac{1}{2}+1$$

$= (1+a^2+b^2+c^2)\begin{vmatrix} 1 & 0 & 0 \\ b^2 & 1 & 0 \\ c^2 & 0 & 1 \end{vmatrix} = 1+a^2+b^2+c^2. \begin{cases} C_2 \to C_2 - C_1 \\ C_3 \to C_3 - C_1 \end{cases}$ $1\frac{1}{2}$

= RHS

17. $y = (\cot^{-1} x)^2 \Rightarrow \frac{dy}{dx} = 2\cot^{-1} x \cdot \left(\frac{-1}{1+x^2}\right)$ 1

$\Rightarrow (1+x^2)\frac{dy}{dx} = -2\cot^{-1} x = -2\sqrt{y}$ $\frac{1}{2}$

squaring both sides, we get

$(1+x^2)^2 \cdot \left(\frac{dy}{dx}\right)^2 = 4y$ $\frac{1}{2}$

differentiating, w.r.t. x,

$2(1+x^2)2x \cdot \left(\frac{dy}{dx}\right)^2 + 2(1+x^2)^2 \cdot \frac{dy}{dx} \cdot \frac{d^2y}{dx^2} = 4 \cdot \frac{dy}{dx}$ $1\frac{1}{2}$

$\Rightarrow 2x(1+x^2)\frac{dy}{dx} + (1+x^2)^2 \frac{d^2y}{dx^2} = 2.$ $\frac{1}{2}$

18. $I = \int \frac{\sin 2x}{(\sin^2 x + 1)(\sin^2 x + 3)} dx$

Put $\sin^2 x = t \Rightarrow \sin 2x\, dx = dt$ $\frac{1}{2}$

$\therefore I = \int \frac{dt}{(t+1)(t+3)} = \int \left(\frac{1/2}{t+1} + \frac{-1/2}{t+3}\right) dt$ $1\frac{1}{2}$

$= \frac{1}{2}\log|t+1| - \frac{1}{2}\log|t+3| + c$ $1\frac{1}{2}$

$= \frac{1}{2}\log(\sin^2 x + 1) - \frac{1}{2}\log(\sin^2 + 3) + c.$ $\frac{1}{2}$

19. RHS $= \int_a^b f(a+b-x)\, dx = -\int_b^a f(t)\, dt$, where a + b – x = t, dx = –dt $\frac{1}{2}$

$= \int_a^b f(t)\, dt = \int_a^b f(x)\, dx$ = LHS $\frac{1}{2}$

Let $I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{1+\sqrt{\tan x}} = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}\,dx$ ...(i) $\frac{1}{2}$

$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\cos(\pi/2-x)}}{\sqrt{\cos(\pi/2-x)}+\sqrt{\sin(\pi/2-x)}}\,dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}\,dx$ ...(ii) $1\frac{1}{2}$

adding (i) and (ii) to get $2I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 1 \cdot dx = x]_{\pi/6}^{\pi/3} = \pi/6.$ $\frac{1}{2}$

$\Rightarrow I = \frac{\pi}{12}$ $\frac{1}{2}$

20. $\frac{dy}{dx} = \frac{x+y}{x-y} = \frac{1+\frac{y}{x}}{1-\frac{y}{x}}$

Put y/x = v so that $\frac{dy}{dx} = v + x\frac{dv}{dx}$ $\frac{1}{2}$

$\therefore\ v + x\frac{dv}{dx} = \frac{1+v}{1-v} \Rightarrow x\frac{dv}{dx} = \frac{1+v}{1-v} - v = \frac{1+v-v+v^2}{1-v}$

$\Rightarrow \int \frac{1-v}{1+v^2}dv = \int \frac{dx}{x} \Rightarrow \int \frac{1}{1+v^2}dv - \frac{1}{2}\int \frac{2v}{1+v^2}dv = \int \frac{dx}{x}$ $1+\frac{1}{2}$

$\Rightarrow \tan^{-1} v = \frac{1}{2}\log|1+v^2| + \log|x| + c$ 1

$\Rightarrow \tan^{-1}\left(\frac{y}{x}\right) = \frac{1}{2}\log\left|\frac{x^2+y^2}{x^2}\right| + \log|x| + c$ 1

or $\tan^{-1}\left(\frac{y}{x}\right) = \frac{1}{2}\log|x^2+y^2| + c$

OR

$(1+x^2)dy + 2xy\,dx = \cot x.dx.$

$\Rightarrow \frac{dy}{dx}+\frac{2x}{1+x^2}\cdot y=\frac{\cot x}{1+x^2}$ 1

I.F. $= e^{\int \frac{2x}{1+x^2}dx} = e^{\log(1+x^2)} = (1+x^2)$ 1

$\therefore$ Solution is, $y\cdot(1+x^2)=\int \cot x\, dx = \log|\sin x|+c$ 1+1

or $y = \frac{1}{1+x^2}\cdot\log|\sin x|+\frac{c}{1+x^2}$

21. Writing the equations of given lines in standard form, as

$\frac{x-5}{5\lambda+2}=\frac{y-2}{-5}=\frac{z-1}{1}; \frac{x}{1}=\frac{y+\frac{1}{2}}{2\lambda}=\frac{z-1}{3}$ $\frac{1}{2}$

lines are perpendicular to each other,

$\Rightarrow (5\lambda+2)\cdot 1+(-5)(2\lambda)+1(3)=0$ $\frac{1}{2}$

$-5\lambda+5=0 \Rightarrow \lambda=1$ $\frac{1}{2}$

$\therefore$ lines are $\frac{x-5}{7}=\frac{y-2}{-5}-\frac{z-1}{1}; \frac{x}{1}=\frac{y+\frac{1}{2}}{2}=\frac{z-1}{3}$ $\frac{1}{2}$

Shortest distance between these lines $= \frac{|(\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2)|}{|\vec{b}_1\times\vec{b}_2|} = \frac{\left|\left(5\hat{i}+\frac{5}{2}\hat{j}\right)\cdot(-17\hat{i}-20\hat{j}+19\hat{k})\right|}{|\vec{b}_1\times\vec{b}_2|}$ 1

$= \frac{135}{|\vec{b}_1\times\vec{b}_2|} \neq 0$ $\frac{1}{2}$

$\therefore$ lines are not intersecting. $\frac{1}{2}$

22. Given $\frac{\vec{b}\cdot\vec{a}}{|\vec{a}|}=\frac{\vec{c}\cdot\vec{a}}{|\vec{a}|} \therefore \vec{b}\cdot\vec{a}=\vec{c}\cdot\vec{a}$ ...(i) $\frac{1}{2}$

$\vec{b}\perp\vec{c} \Rightarrow \vec{b}\cdot\vec{c}=0$ ...(ii) $\frac{1}{2}$

$(|3\vec{a}-2\vec{b}+2\vec{c}|)^2 = 9|\vec{a}|^2+4|\vec{b}|^2+4|\vec{c}|^2-12\,\vec{a}\cdot\vec{b}-8\,\vec{b}\cdot\vec{c}+12\,\vec{a}\cdot\vec{c}$ 1

$= 9(1)^2 + 4(2)^2 + 4(3)^2$ [using (i) and (ii)] 1

$= 9 + 16 + 36 = 61$

$\Rightarrow |3\vec{a} - 2\vec{b} + 2\vec{c}| = \sqrt{61}$ 1

23. Curve $y = \dfrac{x-7}{(x-2)(x-3)}$ cuts at x – axis at the point x = 7, y = 0 i.e. (7, 0) $\frac{1}{2}$

$\dfrac{dy}{dx} = \dfrac{(x^2-5x+6)\cdot 1 - (x-7)(2x-5)}{(x^2-5x+6)^2}$ $\frac{1}{2}$

at (7, 0) $\dfrac{dy}{dx} = \dfrac{20}{(20)^2} = \dfrac{1}{20}$ $\frac{1}{2}$

$\therefore$ Slope of tangent at (7, 0) is $\dfrac{1}{20}$ $\frac{1}{2}$

and slope of Normal at (7, 0) is –20 $\frac{1}{2}$

Equation of tangent at (7, 0) is $y - 0 = \dfrac{1}{20}(x - 7)$

or $x - 20y - 7 = 0$ 1

Equation of Normal at (7, 0) is $y - 0 = -20(x - 7)$

or $20x + y = 140$. $\frac{1}{2}$

SECTION D

24. $f(x) = \sin x + \dfrac{1}{2}\cos 2x \Rightarrow f'(x) = \cos x - \sin 2x$ 1

$f'(x) = 0 \Rightarrow \cos x - 2 \sin x \cos x = 0$

$\Rightarrow \cos x (1 - 2 \sin x) = 0$

$\Rightarrow x = \dfrac{\pi}{2}$ or $x = \dfrac{\pi}{6}$ 1

$x = \dfrac{\pi}{6} \in \left(0, \dfrac{\pi}{2}\right)$ 1

$f''(x) = -\sin x - 2\cos 2x$ 1

$f''(\pi/6) < 0 \Rightarrow x = \frac{\pi}{6}$ is a local maxima. 1

Local Max. Value = $f(\pi/6) = \frac{3}{4}$ 1

Local extreme values do exist at end points x = 0, x = $\frac{\pi}{2}$ but no marks are alotted here for that

25. $A^2 = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$ 2

$A^2 \cdot A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$ 3

or $A \cdot A^2 = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$

$\Rightarrow A^2 = A^{-1}$ 1

OR

Given System of equation can be written as

$\begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 13 \\ -2 \\ -2 \end{bmatrix}$ or $AX = B$ 1

$|A| = 2(0) + 3(-2) + 5(1) = -1 \neq 0$ 1

$\therefore X = A^{-1} \cdot B$

(adj. A) = $\begin{bmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{bmatrix}$ 2

$A^{-1} = -\begin{bmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix}$ $\frac{1}{2}$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix} \begin{bmatrix} 13 \\ -2 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

$\Rightarrow x = 2, y = 2, z = 3.$ $1\frac{1}{2}$

26. Let the events be

E_1 : bag I is selected
E_2 : bag II is selected
A : getting a red ball } 1

$P(E_1) = P(E_2) = \frac{1}{2}$ $\frac{1}{2}$

$P(A/E_1) = \frac{3}{9} = \frac{1}{3}; \; P(A/E_2) = \frac{5}{5+n}$ $\frac{1}{2}+1$

$P(E_2/A) = \frac{3}{5} = \frac{\frac{1}{2}\cdot\frac{5}{5+n}}{\frac{1}{2}\cdot\frac{1}{3}+\frac{1}{2}\cdot\frac{5}{5+n}}$ 2

$\Rightarrow \frac{3}{5} = \frac{15}{5+n+15} \Rightarrow n = 5.$ 1

27. Equation of plane passing through (2, 5, –3), (–2, –3, 5) and (5, 3, –3) is

$$\begin{vmatrix} x-2 & y-5 & z+3 \\ -4 & -8 & 8 \\ 3 & -2 & 0 \end{vmatrix} = 0$$ 1

$\Rightarrow 16(x-2)+24(y-5)+32(z+3)=0$

i.e. $2x + 3y + 4z - 7 = 0$ ...(i) 1

which in vector form can be written as $\vec{r}\cdot(2\hat{i}+3\hat{j}+4\hat{k}) = 7$ 1

Equation of line passing through (3, 1, 5) and (–1, –3, –1) is

$\frac{x-3}{4} = \frac{y-1}{4} = \frac{z-5}{6}$ or $\frac{x-3}{2} = \frac{y-1}{2} = \frac{z-5}{3}$ ...(ii) 1

Any point on (ii) is $(2\lambda+3, 2\lambda+1, 3\lambda+5)$ $\frac{1}{2}$

If this is point of intersection with plane (i), then

$2(2\lambda+3)+3(2\lambda+1)+4(3\lambda+5)-7=0$ $\frac{1}{2}$

$22\lambda+22=0 \Rightarrow \lambda=-1$ $\frac{1}{2}$

$\therefore$ Point of intersection is (1, –1, 2) $\frac{1}{2}$

OR

Equation of plane through the intersection of planes

$\vec{r}\cdot(\hat{i}+\hat{j}+\hat{k})-1=0$ and $\vec{r}\cdot(2\hat{i}+3\hat{j}-\hat{k})+4=0,$ is

$[\vec{r}\cdot(\hat{i}+\hat{j}+\hat{k})-1]+\lambda[\vec{r}\cdot(2\hat{i}+3\hat{j}-\hat{k})+4] = 0$ 1

$\Rightarrow \vec{r}\cdot[(1+2\lambda)\hat{i}+(1+3\lambda)\hat{j}+(1-\lambda)\hat{k}]-1+4\lambda = 0$ ...(i) 1

Plane (i) is || to x–axis $\Rightarrow 1+2\lambda=0 \Rightarrow \lambda=\frac{-1}{2}$ $1\frac{1}{2}$

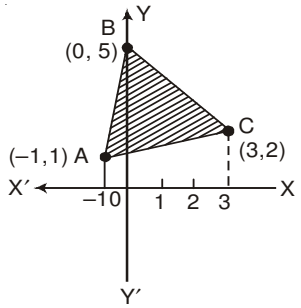
$\therefore$ Equation of plane is $\vec{r}\cdot\left(-\frac{1}{2}\hat{j}+\frac{3}{2}\hat{k}\right)-3=0$ $1\frac{1}{2}$

or $\vec{r}\cdot(-\hat{j}+3\hat{k})-6 = 0$

Distance of this plane from x–axis

$= \frac{|-6|}{\sqrt{(-1)^2+(3)^2}} = \frac{6}{\sqrt{10}}$ units 1

28. Let the points be A (–1, 1), B (0, 5) and C (3, 2)



Correct Figure 1

Equation of AB : $y = 4x+5$

BC : $y = 5-x$

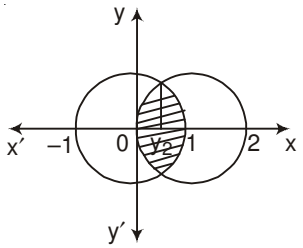
AC : $y = \frac{1}{4}(x+5)$ $1\frac{1}{2}$

Req. Area = $\int_{-1}^{0}(4x+5)dx+\int_{0}^{3}(5-x)dx-\int_{-1}^{3}\frac{1}{4}(x+5)dx$ $1\frac{1}{2}$

$$\therefore\ A = \left[\frac{(4x+5)^2}{8}\right]_{-1}^{0} + \left[\frac{(5-x)^2}{-2}\right]_{0}^{3} - \frac{1}{4}\left[\frac{(x+5)^2}{2}\right]_{-1}^{3}$$ 1

$$= 3 + \frac{21}{2} - 6 = \frac{15}{2}$$ 1

OR



Correct Figure 1

$$(x-1)^2 + y^2 = 1$$

and $x^2 + y^2 = 1 \Rightarrow (x-1)^2 = x^2$

$$\Rightarrow x = \frac{1}{2}$$ 1

$$\therefore \text{ Required area } = 2\left[\int_0^{\frac{1}{2}} \sqrt{1-(x-1)^2}\,dx + \int_{\frac{1}{2}}^{1} \sqrt{1-x^2}\,dx\right]$$ 2

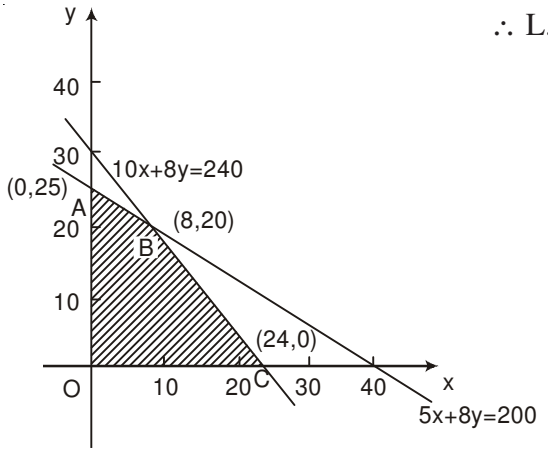
$$= 2\left[\frac{x-1}{2}\sqrt{1-(x-1)^2} + \frac{1}{2}\sin^{-1}(x-1)\right]_0^{\frac{1}{2}} + 2\left[\frac{x}{2}\sqrt{1-x^2} + \frac{1}{2}\sin^{-1}x\right]_{\frac{1}{2}}^{1}$$ 1

$$= 2\left[\frac{-\sqrt{3}}{8} + \frac{\pi}{6}\right] + 2\left[\frac{-\sqrt{3}}{8} + \frac{\pi}{6}\right] = \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right)$$ 1

29. Let number of Souvenirs of type A be x, and that of type B be y.

$\therefore$ L.P.P is maximise $P = 50x + 60y$ $\frac{1}{2}$

such that $5x + 8y \le 200$
$10x + 8y \le 240$
$x, y \ge 0$ $2\frac{1}{2}$



Correct Graph 2

P(at A) = ₹1500

P(at B) = ₹(400 + 1200) = ₹1600

P(at C) = ₹(1200)

$\therefore$ Max Profit = ₹ 1600, when number of Souvenirs of type A = 8 and number of Souvenirs of type B = 20.

QUESTION PAPER CODE 65/3/3
EXPECTED ANSWER/VALUE POINTS

SECTION A

1. DRs are (6, 2, 3) $\therefore$ DC's are $\left(\frac{6}{7},\frac{2}{7},\frac{3}{7}\right)$ $\frac{1}{2}+\frac{1}{2}$

OR

$\frac{x-1}{-3}=\frac{y-7}{p}=\frac{z-3}{2};\frac{x-1}{-3p}=\frac{y-5}{1}=\frac{z-6}{-5}$ $\frac{1}{2}$

$\Rightarrow 9p + p - 10 = 0 \Rightarrow p = 1$ $\frac{1}{2}$

2. $\frac{dy}{dx}-\frac{2}{x}\cdot y = 2x$ $\frac{1}{2}$

$\Rightarrow$ I.F. $= e^{-2\log x} = x^{-2} = \frac{1}{x^2}$ $\frac{1}{2}$

3. $|-2A| = (-2)^3 \cdot |A|$ $\frac{1}{2}$

$= -8 \times 4 = -32$ $\frac{1}{2}$

4. $y = \frac{\pi}{2} \Rightarrow \frac{dy}{dx} = 0$ $\frac{1}{2}+\frac{1}{2}$

SECTION B

5. $B'A' = \begin{bmatrix}4 & 7 & 2\\ 0 & 1 & 2\\ 2 & 4 & 6\end{bmatrix}\begin{bmatrix}3 & 1 & 7\\ 9 & 8 & 5\\ 0 & -2 & 4\end{bmatrix}$ 1

$= \begin{bmatrix}75 & 56 & 71\\ 9 & 4 & 13\\ 42 & 22 & 58\end{bmatrix}$ 1

6. $\int_a^b \frac{\log x}{x}dx = \left[\frac{1}{2}(\log x)^2\right]_a^b$ 1

$= \frac{1}{2}[(\log b)^2 - (\log a)^2]$ 1

7. $y^2 = m(a^2 - x^2) \Rightarrow 2y\frac{dy}{dx} = -2mx$ $\frac{1}{2}$

or $y\frac{dy}{dx} = -mx$ ...(i)

$y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = -m$ ...(ii) $\frac{1}{2}$

form (i) and (ii) we get $y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = \frac{y}{x}\frac{dy}{dx}$ 1

or $xy\frac{d^2y}{dx^2} + x\left(\frac{dy}{dx}\right)^2 - y\frac{dy}{dx} = 0$

8. $\int \frac{\sin(x-a)}{\sin(x+a)}dx = \int \frac{\sin(x+a-2a)}{\sin(x+a)}dx$ $\frac{1}{2}$

$= \int \left[\frac{\sin(x+a)\cdot\cos 2a}{\sin(x+a)} - \frac{\cos(x+a)\sin 2a}{\sin(x+a)}\right]dx$ $\frac{1}{2}$

$= x\cdot\cos 2a - \sin 2a\cdot\log|\sin(x+a)| + c$ $\frac{1}{2}+\frac{1}{2}$

OR

$\int (\log x)^2\cdot 1dx = (\log x)^2\cdot x - \int 2\cdot\log x\cdot\frac{1}{x}\cdot x\,dx$ 1

$= x\cdot(\log x)^2 - \left\{\log x\cdot 2x - \int \frac{1}{x}\cdot 2x dx\right\}$ $\frac{1}{2}$

$= x(\log x)^2 - 2x\log x + 2x + c$ $\frac{1}{2}$

9. A vector perpendicular to both $\vec{a}$ and $\vec{b} = \vec{a}\times\vec{b} = 19\hat{j} + 19\hat{k}$ or $\hat{j} + \hat{k}$ 1

$\therefore$ Unit vector perpendicular to both $\vec{a}$ and $\vec{b} = \frac{1}{\sqrt{2}}(\hat{j} + \hat{k})$ 1

OR

Let $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}, \vec{b} = -2\hat{i} + 3\hat{j} - 4\hat{k}, \vec{c} = \hat{i} - 3\hat{j} + 5\hat{k}$

$\vec{a}, \vec{b}, \vec{c}$ are coplanar if $\vec{a}\cdot\vec{b}\times\vec{c} = 0$ $\frac{1}{2}$

$$\vec{a}\cdot\vec{b}\times\vec{c}=\begin{vmatrix}1 & -2 & 3\\ -2 & 3 & -4\\ 1 & -3 & 5\end{vmatrix}=1(3)+2(-6)+3(3)$$
$$=3-12+9=0$$ $1+\frac{1}{2}$

Hence $\vec{a},\vec{b},\vec{c}$ are coplanar

10. A = {(S, F, M), (S, M, F), (M, F, S), (F, M, S)}
B = {(S, F, M), (M, F, S)} 1

Total number of possible arrangements = 6

$$P(B|A) = \frac{P(B\cap A)}{P(A)}$$

$$= \frac{2/6}{4/6}=\frac{1}{2}$$ 1

11. Given 2 $P(X = x_1) = 3P(X = x_2) = P(X = x_3) = 5P(X = x_4)$

Let $P(X = x_3) = k$, then $P(X = x_1) = \frac{k}{2}, P(X = x_2) = \frac{k}{3}$ and $P(X = x_4) = \frac{k}{5}$ $\frac{1}{2}$

$\therefore\ k+\frac{k}{2}+\frac{k}{3}+\frac{k}{5}=1\Rightarrow k=\frac{30}{61}$ 1

$\therefore$ Probability distribution is

X	x_1	x_2	x_3	x_4
P(X)	$\frac{15}{61}$	$\frac{10}{61}$	$\frac{30}{61}$	$\frac{6}{61}$

$\frac{1}{2}$

OR

(i) P(at least 4 heads) = $P(r \geq 4) = P(4) + P(5)$

$$= {}^5C_4\left(\frac{1}{2}\right)^1\left(\frac{1}{2}\right)^4 + {}^5C_5\left(\frac{1}{2}\right)^5 = 6\left(\frac{1}{2}\right)^5 = \frac{6}{32} \text{ or } \frac{3}{16}$$ 1

(ii) P(at most 4 heads) = $P(r \leq 4) = 1 - P(5)$

$$= 1-\left(\frac{1}{2}\right)^5 = \frac{31}{32}$$ 1

12. Let $e \in \mathbb{R}$ be the identity element.

then $a*e = e*a = a$ 1

$\Rightarrow a^2 + e^2 = e^2 + a^2 = a^2 \Rightarrow e^2 = 0 \Rightarrow e = 0.$ 1

$\therefore$ Identity element is $0 \in \mathbb{R}$

SECTION C

13. $\tan(\sec^{-1}\frac{1}{x}) = \sin(\tan^{-1}2) \Rightarrow \tan\left(\tan^{-1}\frac{\sqrt{1-x^2}}{x}\right) = \sin\left(\sin^{-1}\frac{2}{\sqrt{5}}\right)$ 2

$\Rightarrow \frac{\sqrt{1-x^2}}{x} = \frac{2}{\sqrt{5}} \Rightarrow \frac{1-x^2}{x^2} = \frac{4}{5}$ 1

$\Rightarrow 9x^2 = 5 \Rightarrow x^2 = \frac{5}{9} \Rightarrow x = \frac{\sqrt{5}}{3}, \{x > 0\}$ 1

14. $e^y \cdot (x+1) = 1 \Rightarrow e^y \cdot 1 + (x+1)e^y \cdot \frac{dy}{dx} = 0$ 1

$\Rightarrow \frac{dy}{dx} = -\frac{1}{x+1}$ 1

$\Rightarrow \frac{d^2y}{dx^2} = +\frac{1}{(x+1)^2}$ 1

$\therefore \frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2$ 1

OR

$y = \sin^{-1}\left(\frac{2\cdot 2^x}{1+(2^x)^2}\right) = \sin^{-1}\left(\frac{2t}{1+t^2}\right)$, where $t = 2^x$ 1

$\Rightarrow y = 2\tan^{-1}t$ 1

$\frac{dy}{dt} = \frac{2}{1+t^2}$ and $\frac{dt}{dx} = 2^x \cdot \log 2.$ 1

$\Rightarrow \frac{dy}{dx} = \frac{2}{1+t^2} \cdot 2^x \cdot \log 2 = \frac{2^{x+1}\cdot \log 2}{1+4^x}$ 1

15. $f(x) = 4x^3 - 6x^2 - 72x + 30$

$\Rightarrow f'(x) = 12x^2 - 12x - 72 = 12(x-3)(x+2)$ 1

$f'(x) = 0 \Rightarrow x = -2, x = 3$ 1

$\therefore$ possible intervals are $(-\infty, -2), (-2, 3), (3, \infty)$ $\frac{1}{2}$

$f'(x) < 0$ in $(-2, 3)$

and $f'(x) > 0$ in $(-\infty, -2)$ and $(3, \infty)$

$\Rightarrow f(x)$ is strictly increasing in $(-\infty, -2), (3, \infty)$ or $(-\infty, 2], [3, \infty)$ 1

and strictly decreasing in $(-2, 3)$ or $[-2, 3]$ $\frac{1}{2}$

16. (i) For $a \in Z, (a, a) \in R \because a - a = 0$ is divisible by 2

$\therefore$ R is reflexive ...(i) 1

Let (a, b) $\in$ R for a, b $\in$ Z, then a – b is divisible by 2

$\Rightarrow$ (b – a) is also divisible by 2

$\therefore$ (b, a) $\in$ R $\Rightarrow$ R is symmetric ...(ii) 1

For a, b, c, $\in$ Z, Let (a, b) $\in$ R and (b, c) $\in$ R

$\therefore$ a – b = 2p, p $\in$ Z, and b – c = 2q, q $\in$ Z,

adding, a – c = 2 (p + q) $\Rightarrow$ (a – c) is divisible by 2

$\Rightarrow$ (a, c) $\in$ R, so R is transitive ...(iii) $1\frac{1}{2}$

(i), (ii), and (iii) $\Rightarrow$ R is an equivalence relation. $\frac{1}{2}$

OR

$fof(x) = f\left(\frac{4x+3}{6x-4}\right)$ 1

$= \frac{4\left[\frac{4x+3}{6x-4}\right]+3}{6\left[\frac{4x+3}{6x-4}\right]-4}$ 1

$\Rightarrow \text{fof}(x) = \dfrac{4(4x+3)+3(6x-4)}{6(4x+3)-4(6x-4)} = \dfrac{34x}{34} = x$ 1

Since $\text{fof}(x) = x \Rightarrow \text{fof} = I \Rightarrow f^{-1} = f$ 1

17. Let $\Delta = \begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$

$R_1 \to R_1 - (R_2 + R_3) \Rightarrow \Delta = \begin{vmatrix} 0 & -2c & -2b \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$ 1

$\therefore \Delta = -2\begin{vmatrix} 0 & c & b \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = -\dfrac{2}{b}\begin{vmatrix} 0 & bc & b \\ b & bc+ab & b \\ c & bc & a+b \end{vmatrix}$ $1+\frac{1}{2}$

$C_2 \to C_2 - cC_3$

$\Rightarrow \Delta = -\dfrac{2}{b}\begin{vmatrix} 0 & 0 & b \\ b & ab & b \\ c & -ac & a+b \end{vmatrix}$ 1

$= -\dfrac{2}{b} \cdot b \cdot (-abc - abc) = 4abc.$ $\frac{1}{2}$

18. $y = (\sec^{-1} x)^2,\ x > 0$

$\dfrac{dy}{dx} = 2\sec^{-1} x \cdot \dfrac{1}{x\sqrt{x^2-1}}$ 1

$\Rightarrow x\sqrt{x^2-1}\dfrac{dy}{dx} = 2\sqrt{y}$ $\frac{1}{2}$

squaring both sides, we get

$x^2(x^2-1)\left(\dfrac{dy}{dx}\right)^2 = 4y \quad \text{or} \quad (x^4-x^2)\left(\dfrac{dy}{dx}\right)^2 = 4y$ $\frac{1}{2}$

differentiating w.r.t. x.

$(x^4-x^2)2 \cdot \dfrac{dy}{dx} \cdot \dfrac{d^2y}{dx^2} + (4x^3-2x)\left(\dfrac{dy}{dx}\right)^2 = 4 \cdot \dfrac{dy}{dx}$ 1

$\Rightarrow\ x^2(x^2-1)\dfrac{d^2y}{dx^2}+(2x^3-x)\dfrac{dy}{dx}-2=0$ 1

19. RHS $= \int_a^b f(a+b-x)\,dx = -\int_b^a f(t)\,dt$, where $a + b - x = t$, $dx = -dt$ $\frac{1}{2}$

$= \int_a^b f(t)\,dt = \int_a^b f(x)\,dx = \text{LHS}$ $\frac{1}{2}$

Let $I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \dfrac{dx}{1+\sqrt{\tan x}} = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \dfrac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}\,dx$ ...(i) $\frac{1}{2}$

$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \dfrac{\sqrt{\cos(\pi/2-x)}}{\sqrt{\cos(\pi/2-x)}+\sqrt{\sin(\pi/2-x)}}\,dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \dfrac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}\,dx$ ...(ii) $1\frac{1}{2}$

adding (i) and (ii) to get $2I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 1\cdot dx = x]_{\pi/6}^{\pi/3} = \pi/6.$ $\frac{1}{2}$

$\Rightarrow I = \dfrac{\pi}{12}$ $\frac{1}{2}$

20. $I = \int \dfrac{\sin 2x}{(\sin^2 x+1)(\sin^2 x+3)}\,dx$

Put $\sin^2 x = t \Rightarrow \sin 2x\,dx = dt$ $\frac{1}{2}$

$\therefore\ I = \int \dfrac{dt}{(t+1)(t+3)} = \int\left(\dfrac{1/2}{t+1}+\dfrac{-1/2}{t+3}\right)dt$ $1\frac{1}{2}$

$= \dfrac{1}{2}\log|t+1| - \dfrac{1}{2}\log|t+3| + c$ $1\frac{1}{2}$

$= \dfrac{1}{2}\log(\sin^2 x+1) - \dfrac{1}{2}\log(\sin^2+3) + c.$ $\frac{1}{2}$

21. Writing the equations of given lines in standard form, as

$\dfrac{x-5}{5\lambda+2} = \dfrac{y-2}{-5} = \dfrac{z-1}{1}; \dfrac{x}{1} = \dfrac{y+\frac{1}{2}}{2\lambda} = \dfrac{z-1}{3}$ $\frac{1}{2}$

lines are perpendicular to each other,

$\Rightarrow\ (5\lambda+2)\cdot 1+(-5)(2\lambda)+1(3)=0$ $\frac{1}{2}$

$-5\lambda+5=0\ \Rightarrow\ \lambda=1$ $\frac{1}{2}$

$\therefore$ lines are $\frac{x-5}{7}=\frac{y-2}{-5}-\frac{z-1}{1};\ \frac{x}{1}=\frac{y+\frac{1}{2}}{2}=\frac{z-1}{3}$ $\frac{1}{2}$

Shortest distance between these lines $=\frac{|(\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2)|}{|\vec{b}_1\times\vec{b}_2|}=\frac{\left|\left(5\hat{i}+\frac{5}{2}\hat{j}\right)\cdot(-17\hat{i}-20\hat{j}+19\hat{k})\right|}{|\vec{b}_1\times\vec{b}_2|}$ 1

$=\frac{135}{|\vec{b}_1\times\vec{b}_2|}\neq 0$ $\frac{1}{2}$

$\therefore$ lines are not intersecting. $\frac{1}{2}$

22. Given $\frac{\vec{b}\cdot\vec{a}}{|\vec{a}|}=\frac{\vec{c}\cdot\vec{a}}{|\vec{a}|}\ \therefore\ \vec{b}\cdot\vec{a}=\vec{c}\cdot\vec{a}$ ...(i) $\frac{1}{2}$

$\vec{b}\perp\vec{c}\ \Rightarrow\ \vec{b}\cdot\vec{c}=0$ ...(ii) $\frac{1}{2}$

$(|3\vec{a}-2\vec{b}+2\vec{c}|)^2=9|\vec{a}|^2+4|\vec{b}|^2+4|\vec{c}|^2-12\,\vec{a}\cdot\vec{b}-8\,\vec{b}\cdot\vec{c}+12\,\vec{a}\cdot\vec{c}$ 1

$=9(1)^2+4(2)^2+4(3)^2$ [using (i) and (ii)] 1

$= 9 + 16 + 36 = 61$

$\Rightarrow\ |3\vec{a}-2\vec{b}+2\vec{c}|=\sqrt{61}$ 1

23. $\frac{dy}{dx}=\frac{x+y}{x-y}=\frac{1+\frac{y}{x}}{1-\frac{y}{x}}$

Put y/x = v so that $\frac{dy}{dx}=v+x\frac{dv}{dx}$ $\frac{1}{2}$

$\therefore\ v+x\frac{dv}{dx}=\frac{1+v}{1-v}\ \Rightarrow\ x\frac{dv}{dx}=\frac{1+v}{1-v}-v=\frac{1+v-v+v^2}{1-v}$

$\Rightarrow \int \frac{1-v}{1+v^2}dv = \int \frac{dx}{x} \Rightarrow \int \frac{1}{1+v^2}dv - \frac{1}{2}\int \frac{2v}{1+v^2}dv = \int \frac{dx}{x}$ $1+\frac{1}{2}$

$\Rightarrow \tan^{-1} v = \frac{1}{2}\log|1+v^2| + \log|x| + c$ 1

$\Rightarrow \tan^{-1}\left(\frac{y}{x}\right) = \frac{1}{2}\log\left|\frac{x^2+y^2}{x^2}\right| + \log|x| + c$ 1

or $\tan^{-1}\left(\frac{y}{x}\right) = \frac{1}{2}\log|x^2+y^2| + c$

OR

$(1+x^2)dy + 2xy\,dx = \cot x.dx.$

$\Rightarrow \frac{dy}{dx} + \frac{2x}{1+x^2}\cdot y = \frac{\cot x}{1+x^2}$ 1

I.F. $= e^{\int \frac{2x}{1+x^2}dx} = e^{\log(1+x^2)} = (1+x^2)$ 1

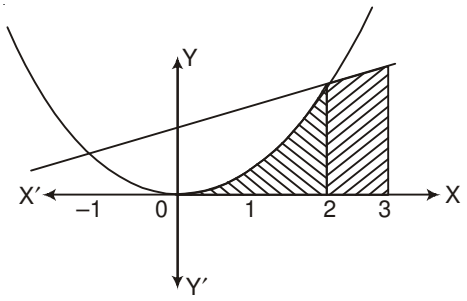
$\therefore$ Solution is, $y\cdot(1+x^2) = \int \cot x\,dx = \log|\sin x| + c$ 1+1

or $y = \frac{1}{1+x^2}\cdot\log|\sin x| + \frac{c}{1+x^2}$

SECTION D

24. $\{(x,y): 0 \le y \le x^2, 0 \le y \le x+2, -1 \le x \le 3\}$

Correct Figure 1



$\text{Area} = \int_{-1}^{2} x^2dx + \int_{2}^{3}(x+2)dx$ 2

$= \frac{x^3}{3}\Big]_{-1}^{2} + \frac{(x+2)^2}{2}\Big]_{2}^{3}$ 2

$= 3 + \frac{9}{2} = \frac{15}{2}$ 1

OR

$I = \lim_{h\to 0} h[f(1) + f(1+h) + f(1+2h) + \cdots + f(1+\overline{n-1})n)]$ 1

where $h = \frac{3}{n}$ or $nh = 3$ $\frac{1}{2}$

$= \lim_{h\to 0} h[(2+e^2) + (2+h+e^{2+2h}) + (2+2h+e^{2+4h}) + \cdots + (2+(n-1)h) + e^{2+2(n-1)h}]$ 1

$= \lim_{h\to 0} h\left[2n + h\cdot\frac{n(n-1)}{2}\right] + \lim_{h\to 0} h\cdot e^2[1+e^{2h}+e^{4h}+\cdots e^{2(n-1)h}]$ 1

$= \lim_{h\to 0}\left[2nh + \frac{nh(nh-h)}{2}\right] + \lim_{h\to 0} \cancel{h}\cdot\frac{e^2}{2\cancel{h}}\frac{e^{2nh}-1}{\frac{e^{2h}-1}{2h}}$ $1\frac{1}{2}$

$= 6 + \frac{9}{2} + \frac{e^2(e^6-1)}{2} = \frac{21}{2} + \frac{e^2(e^6-1)}{2}$ 1

25. p = (prob. of doublet) = 1/6 $\quad\therefore$ q = 5/6 1

X	0	1	2	3	4
P(X)	$\left(\frac{5}{6}\right)^4 = \frac{625}{1296}$	$4\cdot\left(\frac{5}{6}\right)^3\frac{1}{6} = \frac{500}{1296}$	$6\left(\frac{5}{6}\right)^2\left(\frac{1}{6}\right)^2 = \frac{150}{1296}$	$4\frac{5}{6}\left(\frac{1}{6}\right)^3 = \frac{20}{1296}$	$\left(\frac{1}{6}\right)^4 = \frac{1}{1296}$

1

XP(X)	0	$\frac{500}{1296}$	$\frac{300}{1296}$	$\frac{60}{1296}$	$\frac{4}{1296}$
$X^2P(X)$	0	$\frac{500}{1296}$	$\frac{600}{1296}$	$\frac{180}{1296}$	$\frac{16}{1296}$

1

1

Mean $= \Sigma XP(X) = \frac{864}{1296} = \frac{2}{3}$ 1

Variance $= \Sigma X^2\cdot P(X) - [\Sigma X\cdot P(X)]^2 = 1 - \frac{4}{9} = \frac{5}{9}$ 1

26. Equation of plane passing through (2, 5, –3), (–2, –3, 5) and (5, 3, –3) is

$$\begin{vmatrix} x-2 & y-5 & z+3 \\ -4 & -8 & 8 \\ 3 & -2 & 0 \end{vmatrix} = 0$$ 1

$\Rightarrow 16(x-2)+24(y-5)+32(z+3)=0$

i.e. $2x + 3y + 4z - 7 = 0$ ...(i) 1

which in vector form can be written as $\vec{r}\cdot(2\hat{i}+3\hat{j}+4\hat{k})=7$ 1

Equation of line passing through (3, 1, 5) and (–1, –3, –1) is

$\frac{x-3}{4}=\frac{y-1}{4}=\frac{z-5}{6}$ or $\frac{x-3}{2}=\frac{y-1}{2}=\frac{z-5}{3}$ ...(ii) 1

Any point on (ii) is $(2\lambda+3, 2\lambda+1, 3\lambda+5)$ $\frac{1}{2}$

If this is point of intersection with plane (i), then

$2(2\lambda+3)+3(2\lambda+1)+4(3\lambda+4)-7=0$ $\frac{1}{2}$

$22\lambda+22=0 \Rightarrow \lambda=-1$ $\frac{1}{2}$

$\therefore$ Point of intersection is (1, –1, 2) $\frac{1}{2}$

OR

Equation of plane through the intersection of planes

$\vec{r}\cdot(\hat{i}+\hat{j}+\hat{k})-1=0$ and $\vec{r}\cdot(2\hat{i}+3\hat{j}-\hat{k})+4=0,$ is

$[\vec{r}\cdot(\hat{i}+\hat{j}+\hat{k})-1]+\lambda[\vec{r}\cdot(2\hat{i}+3\hat{j}-\hat{k})+4] = 0$ 1

$\Rightarrow \vec{r}\cdot[(1+2\lambda)\hat{i}+(1+3\lambda)\hat{j}+(1-\lambda)\hat{k}]-1+4\lambda = 0$ ...(i) 1

Plane (i) is || to x–axis $\Rightarrow 1+2\lambda=0 \Rightarrow \lambda=\frac{-1}{2}$ $1\frac{1}{2}$

$\therefore$ Equation of plane is $\vec{r}\cdot\left(-\frac{1}{2}\hat{j}+\frac{3}{2}\hat{k}\right)-3 = 0$ $1\frac{1}{2}$

or $\vec{r}\cdot(-\hat{j}+3\hat{k})-6 = 0$

Distance of this plane from x–axis

$= \frac{|-6|}{\sqrt{(-1)^2+(3)^2}}=\frac{6}{\sqrt{10}}$ units 1

27. Let Given surface area of open cylinder be S.

Then $S = 2\pi rh + \pi r^2$

$\Rightarrow h = \dfrac{S - \pi r^2}{2\pi r}$ 1

Volume $V = \pi r^2 h$ $\frac{1}{2}$

$V = \pi r^2\left[\dfrac{S-\pi r^2}{2\pi r}\right] = \dfrac{1}{2}[Sr - \pi r^3]$ 1

$\dfrac{dV}{dr} = \dfrac{1}{2}[S - 3\pi r^2]$ $\frac{1}{2}$

$\dfrac{dV}{dr} = 0 \Rightarrow S = 3\pi r^2$ or $2\pi rh + \pi r^2 = 3\pi r^2$ 1

$\Rightarrow 2\pi rh = 2\pi r^2 \qquad \Rightarrow h = r$ 1

$\dfrac{d^2V}{dr^2} = -6\pi r < 0$ 1

$\therefore$ For volume to be maximum, height = radius

28. $|A| = 1(7) - 1(-3) + 1(-1) = 9$ 1

$(\text{adj } A) = \begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix}$ 2

$\Rightarrow A^{-1} = \dfrac{1}{9}\begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix}$ $\frac{1}{2}$

Given equations can be written as $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 1 & -2 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 11 \\ 0 \end{bmatrix}$

or $AX = B \Rightarrow X = A^{-1}B$ $\frac{1}{2}$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{9}\begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix}\begin{bmatrix} 6 \\ 11 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$ $1\frac{1}{2}$

$\therefore$ x = 1, y = 2, z = 3 $\frac{1}{2}$

OR

Let: $\begin{bmatrix} 2 & 3 & 1 \\ 2 & 4 & 1 \\ 3 & 7 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$ 1

$$R_1 \leftrightarrow R_3 \begin{bmatrix} 3 & 7 & 2 \\ 2 & 4 & 1 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} A$$

$$\begin{matrix} R_1 \to R_1 - R_3 \\ R_2 \to R_2 - R_3 \end{matrix} \begin{bmatrix} 1 & 4 & 1 \\ 0 & 1 & 0 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} A$$

$$R_3 \to R_3 - 2R_1 \begin{bmatrix} 1 & 4 & 1 \\ 0 & 1 & 0 \\ 0 & -5 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 1 & 0 \\ 3 & 0 & -2 \end{bmatrix} A$$

$$R_3 \to R_3 + 5R_2 \begin{bmatrix} 1 & 4 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 1 & 0 \\ -2 & 5 & -2 \end{bmatrix} A$$

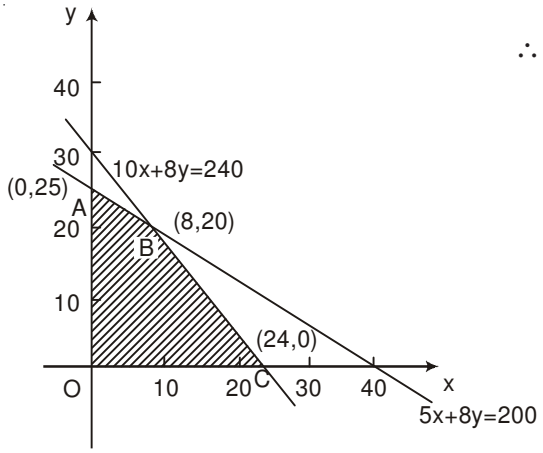
$$\begin{matrix} R_1 \to R_1 - 4R_2 \\ R_3 \to -R_3 \end{matrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -4 & 1 \\ -1 & 1 & 0 \\ 2 & -5 & 2 \end{bmatrix} A$$

$$R_1 \to R_1 - R_3 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 0 \\ 2 & -5 & 2 \end{bmatrix} A$$

4

$$\Rightarrow A^{-1} = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 0 \\ 2 & -5 & 2 \end{bmatrix}$$ 1

29. Let number of Souvenirs of type A be x, and that of type B be y.



$\therefore$ L.P.P is maximise $P = 50x + 60y$ $\frac{1}{2}$

such that $5x + 8y \leq 200$
$10x + 8y \leq 240$
$x, y \geq 0$
$2\frac{1}{2}$

Correct Graph 2

P(at A) = ₹1500

P(at B) = ₹(400 + 1200) = ₹1600

P(at C) = ₹(1200)

$\therefore$ Max Profit = ₹ 1600, when number of Souvenirs of type A = 8 and number of Souvenirs of type B = 20. 1

Strictly Confidential — (For Internal and Restricted Use Only)

Senior School Certificate Examination

March 2019

Marking Scheme — Mathematics (041) 65/4/1, 65/4/2, 65/4/3

General Instructions:

1. You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully. **Evaluation is a 10-12 days mission for all of us. Hence, it is necessary that you put in your best efforts in this process.**
2. Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one's own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. **However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and marks be awarded to them.**
3. The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
4. If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled.
5. If a question does not have any parts, marks must be awarded in the left hand margin and encircled.
6. If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out.
7. No marks to be deducted for the cumulative effect of an error. It should be penalized only once.
8. A full scale of marks 0 to 100 has to be used. Please do not hesitate to award full marks if the answer deserves it.
9. Every examiner has to necessarily do evaluation work for full working hours i.e. 8 hours every day and evaluate 25 answer books per day.
10. Ensure that you do not make the following common types of errors committed by the Examiner in the past:-
    - Leaving answer or part thereof unassessed in an answer book.
    - Giving more marks for an answer than assigned to it.
    - Wrong transfer of marks from the inside pages of the answer book to the title page.
    - Wrong question wise totaling on the title page.
    - Wrong totaling of marks of the two columns on the title page.
    - Wrong grand total.
    - Marks in words and figures not tallying.
    - Wrong transfer of marks from the answer book to online award list.
    - Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.)
    - Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
11. While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as (X) and awarded zero (0) Marks.
12. Any unassessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
13. The Examiners should acquaint themselves with the guidelines given in the Guidelines for spot Evaluation before starting the actual evaluation.
14. Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
15. The Board permits candidates to obtain photocopy of the Answer Book on request in an RTI application and also separately as a part of the re-evaluation process on payment of the processing charges.

QUESTION PAPER CODE 65/4/1
EXPECTED ANSWER/VALUE POINTS

SECTION A

1. $xy = A + 5x \Rightarrow x\dfrac{dy}{dx} + y = 5$ $\frac{1}{2}+\frac{1}{2}$

2. $|2 \text{ adj } A| = 2^3 |A|^{3-1} = 8 \times 81 = 648$ $\frac{1}{2}+\frac{1}{2}$

3. $\cos\theta = \dfrac{3+12-4}{3\times 7} \Rightarrow \theta = \cos^{-1}\left(\dfrac{11}{21}\right)$ $\frac{1}{2}+\frac{1}{2}$

OR

$\dfrac{x}{5/2}+\dfrac{y}{5}+\dfrac{z}{-5}=1, \text{length of } x-\text{intercept} = \dfrac{5}{2}$ 1

4. $\dfrac{dy}{dx} = \dfrac{-\sin e^x}{\cos e^x}\cdot e^x \text{ or } -e^x \cdot \tan e^x$ 1

SECTION B

5. $\displaystyle\int_{-\pi/4}^{0} \tan\left(\frac{\pi}{4}+x\right)dx = \log\left|\sec\left(\frac{\pi}{4}+x\right)\right|_{-\pi/4}^{0}$ $1+\frac{1}{2}$

$= \log\sqrt{2} \text{ or } \dfrac{1}{2}\log 2$ $\frac{1}{2}$

6. $a * b = 2a + b \in R \quad \forall\ a, b \in R$ $\therefore$ * is binary 1

$a * (b * c) = a * (2b + c) = 2a + 2b + c$

$(a * b) * c = (2a + b) * c = 4a + 2b + c$

In genral $a * (b * c) \neq (a * b) * c$ $\therefore$ * is not associative 1

7. Position vector of z = $\dfrac{2(\vec{a}-3\vec{b})-1(3\vec{a}+\vec{b})}{2-1}$ 1

$= -\vec{a} - 7\vec{b}$ 1

OR

$\vec{a}+\vec{b} = 4\hat{i}+\hat{j}-\hat{k}$ $\frac{1}{2}$

$\vec{a}-\vec{b} = -2\hat{i}+3\hat{j}-5\hat{k}$ $\frac{1}{2}$

$(\vec{a}+\vec{b})\cdot(\vec{a}-\vec{b}) = -8+3+5 = 0$ $\frac{1}{2}$

so $(\vec{a}+\vec{b}) \perp (\vec{a}-\vec{b})$ $\frac{1}{2}$

8. Given: A′ = A, B′ = B $\frac{1}{2}$

(AB – BA)′ = (AB)′ – (BA)′ $\frac{1}{2}$

= B′A′ – A′B′ $\frac{1}{2}$

= BA – AB

= –(AB – BA), Hence AB – BA is shew symmetric $\frac{1}{2}$

9. A: card bears odd number
B: Number on the card is greater than 5 } $\frac{1}{2}$

$A \cap B = \{7, 9, 11\}$ $\frac{1}{2}$

$P(A|B) = \frac{P(A\cap B)}{P(B)} = \frac{3/12}{7/2} = \frac{3}{7}$ $\frac{1}{2}+\frac{1}{2}$

10. Required probability = $\frac{{}^3C_2 \times {}^5C_2}{{}^8C_4}$ 1

$= \frac{3}{7}$ 1

OR

$n = 5 \quad p = \frac{1}{3} \quad q = \frac{2}{3}$ $\frac{1}{2}$

$P(X \geq 4) = P(X = 4) + P(X = 5)$

$= 5C_4\left(\frac{1}{3}\right)^4\left(\frac{2}{3}\right)^1 + {}^5C_5\left(\frac{1}{3}\right)^5$ 1

$= \frac{11}{243}$ $\frac{1}{2}$

11. I.F. = e^x $\frac{1}{2}$

Solution is $y.e^x = \int (\cos x - \sin x)\, e^x\, dx + C$ $\frac{1}{2}$

$y.e^x = e^x \cos x + C$ or $y = \cos x + Ce^{-x}$ 1

12. $I = \tan^{-1} x \cdot \frac{x^2}{2} - \int \frac{x^2}{2(1+x^2)} dx$ 1

$= \frac{x^2}{2}\tan^{-1} x - \frac{1}{2}[x - \tan^{-1} x] + C$ 1

OR

$I = \frac{1}{\sqrt{2}} \int \frac{dx}{\sqrt{\frac{5}{2} - 2x - x^2}}$ $\frac{1}{2}$

$= \frac{1}{\sqrt{2}} \int \frac{dx}{\sqrt{\left(\frac{\sqrt{7}}{\sqrt{2}}\right)^2 - (x+1)^2}}$ 1

$= \frac{1}{\sqrt{2}} \sin^{-1}\left[\frac{(x+1)\sqrt{2}}{\sqrt{7}}\right] + C$ $\frac{1}{2}$

SECTION C

13. $R_1 \to R_1 + R_2 + R_3$ & taking 12 + x common

$$(12+x)\begin{vmatrix} 1 & 1 & 1 \\ 4+x & 4-x & 4+x \\ 4+x & 4+x & 4-x \end{vmatrix} = 0$$ $1+\frac{1}{2}$

$C_2 \to C_2 - C_1 \quad C_3 \to C_3 - C_1$

$$(12+x)\begin{vmatrix} 1 & 0 & 0 \\ 4+x & -2x & 0 \\ 4+x & 0 & -2x \end{vmatrix} = 0$$ $1\frac{1}{2}$

$4x^2\ (12 + x) = 0$ $\frac{1}{2}$

$x = 0$ or $x = -12$ $\frac{1}{2}$

14. Given differential equation can be written as

$\frac{dy}{dx} = (1 + x^2)\,(1 + y^2)$ 1

$\int \frac{dy}{1 + y^2} = \int (x^2 + 1)\,dx$ 1

$\tan^{-1} y = \frac{x^3}{3} + x + C$ 1

$x = 0,\ y = 1 \Rightarrow C = \frac{\pi}{4}$ $\frac{1}{2}$

So particular solution is $\tan^{-1} y = \frac{x^3}{3} + x + \frac{\pi}{4}$ $\frac{1}{2}$

OR

Clearly given differential equation can be written as $\frac{dy}{dx} = \frac{y/x}{1 + (y/x)^2}$

Let $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ 1

Given equation becomes

$v + x\frac{dv}{dx} = \frac{v}{1 + v^2}$

$x\frac{dv}{dx} = \frac{-v^3}{1 + v^2}$

$\int \frac{(1 + v^2)}{v^3} dV = -\int \frac{dx}{x}$ 1

$-\frac{1}{2v^2} + \log|v| = -\log|x| + C$ 1

$$-\frac{x^2}{2y^2}+\log|y|=C$$

$x = 0,\ y = 1 \Rightarrow C = 0$ $\frac{1}{2}$

So, particular solution is $\log|y|=\frac{x^2}{2y^2}$ $\frac{1}{2}$

15. Let $x_1, x_2 \in R - \{2\}$ such that $f(x_1) = f(x_2)$

$\Rightarrow \frac{x_1-1}{x_1-2}=\frac{x_2-1}{x_2-2}$ $\frac{1}{2}$

$\Rightarrow x_1x_2 - 2x_1 - x_2 + 2 = x_1x_2 - 2x_2 - x_1 + 2$

$\Rightarrow x_1 = x_2$ 1

So f is one – one

For range let f(x) = y

$$\frac{x-1}{x-2}=y$$ $\frac{1}{2}$

$$x=\frac{2y-1}{y-1}$$ 1

Range of f = R – {1} = co domain B $\frac{1}{2}$

So f is onto.

$f^{-1}(y)=\frac{2y-1}{y-1}$ or $f^{-1}(x)=\frac{2x-1}{x-1}$ $\frac{1}{2}$

OR

For reflexive 1

For symmetric 1

For transitive

Let (a, b) ∈ S & (b, c) ∈ S

|a – b| = 3m, |b – c| = 3n

a – b = ±3m b – c = ±3n

a –c = 3(±m ± n) ⇒ a – c is divisible by 3

⇒ |a – c| is divisible by 3

⇒ (a, c) ∈ S

S is transitive

$1\frac{1}{2}$

As S is reflexive, symmetric & transitive

∴ S is an equivalence reation. $\frac{1}{2}$

16. $I = \int \frac{\cos(x+b+a-b)}{\sin(x+b)}dx$ 1

$= \int [\cot(x+b)\cos(a-b) - \sin(a-b)]dx$ 1

$= \cos(a-b)\log|\sin(x+b)| - x\sin(a-b) + C$ 2

17. $\frac{dx}{dt} = \cos t, \quad \frac{dy}{dt} = p\cos pt \Rightarrow \frac{dy}{dx} = \frac{p\cos pt}{\cos t}$ 1

$\frac{dy}{dx} = \frac{p\sqrt{1-y^2}}{\sqrt{1-x^2}}$ 1

$\Rightarrow (1-x^2)\left(\frac{dy}{dx}\right)^2 = p^2(1-y^2)$ differentiating both sides w.r.t x $\frac{1}{2}$

$\Rightarrow 2(1-x^2)\frac{dy}{dx}\frac{d^2y}{dx^2} - 2x\left(\frac{dy}{dx}\right)^2 = -2p^2y\frac{dy}{dx}$ 1

$\Rightarrow (1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + p^2y = 0$ $\frac{1}{2}$

OR

Let $\theta = \cos^{-1}x^2 \Rightarrow x^2 = \cos\theta$ 1

$\Rightarrow y = \tan^{-1}\left(\frac{\sqrt{1+\cos\theta} - \sqrt{1-\cos\theta}}{\sqrt{1+\cos\theta} + \sqrt{1-\cos\theta}}\right)$

$= \tan^{-1}\left(\dfrac{\cos\frac{\theta}{2} - \sin\frac{\theta}{2}}{\cos\frac{\theta}{2} + \sin\frac{\theta}{2}}\right)$ 1

$= \dfrac{\pi}{4} - \dfrac{1}{2}\theta$ 1

$\therefore \quad \dfrac{dy}{d\theta} = -\dfrac{1}{2}$ 1

18. LHS becomes $\tan^{-1}\dfrac{5}{12} + \tan^{-1}\dfrac{3}{4}$ 1

$= \tan^{-1}\left(\dfrac{\frac{5}{12} + \frac{3}{4}}{1 - \frac{5}{12}\cdot\frac{3}{4}}\right) = \tan^{-1}\dfrac{56}{33}$ 1+1

$= \sin^{-1}\dfrac{56}{65} = \text{RHS}$ 1

19. Let $u = x^{\cos x}$, $v = (\cos x)^{\sin x}$

$y = u + v$

$\Rightarrow \dfrac{dy}{dx} = \dfrac{du}{dx} + \dfrac{dv}{dx}$ ...(1) 1

$\log u = \cos x \log x \Rightarrow \dfrac{1}{u}\dfrac{du}{dx} = \dfrac{\cos x}{x} - \sin x \log x$

$\dfrac{du}{dx} = x^{\cos x}\left(\dfrac{\cos x}{x} - \sin x \log x\right)$ 1

$\log v = \sin x \log \cos x \Rightarrow \dfrac{1}{v}\dfrac{dv}{dx} = -\sin x \tan x + \cos x \log \cos x$

$\Rightarrow \quad \dfrac{dv}{dx} = (\cos x)^{\sin x}(-\sin x \tan x + \cos x \log \cos x)$ 1

So, $\dfrac{dy}{dx} = x^{\cos x}\left(\dfrac{\cos x}{x} - \sin x \log x\right) + (\cos x)^{\sin x}(-\sin x \tan x + \cos x \log \cos x)$ 1

20. Let $a - x = t \Rightarrow -dx = dt$ $\frac{1}{2}$

$$\text{RHS} = -\int_a^0 f(t)\,dt = \int_0^a f(t)\,dt = \int_0^a f(x)\,dx \quad \frac{1}{2}$$

$$I = \int_0^1 x^2(1-x)^n\,dx$$

$$= \int_0^1 (1-x)^2 x^n\,dx \quad 1$$

$$= \int_0^1 [x^n + x^{n+2} - 2x^{n+1}]\,dx \quad \frac{1}{2}$$

$$= \left.\frac{x^{n+1}}{n+1} + \frac{x^{n+3}}{n+3} - \frac{2x^{n+2}}{n+2}\right|_0^1 \quad 1$$

$$= \frac{1}{n+1} + \frac{1}{n+3} - \frac{2}{n+2} \quad \frac{1}{2}$$

21.

$$\left.\begin{aligned}\overrightarrow{BA} &= (x-4)\hat{i} - 6\hat{j} - 2\hat{k}\\ \overrightarrow{BC} &= -\hat{i} + 4\hat{j} + 3\hat{k}\\ \overrightarrow{BD} &= -8\hat{i} - \hat{j} + 3\hat{k}\end{aligned}\right\} \quad 1\frac{1}{2}$$

As points are coplanar

$$\therefore \begin{vmatrix} x-4 & -6 & -2\\ -1 & 4 & 3\\ -8 & -1 & 3\end{vmatrix} = 0 \quad 1$$

$15(x - 4) + 6 \times 21 - 2 \times 33 = 0$ 1

$15x = 0$

$x = 0$ $\frac{1}{2}$

22.

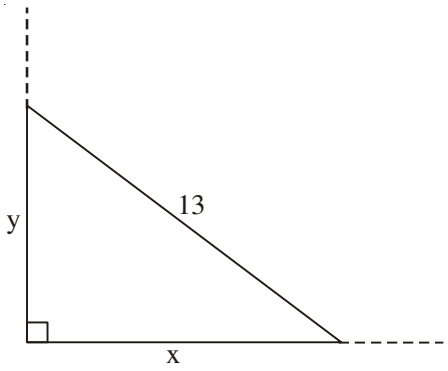


Figure $\frac{1}{2}$

$\frac{dx}{dt} = 2 \text{ cm/sec}$ $\frac{1}{2}$

$y = \sqrt{169 - x^2}$ 1

$\frac{dy}{dt} = -\frac{x}{\sqrt{169 - x^2}} \frac{dx}{dt}$ 1

$\left(\frac{dy}{dt}\right)_{x=5} = \frac{-5}{6} \text{ cm/sec}$ 1

Hence height is decreasing at the rate $\frac{5}{6}$ cm / sec

23. Equation of required plane is

$$\begin{vmatrix} x-3 & y+1 & z-2 \\ 2 & 3 & 2 \\ -4 & 0 & 4 \end{vmatrix} = 0$$ 2

$12(x - 3) - 16(y + 1) + 12(z - 2) = 0$

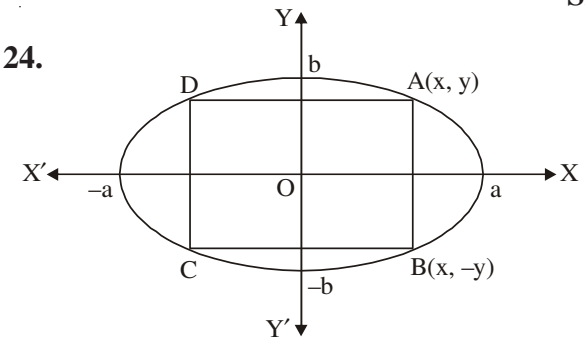
$3x - 4y + 3z = 19$

$\vec{r} \cdot (3\hat{i} - 4\hat{j} + 3\hat{k}) = 19$ 1

Distance from origin = $\frac{19}{\sqrt{34}}$ or $\frac{19\sqrt{34}}{34}$ 1

SECTION D

24.



Correct Figure 1

Let ABCD is a rectangle

$A = 4xy$

$A^2 = 16x^2(a^2 - x^2)\frac{b^2}{a^2} = f(x)$ 1

$f'(x) = \frac{16b^2}{a^2}(2a^2 x - 4x^3)$ 1

$f'(x) = 0 \Rightarrow x = \frac{a}{\sqrt{2}} \Rightarrow y = \frac{b}{\sqrt{2}}$ 1

$f''(x) = \frac{16b^2}{a^2}(2a^2 - 12x^2) = \frac{16b^2}{a^2}(-4a^2) < 0$ at $x = \frac{a}{\sqrt{2}}$

Area is maximum at $x = \frac{a}{\sqrt{2}}, y = \frac{b}{\sqrt{2}}$ 1

Required Area = $4\int_0^{a/\sqrt{2}} \frac{b}{\sqrt{2}} dx$ $\frac{1}{2}$

$= 4\frac{b}{\sqrt{2}}\frac{a}{\sqrt{2}} = 2ab$ $\frac{1}{2}$

Note: Since finding maximum/minimum area using integration is not discussed in prescribed books, so a student who tried to attempt but could not complete may be given full marks.

25. E_1: Selected person is cyclist
E_2: Selected person is scooter driver
E_3: Selected person is car driver
A: insured person met with an accident } 1

$P(E_1) = \frac{3}{18}, \ P(E_2) = \frac{6}{18}, \ P(E_3) = \frac{9}{18}$
$P(A|E_1) = 0.3, P(A|E_2) = 0.05, P(A|E_3) = 0.02$ } 2

$$P(E_1/A) = \frac{P(E_1)\,P(A|E_1)}{P(E_1)\,P(A|E_1) + P(E_2)\,P(A|E_2) + P(E_3)\,P(A|E_3)}$$

$$= \frac{\frac{3}{18}\times\frac{30}{100}}{\frac{3}{18}\times\frac{30}{100} + \frac{6}{18}\times\frac{5}{100} + \frac{9}{18}\times\frac{2}{100}}$$ 2

$= \frac{90}{138}$ or $\frac{15}{23}$ 1

26. $A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}$ we know that A = IA

i.e., $\begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$ 1

$R_1 \to R_1 - R_3$

$\begin{bmatrix} 1 & -4 & 7 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} A$

$R_2 \to R_2 - 3R_1, R_3 \to R_3 - R_1$

$\begin{bmatrix} 1 & -4 & 7 \\ 0 & 14 & -25 \\ 0 & 5 & -9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ -3 & 1 & 3 \\ -1 & 0 & 2 \end{bmatrix} A$

$R_2 \to -(R_2 - 3R_3)$

$\begin{bmatrix} 1 & -4 & 7 \\ 0 & 1 & -2 \\ 0 & 5 & -9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix} A$

$R_1 \to R_1 + 4R_2$

$R_3 \to R_3 - 5R_3$

$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -4 & 11 \\ 0 & -1 & 3 \\ -1 & 5 & -13 \end{bmatrix} A$ 4

$R_1 \to R_1 + R_3$

$R_2 \to R_2 + 2R_3$

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -4 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix} A$

$\Rightarrow A^{-1} = \begin{bmatrix} 0 & 1 & -4 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix}$ 1

OR

$|A| = 67 \neq 0 \qquad \therefore X = A^{-1}B$ $1+\frac{1}{2}$

adj $A = \begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$ 2

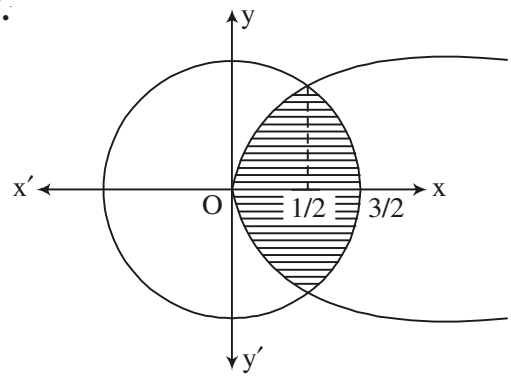
$$A^{-1} = \frac{1}{67}\begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$$ $\frac{1}{2}$

$$\text{So } X = \frac{1}{67}\begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}\begin{bmatrix} -4 \\ 2 \\ 11 \end{bmatrix}$$

$$= \frac{1}{67}\begin{bmatrix} 201 \\ -134 \\ 67 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$$ $1\frac{1}{2}$

$x = 3,\ y = -2,\ z = 1$ $\frac{1}{2}$

27.



Correct Figure 1

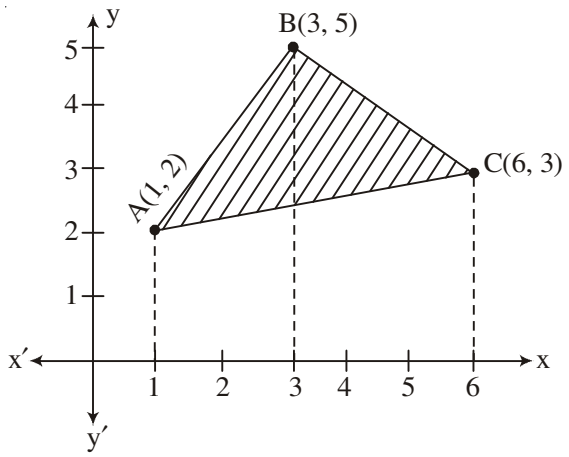
x coordinate of Point of intersection = $\frac{1}{2}$ 1

$$\text{Required Area} = 2\left[\int_0^{\frac{1}{2}} 2\sqrt{x}\,dx + \int_{\frac{1}{2}}^{3/2} \sqrt{\left(\frac{3}{2}\right)^2 - x^2}\,dx\right]$$ 2

$$= 2\left[\frac{4}{3}x^{3/2}\Big|_0^{\frac{1}{2}} + \frac{x}{2}\sqrt{\frac{9}{4} - x^2} + \frac{9}{8}\sin^{-1}\frac{2x}{3}\Big|_{\frac{1}{2}}^{3/2}\right]$$ 1

$$= \frac{\sqrt{2}}{6} + \frac{9\pi}{8} - \frac{9}{4}\sin^{-1}\frac{1}{3}$$ 1

OR



Correct figure 1

Correct points of intersection $1\frac{1}{2}$

$$\text{Required Area} = \int_1^3 \frac{3x+1}{2}dx + \int_3^6 \frac{21-2x}{3}dx - \int_1^6 \frac{x+9}{5}dx$$ 2

$$= \left(\frac{3x^2}{4} + \frac{x}{2}\right)\Big|_1^3 + \left(7x - \frac{x^2}{3}\right)\Big|_3^6 - \left(\frac{x^2}{10} + \frac{9x}{5}\right)\Big|_1^6 \quad 1$$

$$= 7 + 12 - \frac{25}{2}$$

$$= \frac{13}{2} \quad \frac{1}{2}$$

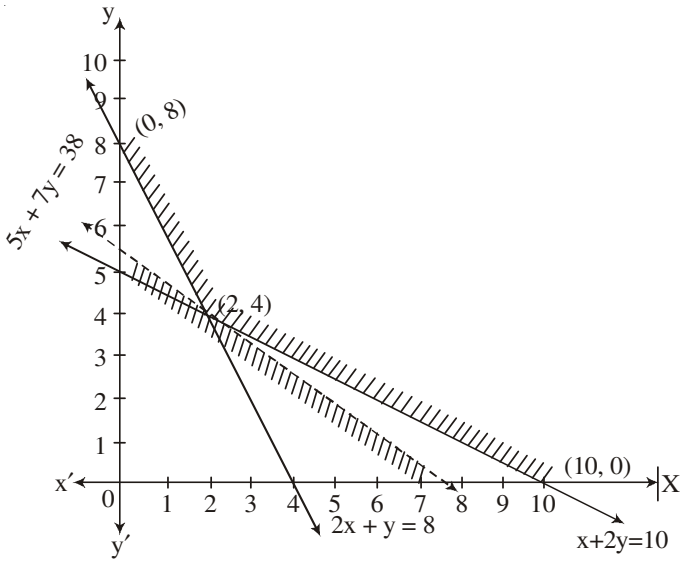
28. Let Quantity of Food I = x kg

Quantity of Food II = y kg

Out L.P.P. is

Minimize Z = 50x + 70y subject to $\frac{1}{2}$

$$\left.\begin{array}{l} 2x + y \geq 8 \\ x + 2y \geq 10 \\ x \geq 0,\ y \geq 0 \end{array}\right\} \quad 2\frac{1}{2}$$



Correct graph 2

Corner points	Value of Z	
(10, 0)	500	
(2, 4)	380	→ minimum
(0, 8)	560	

$\frac{1}{2}$

Consider $50x + 70y < 380$

which has no point common with feasible region

So minimum value of Z = ₹ 380 $\frac{1}{2}$

at x = 2 kg, y = 4 kg

29. Required equation of line is

$\vec{r} = 2\hat{i} + 3\hat{j} + 2\hat{k} + \mu(2\hat{i} - 3\hat{j} + 6\hat{k})$ 2

$\vec{a}_1 = -2\hat{i} + 3\hat{j}, \vec{a}_2 = 2\hat{i} + 3\hat{j} + 2\hat{k}, \quad \vec{b} = 2\hat{i} - 3\hat{j} + 6\hat{k}$

$\vec{a}_2 - \vec{a}_1 = 4\hat{i} + 2\hat{k}$ 1

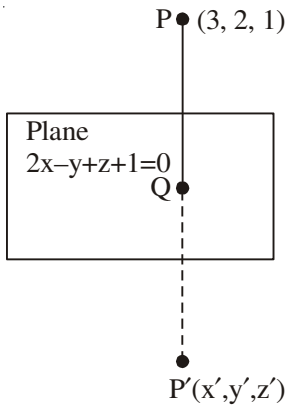
$d = \frac{|(\vec{a}_2 - \vec{a}_1) \times \vec{b}|}{|\vec{b}|}$

$(\vec{a}_2 - \vec{a}_1) \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 0 & 2 \\ 2 & -3 & 6 \end{vmatrix}$

$= 6\hat{i} - 20\hat{j} - 12\hat{k}$ 2

$d = \frac{\sqrt{580}}{7}$ 1

OR



Correct figure 1

Equation of PQ

$\frac{x-3}{2} = \frac{y-2}{-1} = \frac{z-1}{1} = \lambda$ $\frac{1}{2}$

Coodinates of Q $(2\lambda + 3, -\lambda + 2, \lambda + 1)$ $\frac{1}{2}$

As Q lies on plane

$\therefore 4\lambda + 6 + \lambda - 2 + \lambda + 1 = -1$

gives, $\lambda = -1$ 1

Coordinates of Q (1, 3, 0) $\frac{1}{2}$

$PQ = \sqrt{6}$ 1

For unique P′(x′, y′, z′)

$\frac{x'+3}{2}=1,\ \frac{y'+2}{2}=3,\ \frac{z'+1}{2}=0$ 1

$x' = -1\ \ y' = 4\ \ z' = -1$

image is (–1, 4, –1) $\frac{1}{2}$

QUESTION PAPER CODE 65/4/2
EXPECTED ANSWER/VALUE POINTS

SECTION A

1. $|2 \text{ adj } A| = 2^3 |A|^{3-1} = 8 \times 81 = 648$ $\frac{1}{2}+\frac{1}{2}$

2. $\cos\theta = \frac{3+12-4}{3\times 7} \Rightarrow \theta = \cos^{-1}\left(\frac{11}{21}\right)$ $\frac{1}{2}+\frac{1}{2}$

OR

$\frac{x}{5/2}+\frac{y}{5}+\frac{z}{-5}=1, \text{length of } x-\text{intercept} = \frac{5}{2}$ 1

3. $\frac{dy}{dx} = +\frac{\text{cosec}\,(\cot\sqrt{x})\cot(\cot\sqrt{x})\,\text{cosec}^2\sqrt{x}}{2\sqrt{x}}$ 1

4. $\frac{dx}{dy}+\frac{x}{1+y^2}=\frac{\tan^{-1}y}{1+y^2}$ $\frac{1}{2}$

I.F. $= e^{\tan^{-1}y}$ $\frac{1}{2}$

SECTION B

5. A: card bears odd number
B: Number on the card is greater than 5 $\frac{1}{2}$

$A \cap B = \{7, 9, 11\}$ $\frac{1}{2}$

$P(A|B) = \frac{P(A\cap B)}{P(B)} = \frac{3/12}{7/12} = \frac{3}{7}$ $\frac{1}{2}+\frac{1}{2}$

6. Required probability $= \frac{{}^3C_2 \times {}^5C_2}{{}^8C_4}$ 1

$= \frac{3}{7}$ 1

OR

$n = 5 \quad p = \frac{1}{3} \quad q = \frac{2}{3}$ $\frac{1}{2}$

$P(X \geq 4) = P(X = 4) + P(X = 5)$

$= {}^5C_4\left(\frac{1}{3}\right)^4\left(\frac{2}{3}\right)^1 + {}^5C_5\left(\frac{1}{3}\right)^5$ 1

$= \frac{11}{243}$ $\frac{1}{2}$

7. $I = \tan^{-1}x \cdot \frac{x^2}{2} - \int \frac{x^2}{2(1+x^2)}dx$ 1

$= \frac{x^2}{2}\tan^{-1}x - \frac{1}{2}[x - \tan^{-1}x] + C$ 1

OR

$I = \frac{1}{\sqrt{2}}\int \frac{dx}{\sqrt{\frac{5}{2} - 2x - x^2}}$ $\frac{1}{2}$

$= \frac{1}{\sqrt{2}}\int \frac{dx}{\sqrt{\left(\frac{\sqrt{7}}{\sqrt{2}}\right)^2 - (x+1)^2}}$ 1

$= \frac{1}{\sqrt{2}}\sin^{-1}\left[\frac{(x+1)\sqrt{2}}{\sqrt{7}}\right] + C$ $\frac{1}{2}$

8. $\int_{-\pi/4}^{0} \tan\left(\frac{\pi}{4} + x\right)dx = \log\left|\sec\left(\frac{\pi}{4} + x\right)\right|_{-\pi/4}^{0}$ $1+\frac{1}{2}$

$= \log\sqrt{2}$ or $\frac{1}{2}\log 2$ $\frac{1}{2}$

9. Position vector of $z = \frac{2(\vec{a} - 3\vec{b}) - 1(3\vec{a} + \vec{b})}{2-1}$ 1

$= -\vec{a} - 7\vec{b}$ 1

OR

$\vec{a} + \vec{b} = 4\hat{i} + \hat{j} - \hat{k}$ $\frac{1}{2}$

$\vec{a} - \vec{b} = -2\hat{i} + 3\hat{j} - 5\hat{k}$ $\frac{1}{2}$

$(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = -8 + 3 + 5 = 0$ $\frac{1}{2}$

so $(\vec{a} + \vec{b}) \perp (\vec{a} - \vec{b})$ $\frac{1}{2}$

10. Given: A′ = A, B′ = B $\frac{1}{2}$

$(AB - BA)' = (AB)' - (BA)'$ $\frac{1}{2}$

$= B'A' - A'B'$ $\frac{1}{2}$

$= BA - AB$

$= -(AB - BA)$, Hence AB – BA is shew symmetric $\frac{1}{2}$

11. a, b ∈ N ⇒ ab ∈ N ⇒ a + ab ∈ N 1

∴ * is binary

a * (b * c) = a * (b + bc) = a + ab + abc

(a * b) * c = (a + ab) * c = a + ab + ac + abc

In general a * (b * c) ≠ (a * b) * c ∴ * is not associative 1

12. I.F. = $e^{\int 1\,dy} = e^y$ $\frac{1}{2}$

Solution is

$x \cdot e^y = \int e^y (\tan y + \sec^2 y)dy + C$ 1

$x \cdot e^y = e^y \tan y + C$ or $x = \tan y + ce^{-y}$ $\frac{1}{2}$

SECTION C

13. Let $u = x^{\cos x}$, $v = (\cos x)^{\sin x}$

$y = u + v$

$\Rightarrow \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$ ...(1) 1

$\log u = \cos x \log x \Rightarrow \frac{1}{u}\frac{du}{dx} = \frac{\cos x}{x} - \sin x \log x$

$\frac{du}{dx} = x^{\cos x}\left(\frac{\cos x}{x} - \sin x \log x\right)$ 1

$\log v = \sin x \log \cos x \Rightarrow \frac{1}{v}\frac{dv}{dx} = -\sin x \tan x + \cos x \log \cos x$

$\Rightarrow \frac{dv}{dx} = (\cos x)^{\sin x}(-\sin x \tan x + \cos x \log \cos x)$ 1

So, $\frac{dy}{dx} = x^{\cos x}\left(\frac{\cos x}{x} - \sin x \log x\right) + (\cos x)^{\sin x}(-\sin x \tan x + \cos x \log \cos x)$

14. Let $a - x = t \Rightarrow -dx = dt$ $\frac{1}{2}$

$RHS = -\int_a^0 f(t)\,dt = \int_0^a f(t)\,dt = \int_0^a f(x)\,dx$ $\frac{1}{2}$

$I = \int_0^1 x^2(1-x)^n dx$

$= \int_0^1 (1-x)^2 x^n dx$ 1

$= \int_0^1 [x^n + x^{n+2} - 2x^{n+1}]\,dx$ $\frac{1}{2}$

$= \left.\frac{x^{n+1}}{n+1} + \frac{x^{n+3}}{n+3} - \frac{2x^{n+2}}{n+2}\right|_0^1$ 1

$= \frac{1}{n+1} + \frac{1}{n+3} - \frac{2}{n+2}$ $\frac{1}{2}$

15. $\overrightarrow{BA} = (x-4)\hat{i} - 6\hat{j} - 2\hat{k}$

$\overrightarrow{BC} = -\hat{i} + 4\hat{j} + 3\hat{k}$

$\overrightarrow{BD} = -8\hat{i} - \hat{j} + 3\hat{k}$

$1\frac{1}{2}$

As points are coplanar

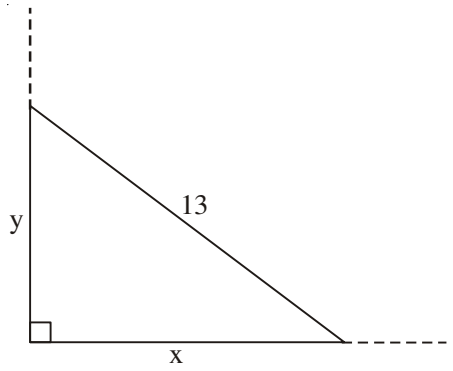
$\therefore \begin{vmatrix} x-4 & -6 & -2 \\ -1 & 4 & 3 \\ -8 & -1 & 3 \end{vmatrix} = 0$ 1

$15(x-4) + 6 \times 21 - 2 \times 33 = 0$ 1

$15x = 0$

$x = 0$ $\frac{1}{2}$

16. Figure $\frac{1}{2}$



$\frac{dx}{dt} = 2\ cm/sec$ $\frac{1}{2}$

$y = \sqrt{169 - x^2}$ 1

$\frac{dy}{dt} = -\frac{x}{\sqrt{169-x^2}}\frac{dx}{dt}$ 1

$\left(\frac{dy}{dt}\right)_{x=5} = \frac{-5}{6}\ cm/sec$ 1

Hence height is decreasing at the rate $\frac{5}{6}\ cm/sec$

17. Equation of required plane is

$\begin{vmatrix} x-3 & y+1 & z-2 \\ 2 & 3 & 2 \\ -4 & 0 & 4 \end{vmatrix} = 0$ 2

$12(x-3) - 16(y+1) + 12(z-2) = 0$

$3x - 4y + 3z = 19$

$\vec{r} \cdot (3\hat{i} - 4\hat{j} + 3\hat{k}) = 19$ 1

Distance from origin = $\frac{19}{\sqrt{34}}$ or $\frac{19\sqrt{34}}{34}$ 1

18. Given differential equation can be written as

$$\frac{dy}{dx} = (1 + x^2)(1 + y^2)$$ 1

$$\int \frac{dy}{1 + y^2} = \int (x^2 + 1)\, dx$$ 1

$$\tan^{-1} y = \frac{x^3}{3} + x + C$$ 1

$$x = 0,\ y = 1 \Rightarrow C = \frac{\pi}{4}$$ $\frac{1}{2}$

So particular solution is $\tan^{-1} y = \frac{x^3}{3} + x + \frac{\pi}{4}$ $\frac{1}{2}$

OR

Clearly given differential equation can be written as $\frac{dy}{dx} = \frac{y/x}{1 + (y/x)^2}$

Let $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ 1

Given equation becomes

$$v + x\frac{dv}{dx} = \frac{v}{1 + v^2}$$

$$x\frac{dv}{dx} = \frac{-v^3}{1 + v^2}$$

$$\int \frac{(1 + v^2)}{v^3} dV = -\int \frac{dx}{x}$$ 1

$$-\frac{1}{2v^2} + \log|v| = -\log|x| + C$$ 1

$$-\frac{x^2}{2y^2} + \log|y| = C$$

$x = 0,\ y = 1 \Rightarrow C = 0$ $\frac{1}{2}$

So, particular solution is $\log|y| = \frac{x^2}{2y^2}$ $\frac{1}{2}$

19. Let $x_1, x_2 \in R - \{2\}$ such that $f(x_1) = f(x_2)$

$\Rightarrow \quad \frac{x_1 - 1}{x_1 - 2} = \frac{x_2 - 1}{x_2 - 2}$ $\frac{1}{2}$

$\Rightarrow \quad x_1x_2 - 2x_1 - x_2 + 2 = x_1x_2 - 2x_2 - x_1 + 2$

$\Rightarrow \quad x_1 = x_2$ 1

So f is one – one

For range let $f(x) = y$

$\frac{x - 1}{x - 2} = y$ $\frac{1}{2}$

$x = \frac{2y - 1}{y - 1}$ 1

Range of f = R – {1} = co domain B $\frac{1}{2}$

So f is onto.

$f^{-1}(y) = \frac{2y - 1}{y - 1}$ or $f^{-1}(x) = \frac{2x - 1}{x - 1}$ $\frac{1}{2}$

OR

For reflexive 1

For symmetric 1

For transitive

Let $(a, b) \in S$ & $(b, c) \in S$

$|a - b| = 3m,\ |b - c| = 3n$

$a - b = \pm 3m \qquad b - c = \pm 3n$

$a - c = 3(\pm m \pm n) \Rightarrow a - c$ is divisible by 3 $1\frac{1}{2}$

$\Rightarrow$ |a – c| is divisible by 3

$\Rightarrow$ (a, c) $\in$ S

S is transitive

As S is reflexive, symmetric & transitive

$\therefore$ S is an equivalence reation. $\frac{1}{2}$

20. $\frac{dx}{dt} = \cos t, \quad \frac{dy}{dt} = p \cos pt \Rightarrow \frac{dy}{dx} = \frac{p \cos pt}{\cos t}$ 1

$$\frac{dy}{dx} = \frac{p\sqrt{1-y^2}}{\sqrt{1-x^2}}$$ 1

$$\Rightarrow (1-x^2)\left(\frac{dy}{dx}\right)^2 = p^2(1-y^2)$$ differentiating both sides w.r.t x $\frac{1}{2}$

$$\Rightarrow 2(1-x^2)\frac{dy}{dx}\frac{d^2y}{dx^2} - 2x\left(\frac{dy}{dx}\right)^2 = -2p^2y\frac{dy}{dx}$$ 1

$$\Rightarrow (1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + p^2y = 0$$ $\frac{1}{2}$

OR

Let $\theta = \cos^{-1} x^2 \Rightarrow x^2 = \cos \theta$ 1

$$\Rightarrow y = \tan^{-1}\left(\frac{\sqrt{1+\cos\theta} - \sqrt{1-\cos\theta}}{\sqrt{1+\cos\theta} + \sqrt{1-\cos\theta}}\right)$$

$$= \tan^{-1}\left(\frac{\cos\frac{\theta}{2} - \sin\frac{\theta}{2}}{\cos\frac{\theta}{2} + \sin\frac{\theta}{2}}\right)$$ 1

$$= \frac{\pi}{4} - \frac{1}{2}\theta$$ 1

$\therefore \quad \frac{dy}{d\theta} = -\frac{1}{2}$ 1

21. $R_1 \to R_1 + R_2 + R_3$

$$\begin{vmatrix} 2x+2y & 2x+2y & 2x+2y \\ y & x+y & x \\ x+y & x & y \end{vmatrix} = k(x^3+y^3) \qquad 1$$

$$2(x+y)\begin{vmatrix} 1 & 1 & 1 \\ y & x+y & x \\ x+y & x & y \end{vmatrix} = k(x^3+y^3) \qquad \frac{1}{2}$$

$C_2 \to C_2 - C_1$, $C_3 \to C_3 - C_1$

$$2(x+y)\begin{vmatrix} 1 & 0 & 0 \\ y & x & x-y \\ x+y & -y & -x \end{vmatrix} = k(x^3+y^3) \qquad 1\frac{1}{2}$$

$2(x + y)(-x^2 + xy - y^2) = k(x^3 + y^3)$ $\frac{1}{2}$

$-2(x^3 + y^3) = k(x^3 + y^3)$

$\therefore \quad k = -2$ $\frac{1}{2}$

22. $I = \dfrac{1}{\cos(b-a)} \int \dfrac{\cos[(x-a)-(x-b)]}{\sin(x-a)\cos(x-b)}\,dx$ 2

$$= \frac{1}{\cos(b-a)} \int \left[\frac{\cos(x-a)\,\cancel{\cos(x-b)}}{\sin(x-a)\,\cancel{\cos(x-b)}} + \frac{\cancel{\sin(x-a)}\,\sin(x-b)}{\cancel{\sin(x-a)}\,\cos(x-b)} \right] dx \qquad 1$$

$$= \frac{1}{\cos(b-a)} [\log|\sin(x-a)| + \log|\sec(x-b)|] + C \qquad 1$$

23. L.H.S. becomes

$$\tan^{-1}\frac{8}{15} + \tan^{-1}\frac{3}{4} \qquad \frac{1}{2}+\frac{1}{2}$$

$$= \tan^{-1}\left(\frac{\frac{8}{15}+\frac{3}{4}}{1-\frac{8}{15}\times\frac{3}{4}} \right) = \tan^{-1}\frac{77}{36} \qquad 1+1$$

$$= \cot^{-1}\frac{36}{77} = \text{RHS} \qquad 1$$

SECTION D

24. $A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}$ we know that $A = IA$

i.e., $\begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$ 1

$R_1 \to R_1 - R_3$

$\begin{bmatrix} 1 & -4 & 7 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} A$

$R_2 \to R_2 - 3R_1, R_3 \to R_3 - R_1$

$\begin{bmatrix} 1 & -4 & 7 \\ 0 & 14 & -25 \\ 0 & 5 & -9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ -3 & 1 & 3 \\ -1 & 0 & 2 \end{bmatrix} A$

$R_2 \to -(R_2 - 3R_3)$

$\begin{bmatrix} 1 & -4 & 7 \\ 0 & 1 & -2 \\ 0 & 5 & -9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix} A$

$R_1 \to R_1 + 4R_2$

$R_3 \to R_3 - 5R_3$

$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -4 & 11 \\ 0 & -1 & 3 \\ -1 & 5 & -13 \end{bmatrix} A$ 4

$R_1 \to R_1 + R_3$

$R_2 \to R_2 + 2R_3$

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -4 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix} A$

$\Rightarrow A^{-1} = \begin{bmatrix} 0 & 1 & -4 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix}$ 1

OR

$|A| = 67 \neq 0 \qquad \therefore\ X = A^{-1}B$ $1+\frac{1}{2}$

$$\text{adj } A = \begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$$ 2

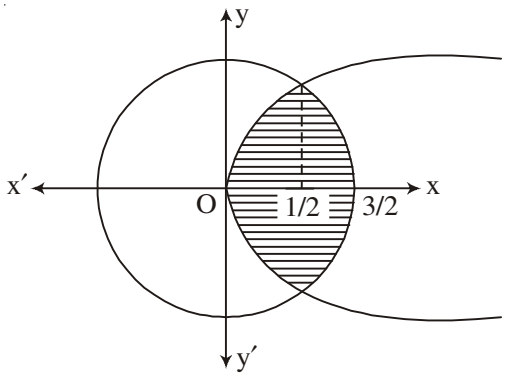
$$A^{-1} = \frac{1}{67}\begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$$ $\frac{1}{2}$

$$\text{So } X = \frac{1}{67}\begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}\begin{bmatrix} -4 \\ 2 \\ 11 \end{bmatrix}$$

$$= \frac{1}{67}\begin{bmatrix} 201 \\ -134 \\ 67 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$$ $1\frac{1}{2}$

$x = 3,\ y = -2,\ z = 1$ $\frac{1}{2}$

25. Correct Figure 1



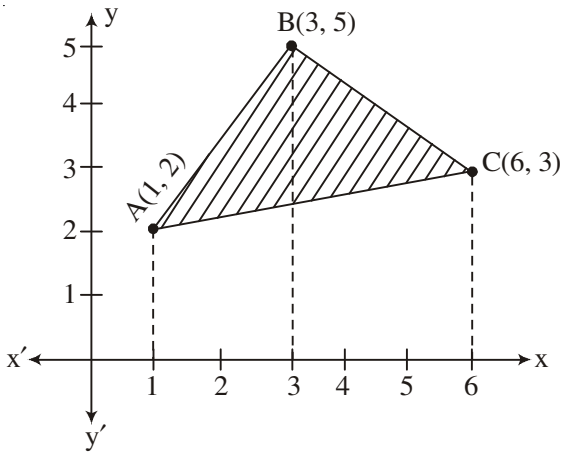
x coordinate of Point of intersection $= \frac{1}{2}$ 1

$$\text{Required Area} = 2\left[\int_0^{\frac{1}{2}} 2\sqrt{x}\,dx + \int_{\frac{1}{2}}^{3/2} \sqrt{\left(\frac{3}{2}\right)^2 - x^2}\,dx\right]$$ 2

$$= 2\left[\frac{4}{3}x^{3/2}\Big|_0^{\frac{1}{2}} + \frac{x}{2}\sqrt{\frac{9}{4} - x^2} + \frac{9}{8}\sin^{-1}\frac{2x}{3}\Big|_{\frac{1}{2}}^{3/2}\right]$$ 1

$$= \frac{\sqrt{2}}{6} + \frac{9\pi}{8} - \frac{9}{4}\sin^{-1}\frac{1}{3}$$ 1

OR



Correct figure 1

Correct points of intersection $1\frac{1}{2}$

Required Area $= \int_1^3 \frac{3x+1}{2}dx + \int_3^6 \frac{21-2x}{3}dx - \int_1^6 \frac{x+9}{5}dx$ 2

$= \left(\frac{3x^2}{4} + \frac{x}{2}\right)\Big|_1^3 + \left(7x - \frac{x^2}{3}\right)\Big|_3^6 - \left(\frac{x^2}{10} + \frac{9x}{5}\right)\Big|_1^6$ 1

$= 7 + 12 - \frac{25}{2}$

$= \frac{13}{2}$ $\frac{1}{2}$

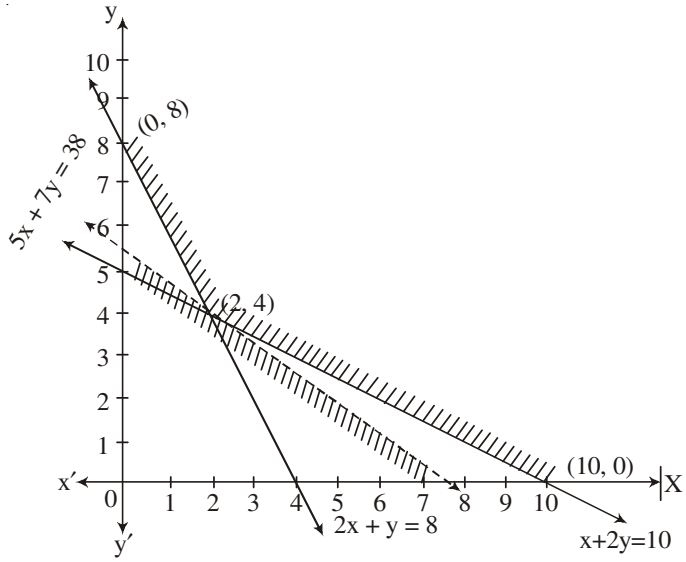
26. Let Quantity of Food I = x kg

Quantity of Food II = y kg

Out L.P.P. is

Minimize Z = 50x + 70y subject to $\frac{1}{2}$

$$\left.\begin{array}{l} 2x + y \geq 8 \\ x + 2y \geq 10 \\ x \geq 0,\ y \geq 0 \end{array}\right\}$$ $2\frac{1}{2}$



Correct Graph 2

Corner points	Value of Z		
(10, 0)	500		
(2, 4)	380	$\rightarrow$ minimum	$\frac{1}{2}$
(0, 8)	560		

Consider $50x + 70y < 380$

which has no point common with feasible region

So minimum value of Z = ₹ 380 $\frac{1}{2}$

at x = 2 kg, y = 4 kg

27. Required equation of line is

$\vec{r} = 2\hat{i} + 3\hat{j} + 2\hat{k} + \mu(2\hat{i} - 3\hat{j} + 6\hat{k})$ 2

$\vec{a}_1 = -2\hat{i} + 3\hat{j}, \vec{a}_2 = 2\hat{i} + 3\hat{j} + 2\hat{k}, \quad \vec{b} = 2\hat{i} - 3\hat{j} + 6\hat{k}$

$\vec{a}_2 - \vec{a}_1 = 4\hat{i} + 2\hat{k}$ 1

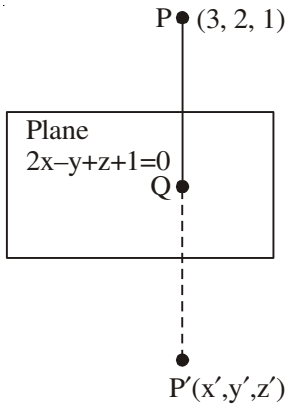
$d = \frac{|(\vec{a}_2 - \vec{a}_1) \times \vec{b}|}{|\vec{b}|}$

$(\vec{a}_2 - \vec{a}_1) \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 0 & 2 \\ 2 & -3 & 6 \end{vmatrix}$

$= 6\hat{i} - 20\hat{j} - 12\hat{k}$ 2

$d = \frac{\sqrt{580}}{7}$ 1

OR



Correct figure 1

Equation of PQ

$\frac{x-3}{2} = \frac{y-2}{-1} = \frac{z-1}{1} = \lambda$ $\frac{1}{2}$

Coodinates of Q $(2\lambda + 3, -\lambda + 2, \lambda + 1)$ $\frac{1}{2}$

As Q lies on plane

$\therefore\ 4\lambda + 6 + \lambda - 2 + \lambda + 1 = -1$

gives, $\lambda = -1$ 1

Coordinates of Q (1, 3, 0) $\frac{1}{2}$

$PQ = \sqrt{6}$ 1

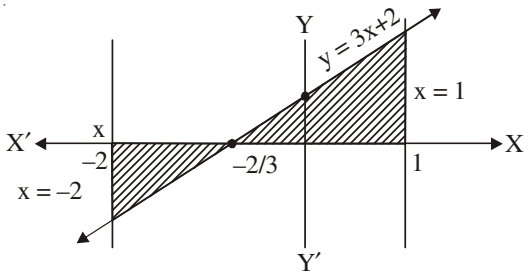
For unique $P'(x', y', z')$

$\frac{x'+3}{2} = 1,\ \frac{y'+2}{2} = 3,\ \frac{z'+1}{2} = 0$ 1

$x' = -1\ \ y' = 4\ \ z' = -1$

image is $(-1, 4, -1)$ $\frac{1}{2}$

28. Correct figure 1



Required Area $= \left|\int_{-2}^{-2/3} (3x+2)\,dx\right| + \int_{\frac{-2}{3}}^{1} (3x+2)\,dx$ 3

$= \left|\left.\frac{+(3x+2)^2}{6}\right|_{-2}^{\frac{-2}{3}} + \left.\frac{(3x+2)^2}{6}\right|_{\frac{-2}{3}}^{1}\right|$ 1

$= \frac{8}{3} + \frac{25}{6}$

$= \frac{41}{6}$ 1

29. E_1: Selected coin has tail on both faces

E_2: Selected coin is biased

E_3: Selected coin is unbiased 1

A: Tail comes up

$P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$ 1

$P(A|E_1) = 1,\ P(A|E_2) = \frac{7}{10},\ P(A|E_3) = \frac{1}{2}$ $1\frac{1}{2}$

$$P(E_1|A) = \frac{P(E_1)\,P(A|E_1)}{P(E_1)\,P(A|E_1) + P(E_2)\,P(A|E_2) + P(E_3)\,P(A|E_3)}$$

$$= \frac{\frac{1}{3}\times 1}{\frac{1}{3}\times 1 + \frac{1}{3}\times\frac{7}{10} + \frac{1}{3}\times\frac{1}{2}}$$ 2

$= \frac{10}{22}$ or $\frac{5}{11}$ $\frac{1}{2}$

QUESTION PAPER CODE 65/4/3
EXPECTED ANSWER/VALUE POINTS

SECTION A

1. $\cos\theta = \dfrac{3+12-4}{3\times 7} \Rightarrow \theta = \cos^{-1}\left(\dfrac{11}{21}\right)$ $\frac{1}{2}+\frac{1}{2}$

OR

$\dfrac{x}{5/2}+\dfrac{y}{5}+\dfrac{z}{-5}=1$, length of $x-$intercept $=\dfrac{5}{2}$ 1

2. $\dfrac{dy}{dx} = \dfrac{-\sin e^x}{\cos e^x}\cdot e^x$ or $-e^x \cdot \tan e^x$ 1

3. $|A\cdot \text{adj } A| = |A|^n = 4^n$ or 16 or 64 $\frac{1}{2}+\frac{1}{2}$

4. $\dfrac{dy}{dx} = A\cos x$ $\frac{1}{2}$

$\Rightarrow \dfrac{dy}{dx} = y\cot x$ $\frac{1}{2}$

SECTION B

5. $I = \tan^{-1}x\cdot\dfrac{x^2}{2} - \displaystyle\int \frac{x^2}{2(1+x^2)}dx$ 1

$= \dfrac{x^2}{2}\tan^{-1}x - \dfrac{1}{2}[x - \tan^{-1}x] + C$ 1

OR

$I = \dfrac{1}{\sqrt{2}}\displaystyle\int \frac{dx}{\sqrt{\frac{5}{2}-2x-x^2}}$ $\frac{1}{2}$

$= \dfrac{1}{\sqrt{2}}\displaystyle\int \frac{dx}{\sqrt{\left(\frac{\sqrt{7}}{\sqrt{2}}\right)^2-(x+1)^2}}$ 1

$= \dfrac{1}{\sqrt{2}}\sin^{-1}\left[\dfrac{(x+1)\sqrt{2}}{\sqrt{7}}\right] + C$ $\frac{1}{2}$

6. I.F. = e^x $\frac{1}{2}$

Solution is $y.e^x = \int (\cos x - \sin x)\, e^x \, dx + C$ $\frac{1}{2}$

$y.e^x = e^x \cos x + C$ or $y = \cos x + Ce^{-x}$ 1

7. $\int_{-\pi/4}^{0} \tan\left(\frac{\pi}{4} + x\right) dx = \log\left|\sec\left(\frac{\pi}{4} + x\right)\right|_{-\pi/4}^{0}$ $1+\frac{1}{2}$

$= \log\sqrt{2}$ or $\frac{1}{2}\log 2$ $\frac{1}{2}$

8. $a, b \in R \Rightarrow 2a \in R \Rightarrow 2a + b \in R$ $\therefore$ * is binary 1

$a * (b * c) = a * (2b + c) = 2a + 2b + c$

$(a * b) * c = (2a + b) * c = 4a + 2b + c$

In general $a * (b * c) \neq (a * b) * c$ $\therefore$ * is not associative 1

9. Position vector of z = $\frac{2(\vec{a} - 3\vec{b}) - 1(3\vec{a} + \vec{b})}{2 - 1}$ 1

$= -\vec{a} - 7\vec{b}$ 1

OR

$\vec{a} + \vec{b} = 4\hat{i} + \hat{j} - \hat{k}$ $\frac{1}{2}$

$\vec{a} - \vec{b} = -2\hat{i} + 3\hat{j} - 5\hat{k}$ $\frac{1}{2}$

$(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = -8 + 3 + 5 = 0$ $\frac{1}{2}$

so $(\vec{a} + \vec{b}) \perp (\vec{a} - \vec{b})$ $\frac{1}{2}$

10. Required probability = $\frac{{}^3C_2 \times {}^5C_2}{{}^8C_4}$ 1

$= \frac{3}{7}$ 1

OR

$n = 5 \quad p = \frac{1}{3} \quad q = \frac{2}{3}$ $\frac{1}{2}$

$P(X \geq 4) = P(X = 4) + P(X = 5)$

$= {}^5C_4\left(\frac{1}{3}\right)^4\left(\frac{2}{3}\right)^1 + {}^5C_5\left(\frac{1}{3}\right)^5$ 1

$= \frac{11}{243}$ $\frac{1}{2}$

11. P(Problem is solved) = 1 – P(Problem is not solved) $\frac{1}{2}$

$= 1 - \frac{2}{3} \times \frac{4}{5}$ 1

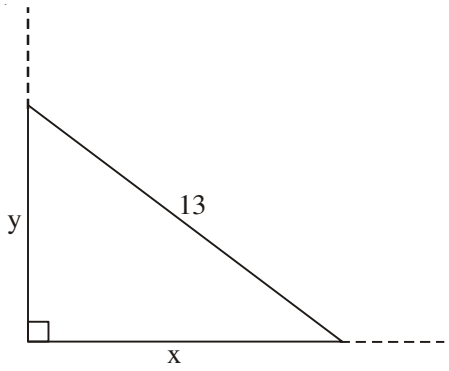
$= \frac{7}{15}$ $\frac{1}{2}$

12. $A + A' = \begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix} + \begin{bmatrix} 2 & 5 \\ 3 & 7 \end{bmatrix} = \begin{bmatrix} 4 & 8 \\ 8 & 14 \end{bmatrix}$ $\frac{1}{2} + \frac{1}{2}$

$(A + A')' = \begin{bmatrix} 4 & 8 \\ 8 & 14 \end{bmatrix} = A + A'$ So $A + A'$ is symmetric 1

SECTION C

13. Figure $\frac{1}{2}$



$\frac{dx}{dt} = 2 \text{ cm/sec}$ $\frac{1}{2}$

$y = \sqrt{169 - x^2}$ 1

$\frac{dy}{dt} = -\frac{x}{\sqrt{169 - x^2}} \frac{dx}{dt}$ 1

$\left(\frac{dy}{dt}\right)_{x=5} = \frac{-5}{6} \text{ cm/sec}$ 1

Hence height is decreasing at the rate $\frac{5}{6} \text{ cm/sec}$

14. LHS becomes $\tan^{-1}\frac{5}{12}+\tan^{-1}\frac{3}{4}$ 1

$= \tan^{-1}\left(\frac{\frac{5}{12}+\frac{3}{4}}{1-\frac{5}{12}\cdot\frac{3}{4}}\right)=\tan^{-1}\frac{56}{33}$ 1+1

$= \sin^{-1}\frac{56}{65}$ = RHS 1

15. Let $a - x = t \Rightarrow -dx = dt$ $\frac{1}{2}$

$\text{RHS} = -\int_a^0 f(t)\,dt=\int_0^a f(t)\,dt=\int_0^a f(x)\,dx$ $\frac{1}{2}$

$I = \int_0^1 x^2(1-x)^n\,dx$

$= \int_0^1 (1-x)^2 x^n dx$ 1

$= \int_0^1 [x^n + x^{n+2} - 2x^{n+1}]\,dx$ $\frac{1}{2}$

$= \frac{x^{n+1}}{n+1}+\frac{x^{n+3}}{n+3}-\frac{2x^{n+2}}{n+2}\Big|_0^1$ 1

$= \frac{1}{n+1}+\frac{1}{n+3}-\frac{2}{n+2}$ $\frac{1}{2}$

16. $\frac{dx}{dt}=\cos t,\ \ \frac{dy}{dt}=p\cos pt \ \Rightarrow\ \frac{dy}{dx}=\frac{p\cos pt}{\cos t}$ 1

$\frac{dy}{dx}=\frac{p\sqrt{1-y^2}}{\sqrt{1-x^2}}$ 1

$\Rightarrow (1-x^2)\left(\frac{dy}{dx}\right)^2 = p^2(1-y^2)$ differentiating both sides w.r.t x $\frac{1}{2}$

$\Rightarrow 2(1-x^2)\frac{dy}{dx}\frac{d^2y}{dx^2} - 2x\left(\frac{dy}{dx}\right)^2 = -2p^2y\frac{dy}{dx}$ 1

$\Rightarrow (1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + p^2y = 0$ $\frac{1}{2}$

OR

Let $\theta = \cos^{-1} x^2 \Rightarrow x^2 = \cos\theta$ 1

$\Rightarrow y = \tan^{-1}\left(\frac{\sqrt{1+\cos\theta} - \sqrt{1-\cos\theta}}{\sqrt{1+\cos\theta} + \sqrt{1-\cos\theta}}\right)$

$= \tan^{-1}\left(\frac{\cos\frac{\theta}{2} - \sin\frac{\theta}{2}}{\cos\frac{\theta}{2} + \sin\frac{\theta}{2}}\right)$ 1

$= \frac{\pi}{4} - \frac{1}{2}\theta$ 1

$\therefore \quad \frac{dy}{d\theta} = -\frac{1}{2}$ 1

17. $I = \int \frac{\cos(x+b+a-b)}{\sin(x+b)}dx$ 1

$= \int [\cot(x+b)\cos(a-b) - \sin(a-b)]dx$ 1

$= \cos(a-b)\log|\sin(x+b)| - x\sin(a-b) + C$ 2

18. Let $x_1, x_2 \in R - \{2\}$ such that $f(x_1) = f(x_2)$

$\Rightarrow \frac{x_1 - 1}{x_1 - 2} = \frac{x_2 - 1}{x_2 - 2}$ $\frac{1}{2}$

$\Rightarrow x_1x_2 - 2x_1 - x_2 + 2 = x_1x_2 - 2x_2 - x_1 + 2$

$\Rightarrow x_1 = x_2$ 1

So f is one – one

For range let $f(x) = y$

$\frac{x-1}{x-2} = y$ $\frac{1}{2}$

$x = \frac{2y-1}{y-1}$ 1

Range of f = R – {1} = co domain B $\frac{1}{2}$

So f is onto.

$f^{-1}(y) = \frac{2y-1}{y-1}$ or $f^{-1}(x) = \frac{2x-1}{x-1}$ $\frac{1}{2}$

OR

For reflexive 1

For symmetric 1

For transitive

Let (a, b) $\in$ S & (b, c) $\in$ S

$|a - b| = 3m$, $|b - c| = 3n$

$a - b = \pm 3m$ $\quad\quad$ $b - c = \pm 3n$

$a - c = 3(\pm m \pm n) \Rightarrow a - c$ is divisible by 3 $1\frac{1}{2}$

$\Rightarrow |a - c|$ is divisible by 3

$\Rightarrow$ (a, c) $\in$ S

S is transitive

As S is reflexive, symmetric & transitive

$\therefore$ S is an equivalence reation. $\frac{1}{2}$

19. Given differential equation can be written as

$\frac{dy}{dx} = (1 + x^2)(1 + y^2)$ 1

$\int \frac{dy}{1+y^2} = \int (x^2 + 1)\, dx$ 1

$\tan^{-1} y = \frac{x^3}{3} + x + C$ 1

$x = 0,\ y = 1 \Rightarrow C = \frac{\pi}{4}$ $\frac{1}{2}$

So particular solution is $\tan^{-1} y = \frac{x^3}{3} + x + \frac{\pi}{4}$ $\frac{1}{2}$

OR

Clearly given differential equation can be written as $\frac{dy}{dx} = \frac{y/x}{1+(y/x)^2}$

Let $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ 1

Given equation becomes

$$v + x\frac{dv}{dx} = \frac{v}{1+v^2}$$

$$x\frac{dv}{dx} = \frac{-v^3}{1+v^2}$$

$$\int \frac{(1+v^2)}{v^3} dV = -\int \frac{dx}{x}$$ 1

$$-\frac{1}{2v^2} + \log|v| = -\log|x| + C$$ 1

$$-\frac{x^2}{2y^2} + \log|y| = C$$

$x = 0,\ y = 1 \Rightarrow C = 0$ $\frac{1}{2}$

So, particular solution is $\log|y| = \frac{x^2}{2y^2}$ $\frac{1}{2}$

20. $R_1 \to R_1 + R_2 + R_3$ & taking 12 + x common

$$(12+x)\begin{vmatrix} 1 & 1 & 1 \\ 4+x & 4-x & 4+x \\ 4+x & 4+x & 4-x \end{vmatrix} = 0$$ $1+\frac{1}{2}$

$C_2 \to C_2 - C_1 \quad C_3 \to C_3 - C_1$

$$(12+x)\begin{vmatrix} 1 & 0 & 0 \\ 4+x & -2x & 0 \\ 4+x & 0 & -2x \end{vmatrix} = 0 \qquad 1\frac{1}{2}$$

$4x^2\ (12 + x) = 0$ $\frac{1}{2}$

$x = 0$ or $x = -12$ $\frac{1}{2}$

21. Required equation of plane is given by

$$\vec{r}\cdot(\hat{i}+2\hat{j}+3\hat{k}) - 4 + \lambda[(\vec{r}\cdot(2\hat{i}-\hat{j}-\hat{k})+5] = 0 \qquad \frac{1}{2}$$

$$\vec{r}\cdot[(1+2\lambda)\hat{i}+(2+\lambda)\hat{j}+(3-\lambda)\hat{k}] = 4 - 5\lambda \qquad 1$$

It is perpendicular to $\vec{r}\cdot(5\hat{i}+3\hat{j}-6\hat{k})+8=0$

So, $5 + 10\lambda + 6 + 3\lambda - 18 + 6\lambda = 0$ 1

$19\lambda = 7$

$\lambda = \frac{7}{19}$ $\frac{1}{2}$

Required equation is

$$\vec{r}\cdot(33\hat{i}+45\hat{j}+50\hat{k}) = 41 \qquad 1$$

22. Here $\overrightarrow{AB} = \hat{i}+(x-2)\hat{j}+4\hat{k},\ \overrightarrow{AC} = \hat{i}-3\hat{k},\ \overrightarrow{AD} = 3\hat{i}+3\hat{j}-2\hat{k}$ $1\frac{1}{2}$

As A, B, C, D are coplanar

$$\therefore \quad \begin{vmatrix} 1 & x-2 & 4 \\ 1 & 0 & -3 \\ 3 & 3 & -2 \end{vmatrix} = 0 \qquad 1$$

$-9 - (2 - x)7 - 4 \times 3 = 0$ 1

$x = 5$ $\frac{1}{2}$

23. Let $u = (\log x)^x \quad v = x^{\log x}$

$y = u + v \Rightarrow \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$ 1

$\log u = x \log x$

$\frac{1}{u}\frac{du}{dx} = \frac{x}{\log x} \cdot \frac{1}{x} + \log \log x$

$\frac{du}{dx} = (\log x)^x \left[\frac{1}{\log x} + \log \log x\right]$ 1

$\log v = (\log x)^2$

$\frac{1}{v}\frac{dv}{dx} = \frac{2 \log x}{x}$

$\therefore \frac{dv}{dx} = x^{\log x}\left(\frac{2 \log x}{x}\right)$ 1

So $\frac{dy}{dx} = (\log x)^x \left[\frac{1}{\log x} + \log \log x\right] + 2x^{(\log x - 1)} \cdot \log x$ 1

SECTION D

24. Required equation of line is

$\vec{r} = 2\hat{i} + 3\hat{j} + 2\hat{k} + \mu(2\hat{i} - 3\hat{j} + 6\hat{k})$ 2

$\vec{a}_1 = -2\hat{i} + 3\hat{j}, \vec{a}_2 = 2\hat{i} + 3\hat{j} + 2\hat{k}, \quad \vec{b} = 2\hat{i} - 3\hat{j} + 6\hat{k}$

$\vec{a}_2 - \vec{a}_1 = 4\hat{i} + 2\hat{k}$ 1

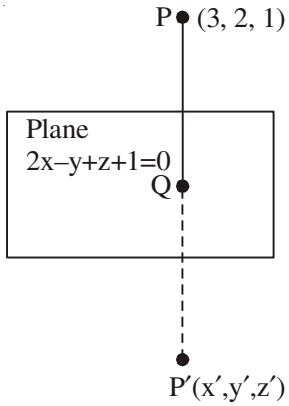
$d = \frac{|(\vec{a}_2 - \vec{a}_1) \times \vec{b}|}{|\vec{b}|}$

$(\vec{a}_2 - \vec{a}_1) \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 0 & 2 \\ 2 & -3 & 6 \end{vmatrix}$

$= 6\hat{i} - 20\hat{j} - 12\hat{k}$ 2

$d = \frac{\sqrt{580}}{7}$ 1

OR



Correct figure 1

Equation of PQ

$\frac{x-3}{2} = \frac{y-2}{-1} = \frac{z-1}{1} = \lambda$ $\frac{1}{2}$

Coodinates of Q $(2\lambda + 3, -\lambda + 2, \lambda + 1)$ $\frac{1}{2}$

As Q lies on plane

$\therefore\ 4\lambda + 6 + \lambda - 2 + \lambda + 1 = -1$

gives, $\lambda = -1$ 1

Coordinates of Q (1, 3, 0) $\frac{1}{2}$

$PQ = \sqrt{6}$ 1

For unique P′(x′, y′, z′)

$\frac{x'+3}{2} = 1,\ \frac{y'+2}{2} = 3,\ \frac{z'+1}{2} = 0$ 1

$x' = -1\ y' = 4\ z' = -1$

image is (–1, 4, –1) $\frac{1}{2}$

25. $A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}$ we know that A = IA

i.e., $\begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$ 1

$R_1 \to R_1 - R_3$

$$\begin{bmatrix} 1 & -4 & 7 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} A$$

$R_2 \to R_2 - 3R_1, R_3 \to R_3 - R_1$

$$\begin{bmatrix} 1 & -4 & 7 \\ 0 & 14 & -25 \\ 0 & 5 & -9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ -3 & 1 & 3 \\ -1 & 0 & 2 \end{bmatrix} A$$

$R_2 \to -(R_2 - 3R_3)$

$$\begin{bmatrix} 1 & -4 & 7 \\ 0 & 1 & -2 \\ 0 & 5 & -9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix} A$$

$R_1 \to R_1 + 4R_2$

$R_3 \to R_3 - 5R_3$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -4 & 11 \\ 0 & -1 & 3 \\ -1 & 5 & -13 \end{bmatrix} A$$

$R_1 \to R_1 + R_3$

$R_2 \to R_2 + 2R_3$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -4 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix} A$$

4

$$\Rightarrow A^{-1} = \begin{bmatrix} 0 & 1 & -4 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix}$$

1

OR

$|A| = 67 \neq 0 \qquad \therefore X = A^{-1}B$ $\quad 1+\frac{1}{2}$

$$\text{adj } A = \begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$$

2

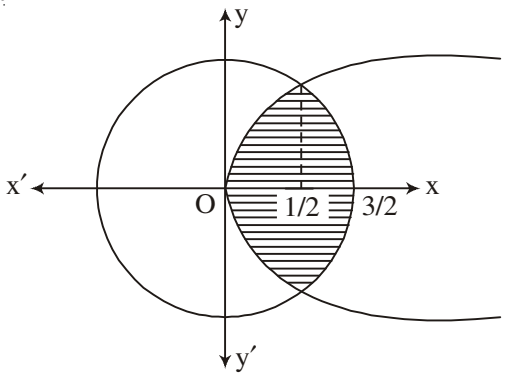
$$A^{-1} = \frac{1}{67}\begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$$ $\frac{1}{2}$

$$\text{So } X = \frac{1}{67}\begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}\begin{bmatrix} -4 \\ 2 \\ 11 \end{bmatrix}$$

$$= \frac{1}{67}\begin{bmatrix} 201 \\ -134 \\ 67 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$$ $1\frac{1}{2}$

$x = 3,\ y = -2,\ z = 1$ $\frac{1}{2}$

26.



Correct Figure 1

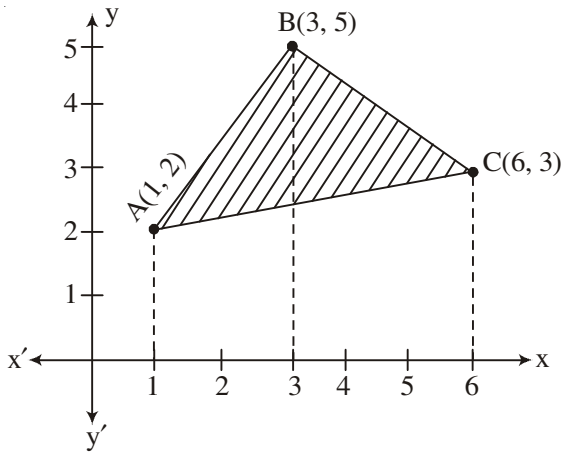
x coordinate of Point of intersection = $\frac{1}{2}$ 1

Required Area = $2\left[\int_0^{\frac{1}{2}} 2\sqrt{x}\,dx + \int_{\frac{1}{2}}^{3/2} \sqrt{\left(\frac{3}{2}\right)^2 - x^2}\,dx\right]$ 2

$= 2\left[\frac{4}{3}x^{3/2}\Big|_0^{\frac{1}{2}} + \frac{x}{2}\sqrt{\frac{9}{4} - x^2} + \frac{9}{8}\sin^{-1}\frac{2x}{3}\Big|_{\frac{1}{2}}^{3/2}\right]$ 1

$= \frac{\sqrt{2}}{6} + \frac{9\pi}{8} - \frac{9}{4}\sin^{-1}\frac{1}{3}$ 1

OR



Correct figure 1

Correct points of intersection $1\frac{1}{2}$

Required Area = $\int_1^3 \frac{3x+1}{2}dx + \int_3^6 \frac{21-2x}{3}dx - \int_1^6 \frac{x+9}{5}dx$ 2

$$= \left(\frac{3x^2}{4}+\frac{x}{2}\right)\Big|_1^3 + \left(7x-\frac{x^2}{3}\right)\Big|_3^6 - \left(\frac{x^2}{10}+\frac{9x}{5}\right)\Big|_1^6 \quad 1$$

$$= 7+12-\frac{25}{2}$$

$$= \frac{13}{2} \quad \frac{1}{2}$$

27. E_1: Selected person is cyclist
E_2: Selected person is scooter driver
E_3: Selected person is car driver
A: insured person met with an accident } 1

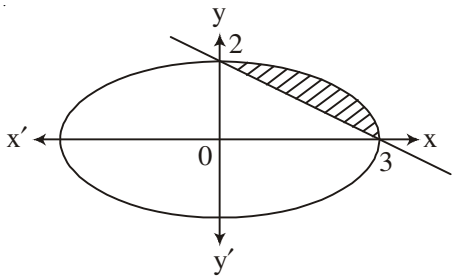
$$P(E_1)=\frac{3}{18},\ P(E_2)=\frac{6}{18},\ P(E_3)=\frac{9}{18}$$
$$P(A|E_1)=0.3,\ P(A|E_2)=0.05,\ P(A|E_3)=0.02 \quad \} \ 2$$

$$P(E_1|A)=\frac{P(E_1)\,P(A|E_1)}{P(E_1)\,P(A|E_1)+P(E_2)\,P(A|E_2)+P(E_3)\,P(A|E_3)}$$

$$= \frac{\frac{3}{18}\times\frac{30}{100}}{\frac{3}{18}\times\frac{30}{100}+\frac{6}{18}\times\frac{5}{100}+\frac{9}{18}\times\frac{2}{100}} \quad 2$$

$$= \frac{90}{138} \text{ or } \frac{15}{23} \quad 1$$

28.



Correct Figure 1

Point of intersection are (3, 0) & (0, 2) 1

Required Area $= \frac{2}{3}\int_0^3 \sqrt{(9-x^2)}\,dx - \frac{2}{3}\int_0^3 (3-x)\,dx$ 2

$$= \frac{2}{3}\left[\frac{x}{2}\sqrt{9-x^2}+\frac{9}{2}\sin^{-1}\frac{x}{3}+\frac{(3-x)^2}{2}\right]_0^3 \quad 1$$

$$= \frac{2}{3}\left[\frac{9\pi}{4}-\frac{9}{2}\right]$$

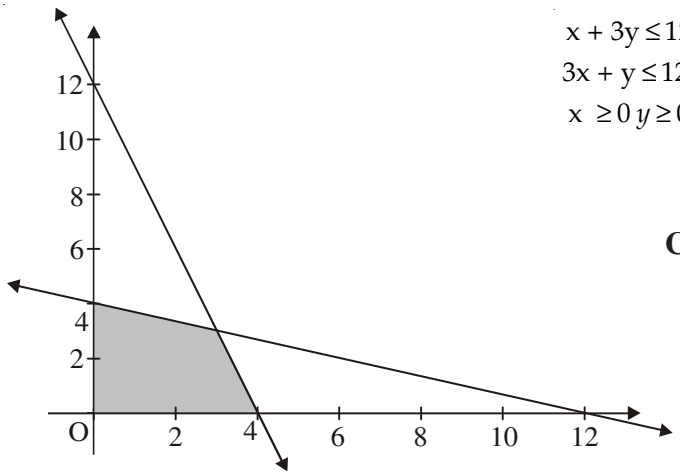
$$= \frac{3\pi}{2}-3 \quad 1$$

29. Let number of packages of nuts = x units

and number of packages of bolts = y units

Our L.P.P is

Maximize profit Z = 35x + 14y $\frac{1}{2}$



$$\left.\begin{array}{l} x + 3y \le 12 \\ 3x + y \le 12 \\ x \ge 0\ y \ge 0 \end{array}\right\}$$ $2\frac{1}{2}$

Correct figure 2

Corner points	**Value of Z**	
(4, 0)	140	
(3, 3)	147	← maximum $\frac{1}{2}$
(0, 4)	56	

Maximum profit = ₹ 147 when x = 3, y = 3 $\frac{1}{2}$

Strictly Confidential — (For Internal and Restricted Use Only)

Senior School Certificate Examination

March 2019

Marking Scheme — Mathematics (041) 65/5/1, 65/5/2, 65/5/3

General Instructions:

1. You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully. **Evaluation is a 10-12 days mission for all of us. Hence, it is necessary that you put in your best efforts in this process.**
2. Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one's own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. **However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and marks be awarded to them.**
3. The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
4. If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled.
5. If a question does not have any parts, marks must be awarded in the left hand margin and encircled.
6. If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out.
7. No marks to be deducted for the cumulative effect of an error. It should be penalized only once.
8. A full scale of marks 0 to 100 has to be used. Please do not hesitate to award full marks if the answer deserves it.
9. Every examiner has to necessarily do evaluation work for full working hours i.e. 8 hours every day and evaluate 25 answer books per day.
10. Ensure that you do not make the following common types of errors committed by the Examiner in the past:-
    - Leaving answer or part thereof unassessed in an answer book.
    - Giving more marks for an answer than assigned to it.
    - Wrong transfer of marks from the inside pages of the answer book to the title page.
    - Wrong question wise totaling on the title page.
    - Wrong totaling of marks of the two columns on the title page.
    - Wrong grand total.
    - Marks in words and figures not tallying.
    - Wrong transfer of marks from the answer book to online award list.
    - Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.)
    - Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
11. While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as (X) and awarded zero (0) Marks.
12. Any unassessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
13. The Examiners should acquaint themselves with the guidelines given in the Guidelines for spot Evaluation before starting the actual evaluation.
14. Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
15. The Board permits candidates to obtain photocopy of the Answer Book on request in an RTI application and also separately as a part of the re-evaluation process on payment of the processing charges.

QUESTION PAPER CODE 65/5/1
EXPECTED ANSWER/VALUE POINTS

SECTION A

1. $\sin\theta = \left|\dfrac{(\hat{i}+\hat{j}+\hat{k})\cdot(3\hat{i}-\hat{j}+2\hat{k})}{\sqrt{1+1+1}\,\sqrt{9+1+4}}\right|$ $\frac{1}{2}$

$\Rightarrow \theta = \sin^{-1}\left(\dfrac{4}{\sqrt{42}}\right)$ $\frac{1}{2}$

OR

Any point on $\dfrac{x+2}{1} = \dfrac{y-5}{3} = \dfrac{z+1}{5} = \lambda$ ...(1) $\frac{1}{2}$

is $(\lambda-2, 3\lambda+5, 5\lambda-1)$

Line (1) cuts yz plane at $\lambda - 2 = 0$ i.e., $\lambda = 2$

hence required point is (0, 11, 9) $\frac{1}{2}$

2. $y = 5e^{7x} + 6e^{-7x} \Rightarrow \dfrac{dy}{dx} = 35e^{7x} - 42e^{-7x}$ $\frac{1}{2}$

$\dfrac{d^2y}{dx^2} = 245e^{7x} + 294e^{-7x} \Rightarrow \dfrac{d^2y}{dx^2} = 49y$ $\frac{1}{2}$

3. $|2\,AA'| = 4|A|\,|A'|$ $\frac{1}{2}$

$= 4 \times 4 \times 4 = 64$ $\frac{1}{2}$

4. Order = 2 1

SECTION B

5. $\begin{bmatrix} 2x-3 & 6 \\ 15 & 2y-4 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$ 1

$\Rightarrow 2x - 3 = 7$ and $2y - 4 = 14 \Rightarrow x = 5,\ y = 9 \Rightarrow x - y = -4$ 1

6. $(y+3x^2)\dfrac{dx}{dy} = x \Rightarrow \dfrac{dy}{dx} - \dfrac{1}{x}y = 3x$ $\frac{1}{2}$

I.F. $= e^{\int -\frac{1}{x}dx} = e^{-\log(x)} = \frac{1}{x}$ $\frac{1}{2}$

Solution is $y \cdot \frac{1}{x} = \int 3x \cdot \frac{1}{x} dx \Rightarrow y = 3x^2 + cx$ 1

7. $\int e^x (\cos x - \sin x) \operatorname{cosec}^2 x \, dx = -\int e^x [\operatorname{cosec} x - \operatorname{cosec} x \cot x] \, dx$ 1

$= -e^x \operatorname{cosec} x + c$ 1

8. Let $P(2, -1, 3)$, $Q(3, -5, 1)$ and $R(-1, 11, 9)$ be three point.

$\overrightarrow{PQ} = \hat{i} - 4\hat{j} - 2\hat{k}$ $\frac{1}{2}$

$\overrightarrow{PR} = -3\hat{i} + 12\hat{j} + 6\hat{k} = -3(\hat{i} - 4\hat{j} - 2\hat{k})$ $\frac{1}{2}$

$\because \overrightarrow{PR} = -3\overrightarrow{PQ}$, since P is common.

Therefore the points P, Q and R are collinear. 1

OR

$(\vec{a} \times \vec{b})^2 = \vec{a}^2 \vec{b}^2 - (\vec{a} \cdot \vec{b})^2$

LHS $= (\vec{a} \times \vec{b})^2$

$= (|\vec{a}||\vec{b}| \sin\theta \, \hat{n})^2 = |\vec{a}|^2 |\vec{b}|^2 \sin^2\theta$ 1

$= |\vec{a}|^2 |\vec{b}|^2 (1 - \cos^2\theta)$

$= |\vec{a}|^2 |\vec{b}|^2 - (|\vec{a}||\vec{b}| \cos\theta)^2$

$= \vec{a}^2 \vec{b}^2 - (\vec{a} \cdot \vec{b})^2$ 1

9. $\int \frac{x-1}{(x-2)(x-3)} dx = \int \left(\frac{-1}{x-2} + \frac{2}{x-3} \right) dx$ 1

$= -\log|x-2| + 2\log|x-3| + C$ 1

OR

$\int \frac{e^x}{\sqrt{5-4e^x-e^{2x}}}\,dx$ Put $e^x = t$ so that $e^x\,dx = dt$ $\frac{1}{2}$

$\int \frac{dt}{\sqrt{5-4t-t^2}} = \int \frac{dt}{\sqrt{3^2-(t+2)^2}}$ 1

$= \sin^{-1}\left(\frac{t+2}{3}\right)+C = \sin^{-1}\left(\frac{e^x+2}{3}\right)+C$ $\frac{1}{2}$

10. $P(B \mid A) = 0.4 \Rightarrow \frac{P(B \cap A)}{P(A)} = 0.4 \Rightarrow P(B \cap A) = 0.24$ $\frac{1}{2}$

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.6 + 0.5 - 0.24 = 0.86$ 1

$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.24}{0.5} = 0.48$ $\frac{1}{2}$

11. (i) $\forall\, a, b \in Z$, $a * b = 2a^2 + b \in Z$ $\therefore$ * is binary 1

(ii) $1, 2 \in Z$, $1 * 2 = 2 \times 1^2 + 2 = 4$, $2 * 1 = 2 \times (2)^2 + 1 = 9$

$1 * 2 \neq 2 * 1$

$\therefore$ * is not commutative. 1

12. $n = 4$, $p = \frac{1}{4}, q = \frac{3}{4}$ $\frac{1}{2}$

P (atleast 3 are diamonds) = P (X = 3) + P(X = 4)

$= {}^4C_3\left(\frac{1}{4}\right)^3\left(\frac{3}{4}\right) + {}^4C_4\left(\frac{1}{4}\right)^4$ 1

$= \left(\frac{1}{4}\right)^4 [12+1] = \frac{13}{256}$ $\frac{1}{2}$

OR

Let E_1 : A coming on time.

E_2 : B coming on time.

$P(\bar{E}_1) = \frac{5}{7}, P(\bar{E}_2) = \frac{3}{7}$ $\frac{1}{2}$

P(only one on time)

$= P(E_1)\,P(\bar{E}_2) + P(\bar{E}_1)\,P(E_2)$

$= \frac{2}{7}\times\frac{3}{7}+\frac{5}{7}\times\frac{4}{7}$ 1

$= \frac{26}{49}$ $\frac{1}{2}$

13. $x^p y^q = (x+y)^{p+q} \Rightarrow p\log x + q\log y = (p+q)\log(x+y)$ $\frac{1}{2}$

Differentaiting w.r.t x, $\frac{p}{x}+\frac{q}{y}\frac{dy}{dx} = \frac{p+q}{x+y}\left(1+\frac{dy}{dx}\right)$ 1

$\Rightarrow \left(\frac{q}{y}-\frac{p+q}{x+y}\right)\frac{dy}{dx} = \frac{p+q}{x+y}-\frac{p}{x}$

$\Rightarrow \left[\frac{qx-py}{y(x+y)}\right]\frac{dy}{dx} = \frac{qx-py}{x(x+y)} \Rightarrow \frac{dy}{dx} = \frac{y}{x}$ 1

Differentiating again w.r.t x

$\frac{d^2y}{dx^2} = \frac{x\frac{dy}{dx}-y}{x^2}$ 1

$\Rightarrow \frac{d^2y}{dx^2} = \frac{x\left(\frac{y}{x}\right)-y}{x^2} \Rightarrow \frac{d^2y}{dx^2} = 0$ $\frac{1}{2}$

14. $\int(\sin x \sin 2x \sin 3x)\,dx = \frac{1}{2}\int(2\sin x \sin 2x)\sin 3x\,dx$ $\frac{1}{2}$

$= \frac{1}{2}\int[\cos(x)-\cos(3x)]\sin 3x\,dx$ 1

$= \frac{1}{4}\int 2\cos x \sin 3x\,dx - \frac{1}{4}\int 2\cos 3x \sin 3x\,dx$

$= \frac{1}{4}\int(\sin 4x+\sin 2x)dx - \frac{1}{4}\int \sin 6x\,dx$ 1

$= -\frac{1}{4}\left(\frac{\cos 4x}{4}+\frac{\cos 2x}{2}\right)+\frac{1}{24}\cos 6x + C$ $1\frac{1}{2}$

15. Let $y = \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$, Put $x = \tan\theta$ $\frac{1}{2}$

$y = \tan^{-1}\left(\frac{3\tan\theta-\tan^3\theta}{1-3\tan^2\theta}\right) \Rightarrow y = \tan^{-1}(\tan 3\theta) = 3\theta$

$y = 3\tan^{-1} x \Rightarrow \frac{dy}{dx} = \frac{3}{1+x^2}$ ...(i) 1

Let $z = \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$, put $x = \sin\phi$ $\frac{1}{2}$

$z = \tan^{-1}\left(\frac{\sin\phi}{\sqrt{1-\sin^2\phi}}\right) \Rightarrow z = \tan^{-1}(\tan\phi) = \phi$

$z = \phi = \sin^{-1} x \Rightarrow \frac{dz}{dx} = \frac{1}{\sqrt{1-x^2}}$ ...(ii) 1

Using (i) & (ii), $\frac{dy}{dz} = \frac{dy/dx}{dz/dx} = \frac{3\sqrt{1-x^2}}{1+x^2}$ 1

OR

$\sqrt{1-x^2}+\sqrt{1-y^2} = a(x-y)$, put $x = \sin\theta$, $y = \sin\phi$ 1

$\sqrt{1-\sin^2\theta}+\sqrt{1-\sin^2\phi} = a(\sin\theta-\sin\phi)$

$\Rightarrow \cos\theta + \cos\phi = a(\sin\theta - \sin\phi)$

$\Rightarrow 2\cos\left(\frac{\theta+\phi}{2}\right)\cos\left(\frac{\theta-\phi}{2}\right) = 2a\cos\left(\frac{\theta+\phi}{2}\right)\sin\left(\frac{\theta-\phi}{2}\right)$ 1

$\Rightarrow \tan\left(\frac{\theta-\phi}{2}\right) = \frac{1}{a}$

$\Rightarrow \frac{\theta-\phi}{2} = \tan^{-1}\left(\frac{1}{a}\right) \Rightarrow \sin^{-1} x - \sin^{-1} y = 2\tan^{-1}\left(\frac{1}{a}\right)$ 1

Differentiating both sides w.r.t x

$$\frac{1}{\sqrt{1-x^2}}-\frac{1}{\sqrt{1-y^2}}\frac{dy}{dx}=0$$ $\frac{1}{2}$

$$\frac{1}{\sqrt{1-y^2}}\frac{dy}{dx}=\frac{1}{\sqrt{1-x^2}} \text{ or } \frac{dy}{dx}=\frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$$ $\frac{1}{2}$

16. $(1+e^{2x})dy+(1+y^2)e^x\,dx=0$

$$\Rightarrow \int\frac{dy}{1+y^2}=-\int\frac{e^x}{1+e^{2x}}dx$$ 1

$$\Rightarrow \tan^{-1}y=-\int\frac{e^x}{1+e^{2x}}dx$$ $\frac{1}{2}$

Put $e^x = t$, so that $e^x\,dx = dt$ $\frac{1}{2}$

$$\tan^{-1}y=-\int\frac{dt}{1+t^2}\Rightarrow \tan^{-1}y=-\tan^{-1}(e^x)+C \quad \text{...(i)}$$ 1

Substituting $y = 1$, when $x = 0$ in equation (i)

$$\tan^{-1}(1)=-\tan^{-1}(1)+C\Rightarrow C=\frac{\pi}{2}$$ $\frac{1}{2}$

Substituting $C=\frac{\pi}{2}$ in equation (i) $\Rightarrow \tan^{-1}y+\tan^{-1}(e^x)=\frac{\pi}{2}$ $\frac{1}{2}$

OR

$$x\frac{dy}{dx}\sin\left(\frac{y}{x}\right)+x-y\sin\left(\frac{y}{x}\right)=0$$

$$\Rightarrow \frac{dy}{dx}=\frac{y}{x}-\frac{1}{\sin\left(\frac{y}{x}\right)} \quad \text{...(i)}$$ $\frac{1}{2}$

Put $\frac{y}{x}=v$ i.e., $y = vx$ in (i) so that $\frac{dy}{dx}=v+x\frac{dv}{dx}$ 1

$$v+x\frac{dv}{dx}=v-\frac{1}{\sin v}$$

$\int \sin v \, dv = \int -\frac{1}{x} dx$ $\frac{1}{2}$

$\Rightarrow -\cos v = -\log|x| + C \Rightarrow \cos\left(\frac{y}{x}\right) = \log|x| + C$ ...(ii) 1

Substituting $y = \frac{\pi}{2}$ when x = 1 in (ii)

$\cos\left(\frac{\pi}{2}\right) = \log 1 + C \Rightarrow C = 0$ $\frac{1}{2}$

Required solution is $\cos\left(\frac{y}{x}\right) = \log|x|$ $\frac{1}{2}$

17. (i) **Reflexive:** $\forall a \in A, |a - a| = 0$ which is even

$\Rightarrow (a, a) \in R$, hence R is reflexive 1

(ii) **Symmetric:** Let $(a, b) \in R \Rightarrow |a - b|$ is even

$\Rightarrow |-(b - a)|$ is even $\Rightarrow |b - a|$ is even

so, $(b, a) \in R$

hence R is symmetric. 1

(iii) **Transitive:** Let $(a, b), (b, c) \in R$

so, $|a - b|$ is even and $|b - c|$ is even

$\Rightarrow a - b = 2\lambda$, $b - c = 2\mu$ where $\lambda, \mu \in Z$

Now, $a - c = (a - b) + (b - c) = 2(\lambda + \mu)$

$\Rightarrow |a - c|$ is even, so $(a, c) \in R$

hence R is transitive. $1\frac{1}{2}$

Since R is reflexive, symmetric and transitive therefore its an equivalence relation $\frac{1}{2}$

OR

Let for $x_1, x_2 \in A$, $f(x_1) = f(x_2)$ $\frac{1}{2}$

$\frac{4x_1+3}{6x_1-4}=\frac{4x_2+3}{6x_2-4}$

$\Rightarrow (4x_1+3)(6x_2-4)=(6x_1-4)(4x_2+3)$

$\Rightarrow 34x_1=34x_2 \Rightarrow x_1=x_2$, hence f is one-one. 1

For any $y \in A$ such that $y=\frac{4x+3}{6x-4}$ there exists x such that

$6xy-4y=4x+3 \Rightarrow (6y-4)x=4y+3$

$\Rightarrow x=\frac{4y+3}{6y-4}, y\in A,\ x=\frac{4y+3}{6y-4}\in A$ 1

$\Rightarrow$ f is onto. 1

Since f is one-one and onto, therefore f^{-1} exists in A

and $f^{-1}(y)=\frac{4y+3}{6y-4}$ or $f^{-1}(x)=\frac{4x+3}{6x-4}$ $\frac{1}{2}$

18. $f(x)=\cos\left(2x+\frac{\pi}{4}\right)\Rightarrow f'(x)=-2\sin\left(2x+\frac{\pi}{4}\right)$ $1\frac{1}{2}$

As $\frac{3\pi}{8}<x<\frac{5\pi}{8}\Rightarrow\frac{3\pi}{4}<2x<\frac{5\pi}{4}$

$\Rightarrow \pi<2x+\frac{\pi}{4}<\frac{3\pi}{2}$ $1\frac{1}{2}$

$\Rightarrow \sin\left(2x+\frac{\pi}{4}\right)<0\Rightarrow f'(x)>0$

$\therefore$ f(x) is increasing in $\left(\frac{3\pi}{8},\frac{5\pi}{8}\right)$ 1

19. Equation of the required plane is

$a(x+1)+b(y-3)+c(z-2)=0$ ...(i) 1

Since plane (i) is perpendicular to planes $x+2y+3z=5$ and $3x+3y+z=0$, therefore

$a+2b+3c=0$ ...(ii)

and $3a+3b+c=0$ ...(iii) 1

Solving (ii) and (iii) we get

$\frac{a}{7} = \frac{b}{-8} = \frac{c}{3}$ 1

Equation of plane is

$7 \cdot (x + 1) - 8 (y - 3) + 3 (z - 2) = 0$

$\Rightarrow 7x - 8y + 3z + 25 = 0$ 1

20. LHS = $\sin^{-1}\frac{4}{5} + \tan^{-1}\frac{5}{12} + \cos^{-1}\frac{63}{65}$

$= \tan^{-1}\frac{4}{3} + \tan^{-1}\frac{5}{12} + \cos^{-1}\frac{63}{65}$ 1

$= \tan^{-1}\left(\frac{\frac{4}{3}+\frac{5}{12}}{1-\frac{4}{3}\times\frac{5}{12}}\right) + \cos^{-1}\frac{63}{65}$ 1

$= \tan^{-1}\left(\frac{63}{16}\right) + \cot^{-1}\left(\frac{63}{16}\right)$ 1

$= \frac{\pi}{2} = \text{RHS}$ 1

21. $\int_1^5 (|x-1| + |x-2| + |x-4|\, dx$

$= \int_1^2 (5-x)dx + \int_2^4 (x+1)dx + \int_4^5 (3x-7)\, dx$ $1\frac{1}{2}$

$= \left[5x - \frac{x^2}{2}\right]_1^2 + \left[\frac{x^2}{2} + x\right]_2^4 + \left[\frac{3x^2}{2} - 7x\right]_4^5$ $1\frac{1}{2}$

$= \frac{7}{2} + 8 + \frac{13}{2} = 18$ 1

22. A(x, 5, –1), B(3, 2, 1), C(4, 5, 5), D(4, 2, –2)

$\left.\begin{aligned}\overrightarrow{BA} &= (x-3)\hat{i} + 3\hat{j} - 2\hat{k} \\ \overrightarrow{BC} &= 1\hat{i} + 3\hat{j} + 4\hat{k} \\ \overrightarrow{BD} &= 1\hat{i} + 0\hat{j} - 3\hat{k}\end{aligned}\right\}$ $1\frac{1}{2}$

$$\begin{vmatrix} x-3 & 3 & -2 \\ 1 & 3 & 4 \\ 1 & 0 & -3 \end{vmatrix} = 0$$ 1

i.e., $(x - 3)(-9) - 3(-7) - 2(-3) = 0$ 1

$\Rightarrow x = 6$ $\frac{1}{2}$

23. $\Delta = 0$

$$\Rightarrow \begin{vmatrix} x & x^2 & x^3 \\ y & y^2 & y^3 \\ z & z^2 & z^3 \end{vmatrix} - \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} = 0$$ 1

$$\Rightarrow xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} - (-1)^2 \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = 0$$ 1

$$\Rightarrow \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} (xyz - 1) = 0$$ $\frac{1}{2}$

$$\Rightarrow \begin{vmatrix} 0 & x-z & x^2-z^2 \\ 0 & y-z & y^2-z^2 \\ 1 & z & z^2 \end{vmatrix} (xyz - 1) = 0$$ $R_1 \to R_1 - R_3$, $R_2 \to R_2 - R_3$ 1

$\Rightarrow (x - y)(y - z)(z - x)(xyz - 1) = 0$

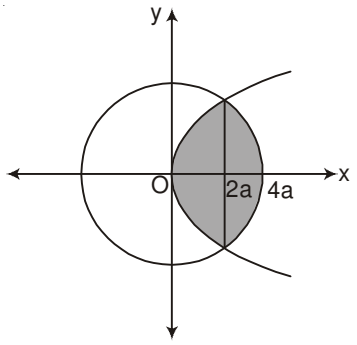
$\because x \neq y \neq z$ therefore $xyz = 1$ $\frac{1}{2}$

SECTION D

24. Correct figure 1

Solving $x^2 + y^2 = 16a^2$ ...(1)

and $y^2 = 6ax$ ...(2) we get



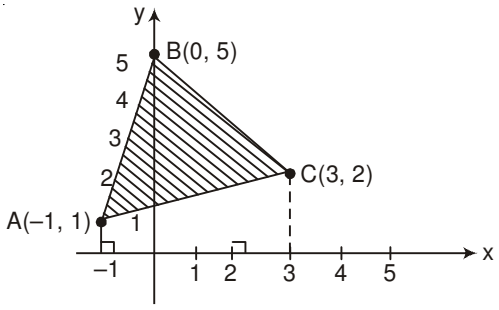
$x = 2a$ (as $-8a$ is not possible) 1

Required Area $= 2\left[\int_0^{2a} \sqrt{6ax}\, dx + \int_{2a}^{4a} \sqrt{(4a)^2 - x^2}\, dx\right]$ 2

$= 2\left[\sqrt{6a}\left(\frac{2}{3}x^{3/2}\right)\right]_0^{2a} + 2\left[\frac{x}{2}\sqrt{(4a)^2 - x^2} + 8a^2 \sin^{-1}\left(\frac{x}{4a}\right)\right]_{2a}^{4a}$ 1

$= \frac{8}{3}\sqrt{12}a^2 + \frac{16}{3}\pi a^2 - 4\sqrt{3}a^2$

$= \frac{4}{3}a^2(4\pi + \sqrt{3})$ 1



OR

Correct figure 1

$4x - y + 5 = 0$ ...(1)

$x + y - 5 = 0$ ...(2)

$x - 4y + 5 = 0$ ...(3)

Coordinates of A(–1, 1), B(0, 5) and C(3, 2) $1\frac{1}{2}$

Required Area $= \int_{-1}^{0} (4x+5)dx + \int_{0}^{3} (5-x)dx - \frac{1}{4}\int_{-1}^{3} (x+5)dx$ 2

$= \left[2x^2 + 5x\right]_{-1}^{0} + \left[5x - \frac{x^2}{2}\right]_0^3 - \frac{1}{4}\left[\frac{x^2}{2} + 5x\right]_{-1}^{3}$ 1

$= \frac{15}{2}$ $\frac{1}{2}$

25. Required equation of the line is

$\vec{r} = 2\hat{i} + \hat{j} - \hat{k} + \mu(2\hat{i} - \hat{j} + \hat{k})$ 2

Let $\vec{a}_1 = 2\hat{i} + \hat{j} - \hat{k}, \vec{a}_2 = \hat{i} + \hat{j}, \vec{b} = 2\hat{i} - \hat{j} + \hat{k}$

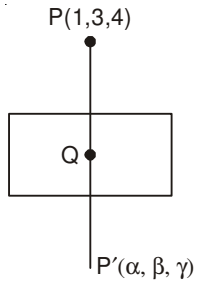
The required distance $= \left|\frac{(\vec{a}_2 - \vec{a}_1) \times \vec{b}}{|\vec{b}|}\right|$

$= \dfrac{|(\hat{i}-\hat{k})\times(2\hat{i}-\hat{j}+\hat{k})|}{|2\hat{i}-\hat{j}+\hat{k}|}$ 1

$(\vec{a}_2-\vec{a}_1)\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & -1 \\ 2 & -1 & 1 \end{vmatrix} = -\hat{i}-3\hat{j}-\hat{k}$ 2

Required distance $= \dfrac{\sqrt{1+9+1}}{\sqrt{4+1+1}} = \dfrac{\sqrt{11}}{\sqrt{6}}$ or $\dfrac{\sqrt{66}}{6}$ 1

OR



Correct figure $\frac{1}{2}$

Equation of line PQ is $\dfrac{x-1}{2} = \dfrac{y-3}{-1} = \dfrac{z-4}{1} = \lambda$ 1

The coordinates of Q are $(2\lambda + 1, -\lambda + 3, \lambda + 4)$ $\frac{1}{2}$

$\because$ Q lies on plane $2x - y + z + 3 = 0$

$\therefore\ 2(2\lambda + 1) - (-\lambda + 3) + (\lambda + 4) + 3 = 0$ 1

$\Rightarrow 6\lambda + 6 = 0$ i.e., $\lambda = -1$ $\frac{1}{2}$

The coordinates of Q are $(-1, 4, 3)$ $\frac{1}{2}$

$PQ = \sqrt{(-1-1)^2 + (4-3)^2 + (3-4)^2} = \sqrt{6}$ 1

Let $P'(\alpha, \beta, \gamma)$ be the image of P.

then $\dfrac{\alpha+1}{2} = -1,\ \dfrac{\beta+3}{2} = 4,\ \dfrac{\gamma+4}{2} = 3$ $\frac{1}{2}$

$\Rightarrow \alpha = -3,\ \beta = 5,\ \gamma = 2$

$\therefore$ the image P′ is $(-3, 5, 2)$ $\frac{1}{2}$

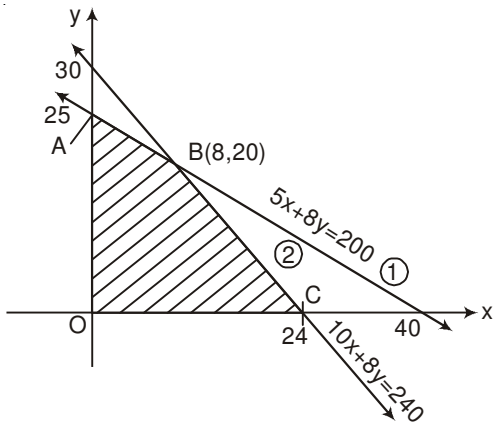
26. Let numbers of souvenirs of type A be x and number of souvenirs of type B be y

$\therefore$ L.P.P is

Maximize P = 100x + 120y $\frac{1}{2}$

Subject to constraints

$5x + 8y \le 200$ ...(1)
$10x + 8y \le 240$ ...(2)
$x, y \ge 0$ } $2\frac{1}{2}$



2

Values at corner points

Points	P
A(0, 25)	3000
B(8, 20)	3200 (Max)
C(24, 0)	2400

1

So, 8 type A Souvenirs and 20 type B Souvenirs should be made to maximize profit.

27. Let E_1 = Student guesses the answer
E_2 = Student copies the answer
E_3 = Student knows the answer
A = Student answers the question correctly. } 1

$$P(E_1) = \frac{1}{4}, P(E_2) = \frac{1}{4}, P(E_3) = 1 - \left(\frac{1}{4} + \frac{1}{4}\right) = \frac{1}{2}$$ $1\frac{1}{2}$

$$P(A \mid E_1) = \frac{1}{4}, P(A \mid E_2) = \frac{3}{4}, P(A \mid E_3) = 1$$ $1\frac{1}{2}$

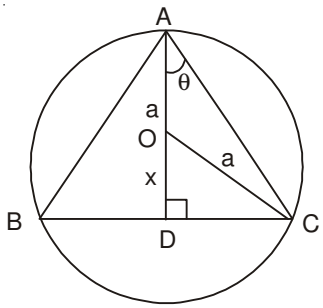
The required probability

$$= P(E_3 \mid A) = \frac{P(E_3) \times P(A \mid E_3)}{\sum_{i=1}^{3} P(E_i) \times P(A \mid E_i)}$$

$$= \frac{\frac{1}{2} \times 1}{\frac{1}{4} \times \frac{1}{4} + \frac{1}{4} \times \frac{3}{4} + \frac{1}{2} \times 1} \qquad 1\frac{1}{2}$$

$$= \frac{1}{\frac{1}{8} + \frac{3}{8} + 1} = \frac{8}{12} = \frac{2}{3} \qquad \frac{1}{2}$$

28. Correct figure 1



$CD = \sqrt{a^2 - x^2}$ $\frac{1}{2}$

Area, $A = \frac{1}{2} \times 2\sqrt{a^2 - x^2}\,(a + x)$ $\frac{1}{2}$

$Z = A^2 = (a - x)\,(a + x)^3$ $\frac{1}{2}$

$\frac{dZ}{dx} = 2(a + x)^2\,(a - 2x)$ $\frac{1}{2}$

$\frac{dZ}{dx} = 0 \Rightarrow x = \frac{a}{2}$ $\frac{1}{2}$

$\frac{d^2Z}{dx^2} = -12(a + x)x$ $\frac{1}{2}$

$\left(\frac{d^2Z}{dx^2}\right)_{x=\frac{a}{2}} = -9a^2 < 0$ $\frac{1}{2}$

$\therefore$ Z is maximum when $x = \frac{a}{2}$

i.e., Area is maximum when $x = \frac{a}{2}$ $\frac{1}{2}$

For maximum area

$$\tan\theta = \frac{CD}{AD} = \frac{\sqrt{a^2 - \frac{a^2}{4}}}{a + \frac{a}{2}} = \frac{1}{\sqrt{3}}$$

$$\theta = \frac{\pi}{6}$$

1

29. Let $A = \begin{bmatrix} 3 & 0 & -1 \\ 2 & 3 & 0 \\ 0 & 4 & 1 \end{bmatrix}$

Then $A = IA$

$$\Rightarrow \begin{bmatrix} 3 & 0 & -1 \\ 2 & 3 & 0 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

1

$$\Rightarrow \begin{bmatrix} 1 & -3 & -1 \\ 2 & 3 & 0 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \qquad (R_1 \to R_1 - R_2)$$

$$\Rightarrow \begin{bmatrix} 1 & -3 & -1 \\ 0 & 9 & 2 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \qquad (R_2 \to R_2 - 2R_1)$$

$$\Rightarrow \begin{bmatrix} 1 & -3 & -1 \\ 0 & 1 & 0 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -2 \\ 0 & 0 & 1 \end{bmatrix} A \qquad (R_2 \to R_2 - 2R_3)$$

$$\Rightarrow \begin{bmatrix} 1 & -3 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -2 \\ 8 & -12 & 9 \end{bmatrix} A \qquad (R_3 \to R_3 - 4R_2)$$

$$\Rightarrow \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 9 & -13 & 9 \\ -2 & 3 & -2 \\ 8 & -12 & 9 \end{bmatrix} A \qquad (R_1 \to R_1 - R_3)$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -4 & 3 \\ -2 & 3 & -2 \\ 8 & -12 & 9 \end{bmatrix} A \qquad (R_1 \to R_2 - 3R_2)$$

4

$\therefore\ A^{-1} = \begin{bmatrix} 3 & -4 & 3 \\ -2 & 3 & -2 \\ 8 & -12 & 9 \end{bmatrix}$ 1

OR

The given system of equations is

$AX = B,$

where $A = \begin{bmatrix} 2 & 3 & 10 \\ 4 & -6 & 5 \\ 6 & 9 & -20 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$

$|A| = 1200 \neq 0$ 1

$\Rightarrow A^{-1}$ exists.

$X = A^{-1}B$ 1

$$\text{adj } A = \begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix}$$ 2

$$A^{-1} = \frac{1}{|A|}\text{adj}A = \frac{1}{1200}\begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix}$$ $\frac{1}{2}$

$$X = A^{-1}B = \frac{1}{1200}\begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix}\begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

$$= \frac{1}{1200}\begin{bmatrix} 600 \\ 400 \\ 240 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{3} \\ \frac{1}{5} \end{bmatrix}$$ 1

$\therefore\ x = \frac{1}{2}, y = \frac{1}{3}, z = \frac{1}{5}$ $\frac{1}{2}$

QUESTION PAPER CODE 65/5/2
EXPECTED ANSWER/VALUE POINTS

SECTION A

1. $2\,|AA'| = 4|A|\,|A'|$ $\frac{1}{2}$

$= 4 \times 4 \times 4 = 64$ $\frac{1}{2}$

2. $\sin\theta = \left|\frac{(\hat{i}+\hat{j}+\hat{k})\cdot(3\hat{i}-\hat{j}+2\hat{k})}{\sqrt{1+1+1}\,\sqrt{9+1+4}}\right|$ $\frac{1}{2}$

$\Rightarrow \theta = \sin^{-1}\left(\frac{4}{\sqrt{42}}\right)$ $\frac{1}{2}$

OR

Any point on $\frac{x+2}{1} = \frac{y-5}{3} = \frac{z+1}{5} = \lambda$ ...(1) $\frac{1}{2}$

is $(\lambda-2, 3\lambda+5, 5\lambda-1)$

Line (1) cuts yz plane at $\lambda - 2 = 0$ i.e., $\lambda = 2$

hence required point is (0, 11, 9) $\frac{1}{2}$

3. $\frac{dy}{dx} = 3A\sin 3x + 3B\cos 3x$ $\frac{1}{2}$

$\frac{d^2y}{dx^2} = 9A\cos 3x - 9B\sin 3x = -9y$ $\frac{1}{2}$

4. Let $y = \cos^{-1}(\sin 2x)$, then $\frac{dy}{dx} = \frac{-1}{\sqrt{1-\sin^2 2x}}(2\cos 2x)$ $\frac{1}{2}$

$= -2$ or 2 $\frac{1}{2}$

SECTION B

5. (i) $\forall\, a, b \in Z$, $a * b = 2a^2 + b \in Z$ $\therefore$ * is binary 1

(ii) $1, 2 \in Z$, $1 * 2 = 2 \times 1^2 + 2 = 4$, $2 * 1 = 2 \times (2)^2 + 1 = 9$

$1 * 2 \neq 2 * 1$

$\therefore$ * is not commutative. 1

6. $n = 4,\ p = \frac{1}{4}, q = \frac{3}{4}$ $\frac{1}{2}$

P (atleast 3 are diamonds) = P (X = 3) + P(X = 4)

$= {}^4C_3\left(\frac{1}{4}\right)^3\left(\frac{3}{4}\right) + {}^4C_4\left(\frac{1}{4}\right)^4$ 1

$= \left(\frac{1}{4}\right)^4 [12+1] = \frac{13}{256}$ $\frac{1}{2}$

OR

Let E_1 : A coming on time.

E_2 : B coming on time.

$P(\bar{E}_1) = \frac{5}{7}, P(\bar{E}_2) = \frac{3}{7}$ $\frac{1}{2}$

P(only one on time)

$= P(E_1)\,P(\bar{E}_2) + P(\bar{E}_1)\,P(E_2)$

$= \frac{2}{7}\times\frac{3}{7} + \frac{5}{7}\times\frac{4}{7}$ 1

$= \frac{26}{49}$ $\frac{1}{2}$

7. $\begin{bmatrix} 2x-3 & 6 \\ 15 & 2y-4 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$ 1

$\Rightarrow 2x - 3 = 7$ and $2y - 4 = 14 \Rightarrow x = 5,\ y = 9 \Rightarrow x - y = -4$ 1

8. $(y + 3x^2)\frac{dx}{dy} = x \Rightarrow \frac{dy}{dx} - \frac{1}{x}y = 3x$ $\frac{1}{2}$

I.F. $= e^{\int -\frac{1}{x}dx} = e^{-\log(x)} = \frac{1}{x}$ $\frac{1}{2}$

Solution is $y \cdot \frac{1}{x} = \int 3x \cdot \frac{1}{x}dx \Rightarrow y = 3x^2 + cx$ 1

9. Let $P(2,-1,3)$, $Q(3,-5,1)$ and $R(-1,11,9)$ be three point.

$\overrightarrow{PQ} = \hat{i} - 4\hat{j} - 2\hat{k}$ $\frac{1}{2}$

$\overrightarrow{PR} = -3\hat{i} + 12\hat{j} + 6\hat{k} = -3(\hat{i} - 4\hat{j} - 2\hat{k})$ $\frac{1}{2}$

$\because$ $\overrightarrow{PR} = -3\overrightarrow{PQ}$, since P is common.

Therefore the points P, Q and R are collinear. 1

OR

$(\vec{a} \times \vec{b})^2 = \vec{a}^2\vec{b}^2 - (\vec{a} \cdot \vec{b})^2$

LHS = $(\vec{a} \times \vec{b})^2$

= $(|\vec{a}||\vec{b}| \sin\theta\, \hat{n})^2 = |\vec{a}|^2 |\vec{b}|^2 \sin^2\theta$ 1

= $|\vec{a}|^2 |\vec{b}|^2 (1 - \cos^2\theta)$

= $|\vec{a}|^2 |\vec{b}|^2 - (|\vec{a}||\vec{b}| \cos\theta)^2$

= $\vec{a}^2\, \vec{b}^2 - (\vec{a} \cdot \vec{b})^2$ 1

10. $\int \frac{x-1}{(x-2)(x-3)}\,dx = \int \left(\frac{-1}{x-2} + \frac{2}{x-3}\right) dx$ 1

= $-\log|x-2| + 2\log|x-3| + C$ 1

OR

$\int \frac{e^x}{\sqrt{5 - 4e^x - e^{2x}}}\,dx$ Put $e^x = t$ so that $e^x\,dx = dt$ $\frac{1}{2}$

$\int \frac{dt}{\sqrt{5 - 4t - t^2}} = \int \frac{dt}{\sqrt{3^2 - (t+2)^2}}$ 1

= $\sin^{-1}\left(\frac{t+2}{3}\right) + C = \sin^{-1}\left(\frac{e^x + 2}{3}\right) + C$ $\frac{1}{2}$

11. $\int e^x \left(\frac{2+\sin 2x}{2\cos^2 x} \right) dx = \int e^x (\sec^2 x + \tan x)\, dx$ 1

$= e^x \cdot \tan x + C$ 1

12. $P(A' \cap B') = P(A') \cdot P(B')$ 1

$= \frac{4}{7} \times \frac{3}{5}$

$= \frac{12}{35}$ 1

SECTION C

13. Equation of the required plane is

$a(x + 1) + b(y - 3) + c(z - 2) = 0$ ...(i) 1

Since plane (i) is perpendicular to planes $x + 2y + 3z = 5$ and $3x + 3y + z = 0$, therefore

$a + 2b + 3c = 0$ ...(ii)
and $3a + 3b + c = 0$ ...(iii) 1

Solving (ii) and (iii) we get

$\frac{a}{7} = \frac{b}{-8} = \frac{c}{3}$ 1

Equation of plane is

$7 \cdot (x + 1) - 8\,(y - 3) + 3\,(z - 2) = 0$

$\Rightarrow 7x - 8y + 3z + 25 = 0$ 1

14. $\int_1^5 (|x-1| + |x-2| + |x-4|\, dx$

$= \int_1^2 (5 - x) dx + \int_2^4 (x + 1) dx + \int_4^5 (3x - 7)\, dx$ $1\frac{1}{2}$

$= \left[5x - \frac{x^2}{2} \right]_1^2 + \left[\frac{x^2}{2} + x \right]_2^4 + \left[\frac{3x^2}{2} - 7x \right]_4^5$ $1\frac{1}{2}$

$= \frac{7}{2} + 8 + \frac{13}{2} = 18$ 1

15. $\Delta = 0$

$$\Rightarrow \begin{vmatrix} x & x^2 & x^3 \\ y & y^2 & y^3 \\ z & z^2 & z^3 \end{vmatrix} - \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} = 0 \qquad 1$$

$$\Rightarrow xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} - (-1)^2 \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = 0 \qquad 1$$

$$\Rightarrow \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} (xyz - 1) = 0 \qquad \frac{1}{2}$$

$$\Rightarrow \begin{vmatrix} 0 & x-z & x^2-z^2 \\ 0 & y-z & y^2-z^2 \\ 1 & z & z^2 \end{vmatrix} (xyz - 1) = 0 \qquad \begin{matrix} R_1 \to R_1 - R_3 \\ R_2 \to R_2 - R_3 \end{matrix} \qquad 1$$

$\Rightarrow (x - y)\,(y - z)\,(z - x)\,(xyz - 1) = 0$

$\because x \neq y \neq z$ therefore $xyz = 1$ $\qquad \frac{1}{2}$

16. LHS $= \sin^{-1}\frac{4}{5} + \tan^{-1}\frac{5}{12} + \cos^{-1}\frac{63}{65}$

$= \tan^{-1}\frac{4}{3} + \tan^{-1}\frac{5}{12} + \cos^{-1}\frac{63}{65}$ $\qquad 1$

$= \tan^{-1}\left(\frac{\frac{4}{3}+\frac{5}{12}}{1-\frac{4}{3}\times\frac{5}{12}}\right) + \cos^{-1}\frac{63}{65}$ $\qquad 1$

$= \tan^{-1}\left(\frac{63}{16}\right) + \cot^{-1}\left(\frac{63}{16}\right)$ $\qquad 1$

$= \frac{\pi}{2} = \text{RHS}$ $\qquad 1$

17. A(x, 5, –1), B(3, 2, 1), C(4, 5, 5), D(4, 2, –2)

$$\left.\begin{aligned}\overrightarrow{BA} &= (x-3)\hat{i}+3\hat{j}-2\hat{k}\\ \overrightarrow{BC} &= 1\hat{i}+3\hat{j}+4\hat{k}\\ \overrightarrow{BD} &= 1\hat{i}+0\hat{j}-3\hat{k}\end{aligned}\right\}$$ $1\frac{1}{2}$

$$\begin{vmatrix} x-3 & 3 & -2\\ 1 & 3 & 4\\ 1 & 0 & -3\end{vmatrix}=0$$ 1

i.e., $(x-3)(-9)-3(-7)-2(-3)=0$ 1

$\Rightarrow x = 6$ $\frac{1}{2}$

18. Let $y=\tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$, Put $x=\tan\theta$ $\frac{1}{2}$

$$y = \tan^{-1}\left(\frac{3\tan\theta-\tan^3\theta}{1-3\tan^2\theta}\right)\Rightarrow y=\tan^{-1}(\tan 3\theta)=3\theta$$

$$y = 3\tan^{-1}x \Rightarrow \frac{dy}{dx}=\frac{3}{1+x^2} \quad ...(i)$$ 1

Let $z=\tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$, put $x=\sin\phi$ $\frac{1}{2}$

$$z = \tan^{-1}\left(\frac{\sin\phi}{\sqrt{1-\sin^2\phi}}\right)\Rightarrow z=\tan^{-1}(\tan\phi)=\phi$$

$$z=\phi=\sin^{-1}x\Rightarrow\frac{dz}{dx}=\frac{1}{\sqrt{1-x^2}} \quad ...(ii)$$ 1

Using (i) & (ii), $\frac{dy}{dz}=\frac{dy/dx}{dz/dx}=\frac{3\sqrt{1-x^2}}{1+x^2}$ 1

OR

$\sqrt{1-x^2}+\sqrt{1-y^2}=a(x-y)$, put $x=\sin\theta$, $y=\sin\phi$ 1

$\sqrt{1-\sin^2\theta}+\sqrt{1-\sin^2\phi}=a(\sin\theta-\sin\phi)$

$\Rightarrow \cos\theta+\cos\phi=a(\sin\theta-\sin\phi)$

$\Rightarrow 2\cos\left(\frac{\theta+\phi}{2}\right)\cos\left(\frac{\theta-\phi}{2}\right)=2a\cos\left(\frac{\theta+\phi}{2}\right)\sin\left(\frac{\theta-\phi}{2}\right)$ 1

$\Rightarrow \tan\left(\frac{\theta-\phi}{2}\right)=\frac{1}{a}$

$\Rightarrow \frac{\theta-\phi}{2}=\tan^{-1}\left(\frac{1}{a}\right)\Rightarrow \sin^{-1}x-\sin^{-1}y=2\tan^{-1}\left(\frac{1}{a}\right)$ 1

Differentiating both sides w.r.t x

$\frac{1}{\sqrt{1-x^2}}-\frac{1}{\sqrt{1-y^2}}\frac{dy}{dx}=0$ $\frac{1}{2}$

$\frac{1}{\sqrt{1-y^2}}\frac{dy}{dx}=\frac{1}{\sqrt{1-x^2}}$ or $\frac{dy}{dx}=\frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$ $\frac{1}{2}$

19. (i) **Reflexive:** $\forall\, a\in A, |a-a|=0$ which is even

$\Rightarrow (a, a)\in R$, hence R is reflexive 1

(ii) **Symmetric:** Let $(a, b)\in R \Rightarrow |a-b|$ is even

$\Rightarrow |-(b-a)|$ is even $\Rightarrow |b-a|$ is even

so, $(b, a)\in R$

hence R is symmetric. 1

(iii) **Transitive:** Let $(a, b), (b, c)\in R$

so, $|a-b|$ is even and $|b-c|$ is even

$\Rightarrow a-b=2\lambda$, $b-c=2\mu$ where $\lambda, \mu\in Z$

Now, $a-c=(a-b)+(b-c)=2(\lambda+\mu)$

$\Rightarrow |a-c|$ is even, so $(a, c)\in R$

hence R is transitive. $1\frac{1}{2}$

Since R is reflexive, symmetric and transitive therefore its an equivalence relation $\frac{1}{2}$

OR

Let for $x_1, x_2 \in A$, $f(x_1) = f(x_2)$ $\frac{1}{2}$

$$\frac{4x_1+3}{6x_1-4}=\frac{4x_2+3}{6x_2-4}$$

$$\Rightarrow (4x_1+3)(6x_2-4)=(6x_1-4)(4x_2+3)$$

$\Rightarrow 34x_1 = 34x_2 \Rightarrow x_1 = x_2$, hence f is one-one. 1

For any $y \in A$ such that $y = \frac{4x+3}{6x-4}$ there exists x such that

$$6xy-4y=4x+3 \Rightarrow (6y-4)x=4y+3$$

$\Rightarrow x = \frac{4y+3}{6y-4}, y \in A, \; x = \frac{4y+3}{6y-4} \in A$ 1

$\Rightarrow$ f is onto. 1

Since f is one-one and onto, therefore f^{-1} exists in A

and $f^{-1}(y) = \frac{4y+3}{6y-4}$ or $f^{-1}(x) = \frac{4x+3}{6x-4}$ $\frac{1}{2}$

20. $(1+e^{2x})dy + (1+y^2)e^x \, dx = 0$

$$\Rightarrow \int \frac{dy}{1+y^2} = -\int \frac{e^x}{1+e^{2x}} dx$$ 1

$$\Rightarrow \tan^{-1} y = -\int \frac{e^x}{1+e^{2x}} dx$$ $\frac{1}{2}$

Put $e^x = t$, so that $e^x \, dx = dt$ $\frac{1}{2}$

$$\tan^{-1} y = -\int \frac{dt}{1+t^2} \Rightarrow \tan^{-1} y = -\tan^{-1}(e^x) + C \quad \text{...(i)}$$ 1

Substituting y = 1, when x = 0 in equation (i)

$$\tan^{-1}(1) = -\tan^{-1}(1) + C \Rightarrow C = \frac{\pi}{2}$$ $\frac{1}{2}$

Substituting $C = \frac{\pi}{2}$ in equation (i) $\Rightarrow \tan^{-1} y + \tan^{-1}(e^x) = \frac{\pi}{2}$ $\frac{1}{2}$

OR

$x\frac{dy}{dx}\sin\left(\frac{y}{x}\right) + x - y\sin\left(\frac{y}{x}\right) = 0$

$\Rightarrow \frac{dy}{dx} = \frac{y}{x} - \frac{1}{\sin\left(\frac{y}{x}\right)}$ ...(i) $\frac{1}{2}$

Put $\frac{y}{x} = v$ i.e., $y = vx$ in (i) so that $\frac{dy}{dx} = v + x\frac{dv}{dx}$ 1

$v + x\frac{dv}{dx} = v - \frac{1}{\sin v}$

$\int \sin v\, dv = \int -\frac{1}{x}\, dx$ $\frac{1}{2}$

$\Rightarrow -\cos v = -\log|x| + C \Rightarrow \cos\left(\frac{y}{x}\right) = \log|x| + C$ ...(ii) 1

Substituting $y = \frac{\pi}{2}$ when $x = 1$ in (ii)

$\cos\left(\frac{\pi}{2}\right) = \log 1 + C \Rightarrow C = 0$ $\frac{1}{2}$

Required solution is $\cos\left(\frac{y}{x}\right) = \log|x|$ $\frac{1}{2}$

21. $y = (\sin x)^x + \sin^{-1}\sqrt{1-x^2}$

Let $A = (\sin x)^x \Rightarrow \log A = x \log \sin x$ $\frac{1}{2}$

$\Rightarrow \frac{1}{A}\frac{dA}{dx} = \log \sin x + x \cot x$ 1

$\Rightarrow \frac{dA}{dx} = (\sin x)^x\ (\log \sin x + x \cot x)$ $\frac{1}{2}$

$B = \sin^{-1}\sqrt{1-x^2} = \cos^{-1} x$

$\frac{dB}{dx} = \frac{-1}{\sqrt{1-x^2}}$ 1

$y = A + B \Rightarrow \frac{dy}{dx} = \frac{dA}{dx} + \frac{dA}{dx} = (\sin x)^x \ (\log \sin x + x \cot x) - \frac{1}{\sqrt{1-x^2}}$ 1

22. $\int \cos 2x \cos 4x \cos 6x \, dx$

$= \frac{1}{2}\int 2\cos 6x \cos 2x \cdot \cos 4x \, dx$ $\frac{1}{2}$

$= \frac{1}{2}\int (\cos 8x + \cos 4x) \cos 4x \, dx$ 1

$= \frac{1}{4}\int (2\cos 8x \cos 4x + 2\cos^2 4x) \, dx$ $\frac{1}{2}$

$= \frac{1}{4}\int (\cos 12x + \cos 4x + 1 + \cos 8x) \, dx$ 1

$= \frac{1}{4}\left(\frac{\sin 12x}{12} + \frac{\sin 4x}{4} + x + \frac{\sin 8x}{8}\right) + C$ 1

23. $f(x) = \sin 2x + \cos 2x$

$f'(x) = 2 \cos 2x - 2 \sin 2x$ 1
$f'(x) = 0 \Rightarrow 2 \cos 2x - 2 \sin 2x = 0$

$\Rightarrow x = \frac{\pi}{8}, \frac{5\pi}{8}$ 1

Possible interval are $\left(0, \frac{\pi}{8}\right), \left(\frac{\pi}{8}, \frac{5\pi}{8}\right), \left(\frac{5\pi}{8}, 2\pi\right)$ 1

$f'(x) < 0$ in $\left(\frac{\pi}{8}, \frac{5\pi}{8}\right) \Rightarrow f(x)$ is decreasing in $\left(\frac{\pi}{8}, \frac{5\pi}{8}\right)$ 1

SECTION D

24. Let $A = \begin{bmatrix} 3 & 0 & -1 \\ 2 & 3 & 0 \\ 0 & 4 & 1 \end{bmatrix}$

Then $A = IA$

$$\Rightarrow \begin{bmatrix} 3 & 0 & -1 \\ 2 & 3 & 0 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$ 1

$$\left.\begin{array}{ll}
\Rightarrow \begin{bmatrix} 1 & -3 & -1 \\ 2 & 3 & 0 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A & (R_1 \to R_1 - R_2) \\
\Rightarrow \begin{bmatrix} 1 & -3 & -1 \\ 0 & 9 & 2 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} A & (R_2 \to R_2 - 2R_1) \\
\Rightarrow \begin{bmatrix} 1 & -3 & -1 \\ 0 & 1 & 0 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -2 \\ 0 & 0 & 1 \end{bmatrix} A & (R_2 \to R_2 - 2R_3) \\
\Rightarrow \begin{bmatrix} 1 & -3 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -2 \\ 8 & -12 & 9 \end{bmatrix} A & (R_3 \to R_3 - 4R_2) \\
\Rightarrow \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 9 & -13 & 9 \\ -2 & 3 & -2 \\ 8 & -12 & 9 \end{bmatrix} A & (R_1 \to R_1 - R_3) \\
\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -4 & 3 \\ -2 & 3 & -2 \\ 8 & -12 & 9 \end{bmatrix} A & (R_1 \to R_2 - 3R_2)
\end{array}\right\}$$ 4

$$\therefore A^{-1} = \begin{bmatrix} 3 & -4 & 3 \\ -2 & 3 & -2 \\ 8 & -12 & 9 \end{bmatrix}$$ 1

OR

The given system of equations is

$AX = B$,

where $A = \begin{bmatrix} 2 & 3 & 10 \\ 4 & -6 & 5 \\ 6 & 9 & -20 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$

$|A| = 1200 \neq 0$ 1

$\Rightarrow A^{-1}$ exists.

$X = A^{-1}B$ 1

$$\text{adj } A = \begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix}$$ 2

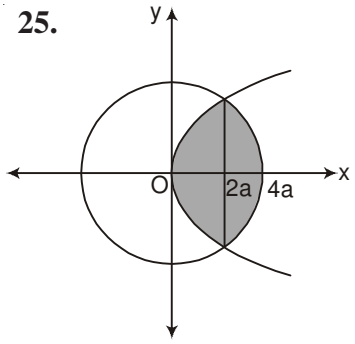
$$A^{-1} = \frac{1}{|A|}\text{adj}A = \frac{1}{1200}\begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix}$$ $\frac{1}{2}$

$$X = A^{-1}B = \frac{1}{1200}\begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix}\begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

$$= \frac{1}{1200}\begin{bmatrix} 600 \\ 400 \\ 240 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{3} \\ \frac{1}{5} \end{bmatrix}$$ 1

$\therefore x = \frac{1}{2}, y = \frac{1}{3}, z = \frac{1}{5}$ $\frac{1}{2}$

25.



Correct figure 1

Solving $x^2 + y^2 = 16a^2$ ...(1)

and $y^2 = 6ax$ ...(2) we get

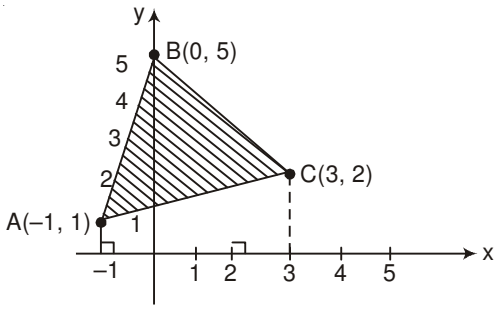
$x = 2a$ (as $-8a$ is not possible) 1

$$\text{Required Area} = 2\left[\int_0^{2a}\sqrt{6ax}\,dx + \int_{2a}^{4a}\sqrt{(4a)^2 - x^2}\,dx\right]$$ 2

$$= 2\left[\sqrt{6a}\left(\frac{2}{3}x^{3/2}\right)\right]_0^{2a} + 2\left[\frac{x}{2}\sqrt{(4a)^2 - x^2} + 8a^2\sin^{-1}\left(\frac{x}{4a}\right)\right]_{2a}^{4a}$$ 1

$$= \frac{8}{3}\sqrt{12}a^2 + \frac{16}{3}\pi a^2 - 4\sqrt{3}a^2$$

$$= \frac{4}{3}a^2(4\pi + \sqrt{3})$$ 1



OR

Correct figure 1

$4x - y + 5 = 0$ ...(1)

$x + y - 5 = 0$ ...(2)

$x - 4y + 5 = 0$ ...(3)

Coordinates of A(–1, 1), B(0, 5) and C(3, 2) $1\frac{1}{2}$

Required Area = $\int_{-1}^{0}(4x+5)dx + \int_{0}^{3}(5-x)dx - \frac{1}{4}\int_{-1}^{3}(x+5)dx$ 2

$= \left[2x^2+5x\right]_{-1}^{0} + \left[5x - \frac{x^2}{2}\right]_{0}^{3} - \frac{1}{4}\left[\frac{x^2}{2}+5x\right]_{-1}^{3}$ 1

$= \frac{15}{2}$ $\frac{1}{2}$

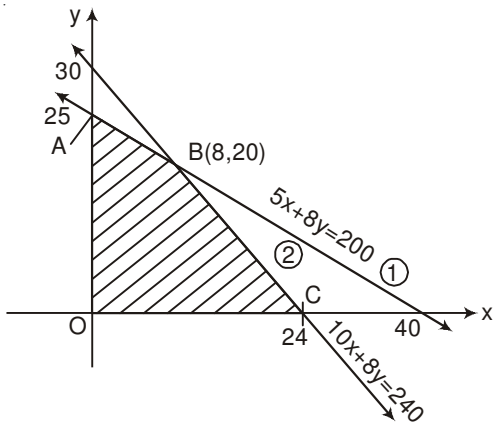
26. Let numbers of souvenirs of type A be x and number of souvenirs of type B be y

$\therefore$ L.P.P is

Maximize P = 100x + 120y $\frac{1}{2}$

Subject to constraints

$5x + 8y \le 200$ ...(1)
$10x + 8y \le 240$ ...(2)
$x, y \ge 0$ } $2\frac{1}{2}$



2

Values at corner points

Points	P
A(0, 25)	3000
B(8, 20)	3200 (Max)
C(24, 0)	2400

1

So, 8 type A Souvenirs and 20 type B Souvenirs should be made to maximize profit.

27. Required equation of the line is

$\vec{r} = 2\hat{i} + \hat{j} - \hat{k} + \mu(2\hat{i} - \hat{j} + \hat{k})$ 2

Let $\vec{a}_1 = 2\hat{i} + \hat{j} - \hat{k}, \vec{a}_2 = \hat{i} + \hat{j}, \vec{b} = 2\hat{i} - \hat{j} + \hat{k}$

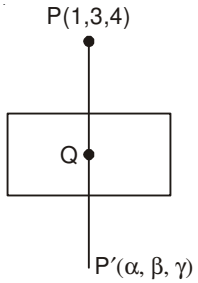
The required distance = $\left|\frac{(\vec{a}_2 - \vec{a}_1) \times \vec{b}}{|\vec{b}|}\right|$

$= \frac{|(\hat{i} - \hat{k}) \times (2\hat{i} - \hat{j} + \hat{k})|}{|2\hat{i} - \hat{j} + \hat{k}|}$ 1

$(\vec{a}_2 - \vec{a}_1) \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & -1 \\ 2 & -1 & 1 \end{vmatrix} = -\hat{i} - 3\hat{j} - \hat{k}$ 2

Required distance = $\frac{\sqrt{1+9+1}}{\sqrt{4+1+1}} = \frac{\sqrt{11}}{\sqrt{6}}$ or $\frac{\sqrt{66}}{6}$ 1

OR



Correct figure $\frac{1}{2}$

Equation of line PQ is $\frac{x-1}{2} = \frac{y-3}{-1} = \frac{z-4}{1} = \lambda$ 1

The coordinates of Q are $(2\lambda + 1, -\lambda + 3, \lambda + 4)$ $\frac{1}{2}$

$\because$ Q lies on plane $2x - y + z + 3 = 0$

$\therefore\ 2(2\lambda + 1) - (-\lambda + 3) + (\lambda + 4) + 3 = 0$ 1

$\Rightarrow 6\lambda + 6 = 0$ i.e., $\lambda = -1$ $\frac{1}{2}$

The coordinates of Q are (–1, 4, 3) $\frac{1}{2}$

$PQ = \sqrt{(-1-1)^2 + (4-3)^2 + (3-4)^2} = \sqrt{6}$ 1

Let P′(α, β , γ) be the image of P.

then $\frac{\alpha+1}{2} = -1,\ \frac{\beta+3}{2} = 4,\ \frac{\gamma+4}{2} = 3$ $\frac{1}{2}$

$\Rightarrow \alpha = -3,\ \beta = 5,\ \gamma = 2$

$\therefore$ the image P′ is (–3, 5, 2) $\frac{1}{2}$

28. Let E_1 : spade card is lost
E_2 : non spade card is lost.
A : Two cards drawn are spade } 1

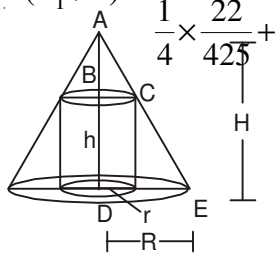
$P(E_1) = \frac{1}{4}, P(E_2) = \frac{3}{4}$ 1

$P(A \mid E_1) = \frac{^{12}C_2}{^{51}C_2} = \frac{22}{425}$ 1

$P(A \mid E_2) = \frac{^{13}C_2}{^{51}C_2} = \frac{26}{425}$ 1

$P(E_1 \mid A) = \frac{\frac{1}{4}\times\frac{22}{425}}{\frac{1}{4}\times\frac{22}{425}+\frac{3}{4}\times\frac{26}{425}} = \frac{11}{50}$ $1\frac{1}{2}+\frac{1}{2}$

29.



Correct figure 1

$\Delta ABC \sim \Delta ADE$

$\frac{H-h}{H} = \frac{r}{R} \Rightarrow r = \frac{HR - hR}{H}$ 1

Now, $A = 2\pi h\left(\frac{HR - hR}{H}\right) = 2\pi hR - \frac{2\pi Rh^2}{H}$ 1

$\frac{dA}{dh} = 2\pi R - \frac{4\pi Rh}{H}$ 1

$\frac{dA}{dh} = 0 \Rightarrow H = 2h$ $\frac{1}{2}$

$\left.\frac{d^2A}{dh^2}\right|_{H=2h} = \frac{-4\pi R}{2h} < 0 \Rightarrow$ area is maximum 1

A is maximum at H = 2h.

$\Rightarrow r = \frac{HR - \frac{HR}{2}}{H} = \frac{R}{2}$ $\frac{1}{2}$

QUESTION PAPER CODE 65/5/3
EXPECTED ANSWER/VALUE POINTS

SECTION A

1. Order = 2 1

2. $\sin\theta = \left|\frac{(\hat{i}+\hat{j}+\hat{k})\cdot(3\hat{i}-\hat{j}+2\hat{k}}{\sqrt{1+1+1}\sqrt{9+1+4}}\right|$ $\frac{1}{2}$

$\Rightarrow \theta = \sin^{-1}\left(\frac{4}{\sqrt{42}}\right)$ $\frac{1}{2}$

OR

Any point on $\frac{x+2}{1}=\frac{y-5}{3}=\frac{z+1}{5}=\lambda$ ...(1) $\frac{1}{2}$

is $(\lambda-2, 3\lambda+5, 5\lambda-1)$

Line (1) cuts yz plane at $\lambda - 2 = 0$ i.e., $\lambda = 2$

hence required point is (0, 11, 9) $\frac{1}{2}$

3. $|\text{adj } A| = |A|^{n-1}$ $\frac{1}{2}$

$|A| = 10$

$|\text{adj } A| = (10)^{2-1} = 10$ $\frac{1}{2}$

4. $\frac{dy}{dx}=\frac{1}{\sqrt{\sec e^{2x}}}\cdot \sec e^{2x}\tan e^{2x}.2e^{2x}$ 1

or $2\sqrt{\sec e^{2x}}\tan e^{2x}\, e^{2x}$

SECTION B

5. $n = 4,\ p = \frac{1}{4}, q = \frac{3}{4}$ $\frac{1}{2}$

P (atleast 3 are diamonds) = P (X = 3) + P(X = 4)

$= {}^4C_3\left(\frac{1}{4}\right)^3\left(\frac{3}{4}\right)+{}^4C_4\left(\frac{1}{4}\right)^4$ 1

$= \left(\frac{1}{4}\right)^4 [12+1] = \frac{13}{256}$ $\frac{1}{2}$

OR

Let E_1 : A coming on time.

E_2 : B coming on time.

$P(\bar{E}_1) = \frac{5}{7}, P(\bar{E}_2) = \frac{3}{7}$ $\frac{1}{2}$

P(only one on time)

$= P(E_1)\, P(\bar{E}_2) + P(\bar{E}_1)\, P(E_2)$

$= \frac{2}{7}\times\frac{3}{7} + \frac{5}{7}\times\frac{4}{7}$ 1

$= \frac{26}{49}$ $\frac{1}{2}$

6. Let $P(2,-1,3)$, $Q(3,-5,1)$ and $R(-1,11,9)$ be three point.

$\overrightarrow{PQ} = \hat{i} - 4\hat{j} - 2\hat{k}$ $\frac{1}{2}$

$\overrightarrow{PR} = -3\hat{i} + 12\hat{j} + 6\hat{k} = -3(\hat{i} - 4\hat{j} - 2\hat{k})$ $\frac{1}{2}$

$\because \overrightarrow{PR} = -3\overrightarrow{PQ}$, since P is common.

Therefore the points P, Q and R are collinear. 1

OR

$(\vec{a}\times\vec{b})^2 = \vec{a}^2\vec{b}^2 - (\vec{a}\cdot\vec{b})^2$

LHS = $(\vec{a}\times\vec{b})^2$

$= (|\vec{a}||\vec{b}|\sin\theta\,\hat{n})^2 = |\vec{a}|^2\,|\vec{b}|^2\sin^2\theta$ 1

$= |\vec{a}|^2\,|\vec{b}|^2\,(1-\cos^2\theta)$

$= |\vec{a}|^2\,|\vec{b}|^2 - (|\vec{a}||\vec{b}|\cos\theta)^2$

$= \vec{a}^2\,\vec{b}^2 - (\vec{a}\cdot\vec{b})^2$ 1

7. $\int \frac{x-1}{(x-2)(x-3)}\,dx = \int\left(\frac{-1}{x-2}+\frac{2}{x-3}\right)dx$ 1

$= -\log|x-2| + 2\log|x-3| + C$ 1

OR

$\int \frac{e^x}{\sqrt{5-4e^x-e^{2x}}}\,dx$ Put $e^x = t$ so that $e^x\,dx = dt$ $\frac{1}{2}$

$\int \frac{dt}{\sqrt{5-4t-t^2}} = \int \frac{dt}{\sqrt{3^2-(t+2)^2}}$ 1

$= \sin^{-1}\left(\frac{t+2}{3}\right) + C = \sin^{-1}\left(\frac{e^x+2}{3}\right) + C$ $\frac{1}{2}$

8. $P(B|A) = 0.4 \Rightarrow \frac{P(B \cap A)}{P(B)} = 0.4 \Rightarrow P(B \cap A) = 0.24$ $\frac{1}{2}$

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.6 + 0.5 - 0.24 = 0.86$ 1

$P(A|B) = \frac{P(A \cap B)}{P(A)} = \frac{0.24}{0.5} = 0.48$ $\frac{1}{2}$

9. $\begin{bmatrix} 2x-3 & 6 \\ 15 & 2y-4 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$ 1

$\Rightarrow 2x - 3 = 7$ and $2y - 4 = 14 \Rightarrow x = 5,\ y = 9 \Rightarrow x - y = -4$ 1

10. $\int e^x(\cos x - \sin x)\,\text{cosec}^2 x\,dx = -\int e^x[\text{cosec } x - \text{cosec } x \cot x]\,dx$ 1

$= -e^x \cos\text{ec } x + c$ 1

11. $y^2 = a(b^2 - x^2)$

$2yy' = -2ax$ $\frac{1}{2}$

$\Rightarrow \frac{yy'}{x} = -a$ $\frac{1}{2}$

$\Rightarrow \frac{(y' \cdot y' + yy'')x - yy'.1}{x^2} = 0$

$\Rightarrow\ x(y')^2 + xyy'' - yy' = 0$ or $x\left(\frac{dy}{dx}\right)^2 + xy\frac{d^2y}{dx^2} - y\frac{dy}{dx} = 0$ 1

12. $a \in N, b \in N \Rightarrow a^b \in N \Rightarrow a * b \in N \Rightarrow *$ is binary operation. 1

In general $a^b \neq b^a$, for $a, b \in N \Rightarrow *$ is not commutative. 1

SECTION C

13. $(1+e^{2x})dy + (1+y^2)e^x\, dx = 0$

$\Rightarrow\ \int\frac{dy}{1+y^2} = -\int\frac{e^x}{1+e^{2x}}dx$ 1

$\Rightarrow\ \tan^{-1}y = -\int\frac{e^x}{1+e^{2x}}dx$ $\frac{1}{2}$

Put $e^x = t$, so that $e^x\, dx = dt$ $\frac{1}{2}$

$\tan^{-1}y = -\int\frac{dt}{1+t^2} \Rightarrow \tan^{-1}y = -\tan^{-1}(e^x) + C$ ...(i) 1

Substituting $y = 1$, when $x = 0$ in equation (i)

$\tan^{-1}(1) = -\tan^{-1}(1) + C \Rightarrow C = \frac{\pi}{2}$ $\frac{1}{2}$

Substituting $C = \frac{\pi}{2}$ in equation (i) $\Rightarrow \tan^{-1}y + \tan^{-1}(e^x) = \frac{\pi}{2}$ $\frac{1}{2}$

OR

$x\frac{dy}{dx}\sin\left(\frac{y}{x}\right) + x - y\sin\left(\frac{y}{x}\right) = 0$

$\Rightarrow \frac{dy}{dx} = \frac{y}{x} - \frac{1}{\sin\left(\frac{y}{x}\right)}$ ...(i) $\frac{1}{2}$

Put $\frac{y}{x} = v$ i.e., $y = vx$ in (i) so that $\frac{dy}{dx} = v + x\frac{dv}{dx}$ 1

$v + x\frac{dv}{dx} = v - \frac{1}{\sin v}$

$\int \sin v \, dv = \int -\frac{1}{x} dx$ $\frac{1}{2}$

$\Rightarrow -\cos v = -\log|x| + C \Rightarrow \cos\left(\frac{y}{x}\right) = \log|x| + C$ ...(ii) 1

Substituting $y = \frac{\pi}{2}$ when x = 1 in (ii)

$\cos\left(\frac{\pi}{2}\right) = \log 1 + C \Rightarrow C = 0$ $\frac{1}{2}$

Required solution is $\cos\left(\frac{y}{x}\right) = \log|x|$ $\frac{1}{2}$

14. (i) **Reflexive:** $\forall a \in A, |a - a| = 0$ which is even

$\Rightarrow (a, a) \in R$, hence R is reflexive 1

(ii) **Symmetric:** Let $(a, b) \in R \Rightarrow |a - b|$ is even

$\Rightarrow |-(b - a)|$ is even $\Rightarrow |b - a|$ is even

so, $(b, a) \in R$

hence R is symmetric. 1

(iii) **Transitive:** Let $(a, b), (b, c) \in R$

so, $|a - b|$ is even and $|b - c|$ is even

$\Rightarrow a - b = 2\lambda$, $b - c = 2\mu$ where $\lambda, \mu \in Z$

Now, $a - c = (a - b) + (b - c) = 2(\lambda + \mu)$

$\Rightarrow |a - c|$ is even, so $(a, c) \in R$

hence R is transitive. $1\frac{1}{2}$

Since R is reflexive, symmetric and transitive therefore its an equivalence relation $\frac{1}{2}$

OR

Let for $x_1, x_2 \in A$, $f(x_1) = f(x_2)$ $\frac{1}{2}$

$\frac{4x_1 + 3}{6x_1 - 4} = \frac{4x_2 + 3}{6x_2 - 4}$

$\Rightarrow (4x_1+3)(6x_2-4) = (6x_1-4)(4x_2+3)$

$\Rightarrow 34x_1 = 34x_2 \Rightarrow x_1 = x_2$, hence f is one-one. 1

For any $y \in A$ such that $y = \frac{4x+3}{6x-4}$ there exists x such that

$6xy - 4y = 4x + 3 \Rightarrow (6y-4)x = 4y+3$

$\Rightarrow x = \frac{4y+3}{6y-4}, y \in A,\ x = \frac{4y+3}{6y-4} \in A$ 1

$\Rightarrow$ f is onto. 1

Since f is one-one and onto, therefore f^{-1} exists in A

and $f^{-1}(y) = \frac{4y+3}{6y-4}$ or $f^{-1}(x) = \frac{4x+3}{6x-4}$ $\frac{1}{2}$

15. $f(x) = \cos\left(2x+\frac{\pi}{4}\right) \Rightarrow f'(x) = -2\sin\left(2x+\frac{\pi}{4}\right)$ $1\frac{1}{2}$

As $\frac{3\pi}{8} < x < \frac{5\pi}{8} \Rightarrow \frac{3\pi}{4} < 2x < \frac{5\pi}{4}$

$\Rightarrow \pi < 2x + \frac{\pi}{4} < \frac{3\pi}{2}$ $1\frac{1}{2}$

$\Rightarrow \sin\left(2x+\frac{\pi}{4}\right) < 0 \Rightarrow f'(x) > 0$

$\therefore$ f(x) is increasing in $\left(\frac{3\pi}{8}, \frac{5\pi}{8}\right)$ 1

16. A(x, 5, –1), B(3, 2, 1), C(4, 5, 5), D(4, 2, –2)

$$\left.\begin{aligned}\overrightarrow{BA} &= (x-3)\hat{i} + 3\hat{j} - 2\hat{k}\\ \overrightarrow{BC} &= 1\hat{i} + 3\hat{j} + 4\hat{k}\\ \overrightarrow{BD} &= 1\hat{i} + 0\hat{j} - 3\hat{k}\end{aligned}\right\}$$ $1\frac{1}{2}$

$$\begin{vmatrix} x-3 & 3 & -2\\ 1 & 3 & 4\\ 1 & 0 & -3\end{vmatrix} = 0$$ 1

i.e., $(x - 3)(-9) - 3(-7) - 2(-3) = 0$ 1

$\Rightarrow x = 6$ $\frac{1}{2}$

17. LHS $= \sin^{-1}\frac{4}{5} + \tan^{-1}\frac{5}{12} + \cos^{-1}\frac{63}{65}$

$= \tan^{-1}\frac{4}{3} + \tan^{-1}\frac{5}{12} + \cos^{-1}\frac{63}{65}$ 1

$= \tan^{-1}\left(\frac{\frac{4}{3}+\frac{5}{12}}{1-\frac{4}{3}\times\frac{5}{12}}\right) + \cos^{-1}\frac{63}{65}$ 1

$= \tan^{-1}\left(\frac{63}{16}\right) + \cot^{-1}\left(\frac{63}{16}\right)$ 1

$= \frac{\pi}{2} = \text{RHS}$ 1

18. $x^p y^q = (x+y)^{p+q} \Rightarrow p\log x + q\log y = (p+q)\log(x+y)$ $\frac{1}{2}$

Differentaiting w.r.t x, $\frac{p}{x} + \frac{q}{y}\frac{dy}{dx} = \frac{p+q}{x+y}\left(1 + \frac{dy}{dx}\right)$ 1

$\Rightarrow \left(\frac{q}{y} - \frac{p+q}{x+y}\right)\frac{dy}{dx} = \frac{p+q}{x+y} - \frac{p}{x}$

$\Rightarrow \left[\frac{qx-py}{y(x+y)}\right]\frac{dy}{dx} = \frac{qx-py}{x(x+y)} \Rightarrow \frac{dy}{dx} = \frac{y}{x}$ 1

Differentiating again w.r.t x

$\frac{d^2y}{dx^2} = \frac{x\frac{dy}{dx} - y}{x^2}$ 1

$\Rightarrow \frac{d^2y}{dx^2} = \frac{x\left(\frac{y}{x}\right) - y}{x^2} \Rightarrow \frac{d^2y}{dx^2} = 0$ $\frac{1}{2}$

19. $\int (\sin x \sin 2x \sin 3x)\,dx = \frac{1}{2}\int (2\sin x \sin 2x)\sin 3x\,dx$ $\frac{1}{2}$

$= \frac{1}{2}\int [\cos(x) - \cos(3x)]\sin 3x\,dx$ 1

$= \frac{1}{4}\int 2\cos x \sin 3x\,dx - \frac{1}{4}\int 2\cos 3x \sin 3x\,dx$

$= \frac{1}{4}\int (\sin 4x + \sin 2x)dx - \frac{1}{4}\int \sin 6x\,dx$ 1

$= -\frac{1}{4}\left(\frac{\cos 4x}{4} + \frac{\cos 2x}{2}\right) + \frac{1}{24}\cos 6x + C$ $1\frac{1}{2}$

20. Let $y = \tan^{-1}\left(\frac{3x - x^3}{1 - 3x^2}\right)$, Put $x = \tan\theta$ $\frac{1}{2}$

$y = \tan^{-1}\left(\frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}\right) \Rightarrow y = \tan^{-1}(\tan 3\theta) = 3\theta$

$y = 3\tan^{-1} x \Rightarrow \frac{dy}{dx} = \frac{3}{1+x^2}$ ...(i) 1

Let $z = \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$, put $x = \sin\phi$ $\frac{1}{2}$

$z = \tan^{-1}\left(\frac{\sin\phi}{\sqrt{1-\sin^2\phi}}\right) \Rightarrow z = \tan^{-1}(\tan\phi) = \phi$

$z = \phi = \sin^{-1} x \Rightarrow \frac{dz}{dx} = \frac{1}{\sqrt{1-x^2}}$ ...(ii) 1

Using (i) & (ii), $\frac{dy}{dz} = \frac{dy/dx}{dz/dx} = \frac{3\sqrt{1-x^2}}{1+x^2}$ 1

OR

$\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$, put $x = \sin\theta$, $y = \sin\phi$ 1

$\sqrt{1-\sin^2\theta}+\sqrt{1-\sin^2\phi}=a(\sin\theta-\sin\phi)$

$\Rightarrow \cos\theta+\cos\phi=a(\sin\theta-\sin\phi)$

$\Rightarrow 2\cos\left(\frac{\theta+\phi}{2}\right)\cos\left(\frac{\theta-\phi}{2}\right)=2a\cos\left(\frac{\theta+\phi}{2}\right)\sin\left(\frac{\theta-\phi}{2}\right)$ 1

$\Rightarrow \tan\left(\frac{\theta-\phi}{2}\right)=\frac{1}{a}$

$\Rightarrow \frac{\theta-\phi}{2}=\tan^{-1}\left(\frac{1}{a}\right)\Rightarrow \sin^{-1}x-\sin^{-1}y=2\tan^{-1}\left(\frac{1}{a}\right)$ 1

Differentiating both sides w.r.t x

$\frac{1}{\sqrt{1-x^2}}-\frac{1}{\sqrt{1-y^2}}\frac{dy}{dx}=0$ $\frac{1}{2}$

$\frac{1}{\sqrt{1-y^2}}\frac{dy}{dx}=\frac{1}{\sqrt{1-x^2}}$ or $\frac{dy}{dx}=\frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$ $\frac{1}{2}$

21. $\begin{vmatrix} a^2 & bc & ac+c^2 \\ a^2+ab & b^2 & ac \\ ab & b^2+bc & c^2 \end{vmatrix}$

$= abc\begin{vmatrix} a & c & a+c \\ a+b & b & a \\ b & b+c & c \end{vmatrix}$ Taking a, b, c common from C_1, C_2, C_3 respectively 1

$= abc\begin{vmatrix} a & c & 0 \\ a+b & b & -2b \\ b & b+c & -2b \end{vmatrix}$ $C_3 \to C_3 - (C_1 + C_2)$ 1

$= abc\,.\,2b\begin{vmatrix} a & c & 0 \\ a+b & b & -1 \\ b & b+c & -1 \end{vmatrix}$

$= 2ab^2c\begin{vmatrix} a & c & 0 \\ a+b & b & -1 \\ -a & c & 0 \end{vmatrix}$ $R_3 \to R_3 - R_2$ 1

Expanding along C_3

$= 2ab^2c \cdot 2ac = 4a^2b^2c^2$ 1

22. $I = \int_1^2 |x^3 - x| \, dx$

$I = \int_{-1}^{0} (x^3 - x)dx - \int_0^1 (x^3 - x)dx + \int_1^2 (x^3 - x)dx$ 2

$I = \left[\frac{x^4}{4} - \frac{x^2}{2}\right]_{-1}^{0} - \left[\frac{x^4}{4} - \frac{x^2}{2}\right]_0^1 + \left[\frac{x^4}{4} - \frac{x^2}{2}\right]_1^2$ 1

$I = -\left(-\frac{1}{4}\right) - \left(-\frac{1}{4}\right) + \frac{9}{4} = \frac{11}{4}$ 1

23. Equation of planes passing through the intersection of given planes

$[\vec{r}.(2\hat{i} + 6\hat{j}) + 12] + \lambda[\vec{r}.(3\hat{i} - \hat{j} + 4\hat{k})] = 0$ 1

$\Rightarrow \vec{r}.[(2 + 3\lambda)\hat{i} + (6 - \lambda)\hat{j} + 4\lambda\hat{k}] + 12 = 0$ $\frac{1}{2}$

Now

$1 = \frac{12}{\sqrt{(2 + 3\lambda)^2 + (6 - \lambda)^2 + (4\lambda)^2}}$ $\frac{1}{2}$

$1 = \frac{144}{26\lambda^2 + 40}$

$\Rightarrow 26\lambda^2 + 40 = 144 \Rightarrow \lambda = \pm 2$ 1

Equation of plane is

$\vec{r}.(8\hat{i} + 4\hat{j} + 8\hat{k}) + 12 = 0$ or $\vec{r}.(2\hat{i} + \hat{j} + 2\hat{k}) + 3 = 0$ $\frac{1}{2}$

and $\vec{r}.(-4\hat{i} + 8\hat{j} - 8\hat{k}) + 12 = 0$ or $\vec{r}.(-\hat{i} + 2\hat{j} - 2\hat{k}) + 3 = 0$ $\frac{1}{2}$

SECTION D

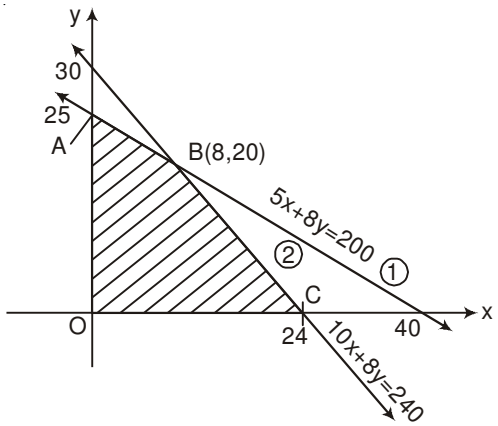
24. Let numbers of souvenirs of type A be x and number of souvenirs of type B be y

$\therefore$ L.P.P is

Maximize P = 100x + 120y $\frac{1}{2}$

Subject to constraints

$5x + 8y \le 200$ ...(1)
$10x + 8y \le 240$ ...(2)
$x, y \ge 0$ } $2\frac{1}{2}$



2

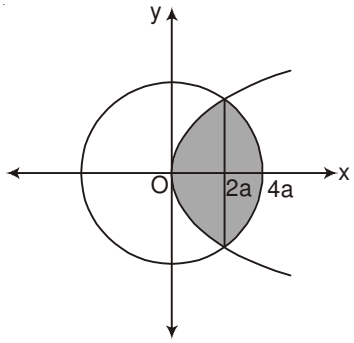
Values at corner points

Points	P
A(0, 25)	3000
B(8, 20)	3200 (Max)
C(24, 0)	2400

1

So, 8 type A Souvenirs and 20 type B Souvenirs should be made to maximize profit.

25. Correct figure 1



Solving $x^2 + y^2 = 16a^2$ ...(1)

and $y^2 = 6ax$ ...(2) we get

$x = 2a$ (as $-8a$ is not possible) 1

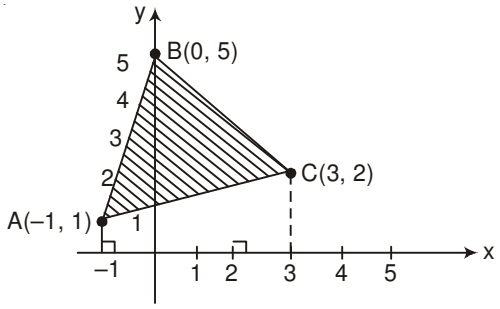
Required Area $= 2\left[\int_0^{2a} \sqrt{6ax}\, dx + \int_{2a}^{4a} \sqrt{(4a)^2 - x^2}\, dx\right]$ 2

$= 2\left[\sqrt{6a}\left(\frac{2}{3}x^{3/2}\right)\right]_0^{2a} + 2\left[\frac{x}{2}\sqrt{(4a)^2 - x^2} + 8a^2 \sin^{-1}\left(\frac{x}{4a}\right)\right]_{2a}^{4a}$ 1

$$= \frac{8}{3}\sqrt{12}a^2 + \frac{16}{3}\pi a^2 - 4\sqrt{3}a^2$$

$$= \frac{4}{3}a^2(4\pi + \sqrt{3}) \qquad 1$$

OR



Correct figure 1

$4x - y + 5 = 0$ ...(1)

$x + y - 5 = 0$ ...(2)

$x - 4y + 5 = 0$ ...(3)

Coordinates of A(–1, 1), B(0, 5) and C(3, 2) $1\frac{1}{2}$

Required Area $= \int_{-1}^{0}(4x+5)dx + \int_{0}^{3}(5-x)dx - \frac{1}{4}\int_{-1}^{3}(x+5)dx$ 2

$$= \left[2x^2 + 5x\right]_{-1}^{0} + \left[5x - \frac{x^2}{2}\right]_{0}^{3} - \frac{1}{4}\left[\frac{x^2}{2} + 5x\right]_{-1}^{3} \qquad 1$$

$$= \frac{15}{2} \qquad \frac{1}{2}$$

26. Required equation of the line is

$$\vec{r} = 2\hat{i} + \hat{j} - \hat{k} + \mu(2\hat{i} - \hat{j} + \hat{k}) \qquad 2$$

Let $\vec{a}_1 = 2\hat{i} + \hat{j} - \hat{k}, \vec{a}_2 = \hat{i} + \hat{j}, \vec{b} = 2\hat{i} - \hat{j} + \hat{k}$

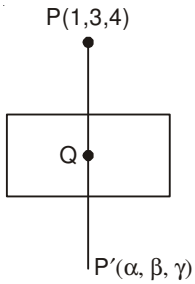
The required distance $= \left|\frac{(\vec{a}_2 - \vec{a}_1)\times\vec{b}}{|\vec{b}|}\right|$

$$= \frac{|(\hat{i} - \hat{k})\times(2\hat{i} - \hat{j} + \hat{k})|}{|2\hat{i} - \hat{j} + \hat{k}|} \qquad 1$$

$$(\vec{a}_2 - \vec{a}_1)\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & -1 \\ 2 & -1 & 1 \end{vmatrix} = -\hat{i} - 3\hat{j} - \hat{k} \qquad 2$$

Required distance = $\frac{\sqrt{1+9+1}}{\sqrt{4+1+1}} = \frac{\sqrt{11}}{\sqrt{6}}$ or $\frac{\sqrt{66}}{6}$ 1

OR



Correct figure $\frac{1}{2}$

Equation of line PQ is $\frac{x-1}{2} = \frac{y-3}{-1} = \frac{z-4}{1} = \lambda$ 1

The coordinates of Q are $(2\lambda + 1, -\lambda + 3, \lambda + 4)$ $\frac{1}{2}$

$\because$ Q lies on plane $2x - y + z + 3 = 0$

$\therefore\ 2(2\lambda + 1) - (-\lambda + 3) + (\lambda + 4) + 3 = 0$ 1

$\Rightarrow 6\lambda + 6 = 0$ i.e., $\lambda = -1$ $\frac{1}{2}$

The coordinates of Q are $(-1, 4, 3)$ $\frac{1}{2}$

$PQ = \sqrt{(-1-1)^2 + (4-3)^2 + (3-4)^2} = \sqrt{6}$ 1

Let $P'(\alpha, \beta, \gamma)$ be the image of P.

then $\frac{\alpha+1}{2} = -1,\ \frac{\beta+3}{2} = 4,\ \frac{\gamma+4}{2} = 3$ $\frac{1}{2}$

$\Rightarrow \alpha = -3,\ \beta = 5,\ \gamma = 2$

$\therefore$ the image P′ is $(-3, 5, 2)$ $\frac{1}{2}$

27. Let $A = \begin{bmatrix} 3 & 0 & -1 \\ 2 & 3 & 0 \\ 0 & 4 & 1 \end{bmatrix}$

Then $A = IA$

$\Rightarrow \begin{bmatrix} 3 & 0 & -1 \\ 2 & 3 & 0 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$ 1

$$\Rightarrow \begin{bmatrix} 1 & -3 & -1 \\ 2 & 3 & 0 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \qquad (R_1 \to R_1 - R_2)$$

$$\Rightarrow \begin{bmatrix} 1 & -3 & -1 \\ 0 & 9 & 2 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \qquad (R_2 \to R_2 - 2R_1)$$

$$\Rightarrow \begin{bmatrix} 1 & -3 & -1 \\ 0 & 1 & 0 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -2 \\ 0 & 0 & 1 \end{bmatrix} A \qquad (R_2 \to R_2 - 2R_3)$$

$$\Rightarrow \begin{bmatrix} 1 & -3 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -2 \\ 8 & -12 & 9 \end{bmatrix} A \qquad (R_3 \to R_3 - 4R_2)$$

$$\Rightarrow \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 9 & -13 & 9 \\ -2 & 3 & -2 \\ 8 & -12 & 9 \end{bmatrix} A \qquad (R_1 \to R_1 - R_3)$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -4 & 3 \\ -2 & 3 & -2 \\ 8 & -12 & 9 \end{bmatrix} A \qquad (R_1 \to R_2 - 3R_2)$$

4

$\therefore A^{-1} = \begin{bmatrix} 3 & -4 & 3 \\ -2 & 3 & -2 \\ 8 & -12 & 9 \end{bmatrix}$ 1

OR

The given system of equations is

$AX = B$,

where $A = \begin{bmatrix} 2 & 3 & 10 \\ 4 & -6 & 5 \\ 6 & 9 & -20 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$

$|A| = 1200 \neq 0$ 1

$\Rightarrow A^{-1}$ exists.

$X = A^{-1}B$ 1

$$\text{adj A} = \begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix}$$ 2

$$A^{-1} = \frac{1}{|A|}\text{adjA} = \frac{1}{1200}\begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix}$$ $\frac{1}{2}$

$$X = A^{-1}B = \frac{1}{1200}\begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix}\begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

$$= \frac{1}{1200}\begin{bmatrix} 600 \\ 400 \\ 240 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{3} \\ \frac{1}{5} \end{bmatrix}$$ 1

$$\therefore\ x = \frac{1}{2}, y = \frac{1}{3}, z = \frac{1}{5}$$ $\frac{1}{2}$

28. Let x be the radius of circle and y be the side of square

$2\pi x + 4y = k$ 1

$A = \pi x^2 + y^2$

$$A = \pi x^2 + \left(\frac{k - 2\pi x}{4}\right)^2 = \frac{16\pi x^2 + k^2 + 4\pi^2 x^2 - 4\pi kx}{16}$$ 1

$$\frac{dA}{dx} = \frac{1}{16}(32\pi x + 8\pi^2 x - 4\pi k)$$ 1

$$\frac{dA}{dx} = 0 \Rightarrow 32\pi x + 8\pi^2 x - 4\pi k = 0$$

$$\Rightarrow\ x = \frac{k}{8 + 2\pi}$$ 1

$$\left.\frac{d^2A}{dx^2}\right]_{x=\frac{k}{8+2\pi}} = \frac{1}{16}[32\pi + 8\pi^2] > 0 \Rightarrow \text{Sum of areas is minimum}$$ 1

$2\pi\left(\frac{k}{8+2\pi}\right)+4y=k \Rightarrow y=\frac{k}{4+\pi} \Rightarrow y=2x$ 1

29. E_1 : ball drawn from first bag
E_2 : ball drawn from second bag
A : both drawn balls are red } 1

$P(E_1)=P(E_2)=\frac{1}{2}$ 1

$P(A \mid E_1)=\frac{5}{8}\times\frac{4}{7}=\frac{20}{56}$ 1

$P(A \mid E_2)=\frac{2}{8}\times\frac{1}{7}=\frac{2}{56}$ 1

$P(E_1 \mid A)=\frac{\frac{1}{2}\cdot\frac{20}{56}}{\frac{1}{2}\cdot\frac{20}{56}+\frac{1}{2}\cdot\frac{2}{56}}$ $1\frac{1}{2}$

$=\frac{\frac{20}{112}}{\frac{22}{112}}=\frac{10}{11}$ $\frac{1}{2}$